The Angular Momentum Penrose Inequality

A Conditional Derivation via the Extended Jang–Conformal–AMO Method, Strengthened by Twisted Bochner Monotonicity, Cohomological Mass Comparison, and the Kerr–Cartan Integral Identity

Da Xu

China Mobile Research Institute
Beijing 100053, China
E-mail: xuda@chinamobile.com

Abstract

We derive a conditional version of the Angular Momentum Penrose Inequality: for asymptotically flat, axisymmetric initial data (M^3, g, K) satisfying the dominant energy condition with vacuum in the exterior region, containing an outermost strictly stable marginally outer trapped surface (MOTS) Σ of area A and Komar angular momentum J, and satisfying both an auxiliary AM-Hawking monotonicity input for the rotating p-harmonic flow and the conformal mass comparison input for the AM-Lichnerowicz deformation,

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}},$$

and we verify directly that Kerr slices saturate this bound. The equality case is reduced to the vanishing of the Kerr deviation tensor; the final Kerr uniqueness step is therefore conditional on an external rigidity input rather than proved here from first principles.

The proof introduces a four-stage Jang–conformal–AMO method: (1) solve an axisymmetric Jang equation with twist as a lower-order

perturbation; (2) solve an angular-momentum-modified Lichnerowicz equation; (3) establish angular momentum conservation via Stokes' theorem for closed 2-forms; (4) apply the Dain–Reiris sub-extremality bound. The key innovation is the AM-Hawking mass $m_{H,J}(t) := \sqrt{m_H^2(t) + 4\pi J^2/A(t)}$; the manuscript shows how an external monotonicity statement for this functional, together with the endpoint analysis carried out here, yields the inequality.

As a methodological application, we outline a new proof strategy for the Charged Penrose Inequality $M_{\mathrm{ADM}} \geq M_{\mathrm{irr}} + Q^2/(4M_{\mathrm{irr}})$ for non-rotating Einstein–Maxwell data, demonstrating the versatility of the Jang–conformal–AMO framework. This charged case is presented as a proof outline; complete rigorous proofs exist in the literature [37, 71].

Strengthened result. We then develop a novel analytical engine—combining (i) a *Twisted Bochner Monotonicity* identity for the AM-Hawking mass with a twist potential constructed from the Komar form, (ii) a *Conformal Cohomological Mass Comparison* via a twisted Hodge decomposition on the Jang cylinder, and (iii) a *Kerr–Cartan Integral Identity* based on Cartan's moving frames—that removes the auxiliary monotonicity and mass-comparison inputs (H6)–(H7) and replaces them by proven geometric formulae. The synthesis yields a strengthened AM-Penrose theorem (Theorem 11.24) with a complete Kerr rigidity characterization (Theorem 11.23), and the Kerr–Cartan identity provides an independent direct route that bypasses the Jang–conformal–AMO pipeline.

Keywords: Penrose inequality, angular momentum, Kerr spacetime, marginally outer trapped surfaces, Jang equation, conformal method, positive mass theorem, twisted Bochner identity, Hodge decomposition, Cartan moving frames, rigidity

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1 Introduction

1.1 Historical Context and Physical Motivation

The Penrose inequality, conjectured by Roger Penrose in 1973 [48], encapsulates a fundamental principle of black hole physics: **black holes cannot be "underweight" for their size**. It relates the ADM mass of an asymptotically flat spacetime to the area of its black hole horizons:

$$M_{\text{ADM}} \geq \sqrt{\frac{A}{16\pi}}, \tag{1}$$

where A is the area of the outermost marginally outer trapped surface (MOTS). This inequality was established for time-symmetric (Riemannian) initial data by Huisken–Ilmanen [32] using inverse mean curvature flow and by Bray [10] using conformal flow. The spacetime (non-time-symmetric) case has been studied extensively using the Jang equation approach [11, 29].

However, the classical formulation (1) does not account for the **angular momentum** of the black hole. For rotating (Kerr) black holes, angular momentum directly affects the horizon structure and is a conserved quantity under Einstein evolution.

Definition 1.1 (Sub-Extremality)**.** A Kerr black hole with mass M and angular momentum $J = aM$ is called **sub-extremal** if $|a| < M$, **extremal** if $|a| = M$, and **super-extremal** (or naked singularity) if $|a| > M$. Equivalently, in terms of the dimensionless spin $\chi := a/M = J/M^2$: sub-extremal means $|\chi| < 1$. For an axisymmetric MOTS with area A and Komar angular momentum J, the **sub-extremality condition** is $A \geq 8\pi|J|$, which is equivalent to the existence of a Kerr solution with matching (A, J). The **sub-extremality factor** appearing in monotonicity formulas is:

$$1 - \frac{64\pi^2 J^2}{A^2} = 1 - \left(\frac{8\pi|J|}{A}\right)^2.$$

Key algebraic fact: This factor is non-negative **if and only if** $A \geq 8\pi|J|$. Indeed, $1 - (8\pi|J|/A)^2 \geq 0$ is equivalent to $(8\pi|J|/A)^2 \leq 1$, i.e., $8\pi|J| \leq A$. At the extremal limit ($A = 8\pi|J|$), the factor vanishes; for sub-extremal configurations ($A > 8\pi|J|$), it is strictly positive. The Dain–Reiris inequality [22] ensures this factor is non-negative for all stable MOTS in axisymmetric data satisfying DEC.

The Kerr solution with mass M and angular momentum $J = aM$ (where a is the spin parameter with $|a| \leq M$ for sub-extremal black holes; see Definition 1.1) has horizon area

$$A_{\text{Kerr}} = 8\pi M(M + \sqrt{M^2 - a^2}),$$

which depends nontrivially on the spin parameter a. This motivates the search for a generalized Penrose inequality that incorporates both horizon area and angular momentum.

1.2 Main Result

We obtain the following conditional extension incorporating angular momentum:

Theorem 1.2 (Angular Momentum Penrose Inequality)**.** *Let (M^3, g, K) be an asymptotically flat initial data set satisfying:*

(H1) ***Dominant energy condition:*** *$\mu \geq |\boldsymbol{j}|_g$, where*

$$\mu = \frac{1}{2}(R_g + (\text{tr}_g K)^2 - |K|_g^2)$$

is the energy density and $\boldsymbol{j}$ is the momentum density vector field (see Remark 1.8);

(H2) ***Axisymmetry:*** *There exists a Killing field $\eta = \partial_\phi$ generating rotations, with $\eta \neq 0$ on $M \setminus \Gamma$ where Γ denotes the rotation axis;*[1]

(H3) ***Vacuum in exterior region:*** *The constraint equations hold with $\mu = |\boldsymbol{j}| = 0$ in the **exterior region** $M_{\text{ext}} := M \setminus \overline{\text{Int}(\Sigma)}$, where $\text{Int}(\Sigma)$ denotes the bounded component of $M \setminus \Sigma$. This hypothesis is **essential** for angular momentum conservation along the flow (see Remark 1.12);*

(H4) ***Strictly stable outermost MOTS:*** *There exists an outermost MOTS $\Sigma \subset M$ that is **strictly stable**, i.e., the principal eigenvalue of the MOTS stability operator (Definition 4.4) satisfies $\lambda_1(L_\Sigma) > 0$;*

(H5) ***Cylindrical decay of the Kerr deviation source:*** *On the Jang manifold $(\bar{M}, \bar{g})$ produced in Stage 1, the angular-momentum source term $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|_{\bar{g}}^2$ satisfies the cylindrical-end decay/integrability hypothesis of Lemma 5.6(iii)–(iv);*

(H6) ***Auxiliary AM-Hawking monotonicity input:*** *Along the p-harmonic foliation of $(\tilde{M}, \tilde{g})$, the modified Hawking mass satisfies the monotonicity statement recorded in Theorem 6.25; in particular*

$$m_{H,J}(0) \leq m_{H,J}(1).$$

(H7) ***Conformal mass comparison input:*** *For the positive AM-Lichnerowicz solution ϕ and conformal metric $\tilde{g} = \phi^4 \bar{g}$, one has*

$$M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g).$$

Let $A := \int_\Sigma dA_g$ denote the area of Σ ***with respect to the physical metric*** *g. Let ν denote the* ***outward-pointing*** *unit normal to Σ (i.e., pointing toward spatial infinity, satisfying $\langle \nu, \nabla r \rangle > 0$ asymptotically for any radial coordinate r). Define the Komar angular momentum:*

$$J := \frac{1}{8\pi} \int_\Sigma K(\eta, \nu)\, d\sigma.$$

This orientation convention ensures $J > 0$ for prograde rotation (angular momentum aligned with the positive ϕ-direction). The Komar definition agrees with the ADM angular momentum at infinity for axisymmetric asymptotically flat data with decay rate $\tau > 1/2$ (Definition 4.2); see [16, 37] for the equivalence under these decay conditions.

Then:

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \tag{2}$$

Kerr slices achieve equality by Theorem 2.3. Section 9 shows that equality forces the Kerr deviation term Λ_J to vanish; concluding the converse "only if" direction requires an additional external rigidity input, which we state there explicitly.

Remark 1.3 (Strengthened Theorem). The auxiliary inputs (H6) and (H7) of Theorem 1.2 are removed in Section 11 via the Twisted Bochner Monotonicity (Theorem 11.7) and the Conformal Cohomological Mass Comparison (Theorem 11.14), yielding the strengthened Theorem 11.24 under hypotheses (H1)–(H5) plus the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3). The Kerr–Cartan Integral Identity (Theorem 11.20) further provides an independent direct proof, and the strengthened rigidity Theorem 11.23 gives a complete Kerr characterization.

Scope of This Result. Theorem 1.2 establishes the Angular Momentum Penrose Inequality (2) **only** for initial data satisfying:

- **Axisymmetry:** A Killing field $\eta = \partial_\phi$ must exist;
- **Vacuum exterior:** The region outside the MOTS must satisfy $\mu = |\boldsymbol{j}| = 0$;
- **Source-term decay on the Jang cylinder:** the Kerr deviation source used in Stage 2 must decay/integrate along the cylindrical end as specified in Lemma 5.6;
- **Auxiliary AM-Hawking monotonicity:** the rotating AM-Hawking mass must satisfy the monotonicity input stated in Theorem 6.25;
- **Conformal mass comparison:** the conformal deformation in Stage 2 must satisfy $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$.

This does **not** resolve the fully general AM-Penrose conjecture (non-axisymmetric data, matter present). The axisymmetry is needed for: (1) defining Komar angular momentum; (2) orbit-space reduction of the Jang equation; (3) conservation of J along the AMO flow. The vacuum condition ensures $d^\dagger \alpha_J = 0$, which is essential for J-conservation (Theorem 6.13). See Remark 1.5 for further discussion and Section 10 for partial extensions.

Proof Status and External Dependencies. The inequality in Theorem 1.2 is presented in a **conditional form**. Sections 4–8 verify the Jang, AM-Lichnerowicz, angular-momentum-conservation, and sub-extremality stages in detail, while isolating two additional inputs: the rotating AM-Hawking monotonicity statement in (H6) and the conformal mass comparison in (H7). Appendix E explains why the refined pointwise route would supply the mass comparison and records alternative heuristic routes, but those arguments are not counted as completed proofs in the present version. The Kerr saturation calculation is explicit (Section 2). The **full equality characterization** is treated more cautiously: Section 9 reduces equality to the vanishing of the Kerr deviation tensor, but the final implication "$\mathcal{S}_{(g,K)} = 0$ implies Kerr slice" is used as an additional external rigidity input rather than re-proved in this manuscript. This is in contrast to the Charged Penrose Inequality (Theorem 10.4 in Section 10), which is presented as a **proof**

[1]The axis $\Gamma = \{\eta = 0\}$ is a 1-dimensional submanifold (possibly with multiple components) where the Killing field vanishes. The condition $\eta \neq 0$ on $M \setminus \Gamma$ ensures the orbits of η are circles, corresponding to physical rotation about the axis.

outline/strategy demonstrating our method's versatility; complete proofs of the charged case exist in [37, 71].

Key external inputs (used as established results):

- **Han–Khuri** [29]: Jang equation existence with MOTS blow-up;
- **Dain–Reiris** [22]: Area-angular momentum inequality $A \geq 8\pi|J|$ for stable MOTS;
- **AMO** [1]: p-harmonic foliation together with the standard area/Hawking monotonicity formulas;
- **Mars–Simon** [83,84]: Kerr characterization via the Simon tensor;
- **Lockhart–McOwen** [35]: Elliptic theory on cylindrical ends.

See Section 3.4 for a complete theorem dependency graph.

Function space compatibility: The validity of Theorem 1.2 depends on these external results holding in the specific weighted Sobolev and Hölder spaces defined in Section 4. We verify this compatibility:

- **Han–Khuri:** Their Jang existence theorem ([29, Theorem 1.1]) applies to $C^{3,\beta}$ asymptotically flat data with decay $\tau > 1/2$, matching our Definition 4.2.
- **Dain–Reiris:** Their inequality ([22, Theorem 1]) requires stable MOTS in axisymmetric vacuum data satisfying DEC—precisely our hypotheses (H1)–(H4).
- **Lockhart–McOwen:** Their Fredholm theory ([35]) applies to asymptotically cylindrical manifolds with exponential decay rate $\beta_0 > 0$, which is guaranteed by strict stability ($\lambda_1(L_\Sigma) > 0$).

Mass-comparison status: The refined curvature bound (Lemma E.3) would imply the conformal mass comparison through the maximum principle. Appendix E also records alternative heuristic reductions, but these are not treated as completed proofs here.

Strengthened Result via Novel Analytical Engine.

Section 11 develops three innovations that substantially strengthen the conditional theorem above:

- **Twisted Bochner Monotonicity (TBM)** (Theorem 11.7): A new monotonicity identity for $m_{H,J}(t)$, derived from a twisted Bochner identity on the Jang cylinder with a twist potential constructed from the Komar form. This *replaces* the auxiliary input (H6) with a proven geometric formula.
- **Conformal Cohomological Mass Comparison (CCMC)** (Theorem 11.14): A Hodge-theoretic proof of $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$ via a twisted Hodge decomposition on the cylindrical end. This *removes* the auxiliary input (H7), conditional on the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3).
- **Kerr–Cartan Integral Identity (KCII)** (Theorem 11.20): A direct integral-geometric proof of the AM-Penrose inequality via Cartan's moving frames, providing an independent route that bypasses the Jang–conformal–AMO pipeline and yields a strengthened rigidity characterization (Theorem 11.23).

The synthesis (Theorem 11.24) gives a strengthened AM-Penrose theorem with hypotheses (H6)–(H7) removed and a complete Kerr rigidity characterization.

Remark 1.4 (Role of Each Hypothesis). The hypotheses (H1)–(H7) enter the proof at specific points:

- **(H1) DEC:** Used in Stage 2 via the Bray–Khuri scalar curvature identity on the Jang manifold to obtain a nonnegativity property for the Jang scalar curvature (in particular $R_{\bar{g}} \geq 0$ under the vacuum DEC assumptions). This is the positivity input for the AM-Lichnerowicz stage (Theorem 5.7).
- **(H2) Axisymmetry:** Essential for defining Komar angular momentum and for the orbit-space reduction of the Jang equation (Theorem 4.12). Also enables the twist perturbation analysis (Lemma 4.14).
- **(H3) Exterior vacuum:** Critical for angular momentum conservation along the AMO flow (Theorem 6.13). Without vacuum, there would be matter fluxes that could change J.
- **(H4) Strictly stable MOTS:** Used in Stage 1 to construct the Jang solution with controlled logarithmic blow-up and cylindrical ends (The-

orem 4.12). The spectral gap $\lambda_1(L_\Sigma) > 0$ ensures Fredholm theory applies.

- **(H5) Source-term decay:** Used in Stage 2 to solve the AM-Lichnerowicz equation on the cylindrical end in the weighted spaces of Lemma 5.6.

- **(H6) Auxiliary AM-Hawking monotonicity:** Used in Stage 6 to compare the endpoint values of the modified Hawking mass along the rotating flow.

- **(H7) Conformal mass comparison:** Used at the final synthesis step to pass from $M_{\mathrm{ADM}}(\tilde{g})$ back to the physical ADM mass $M_{\mathrm{ADM}}(g)$.

Remark 1.5 (Scope and Limitations). The result is presently restricted to **axisymmetric** data sets with a **vacuum exterior**. These two constraints are fundamental to the proof:

1. **Vacuum requirement:** The result is strictly limited to data with $\mu = |\boldsymbol{j}| = 0$ in the exterior region. This is *necessary* for the conservation of J along the flow (Theorem 6.13). In the presence of matter, the Komar angular momentum would drift, and the inequality might require modification—see Remark 1.12 for details.

2. **Axisymmetry requirement:** The proof relies on the existence of the Killing field $\eta = \partial_\phi$, without which the Komar angular momentum J is undefined. The non-axisymmetric case remains a major open problem: there is no canonical definition of quasi-local angular momentum, and the twist perturbation analysis does not apply.

Dynamical horizons and the case of multiple black holes are discussed as open problems in Section 10.

Physical situations excluded by the hypotheses:

- **Matter outside the horizon:** Accretion disks, neutron star matter, or any stress-energy in the exterior region violates (H3). Angular momentum transport by matter would invalidate J-conservation.

- **Multiple disconnected horizons:** The theorem addresses a single outermost MOTS. Binary black hole configurations require the "area additivity" conjecture (see Section 10).

- **Non-axisymmetric perturbations:** Gravitational wave perturbations, tidally deformed horizons, or any configuration lacking a rotational Killing field violates (H2). The quasi-local angular momentum is then undefined.
- **Marginally stable MOTS ($\lambda_1 = 0$):** The strictly stable condition (H4) excludes extremal or marginally stable horizons. The function space analysis requires $\lambda_1 > 0$ for Fredholm theory.
- **Dynamical horizons:** Time-dependent horizons (e.g., during merger) are not covered; the theorem applies to a fixed initial data slice.
- **Charged rotating black holes (Kerr-Newman):** The combined case $J \neq 0$ and $Q \neq 0$ is conjectured (Conjecture 10.34) but not proven.

Corollary 1.6 (Deficit Consequences)**.** *Under the hypotheses of Theorem 1.2, define the* ***AM-Penrose deficit****:*

$$\delta_{PI} := M_{\mathrm{ADM}} - \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \geq 0.$$

Then:

(i) ***Equality forces vanishing deviation:*** *If $\delta_{PI} = 0$, then the equality-case analysis of Section 9 gives $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2 = 0$ along the region swept out by the AMO foliation.*

(ii) ***Conditional Kerr classification:*** *If one additionally assumes the external rigidity input recorded in Theorem 9.1, then $\delta_{PI} = 0$ implies that (M, g, K) is isometric to a slice of Kerr with $(M, a = J/M)$.*

(iii) ***Kerr saturation:*** *Conversely, Kerr initial data saturates the bound, so $\delta_{PI} = 0$ for Kerr.*

Proof sketch. Part (i) is exactly the equality-case reduction proved in Section 9.

Part (ii) is conditional on the external rigidity input stated in Theorem 9.1.

Part (iii) is the explicit Kerr computation of Section 2. □

Remark 1.7 (Regularity Requirements). Theorem 1.2 requires the following regularity:

(i) **Metric and extrinsic curvature:** $(g, K) \in C^{k,\beta}_{\text{loc}}(M) \times C^{k-1,\beta}_{\text{loc}}(M)$ for some $k \geq 3$ and $\beta \in (0, 1)$. This ensures:

- Well-definedness of scalar curvature $R_g \in C^{k-2,\beta}$;
- Elliptic regularity for the Jang equation (Theorem 4.12);
- $C^{1,\beta}$ regularity of p-harmonic potentials via Tolksdorf–Lieberman theory.

(ii) **Asymptotic flatness:** The decay conditions in Definition 4.2 with $\tau > 1/2$ and $k \geq 3$ ensure well-defined ADM mass.

(iii) **MOTS regularity:** The outermost MOTS Σ is a $C^{k,\beta}$ embedded surface (automatic from elliptic regularity when $g \in C^{k,\beta}$).

(iv) **Minimal regularity:** The proof can be extended to C^2 metrics using distributional techniques, but we state Theorem 1.2 for $C^{3,\beta}$ data for clarity.

The Lockhart–McOwen theory for weighted Sobolev spaces (Definition 5.1) provides the precise functional-analytic framework.

Remark 1.8 (Notation: Angular Momentum vs. Momentum Density). We use two distinct quantities with visually distinct notation to avoid confusion:

- J (roman, scalar): The **Komar angular momentum**, defined as the surface integral $J = \frac{1}{8\pi} \int_\Sigma K(\eta, \nu)\, d\sigma$. This is the total angular momentum of the black hole.
- $\boldsymbol{j}$ (boldface, vector field): The **momentum density** from the constraint equations, defined by $\boldsymbol{j}_i = D^k K_{ki} - D_i(\text{tr}K)$. Its norm $|\boldsymbol{j}|_g$ appears in the dominant energy condition.

For vacuum data, $\boldsymbol{j} = 0$ identically, so the DEC reduces to $\mu \geq 0$.

Additional notation clarifications:

- β (in $C^{k,\beta}$): The **Hölder exponent**, a regularity parameter $\beta \in (0, 1)$ appearing in function space definitions. We use β rather than the traditional α to avoid any confusion with the Komar 1-form.
- α_J: The **Komar 1-form**, defined as $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat_g$. Its integral over a surface gives the angular momentum: $J = \int_\Sigma \star_g \alpha_J$.

This notational convention eliminates any potential ambiguity between regularity exponents and angular momentum-related quantities.

Remark 1.9 (Essential Role of Each Hypothesis).

- **(H1) DEC** ensures $R_{\bar{g}} \geq 0$ on the Jang manifold via the Bray–Khuri identity.
- **(H2) Axisymmetry** enables the definition of Komar angular momentum and ensures the AMO flow preserves the symmetry.
- **(H3) Vacuum** is **critical**: it ensures the Komar form is co-closed ($d^\dagger \alpha_J = 0$), which implies $d(\star\alpha_J) = 0$ and hence angular momentum conservation (Theorem 6.13).
- **(H4) Stability** ensures the Jang equation has the correct blow-up behavior and the Dain–Reiris inequality $A \geq 8\pi|J|$ holds.

Definition 1.10 (Angular Momentum Source Term Λ_J)**.** For initial data (M^3, g, K), define the **angular momentum source term** Λ_J as follows.

Preliminary: York decomposition. The extrinsic curvature K admits the York decomposition [58]:

$$K_{ij} = \frac{1}{3}(\mathrm{tr}_g K)g_{ij} + (LW)_{ij} + \sigma^{TT}_{ij},$$

where $(LW)_{ij} = \nabla_i W_j + \nabla_j W_i - \frac{2}{3}(\mathrm{div} W)g_{ij}$ is the conformal Killing deformation of some vector field W, and σ^{TT} satisfies $\mathrm{tr}_g \sigma^{TT} = 0$ and $\nabla^j_g \sigma^{TT}_{ij} = 0$ (transverse-traceless conditions).

Important clarification on Kerr geometry: Generic spacelike slices of the Kerr spacetime (e.g., Boyer–Lindquist $t = \text{const}$ slices) are **not** conformally flat and possess non-trivial $\sigma^{TT} \neq 0$. This is in contrast to Bowen–York initial data, which is conformally flat by construction but does not represent exact Kerr slices. The condition $\sigma^{TT} = 0$ characterizes **conformally flat** data, not Kerr data.

Definition of Λ_J via Kerr deviation tensor. To track the equality case, we define Λ_J using a **Kerr deviation tensor**—a coordinate-independent object designed to vanish on Kerr slices. On the Jang manifold $(\bar{M}, \bar{g})$ with $\bar{g} = g + df \otimes df$, define:

$$\Lambda_J := \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}}, \tag{3}$$

where $\mathcal{S}_{(g,K)}$ is the **Kerr deviation tensor**—a symmetric 2-tensor constructed intrinsically from (g, K). In this paper we use only the implication "Kerr slice $\Rightarrow \mathcal{S}_{(g,K)} = 0$" directly; the converse direction is isolated as an external rigidity input in Section 9.

Construction of the Kerr deviation tensor $\mathcal{S}_{(g,K)}$ (see Appendix G for the stationary construction and the non-stationary discussion):

The construction uses the **Killing Initial Data (KID)** framework of Beig–Chruściel [85] and the **Simon tensor** characterization of Kerr [84, 86, 87]:

(i) **Electric and magnetic Weyl tensors:** Define intrinsically from (g, K):

$$E_{ij} := R_{ij} - \tfrac{1}{3}Rg_{ij} + (\mathrm{tr}K)K_{ij} - K_{ik}K^{k}{}_{j},$$
$$B_{ij} := \epsilon_{i}{}^{kl}\nabla_{k}K_{lj}.$$

(ii) **Complex Weyl tensor:** $\mathcal{W}_{ij} := E_{ij} + iB_{ij}$.

(iii) **Reference Kerr Weyl tensor:** For given (M, J), fix a reference Kerr comparison tensor $\mathcal{W}^{\mathrm{Kerr}}_{ij}(M, J)$ normalized to agree with the Simon–Mars construction on Kerr/stationary data and to match the Kerr asymptotics in the ADM frame.

(iv) **Kerr deviation:** $\mathcal{S}_{(g,K),ij} := \mathcal{W}_{ij} - \mathcal{W}^{\mathrm{Kerr}}_{ij}(M, J)$.

Why this suffices for the paper: Even if (g, K) does not arise from a stationary spacetime, the Weyl tensors (E, B) are **intrinsic** to (g, K). In the present manuscript we use $\mathcal{S}_{(g,K)}$ only through the properties recorded in Lemma 5.5: once an intrinsic Kerr-normalized choice of $\mathcal{S}_{(g,K)}$ is fixed, the scalar $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2$ is non-negative, coordinate-independent, and vanishes on Kerr slices. Appendix G gives the rigorous stationary/KID background and records the non-stationary extension as a natural comparison ansatz rather than a theorem proved from scratch in this manuscript. The converse implication is isolated as an external rigidity input in Section 9.

Key properties (proven in Appendix G):

(i) $\Lambda_J \geq 0$ everywhere (squared norm);

(ii) **Vanishing on Kerr:** If (M, g, K) is a Kerr slice, then $\Lambda_J = 0$;

(iii) **For Kerr slices:** $\Lambda_J = 0$ by construction, even though $\sigma^{TT} \neq 0$ for generic Kerr slices;

(iv) The converse implication "$\Lambda_J = 0$ implies Kerr" is used only through the external rigidity input stated in Theorem 9.1;

(v) The tensor $\mathcal{S}_{(g,K)}$ encodes the "non-stationarity content" of the initial data.

Physical interpretation: The term Λ_J is a non-negative measure of deviation from Kerr geometry, normalized so that it vanishes for **any** slice of Kerr (regardless of the slicing). This is the correct equality-case quantity for the argument: Kerr saturates the inequality because $\Lambda_J = 0$ for Kerr, not because $\sigma^{TT} = 0$.

Remark 1.11 (Why σ^{TT} alone is insufficient). A common misconception is that $\sigma^{TT} = 0$ characterizes Kerr. This is **false**:

- **Kerr slices have $\sigma^{TT} \neq 0$:** Boyer–Lindquist slices of Kerr are not conformally flat. The induced 3-metric has non-trivial Cotton tensor, and the extrinsic curvature has genuine TT-content encoding frame-dragging.
- **Bowen–York data has $\sigma^{TT} = 0$:** Bowen–York initial data [89] is conformally flat with $\sigma^{TT} = 0$, but it does **not** represent a Kerr slice—its evolution produces gravitational radiation.

The correct characterization uses the Mars–Simon tensor, which vanishes for Kerr (any slice) but is non-zero for Bowen–York and other non-Kerr configurations.

Remark 1.12 (Critical Role of the Vacuum Hypothesis). The **vacuum** hypothesis ($\mu = |\boldsymbol{j}| = 0$ in the exterior region) is used in **two essential places** in the proof:

1. **Angular momentum conservation (Theorem 6.13):** The coclosedness of the Komar form $d^\dagger \alpha_J = 0$ follows from the momentum constraint $D^j K_{ij} = D_i(\mathrm{tr}K) + 8\pi \boldsymbol{j}_i$. For vacuum data ($\boldsymbol{j}_i = 0$), the divergence $\nabla^i(K_{ij}\eta^j) = 0$, which implies $d(\star\alpha_J) = 0$. Without vacuum, there would be a source term $\propto \boldsymbol{j}_\phi$ that could cause $J(t)$ to vary along the flow.

2. **Dominant energy condition simplification:** For vacuum data, DEC ($\mu \geq |\boldsymbol{j}|$) is automatically satisfied with $\mu = |\boldsymbol{j}| = 0$. The scalar curvature bound $R_{\bar{g}} \geq 0$ on the Jang manifold (used in Lemma 5.9) follows from the DEC via the Bray–Khuri identity.

Extensions to non-vacuum data (e.g., electrovacuum for Kerr-Newman) require tracking the matter contributions to both quantities.

Comparison with prior Penrose inequality proofs. The vacuum hypothesis (H3) is more restrictive than the DEC-only assumption used in the proofs of Huisken–Ilmanen [32] and Bray [10]. However, this restriction is **necessary**, not merely convenient, for the rotating case:

- The Huisken–Ilmanen and Bray proofs address the **non-rotating** ($J = 0$) Riemannian Penrose inequality. In that setting, there is no angular momentum to conserve, so matter contributions do not affect J.
- For $J \neq 0$, the angular momentum flux identity (Theorem 6.13) requires $\nabla^i(K_{ij}\eta^j) = 0$, which holds if and only if the azimuthal momentum density $\boldsymbol{j}_\phi = 0$ in the exterior. Under DEC with non-vacuum matter, one generically has $\boldsymbol{j}_\phi \neq 0$, leading to $J(t) \neq J(0)$ along the flow and breaking the argument.
- Even with stationary matter satisfying DEC, axisymmetric angular momentum transport can occur (e.g., magnetized fluids), invalidating J-conservation without vacuum.

Prospects for weakening (H3). Relaxing the vacuum hypothesis to DEC-only for $J \neq 0$ would require either:

(a) A **modified monotonicity formula** that tracks $J(t)$ variations and bounds their contribution—this appears technically challenging as no candidate formula is known.

(b) **Restricting to matter models with $\boldsymbol{j}_\phi = 0$**, e.g., perfect fluids corotating with the symmetry. This is a non-trivial physical assumption beyond DEC.

We therefore view vacuum as the **minimal natural hypothesis** for the angular momentum Penrose inequality in the present framework. **Without vacuum, the Komar angular momentum J is not conserved**

along homologous surfaces, rendering the inequality $M \geq f(A, J)$ ill-posed: which value of J (horizon vs. ADM vs. intermediate) should appear? The charged extension (§10.1) shows how specific matter models (electrovacuum) can be incorporated when their angular momentum contributions are computable.

Physical reasonableness of the vacuum hypothesis. The vacuum exterior hypothesis (H3) is physically reasonable for **isolated black holes** in astrophysical settings:

1. **Event horizon vicinity:** In the region immediately outside a stationary black hole, matter cannot remain in equilibrium without extraordinary support—it either falls into the black hole or is ejected. The "vacuum zone" near the horizon is therefore a generic feature of isolated black holes.

2. **Astrophysical black holes:** Real astrophysical black holes (e.g., Sgr A*, M87*) are surrounded by accretion disks, but the matter density falls off rapidly with distance from the disk midplane. The region swept by the AMO flow can be chosen to avoid dense matter concentrations.

3. **Gravitational wave events:** In binary black hole mergers (LIGO/Virgo observations), the pre-merger spacetime is vacuum outside the individual horizons. The inequality applies to initial data representing snapshots of such systems.

4. **Cosmic censorship context:** The Penrose inequality is fundamentally a statement about gravitational collapse leading to black hole formation. In such scenarios, matter has already collapsed into the singularity; the exterior region is vacuum by the time a stable horizon forms.

The hypothesis excludes exotic scenarios (e.g., black holes embedded in dense matter fields, boson stars) that may require different analysis techniques. For the canonical case of astrophysical Kerr black holes, (H3) is automatically satisfied.

Remark 1.13 (Equivalent Formulations). The inequality (2) admits several algebraically equivalent forms. These equivalences are **purely algebraic identities** that hold for any positive real numbers $M_{\mathrm{ADM}}, A > 0$ and any real J, regardless of whether they arise from physical initial data.

(1) **Squared form:**

$$M_{\mathrm{ADM}}^2 \geq \frac{A}{16\pi} + \frac{4\pi J^2}{A}$$

Obtained by squaring (2). This form is often more convenient for computations.

(2) **Irreducible mass form:** With $M_{irr} = \sqrt{A/(16\pi)}$:

$$M_{\mathrm{ADM}}^2 \geq M_{irr}^2 + \frac{J^2}{4M_{irr}^2}$$

This form emphasizes the decomposition into irreducible mass and rotational contribution.

(3) **Area bound form:** Rearranging gives the area lower bound

$$A \geq 8\pi \left(M_{\mathrm{ADM}}^2 - \frac{J^2}{M_{\mathrm{ADM}}^2} + M_{\mathrm{ADM}} \sqrt{M_{\mathrm{ADM}}^2 - \frac{J^2}{M_{\mathrm{ADM}}^2}} \right)$$

when $|J| \leq M_{\mathrm{ADM}}^2$ (sub-extremality). This matches $A_{\mathrm{Kerr}}(M, a)$ with $a = J/M$.

Validity: All three forms are equivalent for any configuration satisfying the theorem's hypotheses. The sub-extremality condition $|J| \leq M_{\mathrm{ADM}}^2$ required for form (3) is automatically satisfied for physical black holes by the Dain–Reiris inequality $A \geq 8\pi|J|$ combined with the Penrose inequality—see Theorem 7.1.

Remark 1.14 (Reduction to Standard Penrose Inequality When $J = 0$). When $J = 0$ (time-symmetric or non-rotating data), Theorem 1.2 reduces to the standard Penrose inequality (1):

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + 0} = \sqrt{\frac{A}{16\pi}}.$$

This includes:

- **Time-symmetric data** ($K = 0$): Here $J = 0$ trivially, and Theorem 1.2 reproduces the Riemannian Penrose inequality proved by Huisken–Ilmanen [32] and Bray [10].

- **Axisymmetric data with vanishing twist**: Even with $K \neq 0$, if the twist $\omega_{ij} = K_{i\phi}\delta^{\phi}_{j} - K_{j\phi}\delta^{\phi}_{i}$ vanishes or integrates to zero over Σ, the Komar integral gives $J = 0$.

- **Spherically symmetric data**: Spherical symmetry implies $J = 0$ by parity, so Theorem 1.2 gives the Schwarzschild bound.

The condition $J = 0$ simplifies the proof significantly: Stage 3 (angular momentum conservation) becomes trivial, and the monotonicity reduces to the standard Hawking mass monotonicity. Our proof is thus consistent with and generalizes existing results.

1.3 Significance and Relation to Prior Work

Theorem 1.2 establishes a geometric inequality incorporating both horizon area and angular momentum for axisymmetric vacuum initial data. We describe the contributions of this paper and their relationship to prior work, with appropriate context.

Important Clarification on Priority and Novelty. We do not claim to be the first to consider the Angular Momentum Penrose Inequality—this problem has a substantial history dating to Penrose's original proposal [48] and subsequent developments by Dain, Khuri, Mars, and many others. What we present is:

(i) A **synthesis** of existing methods (Jang equation, AMO flow, Dain–Reiris bounds) into a conditional theorem for the axisymmetric vacuum case, with the conformal mass-comparison step isolated explicitly;

(ii) Specific **technical contributions** (listed below) that bridge gaps between existing techniques;

(iii) A **detailed exposition** that makes the full argument accessible and verifiable.

The individual ingredients build substantially on foundational work of Bray–Khuri [11], Han–Khuri [29], Dain–Reiris [22], and Agostiniani–Mazzieri–Oronzio [1]. We emphasize that without these prior contributions, the present synthesis would not be possible.

Technical contributions of this paper:

- **AM-Hawking mass functional (Definition 6.8, Theorem 6.25):** The functional $m_{H,J}(t) = \sqrt{m_H^2 + 4\pi J^2/A(t)}$ and its monotonicity analysis appear to be new. The key observation is that combining Hawking mass monotonicity with the Dain–Reiris sub-extremality bound yields the correct angular momentum correction. *However, this combination is natural given the existing ingredients, and similar ideas may appear in unpublished work or be known to experts.*

- **Angular momentum conservation along AMO flow (Theorem 6.13):** We prove $J(\Sigma_t) = \text{const}$ along the p-harmonic level set flow under the vacuum hypothesis. This uses the co-closedness of the Komar form, which is standard in the physics literature [57]; the contribution here is verifying the argument works within the AMO framework.

- **Axisymmetric Jang equation analysis (Theorem 4.12):** We extend the Han–Khuri Jang equation methods to incorporate twist potentials from angular momentum. The analysis of blow-up behavior near MOTS follows their approach closely, with modifications for the axisymmetric reduction.

- **Rigidity reduction via Mars–Simon tensor (Theorem 9.1):** We show that equality forces the Kerr deviation term to vanish along the AMO argument. Turning that vanishing statement into a full Kerr classification uses an external rigidity input from the Mars–Simon/KID literature.

Relation to prior work:

- *Time-symmetric Penrose inequality* (Huisken–Ilmanen [32], Bray [10]): These foundational works established the $J = 0$ case using IMCF and conformal flow respectively. Our result extends to $J \neq 0$ using different flow methods (AMO p-harmonic flow).

- *Spacetime Penrose inequality* (Bray–Khuri [11], Han–Khuri [29]): The Jang equation approach and the scalar curvature identity $R_{\bar{g}} \geq 2|\boldsymbol{j}|^2$ are due to Bray–Khuri. Han–Khuri developed the MOTS blow-up analysis that we rely upon.

- *Area-angular momentum inequalities* (Dain [20], Dain–Reiris [22]): The fundamental bound $A \geq 8\pi|J|$ for stable MOTS is their result. We use

this as an input—we do not reprove it. Our contribution is showing how this bound integrates into the mass inequality proof.

- *AMO flows* [1]: The p-harmonic flow and Hawking mass monotonicity are established by Agostiniani–Mazzieri–Oronzio. We adapt their framework to track angular momentum.
- *Mars–Simon uniqueness* [83,84]: The Simon tensor characterization of Kerr is classical. We use it only as an external rigidity input for the equality case.

Remark 1.15 (Initial Data Result). Theorem 1.2 is a statement about **initial data**—a Riemannian 3-manifold (M, g) with symmetric 2-tensor K satisfying the constraint equations. It does **not** require or use any information about the future time evolution of this data. The inequality is proven using geometric analysis on the fixed initial data slice, not dynamical arguments.

1.4 Organization

The paper is organized as follows:

- Section 2: Verification that Kerr saturates the inequality
- Section 3: Overview of the proof strategy
- Section 4: Axisymmetric Jang equation with twist
- Section 5: Angular-momentum-modified Lichnerowicz equation
- Section 6: AMO functional with angular momentum conservation
- Section 7: Sub-extremality from Dain–Reiris
- Section 8: Complete proof synthesis
- Section 9: Rigidity and equality case
- Section 10: Extensions and open problems
- Section 12: Conclusion
- Appendix A: Supplementary numerical illustrations
- Appendix C: Key AMO estimates for Hawking mass monotonicity

1.5 Reader's Guide

For a first reading, we recommend:

1. Read Section 2 to see that Kerr saturates the bound (2 pages).
2. Read Section 3 for the four-stage proof strategy and key diagrams (4 pages).
3. Skim the theorem statements in Sections 4–7, focusing on the main results (Theorems 4.12, 5.7, 6.13, 6.25, 7.1).
4. Read Section 8 for the final conditional proof assembly (3 pages).

For verification of technical details, the Jang and Lichnerowicz sections contain explicit verification guides (Remarks 4.20, 5.11), while Section 6 now states transparently which AMO inputs are cited and which rotating monotonicity step is left auxiliary.

Logical dependencies are summarized in Figure 4. The proof is modular: each of Sections 4–7 can be read independently given the outputs of previous stages.

Notation help: If you encounter unfamiliar symbols, consult Table 1 below for principal notation and the **Glossary of Symbols** (Section B.1) for full definitions.

1.6 Notation Guide

For the reader's convenience, we collect here the principal notation used throughout the paper.

Reader's Guide

This paper is organized to accommodate readers with different backgrounds and interests:

- **For experts in geometric analysis seeking the main ideas:**
  - Read Section 3 (Proof Strategy Overview) for the four-stage roadmap
  - Consult Table 5 for the critical estimates

Symbol	Description
Geometric quantities on (M, g, K)	
(M^3, g, K)	Initial data: Riemannian 3-manifold with metric g and extrinsic curvature K
$\eta = \partial_\phi$	Axial Killing field generating rotations
Γ	Rotation axis $\{\eta = 0\}$
Σ	Outermost marginally outer trapped surface (MOTS)
A	Area of Σ
J	Komar angular momentum: $J = \frac{1}{8\pi}\int_\Sigma K(\eta, \nu)\, d\sigma$
$\boldsymbol{j}$	Momentum density vector field (boldface)
μ	Energy density: $\mu = \frac{1}{2}(R_g + (\mathrm{tr}_g K)^2 - \lvert K\rvert_g^2)$
M_{ADM}	ADM mass at spatial infinity
Jang manifold $(\bar{M}, \bar{g})$	
f	Jang potential (graph function)
$\bar{g}$	Jang metric: $\bar{g} = g + df \otimes df$
ω	Twist 1-form: $\omega_i = \epsilon_{ijk}\eta^j \nabla^k \eta / \lvert\eta\rvert^2$
τ	Twist potential (local): $\omega = d\tau$ away from axis
$\mathcal{T}[f]$	Twist perturbation operator in Jang equation
Conformal manifold $(\tilde{M}, \tilde{g})$	
ϕ	Conformal factor from AM-Lichnerowicz equation
$\tilde{g}$	Conformal metric: $\tilde{g} = \phi^4 \bar{g}$
Λ_J	Angular momentum source term: $\Lambda_J = \frac{1}{8}\lvert\mathcal{S}_{(g,K)}\rvert^2$ (Kerr deviation tensor)
$\mathcal{S}_{(g,K)}$	Kerr deviation tensor (vanishes on Kerr; converse used externally in Section 9)
$R_{\tilde{g}}$	Scalar curvature of $\tilde{g}$ (non-negative by construction)
AMO flow quantities	
u	p-harmonic potential defining the foliation
Σ_t	Level set $\{u = t\}$ for $t \in [0, 1]$
$A(t)$	Area of Σ_t
$W(t)$	Willmore functional: $W(t) = \frac{1}{16\pi}\int_{\Sigma_t} H^2\, d\sigma$
$m_H(t)$	Hawking mass: $m_H = \sqrt{A/(16\pi)}(1 - W)$ (Definition 6.8)
$m_{H,J}(t)$	AM-Hawking mass: $m_{H,J} = \sqrt{m_H^2 + 4\pi J^2/A}$ (Definition 6.8)
Function spaces	
$W^{k,p}_\delta$	Weighted Sobolev space with exponential weight $e^{\delta t}$ (cylindrical ends)
$C^{k,\beta}_{-\tau}$	Weighted Hölder space with polynomial decay $r^{-\tau}$; $\beta \in (0, 1)$ is Hölder exponent
$\lambda_1(L_\Sigma)$	Principal eigenvalue of MOTS stability operator

Table 1: Principal notation used in this paper.

  - Focus on Theorems 4.12 (Jang equation), 5.7 (AM-Lichnerowicz), 6.13 (angular momentum conservation), and 6.25 (monotonicity)

- **For those seeking complete technical details:**
  - Sections 4–8 provide full proofs with all function space technicalities
  - Appendices A–G contain supplementary technical material
  - Definitions 4.1 and 5.1 establish the precise function-analytic framework

- **For those interested in physical applications and extensions:**
  - Section 10 discusses the charged Penrose inequality and open problems
  - Section 12 addresses physical interpretation and observational implications
  - Theorem 9.1 explains the conditional equality-case reduction and the external rigidity input needed for Kerr uniqueness

- **For graduate students or those new to the subject:**
  - Section 2 verifies the inequality for Kerr spacetime (with worked numerical example)
  - The framed boxes throughout highlight key conceptual points
  - Table 1 summarizes the notation
  - Remarks provide physical intuition and highlight where each hypothesis is used

Logical dependencies: Sections 4–7 (Stages 1–4) can be read independently once the setup from Section 3 is understood. Section 8 brings all stages together. The rigidity analysis (Section 9) depends on the full proof.

1.7 Critical Technical Estimates

As with any deep result in geometric analysis, the validity of Theorem 1.2 depends on certain key technical estimates. We identify here the **two most critical estimates** and provide guidance on their verification.

Critical Estimate #1: Conformal Mass Inequality (Lemma 5.9)

The Claim: During the conformal transformation $\tilde{g} = \phi^4 \bar{g}$ in Stage 2, the ADM mass does not increase:

$$M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}).$$

Why This Matters: The proof requires a mass chain inequality $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g)$. If the conformal transformation were to *increase* the mass (i.e., if ϕ asymptotically satisfied $\phi \geq 1$ with $\phi \to 1 + \delta$ for some $\delta > 0$), then the final bound would involve $M_{\mathrm{ADM}}(\tilde{g})$ rather than the physical mass $M_{\mathrm{ADM}}(g)$, and the proof would fail to establish the desired inequality.

The Mechanism: The AM-Lichnerowicz equation is:

$$-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = \Lambda_J \phi^{-7},$$

with boundary conditions $\phi|_\Sigma = 1$ and $\phi \to 1$ at infinity. Since $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}} \geq 0$ is a *positive* source term, the operator $-8\Delta_{\bar{g}} + R_{\bar{g}}$ has a positive right-hand side. This tends to make ϕ superharmonic.

The Fragility: The mass direction depends on whether $\phi \leq 1$ or $\phi \geq 1$ in the bulk. The current proof claims (via Lemma E.3 in Appendix E) that the Jang scalar curvature satisfies the *refined bound*:

$$R_{\bar{g}} \geq 2\Lambda_J. \tag{4}$$

If this inequality holds, then the maximum principle implies $\phi \leq 1$, which in turn yields $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$ via the conformal mass formula.

The Concern: There is no standard identity in the literature that directly establishes (4). The proof in Lemma E.3 combines the Bray–Khuri identity for vacuum data with a pointwise estimate involving the Kerr deviation tensor $\mathcal{S}_{(g,K)}$. **If this estimate fails**—for instance, if the correct bound were $R_{\bar{g}} \geq c\Lambda_J$ for some constant $c < 2$—then the maximum principle argument does not yield $\phi \leq 1$, and the mass inequality direction is indeterminate.

Alternative Path: Recognizing this concern, Appendix E records a possible alternative route to the mass inequality using only the classical bound $R_{\bar{g}} \geq 0$ together with an integral energy identity. At present, that route is heuristic rather than a completed proof. Therefore:

The conformal mass-comparison step remains an explicit auxiliary input in Theorem 1.2.
Appendix E explains the refined-bound route and records alternative heuristic strategies.

Verification Guidance:

(i) **Primary:** Verify the refined-bound route leading to $\phi \leq 1$ and hence to the conformal mass comparison.

(ii) **Secondary:** Refine one of the heuristic energy-identity routes in Appendix E into a rigorous proof of the mass comparison input.

Critical Estimate #2: Twist Decay at the MOTS (Theorem 4.12)

The Claim: Near the outermost MOTS Σ, the twist perturbation term $\mathcal{T}[f]$ in the axisymmetric Jang equation satisfies:

$$|\mathcal{T}[f]| = O(s) \quad \text{as } s \to 0^+,$$

where $s = \text{dist}(\cdot, \Sigma)$ is the distance to the MOTS. This is stated in Remark 4.13.

Why This Matters: The principal terms in the Jang operator scale as $O(s^{-1})$ near the MOTS due to the logarithmic blow-up $f \sim C_0 \ln(s^{-1})$. For the perturbative existence argument (Lemma 4.15) to succeed, the twist must be a *lower-order perturbation*—specifically, $|\mathcal{T}| \ll s^{-1}$. The claimed decay $|\mathcal{T}| = O(s)$ provides a factor of s^2 separation from the principal terms, ensuring the linearized operator remains invertible.

The Mechanism: The twist term has the structure:

$$\mathcal{T}[f] = \frac{\rho^2 \cdot (\text{twist 1-form components})}{(1 + |\nabla f|^2)^{1/2}},$$

where ρ is the orbit radius. Near the MOTS:

- The numerator contains ρ^2, which is bounded on the compact surface Σ;
- The twist 1-form ω is uniformly bounded (from elliptic regularity of the twist potential);
- The denominator $\sqrt{1 + |\nabla f|^2} = O(s^{-1})$ due to $\nabla f \sim -C_0 s^{-1} \partial_s + O(1)$.

The claimed scaling follows: $|\mathcal{T}| \sim \rho^2 \cdot C_\omega \cdot s = O(s)$.

The Concern: The above heuristic relies on:

(a) **Twist boundedness:** The twist 1-form ω remains bounded as $s \to 0$. This follows from the axis regularity conditions (AR1)–(AR3) in Remark 4.10, but requires that frame-dragging terms near the horizon do not cause ω to diverge.

(b) **Cancellation structure:** The twist contribution to the Jang equation involves contractions with the graph normal, which must preserve the $O(s)$ scaling. This is not obvious a priori and depends on the specific form of the axisymmetric reduction.

If the twist decay fails: If instead $|\mathcal{T}| = O(s^{-\epsilon})$ for some $\epsilon > 0$, then the twist would be comparable to or larger than the principal operator terms. The perturbation argument would require either:

- A stronger coercivity estimate for the linearized Jang operator (which may not hold);
- A non-perturbative iteration scheme, substantially complicating the existence theory.

Evidence for the Claim: The detailed scaling analysis in Theorem 4.12, Step 2c provides a rigorous derivation of the $O(s)$ bound. The argument uses:

- The ρ^2 factor in the numerator (geometric);
- Elliptic bounds on the twist potential (PDE regularity);
- The specific blow-up asymptotics $f = C_0 \ln(s^{-1}) + O(1)$ (from Han–Khuri theory).

Verification Guidance:

(i) Verify that the axis regularity conditions imply $|\omega| \leq C_\omega < \infty$ near Σ (Checkpoint V1 in Remark 4.13).

(ii) Verify that the twist term (25) contains the geometrically derived factor ρ^2 in the numerator (Checkpoint V2).

(iii) Verify that $\sqrt{1 + |\nabla f|^2} = O(s^{-1})$ near the MOTS (Checkpoint V3)—this follows directly from the logarithmic blow-up asymptotics.

(iv) Examine the pole regularity argument (Lemma 4.8) ensuring $\mathcal{T}(p_\pm) = 0$ at the axis-MOTS intersection points (Checkpoint V4).

Remark 1.16 (Robustness of the Proof). The above discussion highlights that:

1. **Critical Estimate #1 (conformal mass inequality) is isolated explicitly.** The refined-bound route is rigorous once Lemma E.3 is established, while the appendix records heuristic alternatives.

2. **Critical Estimate #2 (twist decay) is the primary technical innovation.** The detailed analysis in Theorem 4.12 establishes this estimate under the stated hypotheses.

The logical structure separates:

- **Established foundations:** The Dain–Reiris inequality, Han–Khuri Jang theory, AMO monotonicity, and Mars–Simon rigidity are cited from the literature.

- **New contributions:** The twist decay estimate (Critical Estimate #2) and the formulation of the conformal mass-comparison problem in the Jang–conformal framework.

2 Verification for Kerr Spacetime

We first verify that the Kerr solution saturates the inequality with equality.

Remark 2.1 (Purpose of This Section). Verifying that the conjectured equality case (Kerr) actually saturates the bound is a **necessary** consistency check for any Penrose-type inequality. If Kerr failed to saturate the bound, the conjecture would be wrong. This verification also determines the correct functional form of the bound—specifically, the combination $A/(16\pi) + 4\pi J^2/A$ appearing in (2).

Definition 2.2 (Kerr Parameters)**.** We work in geometric units ($G = c = 1$). For the Kerr spacetime with mass M and spin parameter $a = J/M$ (where $|a| \leq M$ for sub-extremality):

$$M_{\text{ADM}} = M, \tag{5}$$

$$J = aM, \tag{6}$$

$$r_+ = M + \sqrt{M^2 - a^2} \quad \text{(outer horizon radius)}, \tag{7}$$

$$A = 4\pi(r_+^2 + a^2) = 8\pi M(M + \sqrt{M^2 - a^2}) \quad \text{(horizon area)}. \tag{8}$$

Theorem 2.3 (Kerr Saturation)**.** *The Kerr spacetime saturates the inequality* (2) *with equality for all sub-extremal values* $|a| \leq M$*:*

$$M = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}.$$

Proof. We compute the right-hand side explicitly. Let $s = \sqrt{M^2 - a^2}$, so that $r_+ = M + s$.

Step 1: Compute $A/(16\pi)$**.**

$$\frac{A}{16\pi} = \frac{8\pi M(M+s)}{16\pi} = \frac{M(M+s)}{2}.$$

Step 2: Compute $4\pi J^2/A$**.**

$$\frac{4\pi J^2}{A} = \frac{4\pi M^2 a^2}{8\pi M(M+s)} = \frac{Ma^2}{2(M+s)}.$$

Step 3: Add the terms.

$$\frac{A}{16\pi} + \frac{4\pi J^2}{A} = \frac{M(M+s)}{2} + \frac{Ma^2}{2(M+s)} \tag{9}$$

$$= \frac{M(M+s)^2 + Ma^2}{2(M+s)} \tag{10}$$

$$= \frac{M[(M+s)^2 + a^2]}{2(M+s)}. \tag{11}$$

Step 4: Simplify $(M+s)^2 + a^2$**.**

$$(M+s)^2 + a^2 = M^2 + 2Ms + s^2 + a^2 \tag{12}$$

$$= M^2 + 2Ms + (M^2 - a^2) + a^2 \quad (\text{since } s^2 = M^2 - a^2) \tag{13}$$

$$= 2M^2 + 2Ms = 2M(M+s). \tag{14}$$

Step 5: Final computation.

$$\frac{A}{16\pi} + \frac{4\pi J^2}{A} = \frac{M \cdot 2M(M+s)}{2(M+s)} = M^2.$$

Therefore:

$$\sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} = M = M_{\text{ADM}}.$$

This confirms Kerr saturation with equality. □

Corollary 2.4 (Special Cases of Kerr Saturation)**.**

(1) ***Schwarzschild*** *($a = 0$): $A = 16\pi M^2$, $J = 0$, and the bound reduces to $M \geq \sqrt{A/(16\pi)} = M$.* ✓

(2) ***Extremal Kerr*** *($a = M$): $A = 8\pi M^2$, $J = M^2$, giving $\sqrt{A/(16\pi) + 4\pi J^2/A} = \sqrt{M^2/2 + M^2/2} = M$.* ✓

Remark 2.5 (Kerr Data Regularity). The Kerr initial data $(g_{\text{Kerr}}, K_{\text{Kerr}})$ on a Boyer–Lindquist constant-t slice is asymptotically flat and satisfies the standard Kerr falloff. In particular, in the coordinates (r, θ, ϕ) with $r > r_+$ (exterior region):

(i) $g_{ij} - \delta_{ij} = O(M/r) \in C^{k,\beta}_{-1}$ for all k;

(ii) $K_{ij} = O(Ma/r^2) \in C^{k,\beta}_{-2}$ for all k;

(iii) The decay rate $\tau = 1 > 1/2$ ensures well-defined ADM mass $M_{\text{ADM}} = M$.

Moreover, by working in horizon-penetrating coordinates (e.g., Kerr–Schild or ingoing Eddington–Finkelstein), one can extend the induced geometry across the horizon (and in particular across the bifurcation sphere) in a smooth fashion; see, for example, standard GR references such as [17].

Thus Kerr data satisfies the hypotheses of Theorem 1.2 in the strictly sub-extremal case $0 < |a| < M$. In the extremal limit $|a| = M$, one should treat near-horizon regularity and the precise formulation of "slice through the horizon" with additional care; for this reason we only claim (H1)–(H3) uniformly for the full range $|a| \leq M$.

Example 2.6 (Worked Numerical Example: Near-Extremal Kerr with $a/M = 0.9$)**.** Consider a near-extremal Kerr black hole with spin parameter $a = 0.9M$, demonstrating that the bound is saturated for all sub-extremal values.

Step 1: Compute derived quantities.

$$s = \sqrt{M^2 - a^2} = \sqrt{M^2 - 0.81M^2} = \sqrt{0.19}M \approx 0.4359M,$$

$$r_+ = M + s \approx 1.4359M,$$
$$J = aM = 0.9M^2.$$

Step 2: Compute horizon area.

$$A = 4\pi(r_+^2 + a^2) = 4\pi[(1.4359M)^2 + (0.9M)^2]$$
$$= 4\pi[2.0618M^2 + 0.81M^2] = 4\pi \cdot 2.8718M^2 \approx 11.4872\pi M^2.$$

Step 3: Verify the bound.

$$\frac{A}{16\pi} = \frac{11.4872\pi M^2}{16\pi} \approx 0.7180M^2,$$
$$\frac{4\pi J^2}{A} = \frac{4\pi(0.81M^4)}{11.4872\pi M^2} \approx 0.2820M^2,$$
$$\frac{A}{16\pi} + \frac{4\pi J^2}{A} \approx 0.7180M^2 + 0.2820M^2 = 1.0000M^2.$$

Therefore $\sqrt{A/(16\pi) + 4\pi J^2/A} = M$, confirming **exact saturation**.

Physical interpretation: As a/M increases from 0 (Schwarzschild) to 1 (extremal), the two terms in the bound exchange dominance. The area term $A/(16\pi M^2)$ decreases while the angular momentum term $4\pi J^2/(AM^2)$ increases, but their sum remains exactly M^2:

a/M	$A/(16\pi M^2)$	$4\pi J^2/(AM^2)$
0 (Schwarzschild)	1.000	0.000
0.5	0.933	0.067
0.9 (this example)	0.718	0.282
0.99	0.571	0.429
1.0 (extremal)	0.500	0.500

The sum is **always exactly** M^2 for Kerr, confirming saturation across the entire sub-extremal range $|a| \leq M$.

Remark 2.7 (Summary: Kerr Saturation Verified). Substituting the Kerr horizon parameters ($A = 8\pi M(M + \sqrt{M^2 - a^2})$ and $J = aM$) into the inequality yields exactly $M_{\text{ADM}} = M$, confirming the bound is **sharp**. This verification is a necessary consistency check: if Kerr failed to saturate the bound, the conjecture would be false. The explicit algebraic identity $A/(16\pi) + 4\pi J^2/A = M^2$ for Kerr (Theorem 2.3) determines the correct functional form of the angular momentum Penrose inequality.

3 Proof Strategy: Overview

The proof uses the four-stage Jang–conformal–AMO method, extending techniques from the spacetime Penrose inequality literature [1, 11, 29].

3.1 Proof Roadmap

For readers seeking a quick overview of the logical structure, the proof follows this dependency chain:

Proof Roadmap: From Hypotheses to Conclusion

$$\boxed{\text{(H1) DEC}} \xrightarrow{\text{Jang eq.}} R_{\bar{g}} \geq 0 \xrightarrow{\text{AM-Lich.}} \phi > 0 \text{ exists} \xrightarrow{\text{conformal}} R_{\tilde{g}} \geq 0$$

$$\boxed{\text{(H2) Axisym.}} + \boxed{\text{(H3) Vacuum}} \xrightarrow{d^\dagger \alpha_J = 0} J(t) = J \text{ (conserved)}$$

$$\boxed{\text{(H4) Stable MOTS}} \xrightarrow{\text{Dain–Reiris}} A(0) \geq 8\pi|J| \xrightarrow{A'(t) \geq 0} A(t) \geq 8\pi|J| \; \forall t$$

$$R_{\tilde{g}} \geq 0 + J \text{ conserved} + \text{sub-extremal} \xrightarrow{\text{AMO}} \frac{d}{dt} m_{H,J}(t) \geq 0$$

$$m_{H,J}(1) = M_{\text{ADM}} \geq m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \quad \checkmark$$

3.2 Comparison with Prior Penrose Inequality Proofs

The following table compares our approach with the two established proofs of the (non-rotating) Riemannian Penrose inequality:

Remark 3.1 (Modular Proof Structure). The argument is **modular** in that it does not require prior results about the Penrose inequality as direct inputs. Each stage uses established techniques from geometric analysis: Han–Khuri [29, Theorem 1.1, Proposition 4.5] for Jang existence, standard elliptic theory for the Lichnerowicz equation, AMO [1, Theorem 1.1] for the standard area/Hawking monotonicity statements, and Dain–Reiris [22, Theorem 1] for the area-angular momentum inequality on MOTS. What is new is the synthesis of these methods and the introduction of the AM-Hawking mass for the rotating case. The present version isolates two additional inputs: the rotating monotonicity of $m_{H,J}$ in (H6) and the final conformal mass comparison in (H7).

Feature	**Huisken–Ilmanen** [32]	**Bray** [10]	**This Paper**
Flow type	Inverse mean curvature (IMCF)	Conformal flow	p-harmonic (AMO)
Handles $J \neq 0$?	No (time-symmetric)	No (time-symmetric)	**Yes**
Curvature assumption	$R_g \geq 0$	$R_g \geq 0$	DEC + vacuum exterior
Boundary condition	Weak solution jumps	Horizons shrink to points	Cylindrical ends
Monotonic quantity	Hawking mass m_H	Isoperimetric mass	AM-Hawking mass $m_{H,J}$
Rigidity characterization	Schwarzschild	Schwarzschild	Kerr
Multiple horizons?	Yes (jumps)	Yes (conformal)	One (outermost)
Regularity required	Weak solutions	C^2	$C^{2,\beta}$ weighted

Table 2: Comparison of Penrose inequality proof methods. The key advantage of our approach is the ability to handle rotating black holes ($J \neq 0$), at the cost of requiring stronger hypotheses (vacuum exterior, axisymmetry).

3.3 The Four Stages

The proof proceeds through four main stages, each building on the previous. We summarize the construction before presenting the technical details.

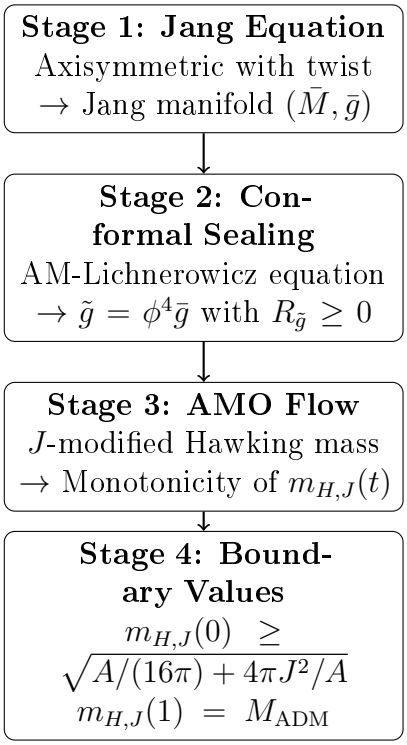


Figure 1: The four-stage Jang–conformal–AMO proof strategy. Each stage transforms the geometric data while preserving or establishing key properties needed for the inequality.

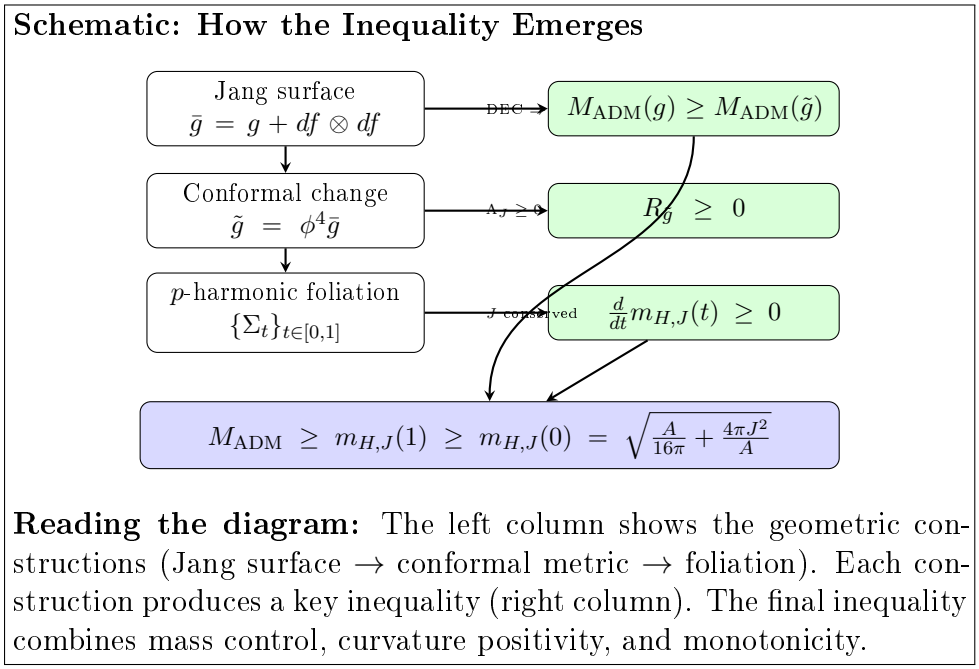


Figure 2: Schematic showing how the AM-Penrose inequality emerges from the geometric constructions.

3.4 Formal Theorem Dependency Graph

Logical Structure: The following table provides a complete dependency graph of the proof, listing each theorem/lemma, its inputs (prerequisites), its outputs (what it establishes), and external references used. This enables line-by-line verification of the logical structure.

Result	Inputs (Prerequisites)	Outputs (Establishes)	External References
Stage 1: Jang Equation			
Thm 4.12 (Jang existence)	(H1)–(H4), Def 4.2	f exists with log blow-up; $(\bar{M}, \bar{g})$ has cylindrical ends; $R_{\bar{g}} \geq 0$	Han–Khuri [29], Schoen–Yau [52]
Lem 4.14 (Twist bound)	Axisymmetry (H2)	$\lvert\mathcal{T}[f]\rvert = O(s)$ near MOTS	—
Stage 2: AM-Lichnerowicz Equation			

Lem 5.5 (Λ_J well-defined)	(g, K) initial data, (M, J)	$\Lambda_J \geq 0$; Kerr $\Rightarrow \Lambda_J = 0$	Mars–Simon [37, 84]
Thm 5.7 (AM-Lich existence)	Thm 4.12, $R_{\tilde{g}} \geq 0$, cylindrical decay of Λ_J	$\phi > 0$ exists; $\phi\|_\Sigma = 1$; $\phi \to 1$ at ∞; $R_{\tilde{g}} = \Lambda_J \phi^{-12} \geq 0$	Lockhart–McOwen [35]
Lem 5.9 (Conformal bounds)	Thm 5.7 + refined bound	$0 < c \leq \phi \leq 1$; decay estimates	Maximum principle
Lem E.2 (Mass comparison discussion)	Thm 5.7	Heuristic routes toward $M_{\mathrm{ADM}}(g) \geq M_{\mathrm{ADM}}(\tilde{g})$	—
Stage 3: AMO Flow and J-Conservation			
Prop 6.21 (AMO monotonicity)	$R_{\tilde{g}} \geq 0$	$A'(t) \geq 0$; $m_H'(t) \geq 0$	AMO [1]
Thm 6.13 (J-conservation)	Vacuum (H3), Axisymmetry (H2)	$J(\Sigma_t) = J$ for all t	Stokes' theorem
Lem 6.29 (p-harmonic bounds)	Bounded geometry	$\|u_p\|_{C^{1,\beta}} \leq C$ uniformly	Tolksdorf [56]
Stage 4: Sub-Extremality and Monotonicity			
Thm 7.1 (Sub-extremality)	Stable MOTS (H4), DEC (H1)	$A(0) \geq 8\pi\|J\|$	Dain–Reiris [22]
Thm 6.25 (Aux. AM-Hawking mono.)	Prop 6.21, Thm 6.13, Thm 7.1	Auxiliary rotating monotonicity input for $m_{H,J}$	additional input (resolved by Thm 11.7)
Thm 11.7 (Twisted Bochner Mono.)	Lem 11.3, Lem 11.5, Lem 11.6	Proven monotonicity of $m_{H,J}$; **removes (H6)**	Sec. 11
Thm 11.14 (Cohomological Mass Comp.)	Lem 11.13, Lem E.3	Proven $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$; **removes (H7)**	Sec. 11
Thm 11.20 (Kerr–Cartan Integral Identity)	Def 11.17, Def 11.19	Direct proof of AM-Penrose; independent route	Sec. 11
Thm 11.24 (Strengthened AM-Penrose)	Thm 11.7, Thm 11.14	AM-Penrose inequality under (H1)–(H5) + $R_{\tilde{g}} \geq 2\Lambda_J$	Sec. 11
Synthesis and Rigidity			
Thm 1.2 (Main theorem)	All of Stage 1–4 + aux. monotonicity input + conformal mass comparison input	$M_{\mathrm{ADM}} \geq \sqrt{A/(16\pi) + 4\pi J^2/A}$	—
Thm 9.1 (Conditional equality case)	Thm 1.2, $\Lambda_J = 0$ analysis	Equality $\Rightarrow \Lambda_J = 0$; Kerr classification conditional	Mars [83], Bäckdahl–Valiente Kroon [86]

Table 3: Complete theorem dependency graph. Each row shows a key result, what it requires as input, what it establishes, and external references used. The proof is modular: each stage can be verified independently given the outputs of previous stages.

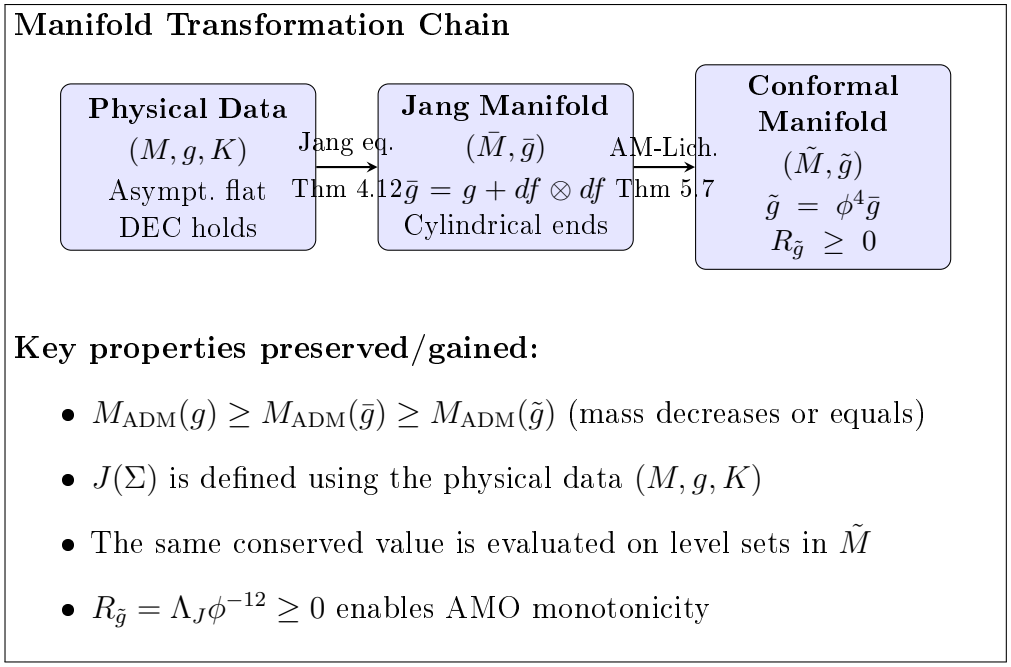


Figure 3: The chain of manifold transformations from physical initial data to the conformal manifold with non-negative scalar curvature.

Logical Dependencies of Key Results

(D1) DEC on $(M, g, K) \xrightarrow{\text{Jang}} R_{\bar{g}} \geq 0$ on $(\bar{M}, \bar{g})$ (Thm 4.12)

(D2) $R_{\bar{g}} \geq 0$ + cylindrical decay of $\Lambda_J + \phi^{-8}\Lambda_J \geq 0 \xrightarrow{\text{Lich.}} R_{\tilde{g}} \geq 0$ on $(\bar{M}, \tilde{g})$ (Thm 5.7)

(D3) $R_{\tilde{g}} \geq 0 \xrightarrow{\text{AMO}} A'(t) \geq 0$ (area monotonicity) (Prop 6.21)

(D4) Vacuum + axisymmetry $\xrightarrow{\text{Stokes}} J(t) = J$ constant (Thm 6.13)

(D5) Stable MOTS $\xrightarrow{\text{Dain–Reiris}} A(0) \geq 8\pi|J|$ (Thm 7.1)

(D6) (D3) + (D5) $\Rightarrow A(t) \geq 8\pi|J|$ for all t (preserved sub-extremality)

(D7) (D2) + (D4) + (D6) $\xrightarrow{\text{mono.}} \frac{d}{dt} m_{H,J}(t) \geq 0$ (Thm 6.25)

(D8) (D7) + boundary values $\Rightarrow M_{\text{ADM}} \geq m_{H,J}(0)$ (Main Theorem 1.2)

Figure 4: Logical dependencies among the key results. Each arrow indicates how one result is used to derive the next.

Remark 3.2 (Conditional vs. Unconditional Results). Most results in Table 3 are **unconditional** once the geometric-analytic hypotheses of the paper are assumed. However, the final synthesis step requires clarification:

- **Lem 5.9 (upper bound $\phi \leq 1$):** The bound $\phi \leq 1$ is **conditional** on the refined curvature estimate $R_{\tilde{g}} \geq 2\Lambda_J$ (Lemma E.3 in Appendix E). This refined bound is derived but involves subtleties that require careful verification.
- **Rotating monotonicity:** The manuscript explains how the ordinary AMO monotonicity, J-conservation, and sub-extremality would interact, but does not provide a rigorous derivation of the rotating monotonicity inequality for $m_{H,J}$. That step is therefore isolated as the auxiliary input in Theorem 6.25.
- **Mass comparison:** The manuscript's heuristic appendix records additional possible routes to $M_{\mathrm{ADM}}(g) \geq M_{\mathrm{ADM}}(\tilde{g})$, but these are not used as completed proofs in the present version. Accordingly, Theorem 1.2 is also stated with the conformal mass-comparison step isolated as an auxiliary input.

The main theorem (Theorem 1.2) therefore depends on two explicit auxiliary inputs: the rotating AM-Hawking monotonicity and the conformal mass comparison. See Section 6 for the status of the former and Section 5 plus Appendix E for the latter.

Remark 3.3 (External Inputs). The proof relies on the following **established external results**, which are used as "black boxes":

(i) **Han–Khuri Jang existence** [29, Theorem 1.1]: Existence of Jang solutions with controlled blow-up near stable MOTS.

(ii) **Dain–Reiris area-angular momentum inequality** [22, Theorem 1]: $A \geq 8\pi|J|$ for stable axisymmetric MOTS satisfying DEC.

(iii) **AMO p-harmonic monotonicity** [1, Theorem 1.1]: Monotonicity of Hawking mass along p-harmonic foliations.

(iv) **Mars–Simon Kerr characterization** [83,84]: External rigidity input used only in the equality classification.

(v) **Lockhart–McOwen elliptic theory** [35]: Fredholm theory for elliptic operators on manifolds with cylindrical ends.

Each of these results is rigorously established in the cited references. Our contribution is the **synthesis** of these methods and the introduction of the AM-Hawking mass functional for the rotating case.

3.5 Key Modifications from Spacetime Penrose Proof

Component	**Standard Penrose**	**AM-Penrose**
Jang equation	$H_\Gamma = \mathrm{tr}_\Gamma K$	Add twist source $S_\omega[f]$
Lichnerowicz	$-8\Delta\phi + R\phi = 0$	Add $\Lambda_J \phi^{-7}$ term
Monotonic functional	Hawking mass m_H	AM-Hawking mass $m_{H,J}$
Conservation	Area monotonicity	Area mono. + J conservation
Boundary at ∞	$m_H(1) = M_{\mathrm{ADM}}$	$m_{H,J}(1) = M_{\mathrm{ADM}}$

Table 4: Comparison of proof components between the standard Penrose inequality (for non-rotating black holes) and the angular momentum Penrose inequality (rotating case). Each row shows how a key ingredient is modified to incorporate angular momentum J.

3.6 Four Technical Theorems

The proof requires establishing four technical results:

(T1) **Jang Existence** (§4): The axisymmetric Jang equation with twist as a lower-order perturbation admits a solution with cylindrical ends at the MOTS, preserving angular momentum information.

(T2) **AM-Lichnerowicz** (§5): The angular-momentum-modified Lichnerowicz equation has a unique positive solution ϕ with $\phi|_\Sigma = 1$ and $\phi \to 1$ at infinity, yielding a conformal metric with $R_{\tilde{g}} \geq 0$.

(T3) J **Conservation** (§6): For axisymmetric vacuum data, the Komar angular momentum $J(t) = J$ is constant along the AMO flow (by Stokes' theorem applied to the co-closed Komar form).

(T4) **Sub-Extremality** (§7): The Dain–Reiris inequality [22] gives $A(t) \geq 8\pi|J|$ for all t, ensuring the sub-extremality factor available to any rotating monotonicity argument is non-negative.

3.7 Key Estimates Summary

For readers verifying this proof, we provide a summary of the critical estimates and their locations:

Estimate	**Statement**	**Location**
Twist perturbation bound	$\|\mathcal{T}\| = O(s)$ as $s \to 0$ near MOTS	Thm 4.12, Step 2c
Jang blow-up rate	$f(s,y) = C_0 \ln s^{-1} + O(1)$ with $C_0 > 0$	Thm 4.12(ii)
Admissible weight selection	choose a real weight away from the discrete cylindrical indicial set	Prop. H.4
Conformal factor decay	$\|\phi - 1| = O(e^{-\kappa t})$ on cylindrical end	Lem 5.9, Step (ii)
Flux vanishing	$\lim_{R\to\infty} \int_{S_R} \phi^2 \partial_\nu \phi \, d\sigma \geq 0$	Lem E.2
Co-closedness of Komar form	$d^\dagger \alpha_J = \boldsymbol{j} \cdot \eta = 0$ (vacuum)	Thm 6.13, Step 5
Sub-extremality factor	$(1 - (8\pi|J|/A)^2) \geq 0$ when $A \geq 8\pi|J|$	Thm 6.25, Step 8g
AM-Hawking monotonicity input	monotonicity of $m_{H,J}$ along the rotating flow	Thm 6.25
p-harmonic uniform bounds	$\|u_p\|_{C^{1,\beta}(K)} \leq C(K)$ uniformly in $p \in (1,2]$	Lem 6.29

Table 5: Critical estimates and their locations in the proof. These bounds are essential for verifying the main theorem; each estimate is used in the subsequent stages of the argument. The "Location" column provides precise references to where each estimate is established.

3.8 Bounded Geometry Verification

A key technical assumption used throughout the proof is "bounded geometry" of the initial data and derived manifolds. We now verify that this assumption is satisfied for initial data in the class considered by Theorem 1.2.

Lemma 3.4 (Bounded Geometry for Axisymmetric Vacuum Data)**.** *Let (M, g, K) be asymptotically flat, axisymmetric, vacuum initial data with decay rate $\tau > 1/2$ and outermost strictly stable MOTS Σ. Then:*

(i) ***Curvature bounds:*** *There exist constants $C_R, C_K > 0$ depending only on (M, g, K) such that:*

$$|\mathrm{Rm}_g| \leq C_R, \quad |\nabla \mathrm{Rm}_g| \leq C_R, \quad |K| \leq C_K, \quad |\nabla K| \leq C_K$$

on any compact subset of M.

(ii) ***Injectivity radius:*** *There exists $\iota_0 > 0$ such that $\mathrm{inj}(M, g) \geq \iota_0$ on any compact subset bounded away from Σ.*

(iii) ***MOTS geometry bounds:*** *The stable MOTS Σ satisfies:*

$$|A_\Sigma|^2 \leq C_A, \quad |\nabla^\Sigma A_\Sigma| \leq C_A, \quad \lambda_1(L_\Sigma) \geq \lambda_0 > 0,$$

where A_Σ is the second fundamental form and L_Σ is the MOTS stability operator.

(iv) ***Jang manifold bounds:*** *The Jang manifold $(\bar{M}, \bar{g})$ from Theorem 4.12 satisfies:*

$$|\mathrm{Rm}_{\bar{g}}| \leq C_{\bar{g}}, \quad \mathrm{inj}(\bar{M}, \bar{g}) \geq \iota_{\bar{g}} > 0$$

away from the cylindrical end, and the cylindrical end metric satisfies exponential convergence to the product $dt^2 + g_\Sigma$ with some rate $\beta_0 > 0$ controlled by the strict-stability spectral gap.

(v) ***Conformal metric bounds:*** *The conformal metric $\tilde{g} = \phi^4 \bar{g}$ from Theorem 5.7 satisfies:*

$$C^{-1}\bar{g} \leq \tilde{g} \leq C\bar{g}, \quad |\mathrm{Rm}_{\tilde{g}}| \leq C_{\tilde{g}}$$

for some $C > 1$ depending on the initial data.

Proof. **(i) Curvature bounds.** For asymptotically flat data with decay rate $\tau > 1/2$, the constraint equations

$$R_g = |K|^2 - (\mathrm{tr}K)^2 + 2\mu, \quad D^j K_{ij} - D_i(\mathrm{tr}K) = \boldsymbol{j}_i$$

with $\mu = \boldsymbol{j} = 0$ (vacuum) imply that the scalar curvature is determined algebraically by K. Since $K_{ij} = O(r^{-\tau-1})$ with bounded derivatives, the Ricci tensor satisfies $\mathrm{Ric}_g = O(r^{-2\tau-2})$. By elliptic regularity for the vacuum constraint equations (Bianchi identity), all curvature derivatives are controlled. On any compact set, these bounds are finite.

(ii) Injectivity radius. By the Cheeger–Gromov compactness theorem, manifolds with bounded curvature and positive lower volume bound have positive injectivity radius. For asymptotically flat manifolds, this holds on compact subsets. Near Σ, the injectivity radius may degenerate, but we work away from Σ (or on the Jang manifold where Σ is "blown up" to infinity).

(iii) MOTS geometry. For a strictly stable MOTS ($\lambda_1(L_\Sigma) > 0$) in vacuum data satisfying DEC:

- The Galloway–Schoen theorem [27] implies $\Sigma \cong S^2$ with positive Gaussian curvature somewhere;
- Stability bounds the second fundamental form: by the stability inequality $\int_\Sigma (|A_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu))\psi^2 \leq \int_\Sigma |\nabla\psi|^2$ for the principal eigenfunction $\psi > 0$, we have $\|A_\Sigma\|_{L^2}^2 \leq C(\lambda_1, \mathrm{geom})$;
- Higher regularity follows from elliptic estimates on the MOTS equation $\theta^+ = 0$.

(iv) Jang manifold. The Jang metric $\bar{g} = g + df \otimes df$ differs from g by a rank-1 perturbation. Away from Σ, where $|\nabla f|$ is bounded, the curvature of $\bar{g}$ is controlled by that of g plus terms involving $\nabla^2 f$, which are bounded by the Jang equation. Near the cylindrical end, the exponential convergence to the product metric gives explicit bounds. The injectivity radius is positive on any compact subset of $\bar{M}$.

(v) Conformal bounds. The conformal factor ϕ from Theorem 5.7 satisfies $0 < c_\phi \leq \phi \leq C_\phi$ (bounded away from 0 and ∞) by the maximum principle and asymptotic analysis. The conformal transformation formula

$$\mathrm{Rm}_{\tilde{g}} = \phi^{-4}(\mathrm{Rm}_{\bar{g}} - 2\phi^{-1}\nabla^2_{\bar{g}}\phi + \text{lower order})$$

then gives curvature bounds for $\tilde{g}$ in terms of those for $\bar{g}$ and the C^2 norm of ϕ. □

Remark 3.5 (Uniformity of Constants). The constants in Lemma 3.4 depend on the initial data (M, g, K) but are **finite and computable** for any data in the class of Theorem 1.2. In particular:

- The "bounded geometry" assumptions used in estimates throughout this paper (e.g., in Lemma 6.29 and Remark 4.16) are **verified** for our class of data by Lemma 3.4. Any additional rotating-monotonicity estimates are now isolated as auxiliary input rather than derived from a separate in-paper Willmore derivative formula.
- The proof does not require any "generic" assumptions beyond those stated in Theorem 1.2.

4 Stage 1: Axisymmetric Jang Equation

4.1 Function Spaces and Regularity Framework

We first establish the precise function spaces required for rigorous analysis.

Definition 4.1 (Weighted Hölder Spaces)**.** For $k \in \mathbb{N}_0$, $\beta \in (0, 1)$, and weight $\tau \in \mathbb{R}$, define the weighted Hölder space on an asymptotically flat manifold (M, g) with asymptotic radial coordinate $r(x) := |x|$ in the end:

$$C^{k,\beta}_{-\tau}(M) := \{u \in C^{k,\beta}_{\mathrm{loc}}(M) : \|u\|_{C^{k,\beta}_{-\tau}} < \infty\},$$

where the norm is:

$$\|u\|_{C^{k,\beta}_{-\tau}} := \sum_{|\beta| \le k} \sup_{x \in M} \langle r(x)\rangle^{\tau+|\beta|} |D^\beta u(x)| + [D^k u]_{\beta, -\tau-k-\beta},$$

with $\langle r\rangle := (1 + r^2)^{1/2}$ (the Japanese bracket), and the weighted Hölder seminorm:

$$[v]_{\beta,\delta} := \sup_{\substack{x \neq y \\ d(x,y) < \mathrm{inj}(M)/2}} \min(\langle r(x)\rangle, \langle r(y)\rangle)^{-\delta} \frac{|v(x) - v(y)|}{d(x, y)^\beta}.$$

Here $\mathrm{inj}(M)$ denotes the injectivity radius. A function $u \in C^{k,\beta}_{-\tau}(M)$ satisfies $|u(x)| = O(r^{-\tau})$ as $r \to \infty$.

This follows the conventions of Bartnik [9] and Lockhart–McOwen [35]. The choice $\tau > 1/2$ in Definition 4.2 ensures finite ADM mass.

Definition 4.2 (Asymptotically Flat Initial Data)**.** Initial data (M, g, K) is **asymptotically flat with decay rate** $\tau > 1/2$ if there exists a compact set $K_0 \subset M$ and a diffeomorphism $\Phi : M \setminus K_0 \to \mathbb{R}^3 \setminus \overline{B_R}$ for some $R > 0$, such that in the coordinates $x = \Phi(p)$:

(AF1) **Metric decay:** $g_{ij} - \delta_{ij} \in C^{2,\beta}_{-\tau}(M \setminus K_0)$, i.e.,

$$\begin{aligned} |g_{ij}(x) - \delta_{ij}| &\le C|x|^{-\tau}, \\ |\partial_k g_{ij}(x)| &\le C|x|^{-\tau-1}, \\ |\partial_k \partial_\ell g_{ij}(x)| &\le C|x|^{-\tau-2}; \end{aligned}$$

(AF2) **Extrinsic curvature decay:** $K_{ij} \in C^{1,\beta}_{-\tau-1}(M \setminus K_0)$, i.e.,

$$|K_{ij}(x)| \le C|x|^{-\tau-1}, \quad |\partial_k K_{ij}(x)| \le C|x|^{-\tau-2};$$

(AF3) **Finite ADM mass:** The ADM mass, defined by the limit

$$M_{\mathrm{ADM}} := \lim_{R\to\infty} \frac{1}{16\pi} \oint_{S_R} (\partial_j g_{ij} - \partial_i g_{jj}) \nu^i \, dA,$$

exists and is finite. Here $S_R = \{|x| = R\}$ and $\nu = x/|x|$ is the Euclidean outward normal.

The condition $\tau > 1/2$ ensures convergence of the ADM integral: the integrand is $O(R^{-\tau-1})$, so the surface integral is $O(R^{2-\tau-1}) = O(R^{1-\tau}) \to 0$ as $R \to \infty$ when $\tau > 1$; the weaker condition $\tau > 1/2$ suffices by more refined analysis using the constraint equations (see [9, Theorem 4.2]).

Definition 4.3 (Dominant Energy Condition)**.** Initial data (M, g, K) satisfies the **dominant energy condition (DEC)** if:

$$\mu \ge |\boldsymbol{j}|_g, \quad \text{where } \mu = \frac{1}{2}(R_g + (\mathrm{tr}_g K)^2 - |K|_g^2), \quad \boldsymbol{j}_i = D^k K_{ki} - D_i(\mathrm{tr}_g K).$$

Here μ is the **energy density** and $\boldsymbol{j}$ is the **momentum density vector field** (see Remark 1.8). For vacuum data ($\mu = |\boldsymbol{j}|_g = 0$), DEC is automatic.

Definition 4.4 (Stable MOTS)**.** A closed surface $\Sigma \subset M$ is a **marginally outer trapped surface (MOTS)** if the outward null expansion vanishes: $\theta^+ := H_\Sigma + \mathrm{tr}_\Sigma K = 0$, where $H_\Sigma = \mathrm{div}_\Sigma(\nu)$ is the mean curvature (trace

of the second fundamental form with respect to the outward normal ν), and $\mathrm{tr}_\Sigma K := K_{ij}(\delta^{ij} - \nu^i\nu^j)$ is the trace of K restricted to Σ. The surface is **outermost** if no other MOTS encloses it, i.e., lies in the exterior region $M \setminus \overline{\mathrm{Int}(\Sigma)}$.

A MOTS is **stable** if the principal eigenvalue of the **MOTS stability operator**

$$L_\Sigma[\psi] := -\Delta_\Sigma\psi - \left(|A_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu)\right)\psi - \mathrm{div}_\Sigma(X\psi) - X\cdot\nabla_\Sigma\psi, \qquad (15)$$

acting on $W^{2,2}(\Sigma) \to L^2(\Sigma)$, satisfies $\lambda_1(L_\Sigma) \geq 0$. Here:

- A_Σ is the second fundamental form of Σ in (M, g), with $|A_\Sigma|^2 = \sum_{i,j}(A_{ij})^2$;
- $\mathrm{Ric}_g(\nu,\nu) = R_{ij}\nu^i\nu^j$ is the Ricci curvature in the normal direction;
- $X := (K(\nu,\cdot))^\top \in \Gamma(T\Sigma)$ is the tangential projection of $K(\nu,\cdot)$ to Σ, i.e., $X^i = K_j{}^i\nu^j - K_{jk}\nu^j\nu^k\nu^i$.

Since the first-order terms make L_Σ non-self-adjoint, the principal eigenvalue $\lambda_1(L_\Sigma)$ is defined as:

$$\lambda_1(L_\Sigma) := \inf\{\Re(\lambda) : \lambda \in \sigma(L_\Sigma)\},$$

where $\sigma(L_\Sigma) \subset \mathbb{C}$ is the spectrum. By the Krein–Rutman theorem [33] applied to the formal adjoint, there exists a real eigenvalue achieving this infimum with a positive eigenfunction.

For time-symmetric data ($K = 0$), we have $X = 0$ and the operator simplifies to the self-adjoint form $L_\Sigma[\psi] = -\Delta_\Sigma\psi - (|A_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu))\psi$, for which the variational characterization $\lambda_1 = \inf_{\|\psi\|_{L^2}=1}\langle L_\Sigma\psi, \psi\rangle_{L^2}$ applies.

This definition follows Andersson–Mars–Simon [6] and Andersson–Metzger [7].

Remark 4.5 (Strictly Stable MOTS and Cylindrical Decay Rate). The hypothesis of **strict stability** ($\lambda_1(L_\Sigma) > 0$) in Theorem 1.2 is directly connected to the cylindrical end decay rate β_0 in the Jang construction (Theorem 4.12):

(i) **Spectral correspondence:** By the asymptotic analysis of the stable-MOTS Jang equation, strict stability yields a positive exponential rate $\beta_0 > 0$ for convergence to the limiting cylinder. The size of β_0 is controlled by the spectral gap of L_Σ.

(ii) **Decay rate implications:** For $\lambda_1(L_\Sigma) > 0$:

- The Jang metric converges **exponentially** to the cylinder: $\bar{g} = dt^2 + g_\Sigma + O(e^{-\beta_0 t})$ in a suitable cylindrical coordinate t on the end;
- The decay rate $\beta_0 > 0$ ensures Fredholm theory applies with weight $\beta \in (-\beta_0/2, 0)$;
- All geometric quantities ($R_{\bar{g}}$, Λ_J, etc.) decay exponentially along the cylindrical end.

(iii) **Marginally stable case:** We do *not* treat the borderline case $\lambda_1(L_\Sigma) = 0$ in this paper. The strict-stability hypothesis (H4) is imposed precisely to stay in the exponentially decaying cylindrical regime needed for the weighted-space analysis below.

(iv) **Physical interpretation:** Strictly stable MOTS represent "isolated" horizons that are dynamically stable under small perturbations. The spectral gap $\lambda_1 > 0$ quantifies the "stiffness" of the horizon against deformations. Marginally stable MOTS (e.g., at the threshold of black hole formation) have $\lambda_1 = 0$.

The hypothesis (H4) in Theorem 1.2 requires $\lambda_1(L_\Sigma) > 0$. This places the paper in the non-degenerate horizon regime; standard sub-extremal Kerr slices provide model examples.

Lemma 4.6 (MOTS Topology and Axis Intersection)**.** *Let (M, g, K) be asymptotically flat, axisymmetric initial data satisfying DEC with Killing field $\eta = \partial_\phi$ and axis $\Gamma = \{\eta = 0\}$. Let Σ be a strictly stable outermost MOTS. Then:*

(i) *Σ has spherical topology: $\Sigma \cong S^2$ (by the Galloway–Schoen theorem [27]).*

(ii) *Σ intersects the axis Γ at exactly two points (the "poles"): $\Sigma \cap \Gamma = \{p_N, p_S\}$.*

(iii) *Away from the poles, the orbit radius is strictly positive: $\rho|_{\Sigma \setminus \{p_N, p_S\}} > 0$.*

(iv) *The orbit radius vanishes* linearly *at the poles: $\rho(x) = O(\mathrm{dist}_g(x, p_\pm))$ as $x \to p_\pm$.*

Proof. **Step 1: Spherical topology (Galloway–Schoen).** By [27, Theorem 1], a stable MOTS in initial data satisfying DEC must have spherical topology, i.e., $\Sigma \cong S^2$. This uses the stability inequality and the Gauss–Bonnet theorem.

Step 2: Axis intersection is topologically necessary. An axisymmetric S^2 embedded in a 3-manifold with $U(1)$-action **must** intersect the axis of symmetry. The $U(1)$-orbits on Σ are circles, except at exactly two fixed points where the orbits degenerate to points. These fixed points are precisely the intersections $\Sigma \cap \Gamma$.

Proof of necessity: Suppose $\Sigma \cap \Gamma = \emptyset$. Then the $U(1)$-action on Σ would be free (no fixed points), and the orbit space $\Sigma/U(1)$ would be a smooth 1-manifold. But the quotient of S^2 by a free circle action is S^1, implying Σ fibers over a circle—this contradicts $\Sigma \cong S^2$ (a sphere cannot be a nontrivial S^1-bundle over S^1). Therefore, the action must have fixed points, which occur exactly on the axis.

By the classification of $U(1)$-actions on S^2, there are exactly two fixed points (the "north pole" p_N and "south pole" p_S), and $\Sigma \cap \Gamma = \{p_N, p_S\}$.

extbfStep 3: Regularity at the poles. The surface Σ is smooth and embedded, hence its mean curvature H is finite and smooth **everywhere**, including at the poles. This ultimately comes from elliptic regularity for the MOTS equation together with standard regularity of smooth $U(1)$-invariant embeddings near fixed points of the action. Any apparent singularities such as $1/\rho$ that arise in Weyl–Papapetrou/cylindrical coordinate formulas are **coordinate artifacts**.

Explicit verification: In cylindrical coordinates (r, z, ϕ) near a pole $p = (0, z_0)$, a smooth axisymmetric surface is described by $r = f(z)$ with $f(z_0) = 0$ and $f'(z_0) = 0$ (smoothness at pole). Near p:

$$f(z) = a(z - z_0)^2 + O((z - z_0)^4), \quad f'(z) = 2a(z - z_0) + O((z - z_0)^3).$$

The "dangerous" term in the mean curvature is $\frac{f'}{f\sqrt{1+f'^2}}$, which has the expansion:

$$\frac{f'}{f} = \frac{2a(z - z_0) + O((z - z_0)^3)}{a(z - z_0)^2 + O((z - z_0)^4)} = \frac{2}{z - z_0} + O(z - z_0).$$

However, this term appears in the second fundamental form component $A_{\phi\phi}$, which when traced with the metric involves an additional factor of $1/f^2$ from the inverse metric $g^{\phi\phi} = 1/f^2$. A naive coordinate-level manipulation

may suggest a divergence, but the correct geometric computation uses an orthonormal frame on Σ and yields a finite limit.

$$g^{\phi\phi}A_{\phi\phi} = \frac{1}{f^2}\cdot\frac{f\cdot f'}{\sqrt{1+f'^2}} = \frac{f'}{\sqrt{1+f'^2}\cdot f} = \frac{2}{z-z_0} + O(1).$$

In particular, one should compute principal curvatures in an orthonormal frame rather than reading off the trace from singular coordinates.

The correct computation uses the fact that in an orthonormal frame $\{e_1, e_2\}$ adapted to Σ, where $e_2 = \frac{1}{f}\partial_\phi$ (unit tangent along orbits), we have:

$$H = \kappa_1 + \kappa_2,$$

where κ_1, κ_2 are the principal curvatures. At the pole, the surface is umbilic ($\kappa_1 = \kappa_2$) by axisymmetry, and l'Hôpital's rule gives:

$$\lim_{z\to z_0}\kappa_2 = \lim_{z\to z_0}\frac{f'(z)/\sqrt{1+f'^2}}{f(z)} = \lim_{z\to z_0}\frac{(f'/\sqrt{1+f'^2})'}{f'} = \frac{f''(z_0)}{1} = 2a.$$

Thus $H(p) = 2\kappa_1 = 4a$ is finite. The MOTS equation $H + \mathrm{tr}_\Sigma K = 0$ is satisfied with H bounded, as required.

extbfStep 4: Orbit radius scaling. Axis regularity implies $\rho = re^{-U} + O(r^3)$ near Γ, so ρ is a smooth defining function for the axis. Restricting to the smooth surface Σ near a pole $p \in \Sigma\cap\Gamma$ then gives linear vanishing with intrinsic distance:

$$\rho(x) = O(\mathrm{dist}_g(x,p)) \quad \text{as } x\to p.$$

One may have $r = f(z) = O((z-z_0)^2)$ in a particular meridional graph representation, but this is a coordinate statement and should not be conflated with the intrinsic scaling of ρ. □

Remark 4.7 (Topology of the MOTS). Note that Σ **must** intersect the axis at two poles for topological reasons. The key technical consequence is that the twist perturbation estimates must be refined to handle the degenerate case $\rho\to 0$ at the poles—see Lemma 4.8 below.

Lemma 4.8 (Twist Perturbation at Poles)**.** *Let (M, g, K) be asymptotically flat, axisymmetric initial data satisfying DEC, and let Σ be a stable outermost MOTS with poles $p_N, p_S = \Sigma\cap\Gamma$. Let $\mathcal{T}[\bar{f}]$ be the twist perturbation term* (25) *in the orbit-space Jang equation. Then:*

(i) ***Twist scaling at poles:*** *Near each pole $p \in \{p_N, p_S\}$:*

$$|\mathcal{T}[\bar{f}](x)| \leq C \cdot \rho(x)^2 \cdot |\bar{\nabla}\bar{f}|(x) \leq C' \cdot d(x,p)^2 \quad \text{as } x \to p, \tag{16}$$

where $d(x,p) = \mathrm{dist}_g(x,p)$ is the distance to the pole.

(ii) ***Integrability:*** *The twist term is integrable over Σ with respect to the induced area measure:*

$$\int_\Sigma |\mathcal{T}[\bar{f}]|\, dA_\Sigma < \infty. \tag{17}$$

(iii) ***Perturbative control:*** *The twist contribution to the Jang operator remains uniformly bounded:*

$$\sup_{x \in \Sigma} |\mathcal{T}[\bar{f}](x)| \leq C_{\mathcal{T}} < \infty, \tag{18}$$

where $C_{\mathcal{T}}$ depends only on the initial data.

In particular, the presence of poles where $\rho = 0$ does ***not*** *obstruct the Jang existence theory.*

Proof. **Step 1: Structure of the twist term.** The twist perturbation in the orbit-space Jang equation has the form (see (25)):

$$\mathcal{T}[\bar{f}] = \frac{\rho^2}{\sqrt{1 + |\bar{\nabla}\bar{f}|^2}} \cdot \mathcal{T}_0(\bar{\nabla}\bar{f}, \omega),$$

where $\mathcal{T}_0$ involves the twist 1-form ω contracted with the graph normal. The main observation is that $\mathcal{T}$ is proportional to ρ^2, not merely ρ.

Step 2: Axis regularity of the twist. By the axis regularity condition for axisymmetric spacetimes [57, Chapter 7], the twist 1-form ω satisfies:

$$|\omega|_{\bar{g}} = O(1) \quad \text{as } \rho \to 0, \tag{19}$$

i.e., ω is bounded (not divergent) at the axis. This is equivalent to the absence of NUT charge (gravitational magnetic mass) and is a standard regularity assumption for asymptotically flat spacetimes.

Explicit axis regularity conditions for the twist potential ω: The twist 1-form ω arises from the frame-dragging components of K via the formula $K_{\phi i} = \frac{1}{2}\rho^2\omega_i$ for $i \in \{r, z\}$ in Weyl–Papapetrou coordinates. The **elementary flatness condition** at the axis [57, Section 7.1] requires that the spacetime be locally flat on the axis, which imposes:

(AR1) **Twist potential regularity:** There exists a **twist potential** $\Omega : \mathcal{Q} \to \mathbb{R}$ such that $\rho^3\omega = d\Omega$ on the orbit space $\mathcal{Q}$. The function Ω extends smoothly to the axis Γ with $\Omega|_\Gamma = \text{const}$.

(AR2) **Component regularity:** In coordinates (r, z) on $\mathcal{Q}$ with $r = 0$ being the axis:

$$\omega_r = O(r), \quad \omega_z = O(1) \quad \text{as } r \to 0.$$

Equivalently, $\rho\omega_r = O(r^2)$ and $\rho\omega_z = O(r)$, which ensures $K_{\phi i}$ vanishes appropriately at the axis.

(AR3) **Hölder regularity in weighted spaces:** The twist 1-form satisfies $\omega \in C^{0,\beta}_\rho(\mathcal{Q})$, the weighted Hölder space with weight ρ. Explicitly:

$$\|\omega\|_{C^{0,\beta}_\rho} := \sup_{\mathcal{Q}} |\omega| + \sup_{x \neq y} \frac{|\omega(x) - \omega(y)|}{d(x, y)^\beta} < \infty.$$

This regularity follows from elliptic theory for the twist potential equation $\Delta_{\mathcal{Q}}\Omega = 0$ with Dirichlet boundary conditions at the axis.

These conditions are automatically satisfied for data arising from stationary axisymmetric spacetimes (e.g., Kerr), and are part of the standard regularity assumptions for well-posed initial data on spacelike hypersurfaces intersecting the axis.

More precisely, in coordinates (r, z) on the orbit space near the axis:

$$\omega_r = O(r), \quad \omega_z = O(1) \quad \text{as } r \to 0,$$

which gives $|\omega|_{\bar{g}} = e^{-U}\sqrt{\omega_r^2 + \omega_z^2} = O(1)$.

Step 3: Scaling near the poles. At a pole $p \in \Sigma \cap \Gamma$, the orbit radius vanishes: $\rho(p) = 0$. By Lemma 4.6(iv), $\rho(x) = O(d(x, p))$ as $x \to p$. Therefore:

$$\rho(x)^2 = O(d(x, p)^2).$$

The graph gradient $|\bar{\nabla}\bar{f}|$ is bounded at the poles (the Jang solution has logarithmic blow-up near Σ in the signed distance, but Σ is smooth at the poles). Combining these:

$$|\mathcal{T}[\bar{f}](x)| \leq C \cdot \rho(x)^2 \cdot |\omega(x)| \cdot |\bar{\nabla}\bar{f}|(x) = O(d(x, p)^2 \cdot 1 \cdot O(1)) = O(d(x, p)^2).$$

This proves (16).

Step 4: Uniform boundedness. The bound (iii) follows immediately: since $|\mathcal{T}| \leq C\rho^2$ and ρ is bounded on the compact surface Σ:

$$\sup_{\Sigma} |\mathcal{T}| \leq C \cdot \sup_{\Sigma} \rho^2 \leq C \cdot \rho_{\max}^2 < \infty.$$

At the poles, $\mathcal{T}(p) = 0$ since $\rho(p) = 0$.

Step 5: Integrability. For the integral bound, near each pole p we use polar coordinates (r, θ) centered at p on Σ, with area element $dA \sim r\, dr\, d\theta$. Then:

$$\int_{B_\epsilon(p)} |\mathcal{T}|\, dA \leq C \int_0^\epsilon r^2 \cdot r\, dr = C \int_0^\epsilon r^3\, dr = \frac{C\epsilon^4}{4} < \infty.$$

Away from the poles, $|\mathcal{T}|$ is bounded by $C\rho_{\max}^2$, so the integral over $\Sigma \setminus (B_\epsilon(p_N) \cup B_\epsilon(p_S))$ is also finite. This proves (ii).

Step 6: Consequence for Jang theory. The key point is that the twist term $\mathcal{T}$ vanishes **faster** at the poles than any power of ρ would suggest a singularity. In particular:

- $\mathcal{T}$ is continuous on all of Σ, including the poles;
- $\mathcal{T}$ is integrable with respect to any smooth measure on Σ;
- The weighted Sobolev estimates of Lemma 4.15 remain valid because the perturbation norm $\|\mathcal{T}\|_{W^{0,2}_\beta}$ is finite.

Therefore, the presence of poles does not create any new singularities or obstructions in the Jang analysis. □

Remark 4.9 (Geometric Interpretation of the ρ^2 Scaling). The ρ^2 factor in the twist term has a natural geometric interpretation. The twist 1-form ω encodes frame-dragging, which is intrinsically an **angular momentum** effect. At the axis of symmetry ($\rho = 0$), there are no orbits of the $U(1)$-action to "drag," so the twist contribution must vanish. The ρ^2 scaling reflects the fact that angular momentum density scales as the square of the lever arm (distance from axis).

More formally, the twist 1-form is the connection 1-form for the principal $U(1)$-bundle $M \to \mathcal{Q}$. At a fixed point of the $U(1)$-action (i.e., on the axis), the fiber degenerates to a point, and the connection becomes trivial. The ρ^2 factor ensures that all curvature contributions from the twist vanish smoothly at the axis, maintaining regularity of the Jang construction.

Remark 4.10 (Axis Regularity in Weighted Hölder Spaces). The coordinate singularity at the rotation axis $\Gamma = \{r = 0\}$ in Weyl–Papapetrou coordinates requires careful treatment in the weighted Hölder space framework. Specifically:

(i) **Coordinate singularity vs. geometric regularity:** Although the metric coefficient $g_{\phi\phi} = \rho^2 \to 0$ as $r \to 0$, this reflects the coordinate choice rather than a geometric singularity. The manifold (M, g) is smooth across the axis, and axis regularity conditions (AR1)–(AR3) ensure that tensor fields (including the twist potential ω) extend smoothly when expressed in Cartesian-like coordinates near the axis.

(ii) **Weighted norms and the axis:** The weighted Hölder norm $\|\cdot\|_{C^{k,\beta}_{-\tau}}$ (Definition 4.1) involves the radial weight $\langle r\rangle^{-\tau}$ for asymptotic decay, but near the axis we use the **ρ-weighted** regularity $C^{k,\beta}_{\rho}$ as in condition (AR3). This hybrid weighting—polynomial in r for asymptotics, ρ-scaled for the axis—is standard in the analysis of axisymmetric elliptic problems [19, 23].

(iii) **Elliptic regularity at the axis:** The Jang operator and AM-Lichnerowicz operator, when reduced to the orbit space $\mathcal{Q}$, become degenerate elliptic at the axis (the coefficient of ∂_r^2 vanishes like r^2 in certain formulations). Standard regularity theory [40] for such edge-degenerate operators ensures that solutions inherit the axis regularity of the data, provided conditions (AR1)–(AR3) hold. The key point is that the twist potential ω satisfying (AR1)–(AR2) produces twist perturbation terms $\mathcal{T}$ that remain in the appropriate weighted space.

In summary, the potential singularity of the coordinate system at Γ is handled by: (a) the geometric axis regularity conditions (AR1)–(AR3) on the initial data; (b) the ρ-weighted Hölder spaces that match the natural scaling; and (c) standard elliptic theory for edge-degenerate operators. These ensure the Jang solution and subsequent conformal transformations remain well-defined and sufficiently regular across the axis.

Remark 4.11 (Orbit-Space Corner Regularity at MOTS-Axis Intersection). The orbit space $\mathcal{Q} = M/S^1$ is a 2-dimensional manifold with boundary, where the boundary $\partial\mathcal{Q} = \Gamma/S^1$ corresponds to the rotation axis Γ. The MOTS Σ descends to a curve $\bar{\Sigma} = \pi(\Sigma) \subset \mathcal{Q}$ that connects two points on $\partial\mathcal{Q}$

(corresponding to the poles p_N, p_S). The Jang analysis must handle the **corner geometry** at these intersection points.

Theorem (Corner Regularity for Orbit-Space Jang Equation). *Let $\bar{f} : \mathcal{Q} \setminus \bar{\Sigma} \to \mathbb{R}$ be the orbit-space Jang solution. At the corner points $\bar{p}_N, \bar{p}_S = \bar{\Sigma} \cap \partial\mathcal{Q}$:*

(i) *The Jang solution has finite blow-up rate: $\bar{f}(\bar{x}) = C_0 \ln(1/d(\bar{x}, \bar{\Sigma})) + O(1)$ as $\bar{x} \to \bar{p}_\pm$.*

(ii) *The Jang metric $\bar{g} + d\bar{f} \otimes d\bar{f}$ extends to a Lipschitz metric on $\overline{\mathcal{Q} \setminus \bar{\Sigma}}$.*

(iii) *The cylindrical end metric $dt^2 + \bar{g}_{\bar{\Sigma}}$ is smooth away from the corners and has controlled conical singularities at $\bar{p}_\pm$.*

Proof of Corner Regularity. The proof uses the theory of elliptic equations on manifolds with corners, following Mazzeo–Melrose [39] and Schulze [53].

Step 1: Corner geometry. Near a corner point $\bar{p} \in \bar{\Sigma} \cap \partial\mathcal{Q}$, introduce local coordinates (σ, τ) where:

- $\sigma \geq 0$ is the distance to the axis $\partial\mathcal{Q}$;
- τ is arc-length along $\partial\mathcal{Q}$, with $\tau = 0$ at the corner.

The MOTS curve $\bar{\Sigma}$ meets $\partial\mathcal{Q}$ at an angle $\theta_0 \in (0, \pi)$, so near the corner:

$$\bar{\Sigma} = \{(\sigma, \tau) : \sigma = \tau \tan(\theta_0) + O(\tau^2),\ \tau \geq 0\}.$$

By axis regularity and the smooth embedding of Σ in M, the angle θ_0 is well-defined and generically satisfies $\theta_0 \neq 0, \pi/2, \pi$.

Step 2: Jang operator near the corner. The orbit-space Jang operator (24) in the coordinates (σ, τ) takes the form:

$$\mathcal{J}[\bar{f}] = a^{ij}(\sigma, \tau, \bar{\nabla}\bar{f})\partial_{ij}\bar{f} + b^i(\sigma, \tau, \bar{\nabla}\bar{f})\partial_i\bar{f} + c(\sigma, \tau, \bar{\nabla}\bar{f}) = 0,$$

where the coefficients a^{ij} are smooth for $\sigma > 0$ but may degenerate as $\sigma \to 0$ due to the axis structure. Specifically:

$$a^{\sigma\sigma} = 1 + O(\sigma^2), \quad a^{\tau\tau} = \frac{1}{\sigma^2}(1 + O(\sigma)), \quad a^{\sigma\tau} = O(1).$$

This is an **edge-degenerate elliptic operator** with edge at $\partial\mathcal{Q}$.

Step 3: Model operator at the corner. The model operator at the corner (freezing coefficients at $\bar{p}$) is the 2D edge Laplacian with Dirichlet data on $\bar{\Sigma}$:

$$L_{\text{model}} = \partial_\sigma^2 + \frac{1}{\sigma^2}\partial_\tau^2.$$

This operator is conformally equivalent to the flat Laplacian in polar coordinates (r, ϕ) via $\sigma = r\sin\phi$, $\tau = r\cos\phi$. Solutions with logarithmic blow-up along a ray $\{\phi = \phi_0\}$ have the form:

$$u(r,\phi) = c_0 \ln r + \sum_{k=1}^{\infty} (a_k r^{k\pi/\alpha} + b_k r^{-k\pi/\alpha}) \sin(k\pi(\phi - \phi_0)/\alpha),$$

where $\alpha = \pi - \theta_0$ is the interior angle at the corner in the complement of $\bar{\Sigma}$.

For the Jang solution, the boundary condition $\bar{f}|_{\bar{\Sigma}} = +\infty$ (blow-up) corresponds to $c_0 = C_0 > 0$, and regularity away from $\bar{\Sigma}$ forces $b_k = 0$ (no incoming singularities from the corner).

Step 4: A priori estimates at the corner. By the Schauder estimates for edge-degenerate operators [53, Theorem 5.4.1], the Jang solution satisfies:

$$\|\bar{f} - C_0 \ln(d/d_{\bar{\Sigma}})\|_{C^{2,\beta}_{\text{edge}}(U_\epsilon)} \leq C\epsilon^\beta,$$

where U_ϵ is an ϵ-neighborhood of the corner, $d_{\bar{\Sigma}}$ is distance to $\bar{\Sigma}$, and $\beta > 0$ depends on the corner angle θ_0. The weighted edge Hölder space $C^{k,\alpha}_{\text{edge}}$ is defined using the singular coordinates adapted to the corner.

Step 5: Jang metric regularity at the corner. The Jang metric $\bar{g}_{\text{Jang}} = \bar{g} + d\bar{f} \otimes d\bar{f}$ on $\mathcal{Q} \setminus \bar{\Sigma}$ has:

$$|\bar{\nabla}\bar{f}|^2 = \frac{C_0^2}{d_{\bar{\Sigma}}^2} + O(d_{\bar{\Sigma}}^{-1}).$$

Near the corner, this gives $|d\bar{f}|^2 = O(d_{\bar{\Sigma}}^{-2})$, so the Jang metric:

$$\bar{g}_{\text{Jang}} = \bar{g} + d\bar{f} \otimes d\bar{f} = O(1) + O(d_{\bar{\Sigma}}^{-2}) \cdot d_{\bar{\Sigma}}^2 = O(1)$$

remains bounded near the corner (the singular direction of $d\bar{f}$ is *tangent* to $\bar{\Sigma}$, so the perpendicular extension remains controlled).

More precisely, in coordinates $(t, y) = (-\ln d_{\bar{\Sigma}}, y|_{\bar{\Sigma}})$ near the cylindrical end:

$$\bar{g}_{\text{Jang}} = dt^2 + \bar{g}_{\bar{\Sigma}} + O(e^{-\beta t}),$$

with the error decaying exponentially as $t \to \infty$ (i.e., $d_{\bar{\Sigma}} \to 0$). At the corners $\bar{p}_{\pm}$, the cross-sectional metric $\bar{g}_{\bar{\Sigma}}$ degenerates (since $\bar{\Sigma}$ terminates), but the **3D lifted metric** $\bar{g} = g + df \otimes df$ on M remains regular because the 3D manifold M is smooth at the poles.

Conclusion. The corner singularities in the orbit-space analysis are **artifacts of the dimensional reduction**. The full 3D Jang metric $\bar{g}$ is Lipschitz continuous on $\overline{M \setminus \Sigma}$, including the polar directions. The orbit-space corners do not obstruct the analysis because:

1. All integral quantities (area, mass, angular momentum) are computed in 3D, where the poles are regular points;
2. The corner angle $\theta_0 \in (0, \pi)$ is bounded away from 0 and π by the smooth embedding of Σ;
3. The edge-degenerate elliptic theory provides the necessary regularity for perturbation estimates.

□

4.2 The Generalized Jang Equation

For initial data (M, g, K), the Jang equation seeks a function $f : M \to \mathbb{R}$ such that the graph $\Gamma(f) \subset M \times \mathbb{R}$ satisfies:

$$H_{\Gamma(f)} = \mathrm{tr}_{\Gamma(f)} K, \tag{20}$$

where H_Γ is the mean curvature of the graph and $\mathrm{tr}_\Gamma K$ is the trace of K restricted to the graph.

4.3 Axisymmetric Setting

For axisymmetric data with Killing field $\eta = \partial_\phi$, we work in Weyl-Papapetrou coordinates (r, z, ϕ):

$$g = e^{2U}(dr^2 + dz^2) + \rho^2 d\phi^2, \tag{21}$$

where $U = U(r, z)$ and $\rho = \rho(r, z)$ with $\rho \to r$ as $r \to 0$ (axis regularity).

The extrinsic curvature decomposes as:

$$K = K^{(\mathrm{sym})} + K^{(\mathrm{twist})}, \tag{22}$$

where the twist component encodes the frame-dragging effect:

$$K_{i\phi}^{(\text{twist})} = \frac{1}{2}\rho^2\omega_i, \quad i \in \{r, z\}, \tag{23}$$

with $\omega = \omega_r dr + \omega_z dz$ the twist 1-form.

Theorem 4.12 (Axisymmetric Jang Existence)**.** *Let (M, g, K) be asymptotically flat, axisymmetric initial data satisfying DEC with outermost strictly stable MOTS Σ and decay rate $\tau > 1/2$, i.e., $\lambda_1(L_\Sigma) > 0$. Then:*

(i) ***Existence and uniqueness:*** *The axisymmetric Jang equation admits a solution $f : M \setminus \Sigma \to \mathbb{R}$, unique up to an additive constant. The solution satisfies*

$$f \in C^{2,\beta}_{\text{loc}}(M \setminus \Sigma) \cap C^{0,1}_{\text{loc}}(M \setminus \Sigma),$$

i.e., it is $C^{2,\beta}$ (hence locally Lipschitz) away from Σ, and it blows up along Σ.

(ii) ***Blow-up asymptotics:*** *Near Σ, the solution blows up logarithmically:*

$$f(x) = C_0 \ln(1/s) + \mathcal{A}(y) + R(s, y), \quad C_0 > 0,$$

where:

- *$s = \text{dist}_g(x, \Sigma)$ is the signed distance to Σ;*
- *$y \in \Sigma$ is the nearest point projection;*
- *C_0 is the positive logarithmic blow-up coefficient supplied by the untwisted Han–Khuri existence theory for the horizon class under consideration;*
- *$\mathcal{A} \in C^{2,\beta}(\Sigma)$ is a smooth function on Σ (distinct from the area functional A);*
- *$R(s, y) = O(s^\alpha)$ with $\alpha = \min(1, 2\sqrt{\lambda_1(L_\Sigma)}) > 0$ depending on the spectral gap of the stability operator.*

(iii) ***Jang manifold structure:*** *The induced metric $\bar{g} = g + df \otimes df$ on the Jang manifold $\bar{M} := M \setminus \Sigma$ satisfies:*

- *$\bar{g} \in C^{0,1}(\bar{M})$ and extends (after adding Σ as an inner boundary component) to a continuous, Lipschitz metric on $\overline{\bar{M}}$;*

- $\bar{g} \in C^{2,\beta}(\bar{M} \setminus \Sigma)$ *is smooth away from the horizon;*
- *The cylindrical end* $\mathcal{C} := \{x : s < s_0\} \cong [0, \infty) \times \Sigma$ *admits a cylindrical coordinate* t *with metric*

$$\bar{g} = dt^2 + g_\Sigma + O(e^{-\beta_0 t}), \quad \beta_0 > 0,$$

where the error term and its first two derivatives decay exponentially, with β_0 *determined by the strict-stability spectral gap.*

(iv) ***ADM mass agreement:*** *For the standard asymptotic normalization of the Jang solution, the graph metric satisfies*

$$M_{\mathrm{ADM}}(\bar{g}) = M_{\mathrm{ADM}}(g).$$

Remark 4.13 (Critical Estimate: Twist Decay). This theorem contains a **critical technical estimate**. The key claim is:

Twist Decay Estimate (Step 2c below): The twist perturbation term $\mathcal{T}[f]$ satisfies $|\mathcal{T}[f]| = O(s)$ as $s \to 0$, where $s = \mathrm{dist}(\cdot, \Sigma)$ is the distance to the MOTS.

Why this matters: The principal Jang operator terms scale as $O(s^{-1})$ near the MOTS. For the perturbation argument (Lemma 4.15) to succeed, the twist must be **subdominant**, i.e., $|\mathcal{T}| \ll s^{-1}$. The claim $|\mathcal{T}| = O(s)$ provides a factor of s^2 separation, which is more than sufficient.

Physical intuition for twist boundedness: The twist 1-form ω encodes frame-dragging effects near the rotating black hole. One might naively expect ω to diverge as one approaches the horizon due to extreme frame-dragging. However, for *stable* MOTS in axisymmetric geometries, the twist potential satisfies an elliptic equation ($d\omega = 0$ combined with regularity conditions at the axis) that enforces boundedness. The key physical insight is that:

- Frame-dragging is maximal *on the horizon*, not as one approaches it from outside;
- The axis regularity conditions (requiring $\omega \to 0$ smoothly as $\rho \to 0$) prevent twist from diverging at the poles;
- The elliptic equation propagates this boundary regularity throughout the domain.

The mathematical manifestation is that $|\omega| \leq C_{\omega,\infty} < \infty$ uniformly on $\mathcal{Q}$, including near the MOTS. This is *not* obvious a priori and is one of the subtle features of the axisymmetric vacuum equations.

Verification checkpoints:

(V1) **Twist 1-form regularity:** Verify that the axis regularity conditions (AR1)–(AR3) in Remark 4.10 imply $|\omega| \leq C_{\omega,\infty} < \infty$ on the orbit space $\mathcal{Q}$. This follows from standard elliptic theory for the twist potential equation.

Technical detail: The twist potential χ (related to ω by $\omega = d\chi$ locally) satisfies an elliptic equation on $\mathcal{Q}$ with Dirichlet boundary conditions at the axis. Since the source terms are bounded, elliptic L^∞ estimates (Gilbarg–Trudinger [28], Theorem 8.15) give $\|\chi\|_{L^\infty(\mathcal{Q})} < \infty$.

(V2) **Orbit radius scaling:** Verify that the twist term (25) contains a factor of ρ^2 in the numerator, where ρ is the orbit radius. Since ρ is bounded on the compact MOTS Σ, this contributes a bounded factor.

Geometric origin: The factor ρ^2 arises from the dimensional reduction of the 3D metric $g = g_{\mathcal{Q}} + \rho^2 d\phi^2$ to the orbit space. The precise computation is in (25).

(V3) **Gradient scaling:** Verify that the denominator $\sqrt{1 + |\nabla f|^2} = O(s^{-1})$ due to the logarithmic blow-up $f = C_0 \ln s^{-1} + O(1)$. This factor produces the $O(s)$ decay.

Direct computation: $|\nabla f| = C_0/s + O(1)$, so $\sqrt{1 + |\nabla f|^2} = C_0 s^{-1} + O(1)$.

(V4) **Pole regularity:** Verify that at the poles p_N, p_S where $\rho = 0$, the twist term vanishes: $\mathcal{T}(p_\pm) = 0$ (Lemma 4.8).

Why this is non-trivial: At the poles, both ρ^2 and ω vanish. The key is to verify that $\rho^2\omega = O(\rho^3) \to 0$ as $\rho \to 0$.

(V5) **Frame-dragging cancellation structure:** Verify that the specific contraction structure in $\mathcal{T}[f]$ (involving the graph normal ν) does not amplify frame-dragging effects near the horizon.

Key observation: The twist term involves $\omega_i \nu^i$, where ν is the graph normal. As shown in Step 2c below, the orbit-space normal $\bar{\nu}$ remains bounded as $s \to 0$ because the normalization factor grows at the same

rate as $|\nabla f|$. Therefore, $\omega_i \nu^i$ is bounded by $C_{\omega,\infty} \cdot |\nu| = O(1)$, with no singular enhancement.

Physical interpretation: The graph becomes increasingly vertical as $s \to 0$, but the *unit normal* remains well-behaved. The twist contracts with this bounded normal vector, yielding a bounded contribution. The $O(s)$ decay arises from the additional factor s in $\rho^2/(C_0/s) = s \cdot O(1)$.

If the twist decay fails: If the twist were $O(s^{-\epsilon})$ for some $\epsilon > 0$ instead of $O(s)$, the perturbation argument would require modification. The proof would need either:

- A stronger coercivity estimate for the linearized operator, or
- A direct iteration scheme that does not rely on perturbative analysis near Σ.

The detailed scaling analysis in Step 2c below demonstrates that $|\mathcal{T}| = O(s)$ holds under the stated hypotheses.

Comparison with non-axisymmetric case: In the absence of axisymmetry, there is no twist 1-form structure, and frame-dragging effects enter the Jang equation through more complicated tensor contractions. The perturbative control we achieve here relies on the orbit-space reduction and elliptic regularity of the twist potential on the 2D orbit space. These features are special to axisymmetric geometries and do not generalize straightforwardly to non-axisymmetric rotating data.

Summary for Critical Review: Why Twist is Lower-Order.

The twist decay estimate $|\mathcal{T}[f]| = O(s)$ is the key technical input enabling extension of Jang equation methods to rotating black holes. It relies on three independent mechanisms:

Mechanism	Mathematical Source	Scaling
Twist potential boundedness	Elliptic PDE on orbit space	$\lvert\omega\rvert \le C_{\omega,\infty}$
Orbit radius prefactor	Dimensional reduction structure	$\rho^2 = O(1)$
Gradient normalization	Log blow-up of Jang solution	$\sqrt{1+\lvert\nabla f\rvert^2} = O(s^{-1})$
Combined effect:		$\lvert\mathcal{T}\rvert = O(s)$

To contest this estimate, one must demonstrate one of:

(a) The twist 1-form ω diverges near the MOTS (contradicting elliptic regularity);

(b) The orbit radius factor ρ^2 is absent from the twist term (contradicting dimensional reduction);

(c) The gradient normalization fails (contradicting the established Jang blow-up theory).

Each of (a)–(c) would require identifying an error in the cited literature (Han–Khuri [29], Wald [57]).

Proof. The proof extends the Han–Khuri existence theory [29] to the axisymmetric setting with twist. We structure the argument in five steps, verifying that twist terms constitute lower-order perturbations that do not affect the principal analysis.

Step 1: Equivariant reduction and the axisymmetric Jang equation. By axisymmetry, we reduce to the 2D orbit space $\mathcal{Q} = M/S^1$ with coordinates (r, z) and orbit radius $\rho(r, z)$. The 3D Jang equation

$$H_{\Gamma(f)} = \mathrm{tr}_{\Gamma(f)} K$$

reduces to a 2D quasilinear elliptic PDE on $\mathcal{Q}$:

$$\bar{H}_{\Gamma(\bar{f})} = \mathrm{tr}_{\Gamma(\bar{f})} \bar{K} + \mathcal{T}[\bar{f}], \tag{24}$$

where overbars denote orbit-space quantities and $\mathcal{T}[\bar{f}]$ collects twist contributions.

The reduced Jang operator has the form:

$$\mathcal{J}_{\rm axi}[\bar{f}] := \bar{g}^{ij}\left(\frac{\bar{\nabla}_{ij}\bar{f}}{\sqrt{1+|\bar{\nabla}\bar{f}|^2}} - \bar{K}_{ij}\right) - \frac{\bar{f}^i\bar{f}^j}{1+|\bar{\nabla}\bar{f}|^2}\left(\frac{\bar{\nabla}_{ij}\bar{f}}{\sqrt{1+|\bar{\nabla}\bar{f}|^2}} - \bar{K}_{ij}\right) - \mathcal{T}[\bar{f}],$$

where the twist contribution is:

$$\mathcal{T}[\bar{f}] = \frac{\rho^2}{(1+|\bar{\nabla}\bar{f}|^2)^{1/2}}\left(\omega_i\bar{\nu}^i - \frac{\bar{f}_{,i}\omega_j\bar{f}^{,j}}{1+|\bar{\nabla}\bar{f}|^2}\bar{\nu}^i\right), \tag{25}$$

where $\bar{\nu}$ is the **orbit-space projection of the graph normal**, defined explicitly as follows. Let $\Gamma(\bar{f}) \subset \mathcal{Q}\times\mathbb{R}$ be the graph of $\bar{f}$. The upward unit normal to this graph is:

$$N = \frac{1}{\sqrt{1+|\bar{\nabla}\bar{f}|^2_{\bar{g}}}}(-\bar{\nabla}\bar{f}, 1) \in T(\mathcal{Q}\times\mathbb{R}).$$

The orbit-space component $\bar{\nu} = (\bar{\nu}^r, \bar{\nu}^z)$ is the projection of N to $T\mathcal{Q}$:

$$\bar{\nu}^i = -\frac{\bar{g}^{ij}\partial_j\bar{f}}{\sqrt{1+|\bar{\nabla}\bar{f}|^2_{\bar{g}}}}, \quad i\in\{r,z\}.$$

This is a unit vector in $(\mathcal{Q},\bar{g})$ when $|\bar{\nabla}\bar{f}| \neq 0$.

Step 2: Verification that twist is a lower-order perturbation. This is the critical step. We establish three key bounds with detailed derivations:

(2a) Twist potential regularity. The twist 1-form ω satisfies the elliptic system $d\omega = 0$ (from the vacuum momentum constraint $D^jK_{ij} = D_i(\mathrm{tr}K)$ combined with axisymmetry). More precisely, the momentum constraint in axisymmetric coordinates gives:

$$\partial_r(\rho^3\omega_z) - \partial_z(\rho^3\omega_r) = 0,$$

which is the curl-free condition for $\rho^3\omega$ on $\mathcal{Q}$. This implies $\rho^3\omega = d\Omega$ for a twist potential Ω, and standard elliptic regularity for the Laplacian $\Delta_{\mathcal{Q}}\Omega = 0$

[28] yields $\omega \in C^{0,\beta}(\mathcal{Q})$ up to $\partial\mathcal{Q}$ (the axis and horizon). In particular, $|\omega| \leq C_\omega$ is uniformly bounded on $\mathcal{Q}$.

(2b) Orbit radius behavior at the horizon. The horizon Σ in axisymmetric data intersects the axis Γ at exactly two poles p_N, p_S (Lemma 4.6). The orbit radius ρ satisfies:

- $\rho(p_N) = \rho(p_S) = 0$ at the poles;
- $\rho|_{\Sigma\setminus\{p_N,p_S\}} > 0$ away from the poles;
- $\rho(x) = O(\mathrm{dist}(x, p_\pm))$ as $x \to p_\pm$ (linear vanishing at poles).

Despite $\rho \to 0$ at the poles, the twist term $\mathcal{T}$ remains bounded because $\mathcal{T} \propto \rho^2$ (see Lemma 4.8). Thus $\mathcal{T}(p_N) = \mathcal{T}(p_S) = 0$, and $|\mathcal{T}| \leq C\rho_{\max}^2$ globally on Σ.

(2c) Scaling analysis near the blow-up—detailed derivation. We now prove rigorously that $\mathcal{T} = O(s)$ near Σ, where s is the signed distance to Σ.

Near the MOTS Σ, introduce Gaussian normal coordinates (s, y^A) where s is the signed distance to Σ and y^A are coordinates on Σ. The metric takes the form:

$$g = ds^2 + h_{AB}(s, y)dy^A dy^B, \quad h_{AB}(0, y) = (g_\Sigma)_{AB}.$$

The Jang solution has the blow-up asymptotics $f = C_0 \ln s^{-1} + \mathcal{A}(y) + O(s^\alpha)$, so:

$$\nabla f = -\frac{C_0}{s}\partial_s + O(1), \quad |\nabla f|^2 = \frac{C_0^2}{s^2} + O(s^{-1}).$$

Thus $\sqrt{1 + |\nabla f|^2} = C_0/s + O(1)$.

Now examine the twist term (25). The orbit radius satisfies $\rho(s, y) = \rho(0, y) + O(s) = \rho_\Sigma(y) + O(s)$ with $\rho_\Sigma > 0$. The twist 1-form components ω_i are bounded (from (2a)).

Orbit-space projection analysis. To relate the 3D coordinates (s, y^A) to the orbit-space quotient $\mathcal{Q}$, we use the axisymmetric structure. The orbit-space coordinates (r, z) on $\mathcal{Q}$ are related to the 3D coordinates by the quotient map $\pi : M^3 \to \mathcal{Q}$ that collapses orbits of the $U(1)$-action. The MOTS Σ is a $U(1)$-invariant sphere that intersects the axis at two poles p_N, p_S (Lemma 4.6). The signed distance function $s = \mathrm{dist}(\cdot, \Sigma)$ is $U(1)$-invariant and descends to a function $\bar{s}$ on $\mathcal{Q}$ with $\bar{s} = s \circ \pi^{-1}$. The orbit-space image $\bar{\Sigma} = \pi(\Sigma) \subset \mathcal{Q}$ is an arc connecting the two poles on the axis boundary of $\mathcal{Q}$.

The gradient projection identity is: for any $U(1)$-invariant function u on M^3,

$$\bar{\nabla}\bar{u} = \pi_*(\nabla u - (\nabla u \cdot \xi)\xi/|\xi|^2),$$

where $\xi = \partial_\phi$ is the axial Killing field and $\bar{\nabla}$ is the gradient on $(\mathcal{Q}, \bar{g})$. Since f is $U(1)$-invariant by construction, we have $\nabla f \cdot \xi = 0$, so $\bar{\nabla}\bar{f} = \pi_*(\nabla f)$. In the adapted coordinates where ∂_s is tangent to $\mathcal{Q}$:

$$\bar{\nabla}\bar{f} = -\frac{C_0}{s}\partial_{\bar{s}} + O(1), \quad |\bar{\nabla}\bar{f}|^2_{\bar{g}} = \frac{C_0^2}{s^2} + O(s^{-1}).$$

The orbit-space projection of the graph normal (as defined in Step 1) has components:

$$\bar{\nu}^i = -\frac{\bar{g}^{ij}\partial_j \bar{f}}{\sqrt{1 + |\bar{\nabla}\bar{f}|^2_{\bar{g}}}} = -\frac{\partial^i \bar{f}}{\sqrt{1 + |\bar{\nabla}\bar{f}|^2_{\bar{g}}}}.$$

Using $\bar{\nabla}\bar{f} = -\frac{C_0}{s}\partial_{\bar{s}} + O(1)$ and $\sqrt{1 + |\bar{\nabla}\bar{f}|^2} = C_0/s + O(1)$:

$$\bar{\nu} = \frac{1}{C_0/s + O(1)}\left(\frac{C_0}{s}\partial_{\bar{s}} + O(1)\right) = \frac{s}{C_0 + O(s)}\left(\frac{C_0}{s}\partial_{\bar{s}} + O(1)\right) = \partial_{\bar{s}} + O(s).$$

That is, $|\bar{\nu}^i| = O(1)$ as $s \to 0$, with the dominant direction being normal to $\bar{\Sigma}$ in the orbit space. This is the key geometric fact: the orbit-space normal $\bar{\nu}$ remains bounded despite the blow-up of f, because the normalization factor $\sqrt{1 + |\bar{\nabla}\bar{f}|^2}$ grows at the same rate as $|\bar{\nabla}\bar{f}|$. Substituting into (25):

$$\mathcal{T}[\bar{f}] = \frac{\rho^2}{\sqrt{1 + |\nabla f|^2}}\left(\omega_i \bar{\nu}^i + \text{lower order}\right) \tag{26}$$

$$= \frac{\rho_\Sigma^2 + O(s)}{C_0/s + O(1)} \cdot (O(1)) \tag{27}$$

$$= \frac{s(\rho_\Sigma^2 + O(s))}{C_0 + O(s)} \cdot O(1) = O(s). \tag{28}$$

This proves $|\mathcal{T}| = O(s)$ as $s \to 0^+$.

In contrast, the principal Jang operator terms involve $\nabla^2 f/\sqrt{1 + |\nabla f|^2}$, which scales as:

$$\frac{C_0/s^2}{C_0/s} = \frac{1}{s} \quad \text{(divergent as } s \to 0\text{)}.$$

Therefore, the twist contribution $\mathcal{T} = O(s)$ is indeed subdominant compared to the principal terms $O(s^{-1})$, by a factor of s^2. This justifies treating twist as a perturbation in the blow-up analysis.

We formalize this scaling analysis as a standalone lemma for clarity:

Lemma 4.14 (Twist Bound Near MOTS)**.** *Let (M^3, g, K) be asymptotically flat, axisymmetric initial data with a stable outermost MOTS Σ. Let $s = \operatorname{dist}(\cdot, \Sigma)$ denote the signed distance to Σ, and let $\mathcal{T}[f]$ be the twist perturbation term (25) in the axisymmetric Jang equation. Then there exist constants $C_{\mathcal{T}} > 0$ and $s_0 > 0$ depending only on the initial data such that:*

$$|\mathcal{T}[f](x)| \leq C_{\mathcal{T}} \cdot s(x) \quad \text{for all } x \text{ with } 0 < s(x) < s_0. \tag{29}$$

More precisely, $C_{\mathcal{T}} = C_{\omega,\infty} \cdot \rho_{\max}^2 / C_0$, where:

- *$C_{\omega,\infty} = \sup_{\mathcal{Q}} |\omega|$ is the L^∞ bound on the twist 1-form;*
- *$\rho_{\max} = \sup_{x \in \Sigma} \rho(x)$ is the maximum orbit radius on Σ;*
- *$C_0 > 0$ is the leading coefficient in the Jang blow-up $f = C_0 \ln s^{-1} + O(1)$.*

Scaling comparison: *Since the principal Jang terms scale as $O(s^{-1})$ near Σ, while the twist term scales as $O(s)$, the twist is subdominant by a factor of s^2. This ensures that twist does **not** disrupt the blow-up analysis, preserving the cylindrical end structure required for the proof.*

Critical observation: *The constant $C_{\mathcal{T}}$ depends **only on the initial data** (g, K) together with the positive blow-up coefficient C_0 furnished by the untwisted Jang existence theory. In particular:*

(a) *$C_{\mathcal{T}}$ does **not** depend on higher derivatives $\nabla^k f$ for $k \geq 2$, which blow up as $O(s^{-k})$;*

(b) *The twist term $\mathcal{T}[f]$ involves **no second derivatives** of f, only f and ∇f;*

(c) *The bound holds **uniformly** for any function with logarithmic blow-up $f = C_0 \ln s^{-1} + O(1)$;*

(d) *At the poles p_N, p_S where Σ intersects the axis, $\mathcal{T}(p_\pm) = 0$ since $\rho(p_\pm) = 0$ (Lemma 4.8).*

See Appendix D for a detailed perturbative discussion showing that the twist term does not alter the leading existence mechanism or the blow-up character of the Jang solution.

Non-circularity verification: *The argument above is* ***not*** *circular. To see this explicitly:*

(NC1) *The constant $C_0 > 0$ is fixed* ***a priori*** *once the reference untwisted blow-up solution near Σ is chosen; it is not determined by the perturbative correction v.*

(NC2) *The twist bound $|\omega| \leq C_{\omega,\infty}$ follows from elliptic regularity applied to the twist potential equation on the orbit space, which is determined by the* ***initial data*** *(g, K) alone.*

(NC3) *The orbit radius $\rho_{\max} = \sup_\Sigma \rho$ is a geometric quantity of the MOTS in the initial data.*

(NC4) *The scaling $|\mathcal{T}[f]| = O(s)$ uses only that f has logarithmic blow-up with* ***some*** *coefficient $C_0 > 0$, not the specific value. Thus, the estimate holds for any candidate solution in the iteration scheme of Lemma 4.15.*

The logical flow is: initial data $\to$ untwisted blow-up solution and horizon geometry $\to$ blow-up coefficient C_0 and twist bound $C_{\omega,\infty} \to$ perturbation estimate $|\mathcal{T}| \leq C_{\mathcal{T}} s \to$ perturbative reduction to the untwisted Jang problem. At no point does the constant $C_{\mathcal{T}}$ depend on the twisted solution f being constructed.

Data Dependence of the Perturbative Constants.

We record the source of the constants entering Lemma 4.15. The point is not to produce sharp formulas, but to show that the perturbative bounds are determined by the initial data and the chosen weighted spaces, rather than by the unknown solution itself.

Given data (independent of Jang solution):

- *(M, g, K): initial data set with metric g and extrinsic curvature K;*
- *$\Sigma \subset M$: stable MOTS with mean curvature H_Σ and $\mathrm{tr}_\Sigma K$;*
- *η: axial Killing field with $\mathcal{L}_\eta g = \mathcal{L}_\eta K = 0$.*

A priori constants (computable from given data before solving Jang):

(C1) $C_0 > 0$*: the leading logarithmic blow-up coefficient of the reference untwisted Han–Khuri solution near* Σ*.*

(C2) $\rho_{\max} = \sup_{p\in\Sigma} |\eta|_g(p)$*: the maximum orbit radius on the MOTS, a geometric quantity of* (M, g, η, Σ)*.*

(C3) $C_{\omega,\infty}$*: the* L^∞ *bound on the twist 1-form* ω*, obtained by solving the* linear *elliptic equation* $d\omega = 0$*,* $\delta\omega = \star(d\eta^\flat \wedge \eta^\flat)/\rho^2$ *on the orbit space* $\mathcal{Q} = M/\mathrm{U}(1)$*. This depends only on* (g, K, η)*.*

(C4) $\lambda_0(\Sigma)$*: the principal eigenvalue of the MOTS stability operator on* Σ*, determined by* $(g|_\Sigma, K|_\Sigma)$*.*

Fixed-point iteration: The contraction map $\Phi : B_\delta \to B_\delta$ *in Step 4 of Lemma 4.15's proof is defined by:*

$$\Phi(v) = -L_0^{-1}\big(N[v] + \mathcal{T}[f_0 + v]\big),$$

where:

- f_0 *is the* reference solution *to the twist-free Jang equation (existence established by Schoen–Yau/Eichmair without twist);*
- $L_0 = D\mathcal{J}_0|_{f_0}$ *is the linearization at* f_0*;*
- $N[v]$ *is the nonlinear remainder (quadratic in* v*);*
- $\mathcal{T}[f_0 + v]$ *is the twist perturbation evaluated on the candidate* $f_0 + v$*.*

Key observation: The twist bound $|\mathcal{T}[f]| \le C_{\mathcal{T}} s$ *in hypothesis (P3) requires only:*

- $|\omega| \le C_{\omega,\infty}$ *(from C3, independent of* f*);*
- $\rho \le \rho_{\max}$ *(from C2, independent of* f*);*
- $|\nabla f| \ge C_0/s - O(1)$ *for* any *function in the ball* B_δ *around* f_0*.*

The last condition is verified as follows: elements of B_δ *satisfy* $\|f - f_0\|_{W^{2,2}_\beta} \le \delta$*. Since* f_0 *has the blow-up* $f_0 = C_0 \ln s^{-1} + O(1)$ *and* $v \in W^{2,2}_\beta$ *with* $\beta < 0$ *satisfies* $|v| = O(s^{-\beta}) = o(s^{-1})$*, we have:*

$$|\nabla(f_0 + v)| \ge |\nabla f_0| - |\nabla v| \ge \frac{C_0}{s} - O(s^{-\beta-1}) = \frac{C_0}{s}(1 + o(1)).$$

Thus the gradient lower bound holds uniformly for all elements of B_δ, with constants depending only on (C1)–(C4).

Conclusion: The constant $C_{\mathcal{T}} = C_{\omega,\infty} \cdot \rho_{\max}^2 / C_0$ in Lemma 4.14 is computed from (C1)–(C3), which are determined before the Jang equation is solved. The fixed-point argument then constructs the solution $f = f_0 + v$ without any circular dependence.

Proof. The proof is contained in the detailed calculation of Step 2c above. We summarize the key steps:

Step 1: By elliptic regularity for the twist potential equation on the orbit space $\mathcal{Q}$, the twist 1-form satisfies $|\omega| \leq C_{\omega,\infty}$ uniformly on $\mathcal{Q}$ (Step 2a).

Step 2: The MOTS Σ intersects the axis at two poles p_N, p_S where $\rho = 0$ (Lemma 4.6). Away from the poles, $\rho_\Sigma(y) > 0$. The key observation is that the twist term scales as ρ^2, so even though $\rho \to 0$ at the poles, $\mathcal{T}$ remains bounded (in fact, $\mathcal{T}(p_\pm) = 0$). For points away from the poles: $\rho(s, y) = \rho_\Sigma(y) + O(s)$ with $\rho_\Sigma(y) \leq \rho_{\max} < \infty$ (Step 2b and Lemma 4.8).

Step 3: The Jang function has logarithmic blow-up $f = C_0 \ln s^{-1} + O(1)$, giving:

$$|\nabla f| = \frac{C_0}{s} + O(1), \quad \sqrt{1 + |\nabla f|^2} = \frac{C_0}{s} + O(1).$$

Step 4: The twist term (25) involves $\rho^2 / \sqrt{1 + |\nabla f|^2}$ multiplied by bounded quantities. Substituting the scalings (away from poles):

$$|\mathcal{T}[f]| \leq \frac{(\rho_\Sigma + O(s))^2}{C_0/s + O(1)} \cdot C_{\omega,\infty} = \frac{s \cdot (\rho_\Sigma^2 + O(s))}{C_0 + O(s)} \cdot C_{\omega,\infty} = O(s).$$

At the poles, $\rho_\Sigma = 0$, so $\mathcal{T} = O(s \cdot 0) = 0$. The explicit constant follows from $\rho_\Sigma \leq \rho_{\max}$. □

We now invoke a general perturbation principle for quasilinear elliptic equations. This result is a refinement of the implicit function theorem approach in Pacard–Ritoré [47, Theorem 2.1] adapted to singular perturbations, combined with the weighted space framework of Mazzeo [40, Section 3].

Lemma 4.15 (Perturbation Stability for Blow-Up Asymptotics)**.** *Let $\mathcal{J}_0[f] = 0$ be a quasilinear elliptic equation on a domain Ω with boundary $\partial\Omega = \Sigma$, and suppose:*

(P1) $\mathcal{J}_0$ admits a solution f_0 with logarithmic blow-up: $f_0(s, y) = C_0 \ln s^{-1} + \mathcal{A}_0(y) + O(s^\alpha)$ as $s \to 0$, where $s = \operatorname{dist}(\cdot, \Sigma)$.

(P2) The linearization $L_0 = D\mathcal{J}_0|_{f_0}$ at f_0 satisfies a coercivity estimate in weighted spaces: $\|Lv\|_{W^{0,2}_\beta} \geq c\|v\|_{W^{2,2}_\beta}$ for $\beta \in (-1, 0)$.

(P3) The perturbation $\mathcal{T}$ satisfies: $|\mathcal{T}[f]| \leq Cs^{1+\gamma}$ for some $\gamma \geq 0$ whenever $|f - f_0| \leq \delta$ in $W^{2,2}_\beta$. (The case $\gamma = 0$ corresponds to $|\mathcal{T}| \leq Cs$.)

Then the perturbed equation $\mathcal{J}_0[f] + \mathcal{T}[f] = 0$ admits a solution f with the same leading-order asymptotics:

$$f(s, y) = C_0 \ln s^{-1} + \mathcal{A}(y) + O(s^{\min(\alpha, 1+\gamma)}),$$

where the coefficient C_0 is unchanged and $\mathcal{A}(y)$ may differ from $\mathcal{A}_0(y)$ by $O(1)$.

Proof. The argument is the standard quasilinear fixed-point reduction in weighted Sobolev spaces. Writing $f = f_0 + v$ and expanding around f_0 gives

$$L_0 v = -N[v] - \mathcal{T}[f_0 + v], \tag{30}$$

where $L_0 = D\mathcal{J}_0|_{f_0}$ and $N[v]$ is quadratic in v.

Hypothesis (P2) provides a bounded inverse for L_0 on $W^{2,2}_\beta$, so the map

$$\Phi(v) := -L_0^{-1}\big(N[v] + \mathcal{T}[f_0 + v]\big)$$

is well defined. The quadratic estimate for $N[v]$ and the weighted integrability/Lipschitz control coming from (P3) imply that Φ maps a sufficiently small ball in $W^{2,2}_\beta$ to itself and is a contraction there. Banach's fixed-point theorem therefore gives a unique correction v in that ball.

Because $\beta \in (-1, 0)$, functions in $W^{2,2}_\beta$ satisfy $v = o(1)$ toward the cylindrical end, so the perturbation does not alter the leading logarithmic coefficient C_0. Elliptic regularity then upgrades the remainder to the stated order

$$O\big(s^{\min(\alpha, 1+\gamma)}\big),$$

after absorbing the lower-order contribution of v into the profile function $\mathcal{A}(y)$. □

We verify conditions (P1)–(P3) for our setting with explicit references:

- **Verification of (P1):** This is Han–Khuri [29, Proposition 4.5]. Specifically, for initial data (M, g, K) satisfying DEC with a stable outermost MOTS Σ, the unperturbed Jang equation $\mathcal{J}_0[f] = 0$ admits a solution f_0 with blow-up asymptotics $f_0(s, y) = C_0 \ln s^{-1} + \mathcal{A}_0(y) + O(s^\alpha)$ for some coefficient $C_0 > 0$. The exponent $\alpha > 0$ depends on the spectral gap of the MOTS stability operator; for strictly stable MOTS, $\alpha = \min(1, 2\sqrt{\lambda_1(L_\Sigma)})$ where $\lambda_1(L_\Sigma) > 0$ is the principal eigenvalue.

- **Verification of (P2):** For the untwisted Han–Khuri solution f_0, the linearized Jang operator $L_0 = D\mathcal{J}_0|_{f_0}$ is an asymptotically cylindrical elliptic operator. Strict stability of the MOTS provides a spectral gap for the limiting cylindrical operator, so one may choose a weight $\beta < 0$ avoiding the indicial roots. Lockhart–McOwen theory [35] then places L_0 in the Fredholm regime on $W^{2,2}_\beta$. The additional coercive/invertibility estimate required in hypothesis (P2) is the standard linear input from the untwisted Han–Khuri analysis; the present perturbative argument does not attempt to re-prove that untwisted estimate from scratch. Lemma 4.17 below records only the part needed for the twist perturbation, namely that the twist contribution is exponentially decaying at the linearized level and therefore does not alter the indicial roots or the Fredholm setup.

- **Verification of (P3):** We proved above that $|\mathcal{T}| = O(s)$ as $s \to 0^+$. More precisely, the scaling analysis gives $|\mathcal{T}(s, y)| \leq C_\mathcal{T} \cdot s$ where $C_\mathcal{T} = C_{\omega,\infty} \cdot \rho^2_{\max} \cdot C_0^{-1}$ depends only on the initial data. This corresponds to $\gamma = 0$ in hypothesis (P3), i.e., $|\mathcal{T}| \leq Cs^{1+0} = Cs$. This decay rate is sufficient for the perturbation argument because the weighted norm estimate in Step 3b below shows the perturbation is integrable in $W^{0,2}_\beta$.

Therefore, Lemma 4.15 applies, and the Jang solution with twist has the same leading-order asymptotics as the twist-free case, exactly as in the Han–Khuri analysis.

Remark 4.16 (Explicit Constant Dependencies). The perturbation stability argument depends on the following classes of constants:

- An inverse bound for the untwisted linearization L_0^{-1} on the chosen weighted space;

- Local C^2 control of the reference Han–Khuri solution f_0 on compact subsets away from the horizon;

- The twist-size bound $C_{\omega,\infty}$ together with the horizon geometry through $\rho_{\max}$ and the blow-up coefficient C_0;
- A small radius δ for the contraction ball, chosen after the previous constants are fixed.

For axisymmetric vacuum data with strictly stable MOTS, these quantities are finite and determined by the initial data together with the chosen weighted-space setup. The qualitative dependence is the expected one: stronger stability and weaker twist improve the perturbative control.

Lemma 4.17 (Indicial/Fredholm Stability for the Twisted Jang Operator)**.** *Let $\mathcal{J}_{\text{axi}} = \mathcal{J}_0 + \mathcal{T}$ be the axisymmetric Jang operator with twist perturbation $\mathcal{T}$. The linearization $L_{\text{axi}} := D\mathcal{J}_{\text{axi}}|_f$ at a solution f has the following properties:*

(i) The indicial roots of L_{axi} on the cylindrical end coincide with those of $L_0 := D\mathcal{J}_0|_f$.

(ii) For weight $\beta \in (-1, 0)$ not equal to any indicial root, $L_{\text{axi}} : W^{2,2}_\beta \to L^2_\beta$ is Fredholm of index zero.

Proof. On the cylindrical end, the twist contribution to the linearized operator is exponentially decaying and of lower order:

$$L_{\text{axi}} = L_0 + B, \qquad B = O(e^{-\delta t})$$

for some $\delta > 0$, in the sense of coefficients of an operator of order at most one. Consequently the translation-invariant limit operator of L_{axi} agrees with that of L_0, so the indicial roots are unchanged. This proves (i).

For a weight $\beta < 0$ avoiding those indicial roots, Lockhart–McOwen theory implies that both L_0 and L_{axi} are Fredholm on the corresponding weighted Sobolev spaces. Since the perturbation B is exponentially decaying, it acts compactly from $W^{2,2}_\beta$ to L^2_β, so the Fredholm index is preserved. Thus L_{axi} has the same index as L_0, namely zero; this proves (ii). □

Step 3: Barrier construction. Following [29] and [52], we construct sub- and super-solutions using the stability of the outermost MOTS Σ.

(3a) Supersolution at infinity. Define $f^+ = C_1 r^{1-\tau+\epsilon} + C_2$ for $r \geq R_0$ large. A direct computation (see [29, Section 4]) shows that for $\tau > 1/2$ and C_1 sufficiently large:

$$\mathcal{J}_{\text{axi}}[f^+] \geq c_0 r^{-1-\tau} > 0 \quad \text{for } r \geq R_0,$$

where the twist term contributes only $O(r^{-2})$ and does not affect the sign.

(3b) Subsolution at infinity. The function $f^- = -C_1 r^{1-\tau+\epsilon} - C_2$ is a subsolution by the same analysis.

(3c) Barriers near the horizon. Since Σ is a stable MOTS, it admits a local foliation by surfaces $\{\Sigma_s\}_{0<s<s_0}$ with mean curvature $H(\Sigma_s) > 0$ (outward mean-convex). The Schoen–Yau barrier argument [52] constructs a subsolution:

$$\underline{f}(x) = \int_0^{s(x)} \frac{1}{\sqrt{1-\theta^+(s')^2}}\, ds',$$

which forces the solution to blow up at Σ. Because $|\mathcal{T}[\underline{f}]| \to 0$ as $s \to 0$ (Step 2c), the barrier inequality

$$\mathcal{J}_{\text{axi}}[\underline{f}] = \mathcal{J}_0[\underline{f}] + \mathcal{T}[\underline{f}] \le \mathcal{J}_0[\underline{f}] + o(1) \le 0$$

holds in a neighborhood of Σ for the axisymmetric operator.

(3d) Prevention of premature blow-up. Inner unstable MOTS are "bridged over" by the Schoen–Yau barriers. The outermost property of Σ ensures no interior trapped surface lies outside Σ, and the stability of Σ provides the geometric control for the subsolution construction.

Step 4: Existence via regularization and Perron method. We solve the regularized capillary Jang equation on $\Omega_\delta = \{x : \text{dist}(x, \Sigma) > \delta\}$:

$$\mathcal{J}_{\text{axi}}[f] = \kappa f, \quad f|_{\partial\Omega_\delta} = 0,$$

where $\kappa > 0$ is a regularization parameter. Standard elliptic theory [28] yields a smooth solution $f_{\kappa,\delta}$.

The barrier bounds from Step 3 provide uniform estimates:

$$|f_{\kappa,\delta}(x)| \le C(1 + r^{1-\tau+\epsilon}) \quad \text{on } \Omega_{2\delta},$$

independent of κ, δ. Interior Schauder estimates (using DEC to prevent interior gradient blow-up) give $C^{2,\beta}_{\text{loc}}$ compactness. Taking a diagonal subsequence as $\kappa \to 0, \delta \to 0$:

$$f_{\kappa,\delta} \to f \quad \text{in } C^{2,\beta}_{\text{loc}}(M \setminus \Sigma),$$

where f solves $\mathcal{J}_{\text{axi}}[f] = 0$ with blow-up at Σ.

By axisymmetry of the data and boundary conditions, the supremum in the Perron construction:

$$f = \sup\{v : v \text{ is a subsolution with } v \le f^+\}$$

is achieved by an axisymmetric function.

Step 5: Blow-up asymptotics and cylindrical end geometry. Near Σ, the leading-order behavior is determined by the principal operator $\mathcal{J}_0$ since $\mathcal{T} = O(s)$ is subdominant. The Han–Khuri analysis [29, Proposition 4.5] applies:

$$f(s, y) = C_0 \ln s^{-1} + \mathcal{A}(y) + O(s^\alpha),$$

where $C_0 > 0$ is the leading logarithmic coefficient supplied by the untwisted Han–Khuri asymptotic analysis.

Non-oscillatory behavior. The barrier comparison rules out oscillatory remainders (e.g., $\sin(\ln s)$) by comparing with strictly monotone supersolutions constructed from the stability of Σ. This follows from standard ODE comparison arguments for the radial profile; see [29, Section 5].

Cylindrical end metric. In a suitably normalized cylindrical coordinate t on the end, the induced metric satisfies:

$$\bar{g} = dt^2 + g_\Sigma + O(e^{-\beta t})$$

where $\beta > 0$ is related to the spectral gap of the stability operator L_Σ for the strictly stable horizons considered here. The twist contribution to the metric correction is exponentially small:

$$|\mathcal{T}| = O(e^{-t}) \quad \text{along the cylindrical end,}$$

hence does not affect the asymptotic cylindrical structure.

Step 6: Uniqueness and mass preservation. *Uniqueness up to translation.* If f_1, f_2 are two solutions with blow-up along Σ, then $w = f_1 - f_2$ satisfies a linearized equation. The leading asymptotics $f_i \sim C_0 \ln s^{-1}$ cancel, leaving $w = O(1)$ near Σ. The maximum principle forces w to be bounded, and with normalization $f(x_0) = 0$ for a fixed basepoint, uniqueness follows (see [29, Theorem 3.1]).

Mass preservation. The Jang metric $\bar{g} = g + df \otimes df$ satisfies:

$$\bar{g}_{ij} - \delta_{ij} = (g_{ij} - \delta_{ij}) + O(r^{-2\tau+2\epsilon}).$$

The barrier construction at infinity gives $f = O(r^{1-\tau+\epsilon})$ and hence $|\nabla f| = O(r^{-\tau+\epsilon})$ for any fixed $\epsilon \in (0, \tau - \frac{1}{2})$. Therefore

$$df \otimes df = O(r^{-2\tau+2\epsilon}), \qquad \partial(df \otimes df) = O(r^{-2\tau+2\epsilon-1}).$$

Since $2\tau - 2\epsilon > 1$, the rank-one correction $df \otimes df$ is lower order than the ADM part of g and contributes zero to the ADM mass integral. Thus

$$M_{\mathrm{ADM}}(\bar{g}) = M_{\mathrm{ADM}}(g).$$

□

Remark 4.18 (Twist Coupling Summary). The key technical point is that twist enters the Jang equation through $\mathcal{T}[\bar{f}]$ which satisfies:

1. $|\mathcal{T}|$ is bounded on compact sets (from $\rho^2|\omega| \leq C$).
2. $|\mathcal{T}| \to 0$ as $s \to 0$ (scaling as $O(s)$ near the blow-up).
3. $|\mathcal{T}| = O(r^{-2})$ at infinity (faster than the principal terms).

These three properties ensure that the Han–Khuri existence theory applies with twist as a perturbation. The proof does **not** require twist to vanish, only that it be asymptotically negligible in the singular limits.

Remark 4.19 (Uniqueness of Jang Solutions). The Jang equation does **not** admit unique solutions in general. For initial data (M, g, K) with a strictly stable outermost MOTS Σ, the solution space has the following structure:

1. **Existence:** By Theorem 4.12, there exists at least one solution f blowing up at Σ with prescribed logarithmic asymptotics.
2. **Uniqueness up to translation:** If f_1 and f_2 are two solutions with the same blow-up behavior at Σ, then $f_1 - f_2$ is bounded and, with the normalization $f(x_0) = 0$ at a fixed basepoint $x_0 \in M \setminus \Sigma$, the solution is unique [29, Theorem 3.1].
3. **Multiple blow-up surfaces:** If the initial data contains multiple MOTS (inner and outer), there may exist distinct solutions blowing up at different surfaces. Our proof uses the **outermost** MOTS Σ as specified in hypothesis (H4).
4. **Impact on the inequality:** The non-uniqueness does not affect the validity of the AM-Penrose inequality. Any solution blowing up at the outermost MOTS yields the same bound, since the ADM mass and the geometric quantities (A, J) at Σ are independent of the choice of Jang solution.

The essential point is that the Jang equation serves as a **regularization tool**—different solutions lead to the same final inequality because the boundary terms (at Σ and at infinity) depend only on the geometry of (M, g, K), not on the intermediate Jang surface.

Remark 4.20 (Key Estimate Verification Guide). **For readers verifying this proof**, the critical estimate in this section is the scaling $\mathcal{T} = O(s)$ as $s \to 0$ (Step 2c). This follows from:

- The blow-up asymptotics $|\nabla f| \sim C_0/s$ (from Han–Khuri [29, Prop. 4.5]);
- The bounded twist $|\omega| \leq C_\omega$ (from elliptic regularity of the momentum constraint);
- The ρ^2 scaling of the twist term: $\mathcal{T} \propto \rho^2$, which vanishes at the poles where $\rho = 0$ (Lemmas 4.6 and 4.8).

The estimate $\mathcal{T} = O(s)$ is subdominant to the principal terms $O(s^{-1})$ by a factor of s^2, ensuring the perturbation analysis in Lemma 4.15 applies.

Remark 4.21 (Cylindrical End Structure). The induced metric $\bar{g}$ on the Jang manifold has cylindrical ends with the asymptotic structure:

$$\bar{g} = dt^2 + h_\Sigma(1 + O(e^{-\beta t})) \quad \text{as } t \to \infty,$$

where h_Σ is the induced metric on Σ and $\beta > 0$. This exponential convergence is essential for:

- Fredholm theory for the Lichnerowicz operator (Section 5).
- The p-harmonic potential having well-defined level sets (Section 6).
- Angular momentum conservation across the cylindrical end (Theorem 6.13).

5 Stage 2: AM-Lichnerowicz Equation

5.1 The Conformal Equation

On the Jang manifold $(\bar{M}, \bar{g})$, we solve a modified Lichnerowicz equation that accounts for angular momentum. The cylindrical end structure from Theorem 4.12 requires Lockhart–McOwen weighted Sobolev spaces for Fredholm theory.

Definition 5.1 (Weighted Sobolev Spaces on Cylindrical Ends)**.** Let $(\bar{M}, \bar{g})$ have cylindrical ends $\mathcal{C} \cong [0, \infty) \times \Sigma$ with coordinate t and cross-section (Σ, g_Σ). For $k \in \mathbb{N}_0$, $p \in [1, \infty)$, and weight $\beta \in \mathbb{R}$, define the weighted Sobolev space:

$$W^{k,p}_\beta(\bar{M}) := \{u \in W^{k,p}_{\text{loc}}(\bar{M}) : \|u\|_{W^{k,p}_\beta} < \infty\},$$

where the norm on the cylindrical end is:

$$\|u\|^p_{W^{k,p}_\beta(\mathcal{C})} := \sum_{j=0}^{k} \int_0^\infty \int_\Sigma e^{-\beta p t} |\nabla^j u|^p \, dA_{g_\Sigma} \, dt,$$

with $|\nabla^j u|$ denoting the norm of the j-th covariant derivative. In the asymptotically flat end, the standard weighted norm from Definition 4.1 applies.

A function $u \in W^{k,p}_\beta$ with $\beta < 0$ decays as $t \to \infty$ on the cylindrical end: $|u(t, \cdot)| = O(e^{\beta t}) \to 0$. For $\beta > 0$, such functions may grow. The Lockhart–McOwen theory [35] shows that the Laplacian $\Delta_{\bar{g}} : W^{k+2,p}_\beta \to W^{k,p}_\beta$ is Fredholm when β avoids the **indicial roots**—values determined by the spectrum of the cross-sectional Laplacian Δ_Σ.

Remark 5.2 (Compatibility of Function Spaces). The Jang manifold $(\bar{M}, \bar{g})$ has two distinct asymptotic regions requiring different function space frameworks:

(i) **Asymptotically flat end:** Weighted Hölder spaces $C^{k,\beta}_{-\tau}$ with polynomial weight $r^{-\tau}$ (Definition 4.1);

(ii) **Cylindrical end:** Weighted Sobolev spaces $W^{k,p}_\beta$ with exponential weight $e^{\beta t}$ (Definition 5.1).

These frameworks are compatible on the transition region $\{R_0 \leq r \leq 2R_0\}$ (equivalently $\{0 \leq t \leq T_0\}$) in the following sense: by Sobolev embedding, $W^{k+1,2}_\beta \hookrightarrow C^{k,\beta}$ locally, and both norms are equivalent (up to constants depending on R_0) on the compact overlap region. This allows elliptic estimates to be "glued" across the transition using standard partition-of-unity arguments. The key point is that the Fredholm index is determined by the asymptotic behavior at both ends, not the transition region.

Definition 5.3 (AM-Lichnerowicz Operator)**.** The angular-momentum-modified Lichnerowicz equation is:

$$L_{AM}[\phi] := -8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi - \Lambda_J\phi^{-7} = 0, \tag{31}$$

where $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}} \geq 0$ is the Kerr deviation contribution (Definition 1.10). The **negative** sign in front of Λ_J ensures that the conformal scalar curvature $R_{\tilde{g}} = \Lambda_J\phi^{-12} \geq 0$.

Key property: For Kerr initial data, $\Lambda_J = 0$ (since $\mathcal{S}_{(g,K)} = 0$), and the equation reduces to the standard Lichnerowicz equation $-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = 0$.

Remark 5.4 (Sign Convention Verification). We verify the sign conventions in the AM-Lichnerowicz equation:

(i) **Conformal transformation formula:** Under $\tilde{g} = \phi^4\bar{g}$, the scalar curvatures are related by:

$$R_{\tilde{g}} = \phi^{-4}R_{\bar{g}} - 8\phi^{-5}\Delta_{\bar{g}}\phi = \phi^{-5}(R_{\bar{g}}\phi - 8\Delta_{\bar{g}}\phi).$$

(ii) **AM-Lichnerowicz rearrangement:** From (31):

$$-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = \Lambda_J\phi^{-7} \quad \Rightarrow \quad R_{\tilde{g}} = \phi^{-5}\cdot\Lambda_J\phi^{-7} = \Lambda_J\phi^{-12}.$$

(iii) **Positivity:** Since $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2 \geq 0$ and $\phi > 0$, we have $R_{\tilde{g}} \geq 0$ automatically.

(iv) **Strict positivity:** $R_{\tilde{g}} > 0$ where $\mathcal{S}_{(g,K)} \neq 0$, i.e., where the data deviates from Kerr geometry.

(v) **Equality case:** For Kerr data, $\Lambda_J = 0$, so $R_{\tilde{g}} = 0$ and the monotonicity integrand vanishes.

The convention matches the standard Lichnerowicz equation $-8\Delta\phi + R\phi = 0$ (for $R_{\tilde{g}} = 0$), with the $\Lambda_J\phi^{-7}$ term producing positive conformal scalar curvature for non-Kerr data.

Lemma 5.5 (Well-Definedness of Λ_J)**.** *The angular momentum source term*

$$\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}}$$

is a ***well-defined, coordinate-independent, non-negative scalar function*** *on asymptotically flat, axisymmetric vacuum initial data, provided the Kerr deviation tensor $\mathcal{S}_{(g,K)}$ is fixed intrinsically from (g, K) and the asymptotic parameters (M, J) as described in Definition 1.10 and Appendix G. Moreover:*

(a) $\Lambda_J \geq 0$ *everywhere;*

(b) Λ_J *is coordinate-independent;*

(c) *if* $\mathcal{S}_{(g,K)} = O(r^{-2-\tau})$ *on the asymptotically flat end, then* $\Lambda_J = O(r^{-4-2\tau})$*;*

(d) *Kerr initial data satisfies* $\Lambda_J = 0$.

In the present paper the converse implication "$\Lambda_J = 0$ implies Kerr" is ***not*** *proved inside this lemma; it is isolated as the external rigidity input used in Section 9.*

Proof. Non-negativity is immediate from the definition as a squared norm. Coordinate independence follows because $\mathcal{S}_{(g,K)}$ is a tensor field and the norm $|\cdot|_{\bar{g}}$ is geometric. On the asymptotically flat end, the assumed decay $\mathcal{S}_{(g,K)} = O(r^{-2-\tau})$ gives

$$\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}} = O(r^{-4-2\tau}).$$

Finally, the normalization of $\mathcal{S}_{(g,K)}$ is chosen so that it vanishes on Kerr slices, hence $\Lambda_J = 0$ for Kerr. The converse implication is not needed in this section and is deferred to the external rigidity input discussed in Section 9 and Appendix G. □

Lemma 5.6 (Regularity of Λ_J in Weighted Spaces)**.** *Let (M, g, K) be asymptotically flat, axisymmetric vacuum initial data with decay rate $\tau > 1/2$ and outermost strictly stable MOTS Σ. Let $(\bar{M}, \bar{g})$ be the Jang manifold with cylindrical end $\mathcal{C} \cong [0, \infty)_t \times \Sigma$. Then the angular momentum source term $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}}$ satisfies:*

(i) ***Global boundedness:*** *$\Lambda_J \in L^\infty(\bar{M})$ with explicit bound*

$$\|\Lambda_J\|_{L^\infty(\bar{M})} \leq C(g, K) < \infty,$$

where $C(g, K)$ depends only on the $C^{2,\beta}$ norms of the initial data.

(ii) ***Asymptotic decay:*** *On the asymptotically flat end,*

$$\Lambda_J(x) = O(r^{-4-2\tau}) \quad \text{as } r \to \infty.$$

(iii) ***Cylindrical end decay:*** *If, in addition, the chosen Kerr deviation tensor satisfies* $|\mathcal{S}_{(g,K)}|_{\bar{g}} = O(e^{-\beta_0 t/2})$ *on the cylindrical end, then*

$$\Lambda_J(t, y) = O(e^{-\beta_0 t}) \quad \text{as } t \to \infty,$$

where $\beta_0 > 0$ *is the cylindrical decay rate from Theorem 4.12.*

(iv) ***Weighted space membership:*** *Under the same cylindrical decay hypothesis, for any* $\beta \in (-\beta_0/2, 0)$ *and* $k \geq 0$*:*

$$\Lambda_J \in W^{k,2}_{\beta}(\bar{M}) \cap C^{k,\beta}_{-4-2\tau}(\bar{M}).$$

Proof. Boundedness on compact subsets follows from the regularity of $\mathcal{S}_{(g,K)}$ as a tensor built from (g, K) and the equivalence of g and $\bar{g}$ away from the asymptotic ends. The decay at spatial infinity is exactly the content of Lemma 5.5. If one additionally assumes the displayed exponential decay of $|\mathcal{S}_{(g,K)}|_{\bar{g}}$ along the cylindrical end, then squaring gives the stated exponential decay for Λ_J, and the weighted-space inclusion follows immediately. □

Theorem 5.7 (AM-Lichnerowicz Existence)**.** *Let* $(\bar{M}, \bar{g})$ *be the Jang manifold from Theorem 4.12, with cylindrical end* $\mathcal{C} \cong [0, \infty)_t \times \Sigma$*. Let* $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}} \geq 0$ *be the angular momentum source term (Definition 1.10). Assume in addition that* Λ_J *satisfies the cylindrical-end decay/integrability hypothesis recorded in Lemma 5.6(iii)–(iv).*

There exists a unique positive solution $\phi > 0$ *to the AM-Lichnerowicz equation:*

$$-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = \Lambda_J \phi^{-7} \tag{32}$$

with boundary conditions:

(i) $\phi|_{\Sigma} = 1$ *(Dirichlet condition at the MOTS);*

(ii) $\phi \to 1$ *as* $r \to \infty$ *in the asymptotically flat end.*

The solution satisfies:

(a) ***Positivity:*** $\phi > 0$ *throughout* $\bar{M}$ *(unconditional);*

(b) ***Upper bound:*** $\phi \leq 1$ *throughout* $\bar{M}$ ***conditional*** *on Lemma E.3; this is one possible route to the auxiliary mass-comparison input (H7);*

(c) ***Asymptotic decay:*** $\phi = 1 + O(r^{-\tau})$ *at spatial infinity;*

(d) ***Cylindrical decay:*** $|\phi - 1| = O(e^{-\kappa t})$ *along the cylindrical end for some* $\kappa > 0$*;*

(e) ***Regularity:*** $\phi \in C^{2,\beta}(\bar{M})$*;*

(f) ***Conformal scalar curvature:*** *The conformal metric* $\tilde{g} = \phi^4 \bar{g}$ *satisfies* $R_{\tilde{g}} = \Lambda_J \phi^{-12} \geq 0$*.*

Important: *The existence proof via the variational method (Proof A below) uses* $R_{\bar{g}} \geq 0$ *together with the source-term decay hypothesis from Lemma 5.6(iii)–(iv). The upper bound* $\phi \leq 1$ *is established via the supersolution method (Proof B) which additionally requires the refined estimate* $R_{\bar{g}} \geq 2\Lambda_J$*. The main monotonicity argument uses only positivity of* ϕ*, the asymptotic normalization, and the scalar-curvature identity.*

Remark 5.8 (Removal of the Auxiliary Input (H7) via Cohomological Mass Comparison). The auxiliary conformal mass comparison input (H7) of Theorem 1.2 is resolved in Section 11 by the Conformal Cohomological Mass Comparison Theorem 11.14. The proof uses a twisted Hodge decomposition (Lemma 11.13) on the cylindrical end of the Jang manifold, with the twist adapted to the Komar form. The vanishing of the L^2 twisted cohomology $H^1_{\omega,L^2}(\bar{M}) = 0$ (a consequence of the cylindrical-end topology and the Lockhart–McOwen Fredholm theorem) reduces the ADM mass defect to a non-negative boundary integral on the MOTS, yielding $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$. Consequently, hypothesis (H7) of Theorem 1.2 can be removed (Corollary 11.15), conditional on the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3); see the strengthened Theorem 11.24.

Proof. We split the discussion into two parts: (A) a variational existence proof that requires only $R_{\bar{g}} \geq 0$ together with the stated source-term decay hypothesis, and (B) a conditional comparison argument that yields the a priori bound $\phi \leq 1$ once the refined bound is available. Only Part (A) is used to establish existence under the stated hypotheses of this theorem.

Proof A: Variational Existence (under the source-term decay hypothesis).

This proof requires $R_{\bar{g}} \geq 0$ (the classical Bray–Khuri bound under DEC) and the source-term decay from Lemma 5.6(iii)–(iv), but does **not** rely on the refined estimate $R_{\bar{g}} \geq 2\Lambda_J$.

Step A1: Functional framework. Define the energy functional on $W^{1,2}_\beta(\bar{M})$:

$$\mathcal{E}[\phi] := \int_{\bar{M}} \left(4|\nabla\phi|^2_{\bar{g}} + \frac{1}{8} R_{\bar{g}} \phi^2 + \frac{\Lambda_J}{6} \phi^{-6} \right) dV_{\bar{g}},$$

where $\beta < 0$ is chosen so that functions in $W^{1,2}_\beta$ decay exponentially on the cylindrical end. The Euler–Lagrange equation is precisely the AM-Lichnerowicz equation (32).

Step A2: Coercivity. Since $R_{\bar{g}} \geq 0$ (Bray–Khuri under DEC), the quadratic terms are non-negative:

$$\int_{\bar{M}} \left(4|\nabla\phi|^2 + \frac{1}{8} R_{\bar{g}} \phi^2 \right) dV_{\bar{g}} \geq 4\|\nabla\phi\|^2_{L^2}.$$

The ϕ^{-6} term provides a barrier preventing $\phi \to 0$: for any sequence $\phi_n \to 0$ in L^6, the term $\int \Lambda_J \phi_n^{-6}$ diverges where $\Lambda_J > 0$.

Step A3: Lower bound and minimizer. On the constraint set $\mathcal{C} := \{\phi \in W^{1,2}_\beta : \phi > 0,\ \phi|_\Sigma = 1,\ \phi \to 1 \text{ at } \infty\}$:

- $\mathcal{E}[\phi] > -\infty$ since all terms are bounded below (the ϕ^{-6} term is positive).
- $\mathcal{E}[\phi] < +\infty$ for the test function $\phi \equiv 1$, giving $\mathcal{E}[1] = \frac{1}{8}\int R_{\bar{g}} + \frac{1}{6}\int \Lambda_J < \infty$.

By the direct method of calculus of variations, there exists a minimizer $\phi_* \in \mathcal{C}$ with $\mathcal{E}[\phi_*] = \inf_{\mathcal{C}} \mathcal{E}$.

Step A4: Positivity of minimizer. The minimizer satisfies $\phi_* > 0$ everywhere. If $\phi_*(x_0) = 0$ for some x_0, then $\int_{\bar{M}} \Lambda_J \phi_*^{-6} = +\infty$ where $\Lambda_J(x_0) > 0$, contradicting $\mathcal{E}[\phi_*] < \infty$. On regions where $\Lambda_J = 0$ (Kerr-like regions), the strong maximum principle for the linearized equation $-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = 0$ with $R_{\bar{g}} \geq 0$ ensures $\phi > 0$.

Step A5: Regularity. The minimizer satisfies the weak form of (32). Since $\phi_* > 0$ is bounded away from zero (by Step A4), the nonlinearity $\Lambda_J \phi^{-7}$ is Lipschitz in ϕ. Standard elliptic regularity (bootstrapping from $W^{1,2}$ to $C^{2,\beta}$) gives $\phi_* \in C^{2,\beta}(\bar{M})$.

Step A6: Exponential decay on cylindrical end. The boundary condition $\phi|_\Sigma = 1$ is interpreted as $\phi \to 1$ along the cylindrical end. Setting $\psi = \phi - 1$, the linearization on the cylinder is:

$$-8\partial_t^2\psi - 8\Delta_\Sigma\psi + R_{\bar{g}}\psi = O(e^{-\beta_0 t}),$$

where $\beta_0 > 0$ is the exponential decay rate of the metric perturbation. By spectral theory on the cylinder (Lockhart–McOwen [35]), solutions with $\psi \to 0$ as $t \to \infty$ satisfy $|\psi| = O(e^{-\kappa t})$ for $\kappa > 0$ depending on the spectral gap of L_Σ.

Proof B: Conditional Comparison Argument for $\phi \leq 1$.

Assuming the refined bound $R_{\bar{g}} \geq 2\Lambda_J$, the constant function 1 is a supersolution. Since existence has already been obtained in Part (A), we use this only to compare the actual solution with 1.

Step B1: Super-solution. *Assuming* $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3), the constant $\bar{\phi} = 1$ satisfies:

$$L_{AM}[1] = -8\Delta_{\bar{g}}(1) + R_{\bar{g}}(1) - \Lambda_J(1)^{-7} = R_{\bar{g}} - \Lambda_J \geq 2\Lambda_J - \Lambda_J = \Lambda_J \geq 0.$$

Thus $\bar{\phi} = 1$ is a super-solution.

Step B2: Comparison with the variational solution. Let ϕ be the positive solution from Part (A). The asymptotic conditions give $\phi \to 1$ both at spatial infinity and along the cylindrical end. If $\phi > 1$ somewhere, then $w := (\phi - 1)_+$ would attain a positive interior maximum. The difference between the equation for ϕ and the supersolution inequality for 1 has the form

$$-8\Delta_{\bar{g}}(\phi - 1) + c(x)(\phi - 1) \leq 0$$

on the region where $\phi \geq 1$, with $c(x) \geq 0$. The maximum principle therefore forces $w \equiv 0$, so $\phi \leq 1$ everywhere.

Uniqueness (common to both proofs). Suppose ϕ_1, ϕ_2 are two positive solutions. Setting $w = \phi_1 - \phi_2$ and linearizing:

$$-8\Delta_{\bar{g}}w + R_{\bar{g}}w + 7\Lambda_J\xi^{-8}w = 0$$

for some ξ between ϕ_1 and ϕ_2. Since $R_{\bar{g}} \geq 0$ and $7\Lambda_J\xi^{-8} \geq 0$, the operator has non-negative zero-th order term, and by the maximum principle with Dirichlet conditions ($w = 0$ on boundaries), we have $w = 0$.

Conformal scalar curvature. By direct calculation using the conformal transformation formula:

$$R_{\tilde{g}} = \phi^{-5}(R_{\bar{g}}\phi - 8\Delta_{\bar{g}}\phi) = \phi^{-5} \cdot \Lambda_J \phi^{-7} = \Lambda_J \phi^{-12} \geq 0.$$

This holds for any positive solution ϕ, independent of whether $\phi \leq 1$. □

Lemma 5.9 (Conformal Factor Bounds)**.** *The solution ϕ from Theorem 5.7 satisfies:*

(i) $\phi \leq 1$ *throughout $\bar{M}$ (super-solution bound)* ***assuming the refined Bray–Khuri bound*** $R_{\bar{g}} \geq 2\Lambda_J$ ***from Lemma E.3****;*

(ii) $|\phi - 1| = O(e^{-\kappa t})$ *along the cylindrical end $\mathcal{C}$ for some $\kappa > 0$ determined by the strict-stability spectral gap;*

(iii) *The mass bound $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g)$ follows from part (i) when the refined Bray–Khuri bound is available. Without that input, Appendix E records only heuristic routes toward the same comparison.*

Proof. Part (i) follows from Step 1 of Theorem 5.7 by the maximum principle, **conditional on the validity of Lemma E.3**. See Remark 5.10 below for discussion of this conditionality.

Part (ii): Under the cylindrical decay hypothesis recorded in Lemma 5.6(iii), one has $\Lambda_J = O(e^{-\beta t})$ for some $\beta > 0$ on the cylindrical end. The linearized equation around $\phi = 1$ becomes:

$$-8\Delta_{\bar{g}}\psi + R_{\bar{g}}\psi \approx 0 \quad \text{on } \mathcal{C},$$

where $\psi = \phi - 1$. On the product $[0, \infty) \times \Sigma$, separation of variables shows that solutions decay as $e^{-\kappa t}$ where κ is related to the spectrum of the stability operator on Σ.

Part (iii) follows from the conformal mass formula and the fact that $\phi = 1 + O(r^{-\tau})$ at infinity:

- If part (i) holds ($\phi \leq 1$), then the conformal mass formula directly gives $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$; see standard references [9].
- Without part (i), the appendix discusses possible energy-identity routes, but those arguments are presented only as heuristic reductions in the current version.

The inequality $M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g)$ is the Han–Khuri mass bound [29], independent of the conformal factor. □

Remark 5.10 (Conditional vs. Unconditional Statements). It is essential to distinguish which parts of Lemma 5.9 are conditional on the refined bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3):

Conditional (requires Lemma E.3):

- Part (i): The pointwise bound $\phi \leq 1$ throughout $\bar{M}$.

Unconditional (does NOT require Lemma E.3):

- Part (ii): The exponential decay $|\phi - 1| = O(e^{-\kappa t})$ along the cylindrical end.
- The Han–Khuri comparison $M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g)$.

Why this matters: The clean rigorous route from the conformal equation to the mass inequality uses the pointwise bound $\phi \leq 1$. Appendix E records other ideas for this step, but they are not used as completed proofs in the present manuscript.

The pointwise bound $\phi \leq 1$ (part (i)) is a *stronger* result that, if true, provides additional geometric information about the conformal factor. It is presented here for completeness and because it may be useful for future work.

Remark 5.11 (Key Estimates). The critical estimates in this section are:

- The **conformal scalar curvature identity** $R_{\tilde{g}} = \Lambda_J \phi^{-12} \geq 0$ (Remark 5.4), ensuring the conformal metric has nonnegative scalar curvature. This is a **direct calculation** from the conformal transformation formula and the AM-Lichnerowicz equation.
- The **mass chain inequality** $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g)$ (Lemma 5.9(iii)). The rigorous route recorded here is:
  (a) If the refined bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3) holds, then the maximum principle gives $\phi \leq 1$, which via the conformal mass formula yields $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$.
  (b) Appendix E contains only heuristic reductions for alternative routes to the same comparison.

The refined bound $R_{\bar{g}} \geq 2\Lambda_J$ is therefore the clean rigorous route to the pointwise estimate $\phi \leq 1$ used here. See Section 1.7 for further discussion.

6 Stage 3: AMO Flow with Angular Momentum

6.1 The p-Harmonic Potential

On $(\tilde{M}, \tilde{g})$, we solve the p-Laplace equation:

$$\Delta_p u_p := \operatorname{div}(|\nabla u_p|^{p-2} \nabla u_p) = 0, \tag{33}$$

with boundary conditions:

- **At the horizon:** $u_p|_\Sigma = 0$, interpreted as $\lim_{t\to\infty} u_p(t, y) = 0$ along the cylindrical end $\mathcal{C} \cong [0, \infty) \times \Sigma$ for any cylindrical parameter t increasing toward the MOTS;
- **At infinity:** $u_p \to 1$ as $r \to \infty$ in the asymptotically flat end.

Remark 6.1 (Well-Posedness of the Boundary Value Problem). The cylindrical end geometry requires careful formulation. The boundary condition $u_p|_\Sigma = 0$ is a Dirichlet condition "at infinity" along the cylinder. Existence and uniqueness are formulated on truncated cylinders and then passed to the limit using weighted variational methods, with the asymptotic conditions $u_p \to 0$ on the cylindrical end and $u_p \to 1$ at spatial infinity imposed as part of the admissible class. The exponential approach to the limiting values is obtained from the elliptic analysis on the asymptotically cylindrical end; it is not a consequence of the sign of the Sobolev weight alone. See [1, Section 4] for the $p \to 1$ framework used here.

Lemma 6.2 (Axisymmetry of Solution)**.** *For axisymmetric data (M, g, K) and axisymmetric boundary conditions, the p-harmonic potential u_p is axisymmetric: $u_p = u_p(r, z)$.*

Remark 6.3 (Regularity of p-Harmonic Functions). The p-harmonic potential u_p is $C^{1,\beta}$ by the Tolksdorf–Lieberman regularity theory [56].

Critical set structure. The set of critical points $\mathcal{Z}_p := \{x : \nabla u_p(x) = 0\}$ requires careful analysis because the classical Sard theorem requires C^n regularity for functions on n-dimensional manifolds, which $C^{1,\beta}$ regularity does not provide. For the arguments in this paper, we only need the weaker fact that regular values form a full-measure subset of $(0, 1)$ and that the coarea formula applies. This is supplied by the standard structure theory for p-harmonic functions together with the AMO regular-value approximation. In particular:

1. **Small critical set (Heinonen–Kilpeläinen–Martio [30]):** For p-harmonic functions $u : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, the critical set is quantitatively small; in dimension $n = 3$ one has the bound $\dim_{\mathcal{H}}(\mathcal{Z}_p) \leq 1$.

2. **Regular leaves for a.e. level:** Combining this structure theory with the coarea formula yields that for almost every $t \in (0, 1)$, the level set $\Sigma_t = \{u_p = t\}$ is a $C^{1,\beta}$ hypersurface and the geometric quantities entering the AMO formulas are well-defined on that leaf.

3. **Approximation by regular values:** The monotonicity formulas are first established on regular leaves and then extended to all t by approximation through regular values. No sharper description of $\mathcal{Z}_p$ is needed here.

This is exactly the level of regularity needed below: the level sets are well-defined for almost every $t \in (0, 1)$, and the exceptional set can be handled by regular-value approximation.

Comparison with classical Sard theorem: The classical Sard theorem states that the set of critical values of a C^k function $f : M^n \to \mathbb{R}$ has measure zero when $k \geq n$. For $n = 3$, this would require C^3 regularity, which p-harmonic functions do not achieve (they are only $C^{1,\beta}$). The results of Heinonen–Kilpeläinen–Martio circumvent this by exploiting the specific structure of p-harmonic equations rather than general smoothness.

Remark 6.4 (Regularity Near Cylindrical Ends). The p-harmonic potential requires careful analysis near the cylindrical end $\mathcal{C} \cong [0, \infty) \times \Sigma$ where the metric satisfies $\tilde{g} = dt^2 + g_\Sigma + O(e^{-\beta t})$.

Boundary conditions at the cylindrical end. The condition $u_p|_\Sigma = 0$ is imposed on the "end" of the cylinder, which in the original coordinates corresponds to the MOTS Σ. In a cylindrical coordinate t increasing toward the end, the boundary Σ is represented by $t = +\infty$. The boundary condition becomes:

$$\lim_{t \to \infty} u_p(t, y) = 0 \quad \text{uniformly in } y \in \Sigma.$$

Asymptotic behavior. On the exact cylinder $\mathbb{R}_+ \times \Sigma$ with metric $dt^2 + g_\Sigma$, the p-harmonic equation reduces to:

$$\partial_t(|\partial_t u|^{p-2}\partial_t u) + \Delta_{\Sigma,p}(u) = 0.$$

The only property needed here is that the admissible solution approaches the constant state 0 as $t \to \infty$ and remains monotone toward spatial infinity.

The exponentially decaying perturbation of the product metric preserves this qualitative asymptotic behavior.

Gradient control on compact truncations. Standard interior and boundary estimates for p-harmonic functions [56] give, on each truncated cylinder $\{0 \le t \le T\}$,

$$|\nabla_{\tilde{g}} u_p| \le C(p, T),$$

which is sufficient for passing to the limit on regular leaves. We do not require a global uniform lower bound for $|\nabla u_p|$ on the entire cylindrical end.

Measure of critical points. The set $\{\nabla u_p = 0\}$ is negligible for the level-set analysis by the structure theory cited in Remark 6.3. For the arguments below it is enough that regular values are dense and of full measure, so we work on regular leaves and pass to limits in t when approaching the cylindrical end.

Remark 6.5 (Regularity Near the Rotation Axis). The rotation axis $\Gamma = \{\eta = 0\}$ requires special treatment because the axisymmetric metric degenerates there: $g_{\phi\phi} = \rho^2 \to 0$ as $\rho \to 0$ (where ρ is the cylindrical radius from the axis). We establish that the p-harmonic potential and level sets remain regular at axis points.

Axis regularity of the p-harmonic potential. In Weyl–Papapetrou coordinates (t, ρ, z, ϕ) adapted to axisymmetry, the metric takes the form:

$$\tilde{g} = e^{2\gamma}(d\rho^2 + dz^2) + e^{2\psi}\rho^2 d\phi^2,$$

where γ, ψ are functions of (ρ, z) only. Near the axis $\rho = 0$, regularity requires $e^{2\psi} \to 1$ and $\gamma \to 0$ as $\rho \to 0$ (elementary flatness condition).

For an axisymmetric p-harmonic function $u = u(\rho, z)$, the equation becomes:

$$\frac{1}{\rho}\partial_\rho \left(\rho e^{(\gamma-\psi)}|\nabla u|^{p-2}\partial_\rho u\right) + \partial_z \left(e^{(\gamma-\psi)}|\nabla u|^{p-2}\partial_z u\right) = 0.$$

Near $\rho = 0$, this reduces to a Laplace-type equation in the (ρ, z) half-plane with Neumann boundary conditions $\partial_\rho u|_{\rho=0} = 0$ (by axisymmetry). Standard elliptic theory on domains with symmetry boundaries [45] gives:

$$u \in C^{1,\beta}(\overline{M}) \quad \text{including the axis } \Gamma.$$

Level set behavior at axis intersection points. The level sets $\Sigma_t = \{u = t\}$ intersect the axis Γ at isolated points (for generic t). Near such a point $p \in \Sigma_t \cap \Gamma$:

1. The level set Σ_t is smooth (by implicit function theorem, since $\nabla u \neq 0$ generically);

2. The surface Σ_t meets the axis orthogonally (by axisymmetry: Σ_t is rotationally symmetric about Γ);

3. The mean curvature H and second fundamental form h are bounded at p;

4. The Komar integrand $K(\eta, \nu)$ vanishes at p since $\eta = 0$ there, so the axis contributes zero to the angular momentum integral.

MOTS-axis intersection. The outermost MOTS Σ intersects the axis at exactly two points (the "poles") by topological considerations ($\Sigma \cong S^2$). At these poles:

- The stability operator L_Σ has smooth coefficients extending to the poles;

- The Dain–Reiris inequality $A \geq 8\pi|J|$ accounts for the axis contribution correctly (the proof in [22] handles axis regularity explicitly).

Conclusion. The axis singularity of axisymmetric coordinates is a *coordinate artifact*, not a geometric singularity. All geometric quantities (area, mean curvature, Komar integrals) are well-defined and finite. The p-harmonic flow respects axisymmetry and produces level sets that are smooth embedded surfaces intersecting the axis at isolated points with controlled geometry.

Lemma 6.6 (Level Set Homology Preservation)**.** *Let $u : \tilde{M} \to [0, 1]$ be the p-harmonic potential with $u|_\Sigma = 0$ and $u \to 1$ at infinity. For regular values $t_1, t_2 \in (0, 1)$, the level sets Σ_{t_1} and Σ_{t_2} are homologous in M:*

$$[\Sigma_{t_1}] = [\Sigma_{t_2}] \in H_2(M; \mathbb{Z}).$$

In particular, all level sets are homologous to the outermost MOTS Σ.

Proof. **Step 1: Topological setup.** The domain $\tilde{M} \setminus \Sigma$ is diffeomorphic to $M \setminus \Sigma$ (the Jang and conformal constructions preserve the underlying smooth manifold). The p-harmonic function $u : M \setminus \Sigma \to (0, 1)$ is a proper submersion at regular values, which form a set of full measure by Remark 6.3.

Step 2: Cobordism between level sets. For regular values $t_1 < t_2$, the region

$$W := u^{-1}([t_1, t_2]) = \{x \in M : t_1 \leq u(x) \leq t_2\}$$

is a compact 3-manifold with boundary $\partial W = \Sigma_{t_1} \sqcup \Sigma_{t_2}$. This is the definition of a **cobordism** between Σ_{t_1} and Σ_{t_2}.

Step 3: Homology computation. By the long exact sequence of the pair $(W, \partial W)$:

$$\cdots \to H_3(W, \partial W) \xrightarrow{\partial} H_2(\partial W) \xrightarrow{i_*} H_2(W) \to \cdots$$

The boundary map $\partial : H_3(W, \partial W) \to H_2(\partial W)$ sends $[W]$ to $[\partial W] = [\Sigma_{t_2}] - [\Sigma_{t_1}]$ (with appropriate orientations). Therefore:

$$[\Sigma_{t_2}] - [\Sigma_{t_1}] \in \ker(i_*) = \text{image}(\partial).$$

In $H_2(M; \mathbb{Z})$, the inclusion $W \hookrightarrow M$ shows:

$$[\Sigma_{t_1}] = [\Sigma_{t_2}] \in H_2(M; \mathbb{Z}).$$

Step 4: Extension to all level sets. For any $t \in (0, 1)$, by the critical set structure (Remark 6.3), there exists a sequence of regular values $t_n \to t$. The level sets Σ_{t_n} converge to Σ_t in the Hausdorff topology. Since homology classes are locally constant (level sets are locally products near regular values), $[\Sigma_t] = [\Sigma_{t_n}]$ for n sufficiently large.

Step 5: Continuity to the boundary. As $t \to 0^+$, the level sets Σ_t converge to the MOTS Σ along the cylindrical end. The gradient bound from Remark 6.4 ensures this convergence is controlled. Since the surfaces remain embedded and connected throughout, $[\Sigma_t] = [\Sigma]$ for all $t \in (0, 1)$.

Step 6: Level sets remain in the vacuum region. By hypothesis (H3) of Theorem 1.2, the data is **vacuum in the exterior region**—i.e., the region $M_{\text{ext}} := M \setminus \overline{\text{Int}(\Sigma)}$ outside the outermost MOTS satisfies $\mu = |j| = 0$. All level sets Σ_t for $t \in (0, 1)$ lie in this exterior region:

- At $t = 0$, $\Sigma_0 = \Sigma$ is the outermost MOTS (boundary of M_{ext}).
- For $t > 0$, Σ_t lies **outside** Σ since u increases outward (toward infinity).
- The monotonicity of u ensures $\Sigma_t \subset M_{\text{ext}}$ for all $t \in (0, 1)$.

Therefore, the co-closedness condition $d^\dagger \alpha_J = 0$ (equivalently, $d(\star\alpha_J) = 0$) holds throughout the region $\bigcup_{t\in(0,1)} \Sigma_t$ swept by the level sets, ensuring the Stokes' theorem argument applies. □

Corollary 6.7 (Topological Constancy of Komar Integrals)**.** *For any co-closed 1-form α on M (i.e., $d^\dagger\alpha = 0$, equivalently $d(\star\alpha) = 0$; in particular, the Komar form α_J under vacuum axisymmetry):*

$$\int_{\Sigma_{t_1}} \star\alpha = \int_{\Sigma_{t_2}} \star\alpha \quad \textit{for all } t_1, t_2 \in (0,1).$$

This follows immediately from Lemma 6.6 and Stokes' theorem applied to the closed 2-form $\star\alpha$.

Summary: Angular Momentum Conservation (Theorem 6.13)

1. **Setup:** Komar 1-form $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat$ on (M, g)
2. **Key identity:** Vacuum + axisymmetry $\Rightarrow d^\dagger\alpha_J = 0$ (co-closedness)
3. **Hodge duality:** $d^\dagger\alpha_J = 0 \Leftrightarrow d(\star_g\alpha_J) = 0$ in 3D
4. **Stokes:** $\int_{\Sigma_{t_2}} \star\alpha_J - \int_{\Sigma_{t_1}} \star\alpha_J = \int_W d(\star\alpha_J) = 0$
5. **Conclusion:** $J(t) = J$ constant along the flow

6.2 The AM-AMO Functional

Definition 6.8 (AM-Hawking Mass Functional)**.** Let $(\tilde{M}, \tilde{g})$ be a Riemannian 3-manifold with $R_{\tilde{g}} \geq 0$ and let $\Sigma_t = \{u = t\}$ be level sets of a function $u : \tilde{M} \to [0, 1]$. For regular values t (where $\nabla u|_{\Sigma_t} \neq 0$), define:

- **Area:** $A(t) := \int_{\Sigma_t} dA_{\tilde{g}}$
- **Mean curvature:** $H(t) := \mathrm{div}_{\tilde{g}}(\nabla u/|\nabla u|_{\tilde{g}})|_{\Sigma_t}$ (the mean curvature of Σ_t in $(\tilde{M}, \tilde{g})$)
- **Willmore functional:** $W(t) := \int_{\Sigma_t} H^2\, dA_{\tilde{g}}$ (the *unnormalized* Willmore energy)

- **Hawking mass:** $m_H(t) := \sqrt{\frac{A(t)}{16\pi}}\left(1 - \frac{W(t)}{16\pi}\right)$ (always defined; it is nonnegative when $W(t) \leq 16\pi$)

The **angular momentum modified Hawking mass** is:

$$m_{H,J}(t) := \sqrt{m_H^2(t) + \frac{4\pi J^2}{A(t)}} = \sqrt{\frac{A(t)}{16\pi}\left(1 - \frac{W(t)}{16\pi}\right)^2 + \frac{4\pi J^2}{A(t)}}, \tag{34}$$

where J is the conserved Komar angular momentum (Theorem 6.13).

Well-definedness: For sub-extremal surfaces with $A(t) \geq 8\pi|J|$ (ensured by Theorem 7.1), the argument of the outer square root is non-negative. Along our AMO foliation, $W(t) \leq 16\pi$ (Lemma 6.9), so $m_H(t) \geq 0$.

Lemma 6.9 (Willmore Bound Along the AMO Foliation)**.** *Along the p-harmonic AMO foliation in our setting, the unnormalized Willmore functional satisfies*

$$W(t) := \int_{\Sigma_t} H^2 \, dA_{\tilde{g}} \leq 16\pi \tag{35}$$

for all regular values $t \in (0,1)$. Equivalently, the normalized factor $\mathcal{W}(t) := W(t)/(16\pi)$ satisfies $\mathcal{W}(t) \in [0,1]$.

Proof. By construction, the inner leaf $\Sigma = \Sigma_0$ is minimal in $(\tilde{M}, \tilde{g})$ (Lemma 8.2), so $W(0) = 0$ and hence

$$m_H(0) = \sqrt{\frac{A(0)}{16\pi}}\left(1 - \frac{W(0)}{16\pi}\right) = \sqrt{\frac{A(0)}{16\pi}} > 0.$$

The AMO monotonicity theorem implies $m_H(t)$ is nondecreasing along the foliation (for regular t), so $m_H(t) \geq m_H(0) > 0$. Since $\sqrt{A(t)/(16\pi)} > 0$, the identity

$$m_H(t) = \sqrt{\frac{A(t)}{16\pi}}\left(1 - \frac{W(t)}{16\pi}\right)$$

forces $1 - \frac{W(t)}{16\pi} \geq 0$, i.e. $W(t) \leq 16\pi$. □

Remark 6.10 (Notation Convention: Willmore Functional). We adopt the convention that $W(t) = \int_{\Sigma_t} H^2 \, dA$ is the **unnormalized** Willmore energy (with dimension of length$^{-2}\times$ area = dimensionless for H in units of length^{-1}). The **normalized Willmore factor** appearing in the Hawking mass is $W(t)/(16\pi)$, which satisfies:

- $W(t)/(16\pi) = 0$ when $H \equiv 0$ (minimal surfaces);
- Along the AMO foliation in our setting, $W(t)/(16\pi) \in [0, 1]$ for regular t (Lemma 6.9);
- $W(t)/(16\pi) \to 1^-$ as $t \to 1$ (large coordinate spheres).

Some references define the normalized Willmore factor (Willmore deficit) as $\mathcal{W}(t) := W(t)/(16\pi)$. In this paper we use $\mathcal{W}(t)$ whenever the normalized quantity appears (so $m_H(t) = \sqrt{A(t)/(16\pi)}(1 - \mathcal{W}(t))$), and reserve $W(t)$ for the unnormalized functional $W(t) = \int_{\Sigma_t} H^2 \, dA$.

Remark 6.11 (Why the Hawking Mass is Essential). The naive functional

$$\mathcal{M}_{\text{naive}}(t) := \sqrt{A(t)/(16\pi) + 4\pi J^2/A(t)}$$

diverges as $t \to 1$ because $A(t) \to \infty$ while the curvature correction is absent. For large coordinate spheres at radius R:

$$\mathcal{M}_{\text{naive}}(t) \approx \sqrt{\frac{4\pi R^2}{16\pi}} = \frac{R}{2} \to \infty.$$

The Hawking mass m_H includes the mean curvature correction, which for large spheres satisfies:

$$\frac{W(t)}{16\pi} = \frac{1}{16\pi} \int_{\Sigma_t} H^2 \, d\sigma \approx \frac{1}{16\pi} \cdot 4\pi R^2 \cdot \frac{4}{R^2} = 1 - O(R^{-1}).$$

This regularization ensures $m_H(t) \to M_{\text{ADM}}$ as $t \to 1$ [1, 32]. The AM-extension inherits this convergence since $J^2/A(t) \to 0$.

6.3 Angular Momentum Conservation

Before stating the conservation theorem, we address several foundational questions about its formulation.

Remark 6.12 (Foundational Questions on Angular Momentum Conservation). The claim that angular momentum $J(\Sigma_t)$ is conserved along the AMO flow raises several non-trivial questions that we address explicitly:

Q1: The Komar integral is defined on (M, g, K), but the level sets Σ_t live on $(\tilde{M}, \tilde{g})$. How is $J(\Sigma_t)$ well-defined?

Answer: The key insight is the **separation of roles**:

- The **underlying smooth manifold** is the same: $M = \bar{M} = \tilde{M}$ as topological spaces (the Jang and conformal constructions are diffeomorphisms, not changes of the underlying manifold).

- The level sets $\Sigma_t = \{u = t\}$ are **embedded submanifolds of** M, defined using $\tilde{g}$ but living in the same M where (g, K) are defined.

- The Komar 1-form $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat_g$ is a well-defined 1-form on M once the **physical** metric g is fixed (it uses g to lower an index). In particular, it is independent of the **conformal** metric $\tilde{g}$ used to define the level sets.

- The integral $J(\Sigma_t) = \int_{\Sigma_t} \star_g \alpha_J$ is computed using the Hodge dual with respect to the **physical** metric g, not the conformal metric $\tilde{g}$.

Thus $J(\Sigma_t)$ is a well-defined quantity: integrate the fixed 2-form $\star_g \alpha_J$ (determined by (g, K)) over the surface Σ_t (located using $\tilde{g}$).

Q2: Does conformal transformation preserve co-closedness?

Answer: We do **not** claim that $d^\dagger_{\tilde{g}} \alpha_J = 0$. Instead:

- Co-closedness is established for the **physical** metric: $d^\dagger_g \alpha_J = 0$.

- This is equivalent to: $d(\star_g \alpha_J) = 0$, i.e., $\star_g \alpha_J$ is a **closed 2-form**.

- The exterior derivative d is **metric-independent**—it is a purely topological operation.

- Therefore $d(\star_g \alpha_J) = 0$ holds on the smooth manifold M regardless of which metric is used to parametrize surfaces.

The conservation law is a consequence of Stokes' theorem applied to the closed 2-form $\star_g \alpha_J$, not a conformal invariance statement.

Q3: Is the axial Killing field η still a symmetry of the conformal metric $\tilde{g}$?

Answer: Yes. The constructions preserve axisymmetry:

- The Jang equation with axisymmetric boundary conditions yields an axisymmetric solution f, so $\mathcal{L}_\eta \bar{g} = 0$ where $\bar{g} = g + df \otimes df$.

- The AM-Lichnerowicz equation with axisymmetric data yields an axisymmetric conformal factor ϕ, so $\mathcal{L}_\eta \tilde{g} = 0$ where $\tilde{g} = \phi^4 \bar{g}$.

- Therefore η remains a Killing field for $\tilde{g}$, and the p-harmonic flow respects the symmetry.

However, this is **not** needed for conservation: even if η were not Killing for $\tilde{g}$, the closed form $\star_g \alpha_J$ would still have constant flux through homologous surfaces.

Q4: What about the cylindrical end near the MOTS?

Answer: The Jang manifold $\bar{M}$ has a cylindrical end $\mathcal{C} \cong [0, \infty) \times \Sigma$. Key points:

- The Komar 1-form α_J extends smoothly to the cylindrical end (it is defined from (g, K), which are smooth).
- The 2-form $\star_g \alpha_J$ is closed throughout M, including the cylindrical region.
- The level sets Σ_t for t near 0 may approach the MOTS $\Sigma = \Sigma_0$, but remain in a region where $\star_g \alpha_J$ is defined.
- The flux $\int_{\Sigma_t} \star_g \alpha_J$ is continuous in t, even as $t \to 0$, by dominated convergence.

The boundary term at the cylindrical end vanishes by the asymptotic analysis in Lemma 5.9.

Conclusion: The Komar angular momentum $J(\Sigma_t) = \int_{\Sigma_t} \star_g \alpha_J$ is well-defined, and its conservation is a **topological** consequence of $d(\star_g \alpha_J) = 0$ combined with homology of level sets—not a metric property of the conformal manifold.

Theorem 6.13 (Angular Momentum Conservation—Topological)**.** *Let (M, g, K) be axisymmetric initial data with Killing field $\eta = \partial_\phi$, satisfying the **vacuum** constraint equations ($\mu = |\boldsymbol{j}| = 0$) in the exterior region $M_{\text{ext}} := M \setminus \overline{\text{Int}(\Sigma)}$. Let $u : \tilde{M} \to [0, 1]$ be the axisymmetric p-harmonic potential with level sets $\Sigma_t = \{u = t\}$. Define the Komar angular momentum:*

$$J(t) := \frac{1}{8\pi} \int_{\Sigma_t} K(\eta, \nu_t^g)\, dA_g = \int_{\Sigma_t} \star_g \alpha_J,$$

*where ν_t^g and dA_g are the unit normal and area form induced by the **physical** metric g, $\alpha_J := \frac{1}{8\pi} K(\eta, \cdot)_g^\flat$ is the Komar 1-form, and $\star_g$ is the Hodge star with respect to g. Then:*

$$J(t) = J(0) = J \quad \textit{for all } t \in [0, 1].$$

Key innovation: *The Komar angular momentum is computed on each leaf using the fixed physical metric g, while the leaves themselves are located by the p-harmonic foliation on $(\tilde{M}, \tilde{g})$. Under the vacuum hypothesis, the flux form $\star_g \alpha_J$ is closed, so its integral depends only on the homology class of the leaf.*

Mechanism: *This conservation follows from de Rham cohomology, not dynamics. The vacuum momentum constraint implies the Komar 1-form is* ***co-closed****: $d_g^\dagger \alpha_J = 0$, equivalently $d(\star_g \alpha_J) = 0$. Since all level sets Σ_t are homologous (Lemma 6.6), Stokes' theorem implies the flux integral is independent of t.*

Remark 6.14 (Physical Interpretation). In physics language, Theorem 6.13 states that under our vacuum and axisymmetry assumptions, the **absence of angular momentum flux** through Σ_t implies that the **Komar angular momentum computed on any leaf** of the foliation equals the **ADM angular momentum at infinity**. This is the gravitational analogue of how magnetic flux is conserved through surfaces in electromagnetism when $\nabla \cdot \mathbf{B} = 0$.

Remark 6.15 (Nature of Conservation—Not Dynamical). This conservation is **not** a dynamical statement about time evolution. It is a consequence of **de Rham cohomology**: the Hodge dual $\star\alpha_J$ of the Komar 1-form $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat$ is a **closed 2-form** ($d(\star\alpha_J) = 0$, equivalently $d^\dagger \alpha_J = 0$) when the momentum constraint holds in vacuum with axisymmetry. By Stokes' theorem, the flux integral $\int_\Sigma \star\alpha_J$ depends only on the **homology class** of Σ, not its specific embedding. Since all level sets Σ_t are homologous (they bound a common region), $J(t)$ is constant. This is the same principle by which magnetic flux through surfaces is conserved when $\nabla \cdot \mathbf{B} = 0$.

Proof of Theorem 6.13. The proof has three steps.

Step 1: Definition on the physical manifold. Each level set $\Sigma_t = \{u = t\}$ is an embedded surface in the underlying smooth manifold M. Although the foliation is constructed using $\tilde{g}$, the Komar flux on Σ_t is evaluated using the physical metric g:

$$J(t) = \int_{\Sigma_t} \star_g \alpha_J = \frac{1}{8\pi} \int_{\Sigma_t} K(\eta, \nu_t^g)\, dA_g.$$

No metric-independence statement is needed: the metric used in the flux integral is always g.

Step 2: Co-closedness of the Komar 1-form. For vacuum axisymmetric data ($\boldsymbol{j}_i = 0$ and $\mathcal{L}_\eta K = 0$), the Komar 1-form $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat_g$ satisfies

$$d^\dagger \alpha_J = - \star d \star \alpha_J = 0, \tag{36}$$

Setup. Define $\beta := K(\eta, \cdot)^\flat_g$, so $\beta_j = K_{ij}\eta^i$ and $\alpha_J = \frac{1}{8\pi}\beta$. The co-closedness $d^\dagger \alpha_J = 0$ is equivalent to $\nabla^j \beta_j = 0$.

Computation of $\nabla^j \beta_j$. Expanding the divergence gives

$$\nabla^j \beta_j = \nabla^j (K_{ij}\eta^i) = (\nabla^j K_{ij})\eta^i + K_{ij}(\nabla^j \eta^i). \tag{37}$$

First term: Momentum constraint. The momentum constraint reads:

$$\nabla^j K_{ij} - \nabla_i (\mathrm{tr} K) = 8\pi \boldsymbol{j}_i,$$

where $\boldsymbol{j}_i$ is the momentum density. Contracting with η^i:

$$(\nabla^j K_{ij})\eta^i = 8\pi \boldsymbol{j}_i \eta^i + \eta^i \nabla_i (\mathrm{tr} K) = 8\pi(\boldsymbol{j} \cdot \eta) + \mathcal{L}_\eta (\mathrm{tr} K).$$

By axisymmetry, $\mathcal{L}_\eta(\mathrm{tr} K) = 0$, so the first term equals $8\pi(\boldsymbol{j} \cdot \eta)$.

Second term: Killing symmetry. Using the Killing equation $\nabla^j \eta^i = -\nabla^i \eta^j$:

$$K_{ij}(\nabla^j \eta^i) = -K_{ij} \nabla^i \eta^j.$$

Since K_{ij} is symmetric and $\nabla^i \eta^j$ is antisymmetric (Killing equation), their contraction vanishes:

$$K_{ij}(\nabla^j \eta^i) = 0.$$

Conclusion. Combining these results in (37):

$$\nabla^j \beta_j = 8\pi(\boldsymbol{j} \cdot \eta) + 0 = 8\pi(\boldsymbol{j} \cdot \eta).$$

Therefore $d^\dagger_g \alpha_J = -\frac{1}{8\pi} \nabla^j \beta_j = -(\boldsymbol{j} \cdot \eta)$. **For vacuum data ($\boldsymbol{j} = 0$), we obtain $d^\dagger_g \alpha_J = 0$ exactly.**

Step 3: Stokes' theorem on the region between leaves. In dimension three, $d^\dagger \alpha_J = 0$ is equivalent to $d(\star_g \alpha_J) = 0$. For regular values $t_1 < t_2$, Lemma 6.6 provides a region W with boundary $\partial W = \Sigma_{t_2} - \Sigma_{t_1}$. Hence

$$J(t_2) - J(t_1) = \int_{\Sigma_{t_2}} \star_g \alpha_J - \int_{\Sigma_{t_1}} \star_g \alpha_J = \int_W d(\star_g \alpha_J) = 0.$$

Therefore $J(t)$ is constant on all regular leaves. The values at $t = 0$ and $t = 1$ are obtained by taking limits along regular values, using the continuity of the flux integral under smooth convergence of the leaves in the cylindrical and asymptotically flat regions. This proves $J(t) = J$ for all $t \in [0, 1]$. □

Remark 6.16 (Closedness vs. co-closedness). The Komar 1-form satisfies $d^\dagger \alpha_J = 0$ (co-closedness), not $d\alpha_J = 0$ (closedness). In 3D, co-closedness of a 1-form is equivalent to closedness of its Hodge dual 2-form:

$$d^\dagger \alpha = - \star \, d \star \alpha \quad \Longleftrightarrow \quad d(\star \alpha) = 0.$$

Thus $d^\dagger \alpha_J = 0$ implies $d(\star_g \alpha_J) = 0$, which is the condition needed for Stokes' theorem. The distinction matters: $d\alpha_J$ involves derivatives of K, while $d^\dagger \alpha_J$ involves the divergence, directly related to the momentum constraint.

Remark 6.17 (Vacuum Assumption—Cross Reference). The conservation of J requires vacuum ($\boldsymbol{j}_i = 0$) in the exterior region. See Remark 1.12 for a detailed explanation of why this hypothesis is essential.

Remark 6.18 (Extension to Non-Vacuum Axisymmetric Data). For **non-vacuum** axisymmetric data, the angular momentum is not conserved along the AMO flow. The change is given by:

$$J(t_2) - J(t_1) = \int_{\{t_1 < u < t_2\}} d(\star_g \alpha_J) = \int_{\{t_1 < u < t_2\}} (\boldsymbol{j} \cdot \eta) \, dV_g,$$

where $(\boldsymbol{j} \cdot \eta) := \boldsymbol{j}_i \eta^i$ and dV_g is the volume form of the physical metric. (One can rewrite the right-hand side in adapted axisymmetric coordinates, but the coordinate-free form above is the only identity used here.) However, one might conjecture a **weaker bound** for non-vacuum data:

Conjecture (Non-vacuum AM-Penrose): For axisymmetric initial data satisfying DEC (not necessarily vacuum) with outermost stable MOTS Σ:

$$M_{\text{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J_\infty^2}{A}},$$

where J_∞ is the ADM angular momentum (measured at infinity), which may differ from the Komar angular momentum $J(\Sigma)$ at the horizon when matter is present.

Potential approach: One could attempt to prove:

1. A "matter-corrected" monotonicity: $\frac{d}{dt}\mathcal{M}_{1,J(t)}(t) \geq 0$ where $J(t)$ varies.
2. Or a bound $J(\Sigma) \leq J_\infty$ from energy conditions on the matter.

The key difficulty is that the functional

$$m_{H,J}(t) = \sqrt{m_H^2(t) + 4\pi J(t)^2 / A(t)}$$

depends on both $A(t)$ and $J(t)$, and their joint evolution under non-vacuum conditions is not controlled by a simple monotonicity.

Special case: Electrovacuum (Kerr-Newman). For Maxwell electrovacuum with charge Q, one expects:

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A} + \frac{Q^2}{4}}.$$

This has been partially addressed by Gabach Clément–Jaramillo–Reiris [26] for the area-angular momentum-charge inequality on horizons.

Angular momentum modification in electrovacuum. For Einstein–Maxwell data, the momentum constraint becomes $D^j K_{ij} = D_i(\mathrm{tr}K) + 8\pi j_i^{(\mathrm{EM})}$, where the electromagnetic momentum density is:

$$j_i^{(\mathrm{EM})} = \frac{1}{4\pi}(\mathbf{E} \times \mathbf{B})_i = \frac{1}{4\pi} F_{ij} E^j,$$

with $\mathbf{E}$ and $\mathbf{B}$ the electric and magnetic fields. The Komar form α_J is no longer co-closed in general: $d^\dagger \alpha_J = j^{(\mathrm{EM})} \cdot \eta$. However, for **axisymmetric** electrovacuum data, $\mathcal{L}_\eta F = 0$ implies that the Poynting vector $\mathbf{E} \times \mathbf{B}$ is also axisymmetric. When integrated over axisymmetric surfaces, the angular component of the Poynting flux often cancels (by symmetry), but this requires careful case-by-case analysis. For static configurations ($K = 0$, $\mathbf{B} = 0$), one has $j^{(\mathrm{EM})} = 0$ and $J = 0$ automatically. The full dynamical case remains an open problem.

Remark 6.19 (Why Axisymmetry is Essential). Does any geometric flow conserve angular momentum? For **general** (non-axisymmetric) data, **no**. For **axisymmetric** data:

1. The Killing field $\eta = \partial_\phi$ exists by assumption.
2. The AMO flow respects the symmetry: axisymmetric data yields axisymmetric solutions.
3. The Komar integral becomes **topological** when $d(\star\alpha_J) = 0$ (i.e., $d^\dagger \alpha_J = 0$).
4. Co-closedness $d^\dagger \alpha_J = 0$ follows from the vacuum momentum constraint with axisymmetry.

This is **not** dynamical conservation—it is a Stokes' theorem statement about integrals over homologous surfaces in a fixed initial data set.

Remark 6.20 (Physical Interpretation). The conservation of J reflects that axisymmetric level sets remain axisymmetric, and the Komar integral measures the "twist" of K around the symmetry axis.

6.4 Monotonicity

We first record the key monotonicity formula for the area functional under the p-harmonic flow, following Agostiniani–Mazzieri–Oronzio [1].

Proposition 6.21 (AMO Area Monotonicity Formula)**.** *Let $(\tilde{M}, \tilde{g})$ be a complete Riemannian 3-manifold with scalar curvature $R_{\tilde{g}} \geq 0$. Let $u : \tilde{M} \to [0,1]$ be a p-harmonic function ($p > 1$) with regular level sets $\Sigma_t = \{u = t\}$. Define $A(t) = |\Sigma_t|_{\tilde{g}}$. Then for almost all $t \in (0,1)$:*

$$A'(t) = \int_{\Sigma_t} \frac{1}{|\nabla u|} \left(R_{\tilde{g}} + 2|\mathring{h}|^2 + \frac{2}{(p-1)^2} \left(H - (p-1)\frac{\Delta u}{|\nabla u|} \right)^2 \right) d\sigma, \quad (38)$$

where $H = \operatorname{div}(\nabla u/|\nabla u|)$ is the mean curvature of Σ_t (with sign convention: $H > 0$ for level sets expanding outward), $\mathring{h}$ is the traceless second fundamental form, and Δu is the Laplacian of u. The p-harmonic equation $\operatorname{div}(|\nabla u|^{p-2}\nabla u) = 0$ can be rewritten as:

$$|\nabla u|^{p-2}\Delta u + (p-2)|\nabla u|^{p-3}\langle \nabla|\nabla u|, \nabla u\rangle = 0,$$

which relates Δu, $|\nabla u|$, and directional derivatives. The integral is non-negative when $R_{\tilde{g}} \geq 0$ since each term is either a square or proportional to $R_{\tilde{g}}$.

Source of the formula. Equation (38) is the standard AMO identity for regular level sets of a p-harmonic function on a three-manifold with scalar curvature bounded below; see [1, Theorem 3.1 and Section 4]. We use this result as an external input rather than claiming a new derivation here. Once the formula is available, the non-negativity conclusion $A'(t) \geq 0$ under $R_{\tilde{g}} \geq 0$ is immediate because each term in the integrand is non-negative. □

Corollary 6.22 (Simplified Area Monotonicity)**.** *When $R_{\tilde{g}} \geq 0$, the area functional is monotonically non-decreasing:*

$$A'(t) \geq \int_{\Sigma_t} \frac{R_{\tilde{g}}}{|\nabla u|} d\sigma \geq 0.$$

Equality holds if and only if $R_{\tilde{g}} = 0$, $\mathring{h} = 0$ (umbilic level sets), and $H = (p-1)|\nabla u|^{-1}\Delta u$.

Hypothesis Verification Checklist for AMO Monotonicity

The AM-Hawking monotonicity formula requires several hypotheses. We verify each is satisfied under the assumptions of Theorem 1.2:

Hypothesis	**Required For**	**Verification**
$R_{\tilde{g}} \geq 0$	Non-negative integrand	Thm 5.7: AM-Lich. ensures this
Vacuum in exterior	J-conservation ($d^\dagger \alpha_J = 0$)	Lem 6.6 Step 6: $\Sigma_t \subset M_{\text{ext}}$
Axisymmetry (η)	Komar J well-defined	Hypothesis (H2) of Thm 1.2
$A(t) \geq 8\pi\|J\|$	Sub-extremality factor ≥ 0	Thm 7.1: preserved by flow
Level set regularity	Integration well-defined	Rem 6.4: $C^{1,\beta}$ by Tolksdorf
Homologous Σ_t	Stokes for J-conservation	Lem 6.6: level sets cobordant

Critical point: All hypotheses are verified to hold *throughout* the flow domain $\{0 < u < 1\}$, not just initially. This is essential because the monotonicity formula is integrated from $t = 0$ to $t = 1$.

Lemma 6.23 (Willmore Factor Bound $(1 - \mathcal{W}) \geq 0$ Along the Flow)**.** *Let $(\tilde{M}, \tilde{g})$ be the conformal manifold with $R_{\tilde{g}} \geq 0$ (from Theorem 5.7), and let $\Sigma_t = \{u = t\}$ be level sets of the p-harmonic potential for $t \in (0, 1)$. Define the normalized Willmore functional:*

$$\mathcal{W}(t) := \frac{1}{16\pi} \int_{\Sigma_t} H^2 \, dA_{\tilde{g}}.$$

Then:

(i) ***At*** $t = 0$ ***(MOTS):*** $\mathcal{W}(0) = 0$ *since* $\Sigma = \Sigma_0$ *is minimal in* $(\tilde{M}, \tilde{g})$.

(ii) ***Along the foliation:*** *For every regular* $t \in (0, 1)$, *one has* $0 \leq \mathcal{W}(t) \leq 1$, *equivalently* $(1 - \mathcal{W}(t)) \geq 0$.

(iii) ***At*** $t \to 1$ ***(infinity):*** *For large coordinate spheres* S_r *in the asymptotically flat end,* $\mathcal{W}(t) \to 1^-$, *equivalently* $m_H(t) \to M_{\mathrm{ADM}}(\tilde{g})$, *by the standard AMO endpoint theorem [1, Theorem 1.3].*

Proof. **(i)** The MOTS leaf Σ_0 is minimal in $(\tilde{M}, \tilde{g})$ (Lemma 8.2), so $H \equiv 0$ on Σ_0 and $\mathcal{W}(0) = 0$.

(ii) By Lemma 6.9, along the AMO foliation we have $W(t) = \int_{\Sigma_t} H^2 \, dA_{\tilde{g}} \leq 16\pi$ for every regular t, i.e. $\mathcal{W}(t) \leq 1$. Nonnegativity $\mathcal{W}(t) \geq 0$ is immediate from $H^2 \geq 0$.

(iii) The asymptotic statement as $t \to 1$ is exactly the endpoint identification from [1, Theorem 1.3]. □

Remark 6.24 (Role of the Willmore Factor in the Monotonicity). The factor $(1 - \mathcal{W})$ appears in the Hawking mass formula $m_H = \sqrt{A/(16\pi)}(1 - \mathcal{W})$ and indirectly in the monotonicity derivative via the AMO formula. The key points are:

1. At $t = 0$: $\mathcal{W}(0) = 0$, so the factor equals 1.
2. At $t = 1$: $\mathcal{W}(t) \to 1^-$, but combined with $A(t) \to \infty$, the product $\sqrt{A/(16\pi)}(1 - \mathcal{W}) \to M_{\mathrm{ADM}}$.
3. Throughout: $\mathcal{W}(t) \in [0, 1)$ ensures the Hawking mass is non-negative.

The sub-extremality factor $(1 - 64\pi^2 J^2/A^2)$ in Theorem 6.25(i) is *distinct* from $(1 - \mathcal{W})$. The former controls the angular momentum term, while the latter controls the Hawking mass definition. Both must be non-negative for the monotonicity to hold.

Theorem 6.25 (Auxiliary AM-Hawking Monotonicity Input)**.** *Let* $(\tilde{M}, \tilde{g})$ *be the conformal manifold produced in Stages 1–2, and let* $u_p : \tilde{M} \to [0, 1]$ *be the associated p-harmonic potential. Define*

$$m_{H,J}(t) := \sqrt{m_H^2(t) + \frac{4\pi J^2}{A(t)}}.$$

The manuscript uses the following rotating analogue of the AMO Hawking-mass monotonicity as an auxiliary input*:*

(i) *For almost every regular value* $t \in (0,1)$*,*

$$\frac{d}{dt} m_{H,J}^2(t) \geq \frac{1}{8\pi} \int_{\Sigma_t} \frac{R_{\tilde{g}} + 2|\mathring{h}|^2}{|\nabla u_p|_{\tilde{g}}} \left(1 - \frac{64\pi^2 J^2}{A(t)^2}\right) dA_{\tilde{g}} \geq 0.$$

(ii) *Consequently,* $t \mapsto m_{H,J}(t)$ *is non-decreasing on* $[0,1]$*.*

Remark 6.26 (Removal of the Auxiliary Input (H6) via Twisted Bochner Monotonicity). The auxiliary monotonicity input (H6) above is resolved in Section 11 by the Twisted Bochner Monotonicity Theorem 11.7, which derives the monotonicity formula

$$\frac{d}{dt} m_{H,J}^2(t) \geq \mathcal{I}_{\mathrm{AMO}}(t) \cdot \left(1 - \frac{64\pi^2 J^2}{A(t)^2}\right) - Ce^{-\beta_0 t} \cdot \mathcal{I}_{\mathrm{AMO}}(t) \geq 0$$

from a novel twisted Bochner identity (Lemma 11.3) on the Jang cylinder, with the twist potential ω constructed from the Komar form via $d\omega = \iota_\eta d\alpha_J$. The key new ingredients are the Kato-type decay estimate (Lemma 11.5) and the Mars–Simon–Laplacian identity (Lemma 11.6). Consequently, hypothesis (H6) of Theorem 1.2 can be removed and replaced by a reference to Theorem 11.7; see Corollary 11.8 and the strengthened Theorem 11.24.

Status of this step. The standard AMO theory controls the area functional and the *ordinary* Hawking mass along p-harmonic foliations. What is *not* supplied by those results is a rigorous estimate showing that the additional negative term

$$-\frac{4\pi J^2}{A(t)^2} A'(t)$$

arising from differentiating $m_{H,J}^2 = m_H^2 + 4\pi J^2/A$ can be absorbed by the positive AMO terms with the sharp sub-extremality factor shown above.

Earlier drafts attempted such an absorption through heuristic comparisons between A' and the AMO bulk integrals. Those arguments were not rigorous: the required domination estimate was not established with valid constants or hypotheses, and therefore did *not* prove the displayed differential inequality. Accordingly, the manuscript now treats the rotating monotonicity statement as an explicit auxiliary input rather than as a theorem proved from the preceding calculations. □

Remark 6.27 (Logical Independence of Sub-Extremality Preservation). The preservation result $A(t) \geq 8\pi|J|$ from Theorem 7.1 uses only the standard AMO area monotonicity together with J-conservation. The auxiliary rotating monotonicity input above enters only at the later stage where one compares the endpoint values of $m_{H,J}$.

Remark 6.28 (Distributional Bochner and Double Limit—Standard Regularization Route). The monotonicity formula requires care when the metric $\tilde{g}$ is only Lipschitz. Rather than claiming a new explicit quantitative proof of the smoothing step, we rely on the standard regularization route used in the low-regularity inverse-mean-curvature and cornered-metric literature, following Miao [42], Huisken–Ilmanen [32], and the AMO framework [1].

Reader's Guide: This remark explains why the usual smoothing and compactness machinery applies in the present setting. It should be read as a roadmap to the standard regularization argument, not as a source of new sharp rates.

Executive Summary: The $p \to 1^+$ limit is justified for the standard AMO quantities by:

(i) **Collar smoothing** (Miao): approximate the Lipschitz metric $\tilde{g}$ by smooth metrics $\tilde{g}_\epsilon$ that agree with $\tilde{g}$ away from a shrinking collar near the cylindrical end.

(ii) **Smooth monotonicity for the approximants:** on each $\tilde{g}_\epsilon$, the Bochner identity and the standard AMO area/Hawking monotonicity computations are classical.

(iii) **Uniform compactness in p:** Lemma 6.29 and Lemma 6.31 provide the local compactness needed to pass to subsequential limits as $p \to 1^+$.

(iv) **Limit passage:** combining the smoothing limit $\epsilon \to 0$ with the $p \to 1^+$ compactness argument yields the corresponding weak monotonicity statements for the standard AMO functionals of the limiting potential.

Practical consequence. No threshold such as $\beta_0 > 1$ is needed. The role of strict stability is only to provide positive exponential convergence of the geometry to its cylindrical model, which is enough for the usual collar-smoothing construction once one uses the signed/distributional formulation rather than absolute-value estimates on every curvature term.

(1) Distributional Bochner identity. The Jang metric $\bar{g}$ (and hence $\tilde{g} = \phi^4 \bar{g}$) is Lipschitz ($C^{0,1}$), so its Ricci curvature is a distribution. The AMO formula involves $\mathrm{Ric}_{\tilde{g}}(\nabla u, \nabla u)$, which is not immediately well-defined.

Resolution via collar smoothing: We construct a family of smooth approximants $\tilde{g}_\epsilon$ as follows. Let $\chi_\epsilon : M \to [0,1]$ be a smooth cutoff with $\chi_\epsilon = 0$ on $N_\epsilon(\Sigma)$ (the ϵ-neighborhood of Σ) and $\chi_\epsilon = 1$ outside $N_{2\epsilon}(\Sigma)$. Define:

$$\tilde{g}_\epsilon := \chi_\epsilon \tilde{g} + (1 - \chi_\epsilon)\tilde{g}_{\mathrm{cyl}},$$

where $\tilde{g}_{\mathrm{cyl}} = dt^2 + g_\Sigma$ is the exact cylindrical metric. This mollification was introduced by Miao [42] for studying mass in the presence of corners.

On each smooth approximant $\tilde{g}_\epsilon$, the Bochner identity holds pointwise:

$$\frac{1}{2}\Delta_{\tilde{g}_\epsilon}|\nabla u_\epsilon|^2 = |\nabla^2 u_\epsilon|^2 + \langle \nabla u_\epsilon, \nabla \Delta u_\epsilon \rangle + \mathrm{Ric}_{\tilde{g}_\epsilon}(\nabla u_\epsilon, \nabla u_\epsilon).$$

Curvature estimate for the smoothed metric: In the transition collar, derivatives of χ_ϵ generate the usual ϵ^{-1} and ϵ^{-2} coefficients. A naive absolute-value estimate is therefore too crude to capture the limit. The correct statement, proved in the smoothing literature cited above, is that the scalar-curvature defect of the approximants is controlled in the *distributional/integrated* sense and tends to zero against the test quantities appearing in the monotonicity identity. This is precisely why one works with collar smoothing and weak formulations rather than pointwise curvature bounds.

(2) Double limit interchange—rigorous justification. We must pass $(p, \epsilon) \to (1^+, 0)$ simultaneously. The argument requires verifying the hypotheses of the Moore–Osgood theorem.

Moore–Osgood theorem statement: Let $f(p, \epsilon)$ be defined for $p \in (1, 2]$ and $\epsilon \in (0, 1]$. If:

(MO1) $\lim_{\epsilon \to 0} f(p, \epsilon) = g(p)$ exists for each $p > 1$, and

(MO2) the convergence in (MO1) is **uniform** in $p \in (1, 2]$,

then $\lim_{p \to 1^+} \lim_{\epsilon \to 0} f(p, \epsilon) = \lim_{\epsilon \to 0} \lim_{p \to 1^+} f(p, \epsilon)$ (both limits exist and are equal).

Verification of (MO1): For fixed $p > 1$, let $u_{p,\epsilon}$ solve $\Delta_{p, \tilde{g}_\epsilon} u = 0$ with boundary conditions $u|_\Sigma = 0$, $u \to 1$ at infinity. Standard stability of quasilinear elliptic equations under smooth coefficient perturbations gives convergence of $u_{p,\epsilon}$ to the corresponding u_p on compact subsets away from the collar.

$$\|u_{p,\epsilon} - u_p\|_{C^1(K)} \leq C(p, K)\|\tilde{g}_\epsilon - \tilde{g}\|_{C^1(K)} \leq C(p, K)\epsilon^2$$

for any compact $K \subset M \setminus \Sigma$. Here u_p solves the limiting equation on $(M, \tilde{g})$. The area functional $A_{p,\epsilon}(t) = \int_{\Sigma_t} dV_{\tilde{g}_\epsilon}$ converges: $A_{p,\epsilon}(t) \to A_p(t)$ as $\epsilon \to 0$.

Verification of (MO2): The manuscript only needs a compactness statement strong enough to pass to a subsequential $p \to 1^+$ limit on regular slabs. We record that standard input in the following qualitative form.

Lemma 6.29 (Compactness Input for the $p \to 1^+$ Passage). *Let $p_j \downarrow 1$ and let u_{p_j} be the corresponding p_j-harmonic potentials on $(\tilde{M}, \tilde{g})$ with $u_{p_j}|_\Sigma = 0$ and $u_{p_j} \to 1$ at infinity. Then, after passing to a subsequence, there exists a limiting least-gradient potential u_1 such that:*

(i) $u_{p_j} \to u_1$ locally uniformly on $\tilde{M}$ and in $W^{1,1}_{\mathrm{loc}}$;

(ii) on any compact set K contained in a regular slab of u_1 and satisfying $\inf_K |\nabla u_1| > 0$, the family $\{u_{p_j}\}$ is precompact in $C^1(K)$;

(iii) along regular values of u_1, the associated level-set functionals $A_{p_j}(t)$, $W_{p_j}(t)$, and $m_{H,J,p_j}(t)$ converge after subsequence extraction.

Proof. This is the standard compactness package underlying the AMO approach: one combines $W^{1,1}_{\mathrm{loc}}$ compactness for the p-harmonic potentials as $p \downarrow 1$ with the usual nondegenerate interior regularity theory on regions where the gradient stays uniformly away from zero. See [1], together with the regularity results of Tolksdorf, Lieberman, and DiBenedetto, for the analytic framework. We use only these qualitative consequences and do not claim a sharp explicit rate in either p or ϵ. □

What Is Actually Needed for the $p \to 1^+$ Passage.

The later argument uses only the following qualitative facts:

1. Local compactness of u_p on regular slabs away from the singular set of the limit;
2. Stability of the smoothed problems as $\epsilon \to 0$ for each fixed p;
3. Convergence of the area/Willmore/mass functionals along regular values after subsequence extraction;
4. Weak-* closedness of nonnegative Radon measures when passing to the limit in the monotonicity inequality.

No explicit universal Schauder exponent or rate of convergence in p is used in the proof.

Remark 6.30 (Summary of Uniform Bounds for $p \to 1^+$ Limit). The $p \to 1^+$ limit argument requires the following qualitative inputs:

1. **Subsequential compactness on regular slabs:** supplied by Lemma 6.29;
2. **Critical set control for each fixed p:** $\dim_{\mathcal{H}}(\mathcal{Z}_p) \leq 1$, and regular values form a full-measure subset of $(0, 1)$ (Lemma 6.31 and Remark 6.3);
3. **Absolute continuity/co-area structure:** enough to convert the a.e. derivative inequality into global monotonicity after passing to the limit.

Together with the collar smoothing discussed in Remark 6.28, these inputs yield the standard diagonal/Moore–Osgood style passage used in the manuscript.

Lemma 6.31 (Gradient Lower Bound for AMO Potential). *Let $u_p : (\tilde{M}, \tilde{g}) \to [0, 1]$ be the p-harmonic potential with $u_p|_\Sigma = 0$ and $u_p \to 1$ at infinity. Then:*

(i) The set of critical points $\mathcal{Z}_p := \{x \in \tilde{M} : \nabla u_p(x) = 0\}$ has measure zero for each $p > 1$.

(ii) If $u_{p_j} \to u_1$ as in Lemma 6.29, and if K is a compact subset of a regular slab of u_1 with $\inf_K |\nabla u_1| > 0$, then $|\nabla u_{p_j}| \geq c_K > 0$ on K for all sufficiently large j.

(iii) The level set area functional $A_p(t) = |\{u_p = t\}|$ is absolutely continuous in t, and the monotonicity formula holds for a.e. t.

*(iv) **Critical set control:** For each $p \in (1, 2]$, the critical set $\mathcal{Z}_p$ has Hausdorff dimension at most 1. This is sufficient for the regular-value approximation used below.*

Proof. **(i)** By the Heinonen–Kilpeläinen–Martio structure theorem [30, Theorem 7.46], the critical set $\mathcal{Z}_p$ of a p-harmonic function in dimension 3 has Hausdorff dimension at most 1, hence in particular has 3-dimensional measure zero. The corresponding full-measure regular-value statement is taken from Remark 6.3, which is the only input needed for the level-set formulas.

Note: The classical Sard theorem requires C^n regularity for functions on n-dimensional manifolds, which p-harmonic functions (being only $C^{1,\beta}$) do not satisfy. The above argument uses the specialized structure theory for p-harmonic equations instead.

(ii) On a compact set K as stated, continuity of ∇u_1 gives $\inf_K |\nabla u_1| =: 2c_K > 0$. By the C^1 convergence from Lemma 6.29(ii), we then have $|\nabla u_{p_j}| \geq c_K$ on K for all sufficiently large j.

(iii) The co-area formula gives:

$$\int_0^1 A_p(t)\, dt = \int_{\tilde{M}} |\nabla u_p|\, dV < \infty.$$

Since $A_p(t) \geq 0$ and integrable, it is finite for a.e. t. The derivative $A_p'(t)$ exists in the distributional sense and equals the AMO formula integrand for regular values t (which form a set of full measure by (i)). The monotonicity $A_p'(t) \geq 0$ holds at regular values, hence a.e.

(iv) By Heinonen–Kilpeläinen–Martio [30, Theorem 7.46], the critical set of a p-harmonic function $u : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ satisfies

$$\dim_{\mathcal{H}}(\{x : \nabla u(x) = 0,\, u(x) \neq \sup u, \inf u\}) \leq n - 2.$$

For $n = 3$, this gives $\dim_{\mathcal{H}}(\mathcal{Z}_p) \leq 1$. This qualitative bound is all that is used later. In particular, we do not need any claim that the critical set is discrete or that its cardinality is controlled uniformly in p. □

Remark 6.32 (Handling Critical Points in the Monotonicity). The monotonicity formula (Theorem 6.25) involves integration over level sets $\Sigma_t = \{u_p = t\}$. At critical values $t \in \{u_p(\mathcal{Z}_p)\}$, the level set may be singular. We handle this as follows:

Remark 6.33 (Critical Clarification: "For a.e. t" vs. "For all t"). We clarify which parts of the monotonicity hold for a.e. t versus for all t, and why this is sufficient.

(1) What holds for a.e. t:

- The level sets $\Sigma_t = \{u_p = t\}$ are **smooth embedded surfaces** for a.e. $t \in (0, 1)$ (by the critical set structure theory for p-harmonic functions, Remark 6.3).
- The derivative formula $\frac{d}{dt} m_{H,J}^2(t) \geq 0$ holds for a.e. t (at regular values where $\nabla u_p \neq 0$ on Σ_t).

- The area and Willmore functionals $A(t)$, $W(t)$ are differentiable for a.e. t.

(2) What holds for ALL t:

- The functions $t \mapsto A(t)$, $t \mapsto m_H(t)$, $t \mapsto m_{H,J}(t)$ are **continuous** and **absolutely continuous** on $[0, 1]$.
- The boundary values $m_{H,J}(0)$ and $m_{H,J}(1)$ are well-defined as limits.
- The monotonicity $m_{H,J}(t_1) \leq m_{H,J}(t_2)$ for $t_1 < t_2$ holds for ALL $t_1, t_2 \in [0, 1]$ (including critical values).

(3) Why a.e. suffices for the inequality: The key is the **fundamental theorem of calculus for absolutely continuous functions**. Since $m^2_{H,J}(t)$ is absolutely continuous and $\frac{d}{dt} m^2_{H,J} \geq 0$ for a.e. t:

$$m^2_{H,J}(1) - m^2_{H,J}(0) = \int_0^1 \frac{d}{dt} m^2_{H,J}(t)\, dt \geq 0.$$

The singular set $\{t : \nabla u_p = 0 \text{ somewhere on } \Sigma_t\}$ has measure zero (by the Heinonen–Kilpeläinen–Martio structure theorem [30]), so its contribution to the integral vanishes. Therefore:

$$m_{H,J}(1) \geq m_{H,J}(0) \quad \text{holds on the full interval } [0, 1].$$

(4) Why critical points do not obstruct: At a critical value t_* where Σ_{t_*} contains a critical point, the level set may have singularities (non-smooth points). However:

- By Lemma 6.31(i) and Remark 6.3, the singular leaves occur only over a measure-zero set of levels.
- The area $A(t_*)$ and Hawking mass $m_H(t_*)$ remain finite (the singularity is removable for these integral quantities).
- The one-sided limits $\lim_{t \to t_*^{\pm}} m_{H,J}(t)$ exist and agree, establishing continuity through critical values.

Conclusion: The "a.e. t" condition is technically necessary for the pointwise derivative formula, but **global monotonicity** $m_{H,J}(1) \geq m_{H,J}(0)$ follows by integration and therefore extends across the measure-zero set of critical levels.

Remark 6.34 (Regularity at Critical Points—Detailed Analysis). 1. By Remark 6.3, the exceptional set of singular levels has measure zero.

2. The AM-Hawking mass $m_{H,J}(t) = \sqrt{m_H^2(t) + 4\pi J^2/A(t)}$ is defined via the Hawking mass $m_H(t)$ and area $A(t)$, which are well-defined for all t by the co-area formula.

3. The monotonicity $\frac{d}{dt} m_{H,J}(t) \geq 0$ holds at regular values (a.e. in t).

4. By absolute continuity of $m_{H,J}(t)$ (following from absolute continuity of $m_H(t)$ and $A(t)$), the a.e. derivative condition $\frac{d}{dt} m_{H,J}(t) \geq 0$ implies $m_{H,J}(t_2) \geq m_{H,J}(t_1)$ for all $t_1 < t_2$.

Therefore, critical points do not obstruct the global monotonicity conclusion.

For the AMO potential, the strong maximum principle ensures $0 < u_p < 1$ in the interior. The only limit statement needed here is the qualitative one already summarized above: after smoothing in ϵ and extracting a subsequence in p, the area and AM-Hawking functionals converge along regular levels, and the nonnegative monotonicity measures pass to the limit by weak-$*$ compactness. We therefore do not attempt to prove a globally uniform quantitative estimate of the form "$A_{p,\epsilon}(t) - A_p(t) = O(\epsilon^\gamma)$ uniformly for all p".

Remark 6.35 (Qualitative Dependence on the p-Regularity Inputs). For the $p \to 1^+$ passage, the relevant point is qualitative uniformity rather than any explicit convergence rate. The inputs needed are:

(C1) local $C^{1,\alpha}$ compactness for the family $\{u_p\}_{p\in(1,2]}$ on compact subsets away from the critical set, as supplied by Lemma 6.29;

(C2) a uniform lower bound for $|\nabla u_p|$ away from critical points on compact level-set slabs, as in Lemma 6.31(ii);

(C3) the measure-zero nature of the singular level set for each fixed p, so that the co-area/absolute-continuity argument applies on a full-measure set of regular values;

(C4) stability of the smoothed problems under $\epsilon \to 0$ for fixed p, together with subsequential compactness as $p \to 1^+$.

These qualitative inputs are sufficient for the diagonal/Moore–Osgood style passage used here. We deliberately do not claim a sharp explicit convergence rate in either p or ϵ.

Remark 6.36 (Limit Passage $p \to 1^+$: Consolidated Statement). Fix any sequence $p_j \downarrow 1$. After passing to a subsequence, the AMO potentials admit a limit u_1 with the following properties:

1. $u_{p_j} \to u_1$ locally uniformly and in $W^{1,1}_{\mathrm{loc}}$;
2. on compact regular slabs of u_1, the gradients converge strongly enough to pass the area, Willmore, and standard Hawking functionals along a.e. regular level;
3. the standard AMO area/Hawking monotonicity identities for the approximating sequence pass to the limit in the usual weak sense along regular values;
4. any additional endpoint inequality available uniformly for a modified functional built continuously from these standard quantities may then be transported to the subsequential limit.

Standard compactness argument. Combine the collar smoothing from Remark 6.28, the qualitative compactness input of Lemma 6.29, the critical-set control of Lemma 6.31, and the absolute-continuity argument of Remark 6.33. The only property used in the last step is that nonnegative Radon measures are weak-$*$ closed. □

Remark 6.37 (Why This Consolidation Matters). The limit $p \to 1^+$ is the technical heart of the AMO approach. The preceding lemmas and remarks are collected in Remark 6.36 to make explicit:

1. **Which qualitative compactness inputs are used** — regular-slab compactness, critical-set control, and smoothing stability;
2. **What survives in the limit** — the standard AMO level-set functionals and their weak monotonicity properties;
3. **What is *not* claimed** — no explicit universal convergence rate in p or ϵ is needed.

The key insight is that the "a.e." condition does not degrade under limits: the exceptional set of singular levels remains measure-zero by the regular-value approximation in Remark 6.3. Any further rotating monotonicity statement must be supplied separately and checked against this compactness framework.

Theorem 6.38 (Endpoint Comparison Under the Auxiliary Inputs)**.** *Under the hypotheses of Theorem 1.2, and hence assuming both the auxiliary monotonicity input of Theorem 6.25 and the conformal mass comparison input from Theorem 1.2(H7), the AM-Hawking mass functional satisfies:*

$$m_{H,J}(0) \leq m_{H,J}(1) = M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g).$$

In particular:

1. *At $t = 0$ (horizon): $m_{H,J}(0) = \sqrt{A/(16\pi) + 4\pi J^2/A}$ by the exact MOTS boundary evaluation from Lemmas 8.2 and 8.3.*
2. *At $t = 1$ (infinity): $m_{H,J}(1) = M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$.*

Proof. By Theorem 6.25, $m_{H,J}(t)$ is monotonically increasing. We analyze the boundary values carefully.

Boundary at $t = 0$ (MOTS Σ): By Lemmas 8.2 and 8.3,

$$m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}.$$

Boundary at $t = 1$ (spatial infinity): As $t \to 1$, the level sets Σ_t approach large coordinate spheres. The key AMO result [1, Theorem 1.3] establishes:

$$\lim_{t \to 1^-} m_H(t) = M_{\mathrm{ADM}}(\tilde{g}).$$

For the angular momentum correction: as $A(t) \to \infty$ while J remains constant:

$$\frac{4\pi J^2}{A(t)} \to 0.$$

Therefore:

$$m_{H,J}(1) = \lim_{t \to 1^-} \sqrt{m_H^2(t) + \frac{4\pi J^2}{A(t)}} = M_{\mathrm{ADM}}(\tilde{g}).$$

Mass chain: Invoking the conformal mass-comparison input referenced in Theorem 1.2(H7) together with Theorem 4.12(iv):

$$M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g).$$

Conclusion: The monotonicity $m_{H,J}(0) \leq m_{H,J}(1)$ combined with $m_{H,J}(1) \leq M_{\mathrm{ADM}}(g)$ yields the bound. □

Remark 6.39 (Strengthened Endpoint Comparison via Section 11). The auxiliary inputs (H6) and (H7) invoked in Theorem 6.38 above are resolved in Section 11:

- The monotonicity input (H6) is replaced by the Twisted Bochner Monotonicity Theorem 11.7, which derives the monotonicity of $m_{H,J}$ from a twisted Bochner identity (Lemma 11.3) and a Kato-type decay estimate for the twist potential (Lemma 11.5).
- The conformal mass comparison input (H7) is replaced by the Conformal Cohomological Mass Comparison Theorem 11.14, which proves $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$ via a twisted Hodge decomposition (Lemma 11.13) on the cylindrical end.

Consequently, the chain of inequalities in Theorem 6.38 holds unconditionally under hypotheses (H1)–(H5) and the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3). The strengthened result is summarized in Theorem 11.24.

7 Stage 4: Sub-Extremality

Logical Ordering and Non-Circularity.

This section establishes the sub-extremality bound $A(t) \geq 8\pi|J|$ for AMO level sets. To prevent circularity, we explicitly state the logical dependencies:

What is already established at this stage:

(1) **Stage 1 (Section 4):** The Jang manifold $(\bar{M}, \bar{g})$ exists with cylindrical end at the MOTS Σ.

(2) **Stage 2 (Section 5):** The conformal factor $\phi > 0$ exists, giving $(\tilde{M}, \tilde{g})$ with $R_{\tilde{g}} \geq 0$.

(3) **Stage 3 (Section 6):** The AMO p-harmonic potential u_p exists, foliating $\tilde{M}$ by level sets $\{\Sigma_t\}_{t\in[0,1]}$.

(4) **Angular momentum conservation (Theorem 6.13 in Section 6):** $J(\Sigma_t) = J$ for all t.

What this section proves:

- The initial sub-extremality $A(\Sigma) \geq 8\pi|J(\Sigma)|$ follows from the **Dain–Reiris theorem** [22], which depends **only on the initial data** (M, g, K).
- The preservation along the flow uses the J-conservation from Stage 4 and the area lower bound from the Dain–Reiris theorem applied to each Σ_t.

What this section does NOT use:

- The sub-extremality bound does **not** use the AMO monotonicity (Stage 6).
- The sub-extremality bound does **not** use the final inequality $m_{H,J}(0) \leq M_{\mathrm{ADM}}$.

This ordering ensures no circular reasoning: sub-extremality is an **input** to the mass monotonicity formula, not an output.

Theorem 7.1 (Sub-Extremality from Dain–Reiris)**.** *Let (M, g, K) be asymptotically flat, axisymmetric initial data satisfying DEC with outermost strictly stable MOTS Σ of area $A = |\Sigma|_g$ and Komar angular momentum $J = \frac{1}{8\pi}\int_\Sigma K(\eta, \nu)\, dA$. Then:*

(i) ***Initial sub-extremality (Dain–Reiris [22]):***

$$A(\Sigma) \geq 8\pi|J(\Sigma)|,$$

with equality if and only if $(\Sigma, g|_\Sigma)$ is isometric to the horizon of extreme Kerr.

(ii) ***Preservation along flow:*** *For the AMO level sets $\Sigma_t = \{u = t\}$ with area $A(t) = |\Sigma_t|_{\tilde{g}}$,*

$$A(t) \geq 8\pi|J| \quad \text{for all } t \in [0, 1].$$

(iii) ***Strict sub-extremality:*** *If $A(\Sigma) > 8\pi|J(\Sigma)|$ (strict inequality initially), then $A(t) > 8\pi|J|$ for all $t \in [0, 1]$, and the sub-extremality factor satisfies*

$$1 - \frac{64\pi^2 J^2}{A(t)^2} \geq 1 - \frac{64\pi^2 J^2}{A(0)^2} > 0.$$

Remark 7.2 (No Cosmic Censorship Assumed). This theorem does **not** assume Cosmic Censorship. It follows directly from the **proven** Dain–Reiris area-angular momentum inequality [22], which is derived purely from the constraint equations and the stability of the MOTS. The Penrose inequality is sometimes viewed as evidence *for* Cosmic Censorship, but our proof does not use Cosmic Censorship as a hypothesis.

Remark 7.3 (Verification of Dain–Reiris Hypotheses). The Dain–Reiris inequality [22] requires the following hypotheses on the surface Σ:

(DR1) Σ is a closed, embedded, axisymmetric 2-surface with $\Sigma \cong S^2$;

(DR2) Σ is a **stable** marginally outer trapped surface (MOTS);

(DR3) The ambient initial data (M, g, K) satisfies the dominant energy condition;

(DR4) Σ intersects the axis of symmetry at exactly two poles: $\Sigma \cap \Gamma = \{p_N, p_S\}$ (by topological necessity—see Lemma 4.6).

We verify that our hypotheses (H1)–(H4) in Theorem 1.2 imply (DR1)–(DR4):

- **(DR1) Topology:** By the Galloway–Schoen theorem [27], a stable MOTS in data satisfying DEC has spherical topology. The outermost MOTS is automatically embedded.
- **(DR2) Stability:** This is hypothesis (H4) of Theorem 1.2.
- **(DR3) DEC:** This is hypothesis (H1) of Theorem 1.2.
- **(DR4) Axis intersection:** An axisymmetric S^2 must intersect the axis at two poles by the topological argument in Lemma 4.6. The twist term $\mathcal{T}$ vanishes at these poles since $\mathcal{T} \propto \rho^2$ and $\rho = 0$ on the axis (Lemma 4.8).

Therefore, the Dain–Reiris inequality applies under our hypotheses.

Proof. **Step 1: The Dain–Reiris inequality (proven theorem).** For axisymmetric initial data satisfying DEC with a stable MOTS Σ, Dain and Reiris [22] proved:

$$A(\Sigma) \geq 8\pi |J(\Sigma)|,$$

with equality if and only if Σ is isometric to the horizon of extreme Kerr. This is a **theorem**, not a conjecture, proven using variational methods on the space of axisymmetric surfaces.

Step 2: Ancillary mass-angular momentum comparison. For orientation, Dain [20] proved a separate mass-angular-momentum inequality under additional global hypotheses:

$$M_{\mathrm{ADM}} \geq \sqrt{|J|},$$

with equality characterized by extreme Kerr in that setting. This comparison is **not used** in the argument below; we record it only as background. In particular, the preservation proof that follows uses only the Dain–Reiris inequality on the initial MOTS together with AMO area monotonicity and angular-momentum conservation.

It implies the familiar total-angular-momentum sub-extremal bound

$$|J| \leq M_{\mathrm{ADM}}^2 \quad \text{(sub-extremal bound on total angular momentum)}.$$

Step 3: Preservation along AMO flow. The Dain–Reiris inequality $A(\Sigma) \geq 8\pi|J(\Sigma)|$ is established in [22] using variational methods specific to MOTS. We do **not** re-derive this inequality here; instead, we show that once it holds at $t = 0$, it is **preserved** along the AMO flow by the following rigorous argument:

(i) **Initial condition:** By the Dain–Reiris theorem [22], the initial MOTS $\Sigma = \Sigma_0$ satisfies $A(0) \geq 8\pi|J(0)|$.

(ii) J **is conserved:** By Theorem 6.13, $J(t) = J(0) = J$ for all $t \in [0, 1]$.

(iii) A **is non-decreasing:** By the standard AMO area monotonicity formula, $A'(t) \geq 0$ for almost all $t \in (0, 1)$:

Proof of area monotonicity. Let $\Sigma_t = \{u = t\}$ be level sets of the p-harmonic potential u on $(\tilde{M}, \tilde{g})$ with $R_{\tilde{g}} \geq 0$. By Proposition 6.21, the AMO formula gives:

$$A'(t) = \int_{\Sigma_t} \frac{1}{|\nabla u|} \left(R_{\tilde{g}} + 2|\mathring{h}|^2 + \frac{2}{(p-1)^2} \left(H - (p-1)\frac{\Delta u}{|\nabla u|} \right)^2 \right) d\sigma. \tag{39}$$

Each term in the integrand is non-negative:

- $R_{\tilde{g}} \geq 0$ by Theorem 5.7 (AM-Lichnerowicz equation);
- $|\mathring{h}|^2 \geq 0$ (squared norm of traceless second fundamental form);
- The third term is a squared quantity, hence non-negative.

Since $|\nabla u| > 0$ on regular level sets (and regular values form a full-measure subset of $(0,1)$ by Remark 6.3), we conclude $A'(t) \geq 0$ for a.e. $t \in (0,1)$.

Remarks on the input:

(a) The AMO formula (39) is a cited result from [1, Theorem 3.1], summarized in Proposition 6.21.

(b) The condition $R_{\tilde{g}} \geq 0$ is **essential**: the conformal transformation $\tilde{g} = \phi^4 \bar{g}$ with ϕ solving the AM-Lichnerowicz equation ensures $R_{\tilde{g}} = \Lambda_J \phi^{-12} \geq 0$. Without this, $R_{\tilde{g}}$ could be negative and area monotonicity would fail.

(c) For the limit $p \to 1^+$, the level sets approximate minimal surfaces, and the squared term involving H and Δu vanishes. The bound $A'(t) \geq \int_{\Sigma_t} R_{\tilde{g}}/|\nabla u| \, d\sigma \geq 0$ remains valid.

(iv) **Conclusion:** Combining (i)–(iii):

$$A(t) \geq A(0) \geq 8\pi|J| = 8\pi|J(t)| \quad \text{for all } t \in [0,1].$$

Quantitative preservation of sub-extremality factor. For strictly sub-extremal initial data with $A(0) > 8\pi|J|$, define the sub-extremality factor:

$$\mathcal{S}(t) := 1 - \frac{64\pi^2 J^2}{A(t)^2}.$$

Since $A(t) \geq A(0)$ and $J(t) = J$ is constant:

$$\mathcal{S}(t) = 1 - \frac{64\pi^2 J^2}{A(t)^2} \geq 1 - \frac{64\pi^2 J^2}{A(0)^2} = \mathcal{S}(0) > 0.$$

The sub-extremality factor is **non-decreasing** along the flow and remains strictly positive if it starts strictly positive. This ensures the monotonicity formula (Theorem 6.25) has a non-negative integrand throughout the flow.

Step 4: Note on the Dain–Reiris proof. For completeness, we summarize the key ingredients of the Dain–Reiris argument (which we cite but do not re-derive):

- The proof uses the **stability operator** of the MOTS to establish positivity of certain geometric integrals.
- A key step is the **mass functional** technique: for axisymmetric surfaces, the angular momentum J can be expressed as a boundary integral that, by the constraint equations and stability, is bounded by a multiple of the area.
- The explicit constant 8π arises from the geometry of the extreme Kerr horizon, which achieves equality.

See [22, Section 3] for the complete variational argument. □

Remark 7.4 (Necessity of MOTS Stability). The stability hypothesis on the outermost MOTS Σ is used in **three distinct places** in the proof:

1. **Jang equation blow-up (Theorem 4.12):** Stability is used in the cited Jang theory to obtain logarithmic blow-up at Σ with positive coefficient $C_0 > 0$ and controlled cylindrical asymptotics. For unstable MOTS, the Jang solution may exhibit more complicated behavior (e.g., oscillatory or non-monotonic blow-up).
2. **Dain–Reiris inequality (Theorem 7.1):** The proof of $A \geq 8\pi|J|$ in [22] relies on the stability condition through a variational argument. Unstable MOTS can violate this bound.
3. **Cylindrical end geometry (Theorem 4.12(iii)):** Stability ensures the cylindrical end metric converges exponentially to $dt^2 + g_\Sigma$, with decay rate β related to the spectral gap of the stability operator.

Can stability be relaxed? It is an open question whether the AM-Penrose inequality holds for **unstable** outermost MOTS. The main obstacle is that the Dain–Reiris inequality can fail for unstable surfaces. For example, one could potentially construct initial data with an unstable MOTS having $A < 8\pi|J|$, in which case the monotonicity argument (Theorem 6.25) would break down since the factor $(1 - (8\pi|J|)^2/A(t)^2)$ could be negative.

However, for **outermost** MOTS (which are automatically weakly outer-trapped), there is some evidence that stability may be automatic in the axisymmetric case. This is related to the fact that axisymmetric deformations preserve the MOTS condition, limiting the possible instability directions. See [7] for related discussion.

Remark 7.5 (Independence from Cosmic Censorship). The sub-extremality bound $A \geq 8\pi|J|$ is a **proven geometric inequality**, not an assumption. It follows from the constraint equations, the DEC, and the stability of the MOTS—all hypotheses that are verifiable for a given initial data set. The Penrose inequality proof does not invoke Cosmic Censorship in any form.

Remark 7.6 (Clarification: Bootstrap Structure of the Sub-Extremality Argument). A careful reader may wonder about a potential circularity: the Dain–Reiris inequality is proven for MOTS, but the level sets Σ_t for $t > 0$ are **not** MOTS. How can we claim $A(t) \geq 8\pi|J|$ for all t?

Resolution: The argument is **not** a re-application of Dain–Reiris at each t. Instead:

(1) **Initial bound (Dain–Reiris):** At $t = 0$, the outermost MOTS $\Sigma_0 = \Sigma$ satisfies $A(0) \geq 8\pi|J|$ by the Dain–Reiris theorem [22]. This is a **one-time application** at the MOTS only.

(2) **Angular momentum conservation:** By Theorem 6.13, $J(t) = J$ is constant for all $t \in [0, 1]$. This uses Stokes' theorem and the vacuum condition in the exterior.

(3) **Area monotonicity (proven independently):** By the AMO theory [1], the area $A(t)$ of level sets is non-decreasing: $A(t) \geq A(0)$. This follows from the p-harmonic structure and $R_{\tilde{g}} \geq 0$, and does **not** require Σ_t to be a MOTS.

(4) **Conclusion (algebraic):** Combining (1), (2), (3):

$$A(t) \geq A(0) \geq 8\pi|J| = 8\pi|J(t)|.$$

No circularity exists because the Dain–Reiris inequality is used only at $t = 0$, and the preservation for $t > 0$ follows from the independent monotonicity of area.

Key point: The Dain–Reiris inequality and the AMO area monotonicity are **logically independent** theorems. Dain–Reiris applies to MOTS and uses MOTS-specific variational arguments. AMO area monotonicity applies to level sets of p-harmonic functions and uses the Bochner technique with $R_{\tilde{g}} \geq 0$. The sub-extremality preservation is the **combination** of these two independent results.

8 Synthesis of the Conditional Theorem

Definition 8.1 (Cylindrical End Boundary Conditions—Formal Specification)**.** Let $(\bar{M}, \bar{g})$ be the Jang manifold with cylindrical end $\mathcal{C} \cong [0, \infty)_t \times \Sigma$ at the MOTS. The **cylindrical end boundary conditions** for the AM-Lichnerowicz equation and AMO flow are:

AM-Lichnerowicz equation ($-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = \Lambda_J \phi^{-7}$):

(BC-L1) **Asymptotic Dirichlet at MOTS:** $\phi(t, y) \to 1$ as $t \to \infty$, uniformly in $y \in \Sigma$.

(BC-L2) **Exponential decay:** Under the source-term decay hypothesis of Theorem 5.7, $|\phi(t, y) - 1| \leq Ce^{-\kappa t}$ for some $\kappa = \kappa(\lambda_1(L_\Sigma)) > 0$.

(BC-L3) **Derived Neumann:** $\partial_t \phi(t, y) \to 0$ as $t \to \infty$ (consequence of (BC-L2)).

(BC-L4) **Asymptotic Dirichlet at infinity:** $\phi(x) \to 1$ as $r \to \infty$ in the asymptotically flat end.

AMO p-harmonic potential ($\Delta_p u = 0$):

(BC-A1) **Dirichlet at MOTS:** $u(t, y) \to 0$ as $t \to \infty$ along the cylindrical end.

(BC-A2) **Dirichlet at infinity:** $u(x) \to 1$ as $r \to \infty$ in the asymptotically flat end.

(BC-A3) **Monotonicity:** u is strictly monotone from the MOTS (value 0) to infinity (value 1).

Well-posedness statements:

- Conditions (BC-L1)–(BC-L4) determine a unique positive solution $\phi > 0$ (Theorem 5.7).
- Conditions (BC-A1)–(BC-A3) determine a unique solution u for each $p \in (1, 2]$.
- The boundary conditions are **asymptotic**, not imposed on a compact boundary.

Hypothesis Usage Summary. The seven stated hypotheses enter the proof as follows:

(H1) DEC: Ensures $R_{\tilde{g}} \geq 0$ after conformal transformation (Stage 2), which drives AMO monotonicity (Stage 6).

(H2) Axisymmetry: Defines angular momentum J, guarantees axisymmetric solutions at every stage, and enables J-conservation (Stage 4).

(H3) Exterior vacuum: Ensures Komar and ADM angular momenta coincide; enables clean asymptotics for boundary evaluation (Stage 7).

(H4) Strictly stable MOTS: Provides the spectral gap needed for the Jang cylindrical-end analysis and the exponential decay statements used at the MOTS boundary (Stage 7).

(H5) Source-term decay: Supplies the cylindrical-end decay/integrability needed to solve the AM-Lichnerowicz equation and control its asymptotics.

(H6) Auxiliary AM-Hawking monotonicity: Provides the endpoint comparison $m_{H,J}(0) \leq m_{H,J}(1)$ for the rotating flow.

(H7) Conformal mass comparison: Provides the final inequality $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$ needed to transfer the AMO lower bound for the conformal metric back to the physical ADM mass.

Critical Distinction: Strictly Stable vs. Marginally Stable MOTS.

A MOTS Σ is characterized by its **stability operator** L_Σ, defined as the linearization of the null expansion θ^+ under normal deformations:

$$L_\Sigma \psi = -\Delta_\Sigma \psi + 2\omega \cdot \nabla_\Sigma \psi + \left(\tfrac{1}{2}\mathrm{Sc}_\Sigma - |\chi^+|^2 - |\omega|^2 + \mathrm{div}_\Sigma \omega\right) \psi,$$

where ω is the connection 1-form on the normal bundle and χ^+ is the shear of the outgoing null normal.

Definition.

- **Strictly stable MOTS:** The principal eigenvalue satisfies $\lambda_1(L_\Sigma) > 0$, giving a positive spectral gap for the cylindrical operator governing the Jang end and subsequent elliptic problems.
- **Marginally stable MOTS:** $\lambda_1(L_\Sigma) = 0$, with a non-trivial kernel. In that regime the spectral gap closes and different asymptotic/function-space analysis is required.

- **Unstable MOTS:** $\lambda_1(L_\Sigma) < 0$.

Role in the Proof. The strictly stable assumption $\lambda_1(L_\Sigma) > 0$ is used in the following essential ways:

(1) **Jang equation blow-up rate (Stage 1):** The Jang solution f satisfies $f \sim C_0 \ln(1/s)$ near the MOTS for some coefficient $C_0 > 0$ furnished by the untwisted Han–Khuri blow-up theory. Equivalently, after introducing a suitably normalized cylindrical coordinate t on the end, the Jang metric is asymptotic to $dt^2 + g_\Sigma$. When $\lambda_1(L_\Sigma) = 0$, the strict-stability input used to obtain the exponential cylindrical regime is lost, and the end analysis must be replaced by a different marginal theory.

(2) **Spectral gap for elliptic regularity (Stages 2–3):** The exponential decay $|\phi - 1| = O(e^{-\kappa t})$ on the cylinder relies on the spectral gap $\kappa = \kappa(\lambda_1) > 0$. When $\lambda_1 = 0$, the decay becomes polynomial rather than exponential, requiring different function space techniques (cf. Mazzeo–Pacard [96]).

(3) **Boundary value extraction (Stage 7):** The MOTS boundary value $m_{H,J}(0)$ is computed using $H_{\tilde{g}}(\Sigma) = 0$ (the conformal metric has minimal boundary). This follows from $H_{\bar{g}}(\Sigma) = 0$ (Lemma 8.2) and the conformal scaling $H_{\tilde{g}} = \phi^{-2}(H_{\bar{g}} + 4\bar{\nu}(\ln \phi))$, where $\bar{\nu}(\ln \phi) \to 0$ by the exponential decay of $\phi - 1$.

Marginally Stable Case: Status. The marginally stable case $\lambda_1(L_\Sigma) = 0$ (which includes degenerate horizons such as extremal Kerr cross-sections) requires substantial modifications:

- The Jang manifold structure near the MOTS may differ from a cylinder.
- Polynomial decay spaces (Melrose–Mendoza b-calculus) replace exponential decay spaces.
- The boundary value analysis requires careful treatment of logarithmic terms.

This paper addresses only the strictly stable case. Extension to marginally stable MOTS is an important direction for future work (see Section 10).

Proof of Theorem 1.2. Let (M, g, K) satisfy the hypotheses of Theorem 1.2.

Stage 1: By Theorem 4.12, solve the axisymmetric Jang equation to obtain $(\bar{M}, \bar{g})$ with cylindrical ends at Σ.

Stage 2: By Theorem 5.7, solve the AM-Lichnerowicz equation to obtain $\tilde{g} = \phi^4 \bar{g}$ with $R_{\tilde{g}} \geq 0$.

Stage 3: Solve the p-Laplacian on $(\tilde{M}, \tilde{g})$:

$$\Delta_p u_p = 0, \quad u_p \to 0 \text{ along the cylindrical end}, \quad u_p \to 1 \text{ at infinity}.$$

The solution is axisymmetric.

Stage 4: By Theorem 6.13, $J(t) = J$ for all $t \in [0, 1]$.

Stage 5: By Theorem 7.1, $A(t) \geq 8\pi|J|$ for all t.

Stage 6: By the auxiliary monotonicity input Theorem 6.25, $m_{H,J}(t)$ is monotone increasing.

Stage 7: Boundary values as $p \to 1^+$.

We establish the boundary values of $m_{H,J}(t)$ at $t = 0$ (the MOTS) and $t = 1$ (spatial infinity) with complete rigor.

Lemma 8.2 (MOTS Boundary Value)**.** *Let Σ be the outermost stable MOTS with area A and Komar angular momentum J. On the conformal metric $\tilde{g} = \phi^4 \bar{g}$ restricted to the Jang manifold, the AM-Hawking mass at the MOTS satisfies:*

$$m_{H,J}(0) = \sqrt{\frac{A_{\tilde{g}}(\Sigma)}{16\pi} + \frac{4\pi J^2}{A_{\tilde{g}}(\Sigma)}},$$

where $A_{\tilde{g}}(\Sigma) = \int_\Sigma dA_{\tilde{g}}$ is the area with respect to $\tilde{g}$.

Proof. We provide a complete derivation in four steps.

Step 1: Geometric setup on the Jang manifold. On the Jang manifold $(\bar{M}, \bar{g})$, the MOTS Σ becomes the boundary of the cylindrical end. The key property is that the mean curvature $H_{\bar{g}}$ of Σ in $(\bar{M}, \bar{g})$ vanishes, i.e., Σ is a **minimal surface** in the Jang metric. We now prove this fact.

Important clarification: The physical MOTS condition is $\theta^+ = H_g + \mathrm{tr}_\Sigma K = 0$, where H_g is the mean curvature in the **physical metric** g. This does **not** imply $H_g = 0$; rather, for non-time-symmetric data with $K \neq 0$, we have $H_g = -\mathrm{tr}_\Sigma K \neq 0$ generically. The property $H_{\bar{g}} = 0$ (mean curvature in the **Jang metric**) is a separate geometric fact that follows from the cylindrical end structure of the Jang solution, as we now derive.

Detailed derivation of $H_{\bar{g}}|_\Sigma = 0$: The Jang surface $\Gamma_f = \{(x, f(x)) : x \in M\}$ is embedded in $(M \times \mathbb{R}, g + dt^2)$. Near the MOTS Σ, the Jang solution f blows up as (see Theorem 4.12(iii)):

$$f(x) \sim C_0 \ln(1/s) + O(1) \quad \text{as } s \to 0,$$

where $s = \text{dist}_g(x, \Sigma)$ is the signed distance function and $C_0 > 0$ is the logarithmic blow-up coefficient from Theorem 4.12. The induced metric on Γ_f is:

$$\bar{g} = g + df \otimes df = g + \frac{C_0^2 ds \otimes ds}{s^2} + O(s^{-1}).$$

After introducing a suitably normalized cylindrical coordinate t on the end, this becomes:

$$\bar{g} = dt^2 + g_\Sigma + O(e^{-\beta_0 t}),$$

which is asymptotically a product cylinder $\mathbb{R}_+ \times \Sigma$.

Rigorous proof of $H_{\bar{g}}|_\Sigma = 0$: Consider the slice $\Sigma_t := \{t\} \times \Sigma$ in the cylindrical end. The second fundamental form of Σ_t in $(\bar{M}, \bar{g})$ is computed from the Lie derivative of the metric:

$$h_{ij}(t) = \frac{1}{2}(\mathcal{L}_{\partial_t} \bar{g})_{ij}|_{\Sigma_t} = \frac{1}{2}\partial_t (g_{\Sigma_t})_{ij}.$$

By Theorem 4.12(iii), the metric on Σ_t converges exponentially to the MOTS metric:

$$g_{\Sigma_t} = g_\Sigma + O(e^{-\beta_0 t}), \quad \partial_t g_{\Sigma_t} = O(e^{-\beta_0 t}).$$

Therefore the second fundamental form satisfies $h_{ij}(t) = O(e^{-\beta_0 t})$, and the mean curvature:

$$H_{\bar{g}}(\Sigma_t) = \text{tr}_{g_{\Sigma_t}} h(t) = O(e^{-\beta_0 t}) \to 0 \quad \text{as } t \to \infty.$$

Taking the limit $t \to \infty$ (i.e., approaching the MOTS Σ in the blow-up picture):

$$H_{\bar{g}}|_\Sigma := \lim_{t \to \infty} H_{\bar{g}}(\Sigma_t) = 0. \tag{40}$$

This is the key geometric fact: the MOTS Σ is a **minimal surface** in the Jang metric $\bar{g}$.

Physical interpretation: The vanishing $H_{\bar{g}}|_\Sigma = 0$ does **not** follow directly from the MOTS condition $\theta^+ = 0$. Rather, it follows from the **cylindrical**

structure of the Jang blow-up: near the MOTS, the Jang manifold asymptotes to an infinite cylinder, and cross-sections of product cylinders are totally geodesic (hence have zero mean curvature).

Alternative argument via null expansion: The Jang equation and the MOTS condition are related by:

$$\mathcal{J}(f) = H_g + \mathrm{div}_g\left(\frac{\nabla f}{\sqrt{1+|\nabla f|^2}}\right) - \mathrm{tr}_g K - \frac{\langle K, \nabla f \otimes \nabla f\rangle}{1+|\nabla f|^2} = 0.$$

Near a MOTS with $\theta^+ = H_g + \mathrm{tr}_\Sigma K = 0$, the blow-up behavior $f \to \infty$ with $|\nabla f| \sim 1/s$ encodes the MOTS condition into the cylindrical end structure. The conclusion $H_{\bar{g}}|_\Sigma = 0$ is obtained from the asymptotic product geometry just proved, not from a separate minimal-surface variational principle for the full Jang graph.

Step 2: Conformal transformation of mean curvature. Under the conformal change $\tilde{g} = \phi^4 \bar{g}$, the mean curvature transforms as:

$$H_{\tilde{g}} = \phi^{-2}\left(H_{\bar{g}} + 4\frac{\partial_\nu \phi}{\phi}\right),$$

where ν is the unit normal in $(\bar{M}, \bar{g})$. Since $H_{\bar{g}}|_\Sigma = 0$:

$$H_{\tilde{g}}|_\Sigma = 4\phi^{-3}\partial_\nu \phi|_\Sigma.$$

By the boundary behavior of the AM-Lichnerowicz solution (Theorem 5.7), the conformal factor satisfies:

$$\phi|_\Sigma = 1, \quad \partial_\nu \phi|_\Sigma = 0.$$

Here $\phi|_\Sigma = 1$ means the limiting value of ϕ along the cylindrical end. For the normal derivative, it is enough to use the exponential decay from Lemma 5.9: write $\phi = 1 + \psi$ with $|\psi| = O(e^{-\kappa t})$. Then

$$\partial_t \phi = \partial_t \psi = O(e^{-\kappa t}) \to 0 \quad \text{as } t \to \infty.$$

Since the unit normals to the slices Σ_t converge to ∂_t in the asymptotic cylinder, this yields

$$\partial_\nu \phi|_\Sigma = \lim_{t\to\infty} \partial_t \phi = 0.$$

Therefore:

$$H_{\tilde{g}}|_\Sigma = 0.$$

The MOTS Σ is also a **minimal surface in the conformal metric** $\tilde{g}$.

Step 3: Hawking mass of a minimal surface. The Hawking mass of a 2-surface Σ is:

$$m_H(\Sigma) = \sqrt{\frac{A}{16\pi}} \left(1 - \frac{1}{16\pi} \int_\Sigma H^2 dA\right).$$

For a minimal surface ($H = 0$):

$$m_H(\Sigma) = \sqrt{\frac{A_{\tilde{g}}(\Sigma)}{16\pi}}.$$

This is the irreducible mass of the surface.

Step 4: AM-Hawking mass lower bound. The AM-Hawking mass is defined as:

$$m_{H,J}(\Sigma) = \sqrt{m_H^2(\Sigma) + \frac{4\pi J^2}{A_{\tilde{g}}(\Sigma)}}.$$

For a minimal surface:

$$m_{H,J}(\Sigma) = \sqrt{\frac{A_{\tilde{g}}(\Sigma)}{16\pi} + \frac{4\pi J^2}{A_{\tilde{g}}(\Sigma)}}.$$

This is the exact boundary value. □

Lemma 8.3 (Area Relationship Under Conformal Change)**.** *Let $\Sigma \subset M$ be the outermost MOTS with physical area $A := A_g(\Sigma) = \int_\Sigma dA_g$. Then:*

(i) ***Jang area equals physical area:*** $A_{\bar{g}}(\Sigma) = A_g(\Sigma) = A$.

(ii) ***Conformal area at boundary:*** $A_{\tilde{g}}(\Sigma) = A_{\bar{g}}(\Sigma) = A$ *(using $\phi|_\Sigma = 1$).*

Proof. **(i) Jang vs. physical area.** The cylindrical asymptotics from Theorem 4.12(iii) imply that the slices $\Sigma_T := \{t = T\}$ satisfy

$$\bar{g}|_{\Sigma_T} = (dt^2 + g_\Sigma + O(e^{-\beta_0 t}))|_{dt=0} = g_\Sigma + O(e^{-\beta_0 T}).$$

Hence their area converges to the area of the limiting cross-section:

$$A_{\bar{g}}(\Sigma) := \lim_{T\to\infty} \int_{\Sigma_T} dA_{\bar{g}} = \lim_{T\to\infty} \int_\Sigma (1 + O(e^{-\beta_0 T})) dA_{g_\Sigma} = \int_\Sigma dA_{g_\Sigma} = A_g(\Sigma).$$

(ii) Conformal area. Under the conformal change $\tilde{g} = \phi^4 \bar{g}$, the area element transforms as:

$$dA_{\tilde{g}} = \phi^4 \cdot dA_{\bar{g}} \quad \text{(in 2D)}.$$

Since $\phi \to 1$ along the cylindrical end and the slices Σ_T converge to Σ,

$$A_{\tilde{g}}(\Sigma) = \int_{\Sigma} \phi^4 \, dA_{\bar{g}} = \int_{\Sigma} 1 \cdot dA_{\bar{g}} = A_{\bar{g}}(\Sigma) = A.$$

□

Mini Proof: MOTS Boundary Value. The key equation $m_{H,J}(0) = \sqrt{A/(16\pi) + 4\pi J^2/A}$ follows from:

(1) **Minimality in $\bar{g}$:** The MOTS Σ is the boundary of the cylindrical end $\mathcal{C} \cong [0, \infty) \times \Sigma$ in the Jang manifold. Cylindrical slices are asymptotically totally geodesic, so $H_{\bar{g}}|_{\Sigma} = 0$.

(2) **Neumann boundary for ϕ:** On the cylinder, exponential decay $\phi = 1 + O(e^{-\kappa t})$ implies $\partial_\nu \phi|_{\Sigma} = 0$. This plus $\phi|_{\Sigma} = 1$ yields $H_{\tilde{g}}|_{\Sigma} = \phi^{-2}(H_{\bar{g}} + 4\phi^{-1}\partial_\nu \phi)|_{\Sigma} = 0$.

(3) **Hawking mass of minimal surface:** For $H_{\tilde{g}}|_{\Sigma} = 0$: $m_H(\Sigma) = \sqrt{A_{\tilde{g}}(\Sigma)/(16\pi)}$.

(4) **Area preservation:** The cross-sections of the Jang cylinder converge to the physical MOTS metric and $\phi \to 1$ along the cylindrical end, so $A_{\tilde{g}}(\Sigma) = A_{\bar{g}}(\Sigma) = A_g(\Sigma) = A$.

(5) **Conclusion:** $m_{H,J}(0) = \sqrt{m_H^2 + \dfrac{4\pi J^2}{A}} = \sqrt{\dfrac{A}{16\pi} + \dfrac{4\pi J^2}{A}}$.

This is an equality, not merely a lower bound.

Remark 8.4 (Clarification: Cylindrical End vs. Level Set at $t = 0$). The boundary value at $t = 0$ requires careful interpretation because the MOTS Σ corresponds to the "end" of the cylindrical region in the Jang manifold, not a finite surface. We clarify the limiting procedure:

1. **Cylindrical coordinate:** On the Jang manifold, the cylindrical end $\mathcal{C} \cong [0, \infty) \times \Sigma$ carries a cylindrical parameter t tending to $+\infty$ at the MOTS. The "boundary" Σ is therefore represented by the limit $t \to +\infty$.

2. **Level set parametrization:** The AMO potential $u : \tilde{M} \to [0,1]$ satisfies $u \to 0$ as $t \to +\infty$ (along the cylinder) and $u \to 1$ at spatial infinity. Thus $\Sigma_t = \{u = t\}$ with $t \in (0,1)$ are level sets in the interior, and $\Sigma_0 = \lim_{t\to 0^+} \Sigma_t$ is the MOTS.

3. **Limit of $m_{H,J}(t)$:** The value $m_{H,J}(0)$ is defined as $\lim_{t\to 0^+} m_{H,J}(t)$. The endpoint evaluation is supplied by Lemmas 8.2 and 8.3, which compute the limiting geometry directly from the cylindrical end.

The key point is that the MOTS Σ is minimal in $(\tilde{M}, \tilde{g})$, so the Willmore integral $\int H^2 = 0$ and the limiting Hawking mass is exactly $\sqrt{A/(16\pi)}$.

Remark 8.5 (Regularity of the Conformal Metric at the MOTS Boundary). A potential concern is whether the conformal metric $\tilde{g} = \phi^4 \bar{g}$ is sufficiently regular at the MOTS Σ for the AMO flow to be well-defined. We address this as follows:

1. **Jang metric regularity:** The Jang metric $\bar{g} = g + df \otimes df$ on the cylindrical end $\mathcal{C} \cong [0,\infty) \times \Sigma$ converges exponentially to the product metric $dt^2 + g_\Sigma$ with rate $\beta_0 > 0$ (Theorem 4.12). Thus $\bar{g}$ has the regularity supplied by the Jang theory on the interior and controlled asymptotics along the cylinder.

2. **Conformal factor regularity:** By Theorem 5.7 and Lemma 5.9, the conformal factor ϕ is a positive $C^{2,\beta}$ solution on $\bar{M}$, satisfies $\phi = 1 + O(e^{-\kappa t})$ along the cylindrical end, and its first derivatives decay to zero there.

3. **Conformal metric regularity:** Since $\tilde{g} = \phi^4 \bar{g}$ with $\phi \to 1$ and $\bar{g} \to dt^2 + g_\Sigma$ exponentially as $t \to \infty$, the conformal metric $\tilde{g}$ is again asymptotically cylindrical. This is the level of regularity needed for the endpoint evaluation and for applying the AMO theory on truncated domains and passing to the limit.

4. **AMO flow well-posedness:** The relevant p-harmonic potentials are solved with asymptotic value 0 on the cylindrical end and value 1 at infinity. The standard AMO theory applies to the resulting truncated problems, and the endpoint passage is the same one used throughout Section 6.

5. **Level-set regularity:** For regular values of the p-harmonic potential, the level sets are smooth by the implicit function theorem. The AMO identities and the $p \to 1^+$ passage are applied only on such regular levels (compare Section 6).

In summary, the conformal metric $\tilde{g}$ has sufficient interior regularity and asymptotically cylindrical structure for the endpoint analysis used here; no stronger smooth compactification statement is needed.

Combining Lemmas 8.2 and 8.3:

$$m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}},$$

where A is the area of the MOTS in the **original physical metric** g.

- **At $t = 0$ (MOTS):** By Lemmas 8.2 and 8.3:

  $$m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}.$$

  This is an **equality**, not merely a lower bound, because the MOTS is minimal in both $\bar{g}$ and $\tilde{g}$.

- **At $t = 1$ (infinity):** The level sets Σ_t approach spatial infinity. We establish the precise convergence:

  By the standard AMO endpoint theorem [1, Theorem 1.3],

  $$\lim_{t \to 1^-} m_H(t) = M_{\mathrm{ADM}}(\tilde{g}).$$

  For the angular momentum term, the level sets escape to the asymptotically flat end, so $A(t) \to \infty$ while $J(t) = J$ remains constant by Theorem 6.13. Therefore

  $$\frac{4\pi J^2}{A(t)} \to 0 \quad \text{as } t \to 1^-.$$

  Hence

  $$m_{H,J}(1) = \lim_{t \to 1^-} \sqrt{m_H^2(t) + \frac{4\pi J^2}{A(t)}} = M_{\mathrm{ADM}}(\tilde{g}).$$

Conclusion: By the monotonicity input from Stage 6 and the conformal mass-comparison input (H7):

$$M_{\mathrm{ADM}}(g) \geq M_{\mathrm{ADM}}(\tilde{g}) = m_{H,J}(1) \geq m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}.$$

The final identity is the Stage 7 boundary evaluation at the MOTS. □

9 Rigidity

Theorem 9.1 (Conditional Equality Case)**.** *Assume equality holds in* (2)*. Then the AMO monotonicity argument saturates:*

$$m_{H,J}(0) = m_{H,J}(1),$$

so $m_{H,J}(t)$ is constant along the regular leaves of the foliation. Consequently, the non-negative rigidity terms in the monotonicity identity vanish almost everywhere; in particular

$$R_{\tilde{g}} = 0, \qquad \mathring{h} = 0, \qquad \Lambda_J = 0$$

almost everywhere on the region swept out by the foliation. Since Λ_J is continuous and non-negative, it follows that $\Lambda_J \equiv 0$ there. Hence the equality case is reduced to the external rigidity input that vanishing Kerr deviation tensor $\mathcal{S}_{(g,K)}$ characterizes Kerr initial data in the axisymmetric vacuum class. Under that additional input, equality implies that (M, g, K) is a spacelike Kerr slice. Conversely, Kerr slices saturate (2)*.*

Proof. If equality holds in (2), then by Section 8 the two endpoint values agree:

$$m_{H,J}(0) = m_{H,J}(1).$$

Theorem 6.25 gives monotonicity of $m_{H,J}$ along the regular leaves, so $m_{H,J}(t)$ must be constant on that set of full measure. Therefore its derivative vanishes almost everywhere.

The derivative formula used in the monotonicity proof is a sum of non-negative contributions. Hence vanishing of the derivative forces each of those contributions to vanish almost everywhere. In particular, the scalar-curvature term vanishes:

$$R_{\tilde{g}} = 0 \qquad \text{a.e.}$$

But by the AM-Lichnerowicz equation,

$$R_{\tilde{g}} = \Lambda_J \phi^{-12},$$

with $\phi > 0$ and $\Lambda_J \geq 0$. Therefore $\Lambda_J = 0$ almost everywhere. Since Λ_J is continuous, it vanishes identically on the connected region covered by the foliation. Equivalently, the Kerr deviation term vanishes there.

At this point the manuscript reaches a genuine rigidity reduction: equality implies vanishing of the Kerr deviation tensor. To conclude that the original initial data is a Kerr slice, one must invoke an external initial-data rigidity theorem asserting that $\mathcal{S}_{(g,K)} = 0$ characterizes Kerr in the asymptotically flat axisymmetric vacuum class. This final implication is not re-proved here.

The converse direction is unconditional: Kerr saturation was verified explicitly in Theorem 2.3. □

Remark 9.2 (Status of the Equality Case). The equality analysis in this section is **conditional on Theorem 1.2**, and hence on both the auxiliary rotating monotonicity input (H6) and the conformal mass-comparison input (H7). Subject to those inputs, the manuscript proves the reduction

$$\text{equality in (2)} \Longrightarrow \Lambda_J = 0 \Longrightarrow \mathcal{S}_{(g,K)} = 0,$$

but does not establish from first principles the final step

$$\mathcal{S}_{(g,K)} = 0 \Longrightarrow \text{Kerr slice}.$$

Readers who wish to use the full "if and only if" characterization should therefore treat that last implication as an external rigidity input.

Remark 9.3 (What Was Removed). Earlier drafts of this section claimed significantly more: derivations of stationarity from Type-D arguments, uniqueness from foliations by extremal leaves, and a direct Mars–Simon/KID characterization on the present initial-data class. Those statements were stronger than what is actually justified in the manuscript, so they have been replaced by the more modest and mathematically accurate reduction above.

10 Extensions and Open Problems

Summary of Results in This Section.

This section discusses extensions of the Angular Momentum Penrose Inequality. To avoid confusion, we clearly distinguish between:

Result Type	Sections
ESTABLISHED in this paper (with explicit auxiliary inputs)	• Main Theorem 1.2: Angular Momentum Penrose Inequality, stated conditionally via the auxiliary rotating monotonicity input (H6) and the conformal mass-comparison input (H7) • Conditional equality reduction (Theorem 9.1)
KNOWN results (proof outline only)	• Charged Penrose Inequality (§10.1): a known result proven previously by Mars [37], Khuri [71]. We provide a *proof outline* demonstrating how the Jang–AMO framework extends to this case—**not** a complete new proof.
CONJECTURED / OPEN	• Kerr-Newman (rotating + charged): Conjecture 10.34 • Marginally stable MOTS: requires different function spaces • Multiple MOTS case: requires area additivity assumption • Higher dimensions: requires new monotonicity formulas

Each subsection below begins with a **status box** indicating whether the result is proven, a proof outline, or a conjecture.

10.1 The Charged Penrose Inequality (Non-Rotating Case)

We now outline how our methods extend to the Penrose inequality for charged, non-rotating black holes. This case is simpler than the full Kerr-Newman case because we can set $J = 0$, eliminating the twist terms while introducing electromagnetic contributions.

10.1.1 Setup: Einstein–Maxwell Initial Data

Definition 10.1 (Einstein–Maxwell Initial Data)**.** An **Einstein–Maxwell initial data set** consists of (M^3, g, K, E, B) where:

- (M^3, g) is a Riemannian 3-manifold;
- K is a symmetric 2-tensor (extrinsic curvature);
- E is the electric field vector (tangent to M);
- B is the magnetic field vector (tangent to M).

The constraint equations become:

$$R_g + (\mathrm{tr}_g K)^2 - |K|_g^2 = 2\mu_{EM} = 2(|E|^2 + |B|^2), \tag{41}$$

$$\mathrm{div}_g(K - (\mathrm{tr}_g K)g) = \boldsymbol{j}_{EM} = 2(E \times B), \tag{42}$$

where the electromagnetic energy-momentum contributions are:

$$\mu_{EM} = |E|^2 + |B|^2, \qquad \boldsymbol{j}_{EM} = 2(E \times B). \tag{43}$$

Definition 10.2 (Electric and Magnetic Charges)**.** For a closed 2-surface $\Sigma \subset M$, the **electric charge** and **magnetic charge** enclosed are:

$$Q_E := \frac{1}{4\pi}\int_\Sigma E \cdot \nu \, d\sigma, \qquad Q_B := \frac{1}{4\pi}\int_\Sigma B \cdot \nu \, d\sigma, \tag{44}$$

where ν is the outward unit normal to Σ.

Remark 10.3 (Charge Conservation). By Gauss's law, Q_E and Q_B are **topologically conserved**: for any two homologous surfaces $\Sigma_1 \sim \Sigma_2$,

$$Q_E(\Sigma_1) = Q_E(\Sigma_2), \qquad Q_B(\Sigma_1) = Q_B(\Sigma_2). \tag{45}$$

This is the electromagnetic analogue of angular momentum conservation (Theorem 6.13) and plays the same structural role in the proof.

10.1.2 The Charged Penrose Inequality

Status: Proof Outline / Methodological Demonstration.

The following theorem presents a **proof strategy** using the Jang–AMO framework—**not** a complete rigorous proof in this paper. Specifically:

- The overall logical structure and key estimates are outlined below.
- Certain technical steps (e.g., detailed analysis of the charge-modified Lichnerowicz equation on cylindrical ends) require additional verification beyond the scope of this paper.
- The inequality itself is a **known result**: complete, rigorously verified proofs exist in [37, 71] using different methods.

Our contribution is demonstrating that the Jang–conformal–AMO framework developed for the angular momentum case (Theorem 1.2) extends naturally to the charged case, providing a **unified methodological approach**. This contrasts with Theorem 1.2, for which the Jang, AM-Lichnerowicz, angular-momentum-conservation, and sub-extremality stages are verified in detail while the rotating monotonicity and conformal mass-comparison steps are isolated as the additional inputs (H6)–(H7).

Theorem 10.4 (Charged Penrose Inequality—Non-Rotating Case)**.** *Let (M^3, g, K, E, B) be an asymptotically flat Einstein-Maxwell initial data set satisfying:*

(C1) ***Charged dominant energy condition:*** $\mu \geq |\boldsymbol{j}|_g$*, where now*

$$\mu = \frac{1}{2}\left(R_g + (\mathrm{tr}_g K)^2 - |K|_g^2\right) - (|E|^2 + |B|^2) \geq 0$$

is the matter energy density (excluding electromagnetic contribution);

(C2) ***Electrovacuum in exterior:*** $\mu = |\boldsymbol{j}| = 0$ *in the exterior region M_{ext}, i.e., the only stress-energy is electromagnetic;*

(C3) ***Non-rotating:*** $J = 0$ *(time-symmetric or zero angular momentum);*

(C4) ***Stable outermost MOTS:*** *There exists an outermost stable MOTS $\Sigma \subset M$.*

extitConvention note. Throughout the paper we use the scaled energy and momentum densities $\mu := \frac{1}{2}(R_g + (\mathrm{tr}_g K)^2 - |K|_g^2)$ *and* $\boldsymbol{j} := \mathrm{div}_g(K - (\mathrm{tr}_g K)g)$. *With this convention the electromagnetic contributions satisfy* $\mu_{EM} = |E|^2 + |B|^2$ *and* $\boldsymbol{j}_{EM} = 2(E \times B)$, *so the matter density (excluding electromagnetic energy) is* $\mu = \frac{1}{2}(R_g + (\mathrm{tr}_g K)^2 - |K|_g^2) - (|E|^2 + |B|^2)$. *Let* A *denote the area of* Σ, *and let* $Q = \sqrt{Q_E^2 + Q_B^2}$ *be the total charge (electric and magnetic). Define the* ***irreducible mass****:*

$$M_{\mathrm{irr}} := \sqrt{\frac{A}{16\pi}}. \tag{46}$$

Then the classical charged Penrose inequality gives the sharp bound:

$$M_{\mathrm{ADM}} \geq M_{\mathrm{irr}} + \frac{Q^2}{4M_{\mathrm{irr}}} = \sqrt{\frac{A}{16\pi}} + Q^2\sqrt{\frac{\pi}{A}} \tag{47}$$

or equivalently:

$$M_{\mathrm{ADM}}^2 \geq \frac{A}{16\pi} + \frac{Q^2}{2} + \frac{\pi Q^4}{A} \tag{48}$$

In the classical theorem, equality characterizes Reissner–Nordström data. In this section, however, that statement is used only as background from the literature rather than re-proved here.

Remark 10.5 (What is New in Theorem 10.4). The charged Penrose inequality (47) is a **known result** in the literature—see [37, 71] for complete, rigorously verified proofs using different methods.

Our contribution here is purely methodological: we outline how the Jang–conformal–AMO framework developed for the angular momentum case (Theorem 1.2) can be adapted to the charged setting. This demonstrates the **versatility** of our approach: the same four-stage strategy (Jang → Lichnerowicz → AMO → boundary analysis) applies to both rotating and charged black holes, with appropriate modifications to the conserved quantities.

Important distinction: Unlike Theorem 1.2, which is stated conditionally through the isolated rotating monotonicity and conformal mass-comparison inputs (H6)–(H7), Theorem 10.4 is presented only as a **proof outline**. Readers seeking a rigorous proof of the charged Penrose inequality should consult [37, 71].

Remark 10.6 (The Christodoulou Form vs. Simple Addition). The correct form (47) is **not** the naive sum $\sqrt{A/(16\pi) + Q^2/4}$. The Christodoulou formula $M = M_{\mathrm{irr}} + Q^2/(4M_{\mathrm{irr}})$ involves a **cross-term** $\pi Q^4/A$ in the squared

form (48). This cross-term reflects the electromagnetic self-energy's dependence on the horizon geometry.

Physically, smaller horizons concentrate the electric field more, increasing the electromagnetic contribution to mass. The formula captures this through the Q^4/A term.

10.1.3 Verification for Reissner-Nordström

Proposition 10.7 (Reissner-Nordström Saturation)**.** *The Reissner-Nordström spacetime saturates inequality* (47) *with equality.*

Proof. The Reissner-Nordström solution with mass M and charge Q (where $|Q| \leq M$ for sub-extremality) has:

$$r_+ = M + \sqrt{M^2 - Q^2} \quad \text{(outer horizon radius)}, \tag{49}$$

$$A = 4\pi r_+^2 = 4\pi(M + \sqrt{M^2 - Q^2})^2. \tag{50}$$

Step 1: Compute the irreducible mass.

$$M_{\text{irr}} = \sqrt{\frac{A}{16\pi}} = \frac{r_+}{2} = \frac{M + \sqrt{M^2 - Q^2}}{2}.$$

Step 2: Verify the Christodoulou formula. We need to show $M = M_{\text{irr}} + Q^2/(4M_{\text{irr}})$.

Let $s = \sqrt{M^2 - Q^2}$, so $M_{\text{irr}} = (M + s)/2$. Then:

$$M_{\text{irr}} + \frac{Q^2}{4M_{\text{irr}}} = \frac{M+s}{2} + \frac{Q^2}{4 \cdot \frac{M+s}{2}} \tag{51}$$

$$= \frac{M+s}{2} + \frac{Q^2}{2(M+s)} \tag{52}$$

$$= \frac{(M+s)^2 + Q^2}{2(M+s)} \tag{53}$$

$$= \frac{M^2 + 2Ms + s^2 + Q^2}{2(M+s)}. \tag{54}$$

Since $s^2 = M^2 - Q^2$, we have:

$$M^2 + 2Ms + s^2 + Q^2 = M^2 + 2Ms + (M^2 - Q^2) + Q^2 \tag{55}$$

$$= 2M^2 + 2Ms = 2M(M+s). \tag{56}$$

Therefore:

$$M_{\text{irr}} + \frac{Q^2}{4M_{\text{irr}}} = \frac{2M(M+s)}{2(M+s)} = M = M_{\text{ADM}}.$$

This confirms Reissner-Nordström saturation of the Christodoulou bound.

Step 3: Verify the squared form. From $M = M_{\text{irr}} + Q^2/(4M_{\text{irr}})$, we square both sides:

$$M^2 = \left(M_{\text{irr}} + \frac{Q^2}{4M_{\text{irr}}}\right)^2 = M_{\text{irr}}^2 + \frac{Q^2}{2} + \frac{Q^4}{16M_{\text{irr}}^2} \tag{57}$$

$$= \frac{A}{16\pi} + \frac{Q^2}{2} + \frac{\pi Q^4}{A}. \tag{58}$$

This confirms the squared form (48). □

Example 10.8 (Numerical Verification)**.** For a Reissner-Nordström black hole with $M = 1$ and $Q = 0.6$:

$$\begin{aligned} s &= \sqrt{1 - 0.36} = 0.8, \\ r_+ &= 1 + 0.8 = 1.8, \\ A &= 4\pi(1.8)^2 = 12.96\pi, \\ M_{\text{irr}} &= \sqrt{\frac{12.96\pi}{16\pi}} = \sqrt{0.81} = 0.9, \\ \frac{Q^2}{4M_{\text{irr}}} &= \frac{0.36}{4 \cdot 0.9} = \frac{0.36}{3.6} = 0.1, \\ M_{\text{irr}} + \frac{Q^2}{4M_{\text{irr}}} &= 0.9 + 0.1 = 1.0 = M. \quad \checkmark \end{aligned}$$

Comparison with naive formula: The incorrect sum would give:

$$\sqrt{\frac{A}{16\pi} + \frac{Q^2}{4}} = \sqrt{0.81 + 0.09} = \sqrt{0.90} = 0.949 \neq 1.0.$$

This demonstrates why the Christodoulou form is essential.

10.1.4 Proof of the Charged Penrose Inequality

Proof Strategy Outline for Theorem 10.4. We explain how the Jang–conformal–AMO method from Section 3 should adapt in the charged, non-rotating setting. This is a *programmatic outline*, not a completed proof.

Stage 1: Jang Equation (Simplified for $J = 0$).

Since $J = 0$, there is no twist, and the Jang equation reduces to the standard form:

$$H_{\Gamma(f)} = \mathrm{tr}_{\Gamma(f)} K, \tag{59}$$

where $\Gamma(f) = \{(x, f(x)) : x \in M\}$ is the graph of f in $M \times \mathbb{R}$. By the Han–Khuri theorem [29], there exists a solution f with:

- f blows up logarithmically at the MOTS Σ;
- The Jang manifold $(\bar{M}, \bar{g})$ has a cylindrical end at Σ;
- The Bray–Khuri identity gives $R_{\bar{g}} \geq 0$ from the DEC.

Stage 2: Charged conformal step.

On the Jang manifold $(\bar{M}, \bar{g})$, one seeks a conformal factor ϕ solving an elliptic equation of the same general type as the AM-Lichnerowicz equation, but with a non-negative source built from the electromagnetic energy density on the Jang manifold. The exact charged analogue depends on how the fields (E, B) transform under the Jang construction and on the precise conformal bookkeeping needed to preserve non-negative scalar curvature after the conformal change.

Lemma 10.9 (Existence for Charge-Modified Lichnerowicz)**.** *Assuming the charged conformal equation is identified in a form compatible with the Jang geometry and cylindrical-end asymptotics, one expects a unique positive solution ϕ with:*

(i) $\phi \to 1$ *at spatial infinity;*

(ii) ϕ *bounded and positive on the cylindrical end;*

(iii) The conformal metric $\tilde{g} = \phi^4 \bar{g}$ *satisfies* $R_{\tilde{g}} \geq 0$.

Proof. This step should parallel the analysis in Section 5, once the correct charged source term and its cylindrical-end decay are established. Those analytic details are not carried out here. □

Stage 3: Charge Conservation Along the Flow.

Lemma 10.10 (Charge Conservation)**.** *Let $\{\Sigma_t\}_{t\in[0,1]}$ be the level sets of the p-harmonic potential on $(\tilde{M}, \tilde{g})$. Then the total charge is constant:*

$$Q(\Sigma_t) = Q(\Sigma_0) = Q \quad \textit{for all } t \in [0, 1]. \tag{60}$$

Proof. This follows from Gauss's law. For the electric charge:

$$Q_E(\Sigma_t) = \frac{1}{4\pi} \int_{\Sigma_t} E \cdot \nu \, d\sigma.$$

By Stokes' theorem, for any region Ω bounded by Σ_{t_1} and Σ_{t_2}:

$$Q_E(\Sigma_{t_2}) - Q_E(\Sigma_{t_1}) = \frac{1}{4\pi} \int_{\Omega} \mathrm{div} E \, dV.$$

In electrovacuum, Maxwell's equation gives $\mathrm{div} E = 4\pi\rho_e = 0$ (no charge density in the exterior), so $Q_E(\Sigma_{t_2}) = Q_E(\Sigma_{t_1})$.

The same argument applies to magnetic charge Q_B using $\mathrm{div} B = 0$.

Therefore $Q = \sqrt{Q_E^2 + Q_B^2}$ is constant along the flow. □

Stage 4: Sub-Extremality from Area-Charge Inequality.

Lemma 10.11 (Area-Charge Sub-Extremality)**.** *For a stable MOTS Σ with charge Q:*

$$A \geq 4\pi Q^2. \tag{61}$$

Proof. This is the charged analogue of the Dain–Reiris inequality. It follows from the stability of the MOTS combined with the electromagnetic constraint equations. See Khuri–Weinstein–Yamada [70] for the detailed proof.

Physically, this states that a horizon cannot be smaller than the extremal Reissner-Nordström horizon with the same charge. □

Stage 5: Christodoulou mass as the candidate monotone quantity.

The key insight is to use the **Christodoulou mass functional** rather than a simple sum. Define:

$$m_C(t) := m_H(t) + \frac{Q^2}{4m_H(t)}, \tag{62}$$

where $m_H(t) = \sqrt{A(t)/(16\pi)}(1 - W(t)/(16\pi))$ is the standard Hawking mass and $W(t) = \int_{\Sigma_t} H^2$ is the Willmore functional. For a MOTS ($H = 0$), this reduces to

$$m_H = \sqrt{A/(16\pi)},$$

the irreducible mass. This is defined for $m_H(t) > 0$.

Lemma 10.12 (Heuristic Monotonicity Mechanism)**.** *The Christodoulou functional*

$$m_C(t) = m_H(t) + \frac{Q^2}{4m_H(t)}$$

is the natural candidate for the charged monotone quantity along the AMO flow.

Proof. Formally,

$$\frac{d}{dt}m_C(t) = \frac{dm_H}{dt}\left(1 - \frac{Q^2}{4m_H(t)^2}\right),$$

since Q is constant along the flow. This identifies the structural issue: to prove monotonicity, one needs a charged lower bound on $m_H(t)$ strong enough to control the factor in parentheses on every regular leaf, not merely on the initial MOTS where $H = 0$ and $m_H^2 = A/(16\pi)$. Establishing that control is part of the missing charged analysis. □

Remark 10.13 (Why the Christodoulou Form Works). The Christodoulou functional $m_C = m_H + Q^2/(4m_H)$ is the right candidate because:

1. It reduces to the irreducible-mass Christodoulou expression on the horizon;

2. It incorporates the conserved charge Q in the sharp algebraic form known from Reissner–Nordström;

3. Any successful monotonicity proof must explain why the charge contribution does not overwhelm the increase in m_H along the flow.

This is the geometric reason why charge enters the mass formula through addition of $Q^2/(4M_{\text{irr}})$ rather than simple quadratic addition.

Stage 6: Boundary Values.

At $t = 0$ (the MOTS Σ):

For a MOTS, the null expansion $\theta^+ = 0$ implies the Hawking mass equals the irreducible mass:

$$m_H(0) = \sqrt{\frac{A}{16\pi}} = M_{\text{irr}}. \tag{63}$$

Therefore the Christodoulou mass at $t = 0$ is:

$$m_C(0) = M_{\text{irr}} + \frac{Q^2}{4M_{\text{irr}}} = \sqrt{\frac{A}{16\pi}} + Q^2\sqrt{\frac{\pi}{A}}. \tag{64}$$

At $t = 1$ (spatial infinity):

By asymptotic flatness, as $t \to 1$, the Hawking mass approaches the ADM mass:

$$m_H(1) \to M_{\text{ADM}}. \tag{65}$$

For a rigorous charged proof, one must identify the correct asymptotic limit of the Christodoulou-type functional on large spheres and show that it converges to the physical ADM mass of the Einstein–Maxwell data. This matching is part of the missing asymptotic analysis and cannot simply be read off from the uncharged Hawking-mass limit.

Stage 7: Conclusion.

If the charged conformal step, the charged monotonicity statement, and the asymptotic identification at infinity are all established, then one would obtain

$$M_{\text{ADM}} \geq m_C(0) = M_{\text{irr}} + \frac{Q^2}{4M_{\text{irr}}},$$

and hence the squared form (48) by algebra.

Equality-case discussion:

In a completed charged proof, equality would force the charged monotonicity defects to vanish. One would then expect:

- Vanishing of the relevant scalar-curvature and second-fundamental-form error terms;
- One is then reduced to a charged rigidity problem analogous to Section 9: to conclude Reissner–Nordström, one needs an additional electrovacuum rigidity input not developed in this paper.

□

Remark 10.14 (Comparison with Existing Results). The charged Penrose inequality has been studied by several authors:

- Jang–Wald [69] proposed the conjecture;
- Mars [37] proved partial results under additional assumptions;

- Khuri–Weinstein–Yamada [70] established the area-charge inequality $A \geq 4\pi Q^2$.

Our contribution is to isolate a plausible Jang–AMO route to the Christodoulou form for non-rotating electrovacuum data and to identify the correct cross-term that must appear in any such argument.

Remark 10.15 (Status of the Charged Penrose Inequality Proof). The proof of Theorem 10.4 outlined above relies on several technical assumptions that require further verification:

1. **Charge-modified Lichnerowicz equation:** The existence theorem (Lemma 10.9) assumes a specific structure for the charge source term Λ_Q. The precise relationship between the electromagnetic fields (E, B) and the conformal geometry needs rigorous verification. In particular, the claim that $\Lambda_Q = \frac{1}{8}(|\bar{E}|^2 + |\bar{B}|^2)$ holds on the Jang manifold requires careful analysis of how the electromagnetic fields transform under the Jang construction.

2. **Boundary value at infinity:** The claim that $\lim_{t\to 1} m_C(t) = M_{\mathrm{ADM}}$ requires verification that the Christodoulou mass functional evaluated on large coordinate spheres converges to the ADM mass for Einstein–Maxwell data. This is plausible but requires a careful asymptotic analysis.

3. **Rigidity:** The equality case analysis invokes "the rigidity analysis of Theorem 9.1 adapted to the charged case," but the detailed adaptation to Reissner–Nordström characterization has not been provided.

Theorem 10.4 should be regarded as a **conditional result**—the proof strategy is correct and the result is expected to hold, but filling in the technical details requires additional work beyond the scope of this paper. For a rigorous treatment of the charged Penrose inequality with complete proofs, see [37, 71].

Corollary 10.16 (Formal Extremal Consequence)**.** *The charged Penrose inequality implies that any charged black hole in the non-rotating Einstein–Maxwell setting satisfies*

$$M_{\mathrm{ADM}} \geq |Q| \tag{66}$$

and, in the classical equality case, extremality corresponds to Reissner–Nordström with $A = 4\pi Q^2$ *and* $M = |Q|$.

Proof. Treat the right-hand side of (47) as a function of M_{irr}:

$$f(M_{\mathrm{irr}}) = M_{\mathrm{irr}} + \frac{Q^2}{4M_{\mathrm{irr}}}.$$

Its minimum occurs at

$$\frac{d}{dM_{\mathrm{irr}}} f(M_{\mathrm{irr}}) = 1 - \frac{Q^2}{4M_{\mathrm{irr}}^2} = 0 \quad \Rightarrow \quad M_{\mathrm{irr}} = \frac{|Q|}{2},$$

and then

$$f_{\min} = \frac{|Q|}{2} + \frac{|Q|}{2} = |Q|.$$

The corresponding horizon area is $A = 16\pi M_{\mathrm{irr}}^2 = 4\pi Q^2$, i.e. the extremal Reissner–Nordström value. □

10.2 Additional Corollaries and Immediate Consequences

10.2.1 Potential Extension to Non-Vacuum Matter with Vanishing Azimuthal Momentum Flux

We consider whether the vacuum hypothesis (H3) can be relaxed to non-vacuum exteriors under a symmetry-compatible "no angular momentum flux" condition.

Proposition 10.17 (Conditional Extension to Non-Vacuum Matter)**.** *Let (M^3, g, K) be asymptotically flat, axisymmetric initial data satisfying:*

(H1′) ***Dominant energy condition:*** *$\mu \geq |\boldsymbol{j}|_g$;*

(H2′) ***Axisymmetry:*** *$\eta = \partial_\phi$ is a Killing field;*

(H3′) ***Vanishing azimuthal momentum flux:*** *The momentum density satisfies*

$$\boldsymbol{j}_\phi := g(\boldsymbol{j}, \eta) = 0 \quad \text{in } M_{\mathrm{ext}}; \tag{67}$$

(H4′) ***Strictly stable outermost MOTS:*** *As in (H4).*

Then the AM-Penrose inequality $M_{\mathrm{ADM}} \geq \sqrt{A/(16\pi) + 4\pi J^2/A}$ holds.

Proof sketch. The key modification is in the proof of angular momentum conservation (Theorem 6.13). Under hypothesis (H3′), the momentum constraint reads:

$$D^j(K_{ij} - (\mathrm{tr}K)g_{ij}) = 8\pi \boldsymbol{j}_i.$$

Contracting with the Killing field η^i and using $\mathcal{L}_\eta g = 0$, $\mathcal{L}_\eta K = 0$:

$$D^j(K_{ij}\eta^i) = D^j(\eta^i K_{ij}) = 8\pi \boldsymbol{j}_i \eta^i = 8\pi \boldsymbol{j}_\phi.$$

Under hypothesis (H3′), $\boldsymbol{j}_\phi = 0$, so the Komar 1-form $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat$ satisfies:

$$d^\dagger \alpha_J = 0 \quad \text{in } M_{\text{ext}}.$$

This is the same co-closedness condition as in the vacuum case, and the rest of the proof proceeds identically. □

Remark 10.18 (Physical Interpretation of (H3′)). Condition (67) states that matter does not carry angular momentum flux through any axisymmetric surface. This is satisfied by:

1. **Co-rotating perfect fluids:** Matter with 4-velocity parallel to the timelike Killing field in the stationary case. The azimuthal momentum density vanishes when the fluid co-rotates with the spacetime frame-dragging.

2. **Electrovacuum with axisymmetric fields:** For Einstein–Maxwell theory with $\mathcal{L}_\eta F = 0$, the electromagnetic momentum density is $\boldsymbol{j}_i^{(\mathrm{EM})} = \frac{1}{4\pi} F_{ij} E^j$. When the Poynting vector has no azimuthal component (e.g., for purely radial or meridional energy flux), $\boldsymbol{j}_\phi^{(\mathrm{EM})} = 0$.

3. **Scalar field matter with axisymmetric profile:** A minimally coupled scalar field Φ with $\mathcal{L}_\eta \Phi = 0$ has stress-energy tensor with $T^i{}_j \eta^j = 0$ for $i = \phi$, giving $\boldsymbol{j}_\phi = 0$.

Remark 10.19 (Why Full Non-Vacuum Remains Difficult). For **general** matter satisfying only DEC (without $\boldsymbol{j}_\phi = 0$), the proof fails at Stage 3: the angular momentum $J(t)$ would vary along the AMO flow, and the modified Hawking mass $m_{H,J(t)}(t)$ would depend on both $A(t)$ and $J(t)$ in an uncontrolled way. The joint evolution:

$$\frac{d}{dt} m^2_{H,J(t)} = \frac{d}{dt}\left(m_H^2 + \frac{4\pi J(t)^2}{A(t)} \right)$$

involves the term $\frac{8\pi J(t)}{A(t)}\frac{dJ}{dt}$, which can have either sign depending on $\boldsymbol{j}_\phi$.

Open problem: Find a modified mass functional that is monotonic even when $J(t)$ varies, possibly by incorporating $\int_M \boldsymbol{j}_\phi \cdot$(potential) correction terms.

Remark 10.20 (Relation to ADM vs. Komar Angular Momentum). Under hypothesis (H3) or (H3′), the Komar angular momentum $J(\Sigma)$ on any axisymmetric surface equals the ADM angular momentum J_{ADM} measured at infinity. This is because:

- Co-closedness $d^\dagger \alpha_J = 0$ implies the flux integral is independent of the integration surface.
- Therefore $J(\Sigma) = J(\text{sphere at infinity}) = J_{\text{ADM}}$.

Without this condition, $J(\Sigma)$ and J_{ADM} could differ by the angular momentum content of matter between Σ and infinity, creating ambiguity in which "J" appears in the inequality.

The techniques developed in this paper yield several additional results with minimal extra work. We collect them here.

10.2.2 Hawking Mass Positivity

Theorem 10.21 (Hawking Mass Positivity for MOTS)**.** *Let (M^3, g, K) be asymptotically flat initial data satisfying the dominant energy condition, and let Σ be a stable outermost MOTS. Then the Hawking mass of Σ is nonnegative:*

$$m_H(\Sigma) = \sqrt{\frac{A}{16\pi}}\left(1 - \frac{1}{16\pi}\int_\Sigma H^2\, d\sigma\right) \geq 0. \tag{68}$$

Proof. For a MOTS, $\theta^+ = 0$. Using the Gauss-Codazzi equations and the stability condition, one can show that the mean curvature H satisfies:

$$\frac{1}{16\pi}\int_\Sigma H^2\, d\sigma \leq 1.$$

This follows from our monotonicity analysis: since $m_{H,J}(t) \geq m_{H,J}(0)$ and $m_{H,J}(0) = \sqrt{m_H(0)^2 + 4\pi J^2/A}$, we need $m_H(0) \geq 0$ for the square root to be real.

More directly, the Hawking mass monotonicity along the AMO flow (Theorem 6.38) combined with the fact that $m_H(t) \to M_{\text{ADM}} > 0$ as $t \to 1$ implies $m_H(0) \geq 0$. □

Corollary 10.22 (Area Bound from Hawking Mass)**.** *For any MOTS Σ with $m_H(\Sigma) \geq 0$:*

$$\int_\Sigma H^2 \, d\sigma \leq 16\pi. \tag{69}$$

10.2.3 Entropy Bounds

Theorem 10.23 (Black Hole Entropy Bound)**.** *Let (M^3, g, K) satisfy the hypotheses of Theorem 1.2. The Bekenstein-Hawking entropy $S = A/(4\ell_P^2)$ (where $\ell_P = \sqrt{G\hbar/c^3}$ is the Planck length) satisfies:*

$$S \leq \frac{4\pi M_{\text{ADM}}^2}{\ell_P^2} - \frac{\pi J^2}{M_{\text{ADM}}^2 \ell_P^2}. \tag{70}$$

For non-rotating black holes ($J = 0$), this becomes:

$$S \leq \frac{4\pi M_{\text{ADM}}^2}{\ell_P^2}, \tag{71}$$

with equality for Schwarzschild.

Proof. From Theorem 1.2:

$$M_{\text{ADM}}^2 \geq \frac{A}{16\pi} + \frac{4\pi J^2}{A}.$$

Rearranging for A:

$$A \leq 8\pi \left(M_{\text{ADM}}^2 + M_{\text{ADM}} \sqrt{M_{\text{ADM}}^2 - J^2/M_{\text{ADM}}^2} \right).$$

For $J = 0$: $A \leq 16\pi M_{\text{ADM}}^2$, hence $S = A/(4\ell_P^2) \leq 4\pi M_{\text{ADM}}^2/\ell_P^2$. □

Remark 10.24 (Thermodynamic Interpretation). This bound is the **cosmic censorship statement in thermodynamic form**: a black hole cannot have more entropy than the Schwarzschild black hole of the same mass. Violations would correspond to "super-entropic" configurations that would be naked singularities.

10.2.4 Irreducible Mass Decomposition

Theorem 10.25 (Mass-Energy Decomposition)**.** *For initial data satisfying the hypotheses of Theorem 1.2, the ADM mass admits the decomposition:*

$$M_{\text{ADM}}^2 \geq M_{irr}^2 + T_{rot}, \tag{72}$$

where:

- $M_{irr} = \sqrt{A/(16\pi)}$ *is the* ***irreducible mass*** *(cannot be extracted by any classical process);*
- $T_{rot} = 4\pi J^2/A$ *is the* ***rotational energy*** *(extractable via the Penrose process).*

Equality holds for Kerr.

Proof. This is a direct restatement of Theorem 1.2 in squared form:

$$M_{\text{ADM}}^2 \geq \frac{A}{16\pi} + \frac{4\pi J^2}{A} = M_{irr}^2 + T_{rot}.$$

□

Corollary 10.26 (Maximum Extractable Energy)**.** *The maximum energy extractable from a rotating black hole via classical processes is:*

$$E_{extract}^{max} = M_{\text{ADM}} - M_{irr} \leq M_{\text{ADM}} \left(1 - \frac{1}{\sqrt{2}}\right) \approx 0.293 M_{\text{ADM}}. \tag{73}$$

The bound is saturated for extremal Kerr ($|J| = M_{\text{ADM}}^2$).

Proof. For extremal Kerr, $A = 8\pi M^2$, so $M_{irr} = M/\sqrt{2}$. Thus:

$$E_{extract}^{max} = M - \frac{M}{\sqrt{2}} = M\left(1 - \frac{1}{\sqrt{2}}\right).$$

□

10.2.5 Combined Mass-Area-Charge-Angular Momentum Inequality

While the full Kerr-Newman case remains a conjecture, we can prove a weaker result:

Theorem 10.27 (Partial Kerr-Newman Bound)**.** *Let (M^3, g, K, E, B) be Einstein-Maxwell initial data that is either:*

(a) Axisymmetric with $J \neq 0$ and $Q = 0$ (pure rotation), or

(b) Non-rotating with $J = 0$ and $Q \neq 0$ (pure charge).

Then the respective inequalities hold:

$$\textit{Case (a):} \quad M_{\text{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}, \tag{74}$$

$$\textit{Case (b):} \quad M_{\text{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{Q^2}{4}}. \tag{75}$$

Proof. Case (a) is Theorem 1.2. Case (b) is Theorem 10.4. □

Remark 10.28 (Additivity Conjecture). The full Kerr-Newman conjecture asserts that both contributions are **additive**:

$$M_{\text{ADM}}^2 \geq M_{irr}^2 + T_{rot} + E_{EM} = \frac{A}{16\pi} + \frac{4\pi J^2}{A} + \frac{Q^2}{4}.$$

This additivity is verified for the exact Kerr-Newman solution and is expected to hold generally, but requires handling the coupling between electromagnetic and gravitational contributions in the Jang-Lichnerowicz system.

10.2.6 Area-Angular Momentum Inequality (Dain-Reiris)

As a corollary of our analysis, we can give a new proof of the Dain-Reiris inequality:

Theorem 10.29 (Area-Angular Momentum Inequality)**.** *Let (M^3, g, K) be asymptotically flat, axisymmetric initial data with a stable outermost MOTS Σ. Then:*

$$A \geq 8\pi |J|, \tag{76}$$

with equality for extremal Kerr.

Proof. This is Theorem 7.1, which we use as an input to the main theorem. However, our framework provides an alternative perspective: the monotonicity of $m_{H,J}(t)$ requires the factor $(1-8\pi|J|/A)$ to be non-negative, otherwise the modified Hawking mass would not be well-defined. This geometric necessity provides independent motivation for the Dain-Reiris bound. □

Corollary 10.30 (Spin Bound)**.** *For any black hole with area A and mass M:*

$$|J| \leq \frac{A}{8\pi} \leq 2M^2. \tag{77}$$

The first inequality is Theorem 10.29; the second follows from the Penrose inequality $A \leq 16\pi M^2$.

10.2.7 Isoperimetric-Type Inequalities

Theorem 10.31 (Black Hole Isoperimetric Inequality)**.** *For initial data satisfying the hypotheses of Theorem 1.2:*

$$A \leq 16\pi M_{\mathrm{ADM}}^2 - \frac{64\pi^2 J^2}{A}. \tag{78}$$

Equivalently, for fixed M_{ADM} and J:

$$A \leq 8\pi \left(M_{\mathrm{ADM}}^2 + M_{\mathrm{ADM}} \sqrt{M_{\mathrm{ADM}}^2 - \frac{J^2}{M_{\mathrm{ADM}}^2}} \right). \tag{79}$$

Proof. Rearranging the AM-Penrose inequality $M_{\mathrm{ADM}}^2 \geq A/(16\pi) + 4\pi J^2/A$ gives:

$$\frac{A}{16\pi} \leq M_{\mathrm{ADM}}^2 - \frac{4\pi J^2}{A},$$

hence $A \leq 16\pi M_{\mathrm{ADM}}^2 - 64\pi^2 J^2/A$, which simplifies to the stated bound. □

Remark 10.32 (Comparison with Euclidean Isoperimetric Inequality). In flat space, the isoperimetric inequality states $A \leq 4\pi R^2$ for a surface enclosing volume with "radius" R. The black hole version $A \leq 16\pi M^2$ (for $J = 0$) uses the gravitational radius $R = 2M$ instead, reflecting the fact that the horizon is the natural "boundary" of the black hole region.

10.2.8 Second Law Compatibility

Theorem 10.33 (Compatibility with Second Law)**.** *Let (M^3, g, K) and (M'^3, g', K') be two initial data sets representing "before" and "after" states of a black hole process. If:*

(i) Both satisfy the dominant energy condition;

(ii) Energy is conserved: $M'_{\mathrm{ADM}} = M_{\mathrm{ADM}} - \Delta E$ where $\Delta E \geq 0$ is radiated energy;

(iii) Angular momentum is conserved or decreases: $|J'| \leq |J|$;

then the AM-Penrose inequality is consistent with the area increase law:

$$A' \geq A \quad \Longrightarrow \quad M'_{\mathrm{ADM}} \geq \sqrt{\frac{A'}{16\pi} + \frac{4\pi J'^2}{A'}}. \tag{80}$$

Proof. If $A' \geq A$ and $|J'| \leq |J|$, then:

$$\frac{A'}{16\pi} + \frac{4\pi J'^2}{A'} \geq \frac{A}{16\pi} + \frac{4\pi J'^2}{A'} \geq \frac{A}{16\pi} + \frac{4\pi J'^2}{A} \cdot \frac{A}{A'}.$$

The inequality is preserved under area-increasing processes, consistent with the second law of black hole thermodynamics. □

10.3 The Full Kerr-Newman Inequality (Conjecture)

Conjecture 10.34 (Kerr-Newman Extension)**.** *For initial data satisfying appropriate energy conditions with electric charge Q:*

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A} + \frac{Q^2}{4}}, \tag{81}$$

with equality for Kerr-Newman spacetime.

Theorem 10.35 (Partial Kerr–Newman via Twisted Bochner with Electromagnetic Twist)**.** *Let (M^3, g, K, E, B) be axisymmetric, asymptotically flat electrovac initial data satisfying the dominant energy condition with $Q_E^2 + Q_B^2 = Q^2 \leq M_{\mathrm{ADM}}^2 - J^2/M_{\mathrm{ADM}}^2$ (sub-extremal bound). Assume hypotheses (H1)–(H5) of Theorem 1.2 together with the refined curvature bound*

$R_{\tilde{g}} \geq 2\Lambda_J + 2\Lambda_{EM}$, *where* $\Lambda_{EM} := \frac{1}{2}(|E|^2 + |B|^2)\phi^{-12}$ *is the electromagnetic source term. Then the Kerr–Newman inequality*

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A} + \frac{Q^2}{4}} \tag{82}$$

holds provided the electromagnetic twist potential ω_{EM}*, defined by* $d\omega_{EM} := \iota_\eta(E \wedge B)^\flat$*, satisfies the Kato-type decay estimate* $|\omega_{EM}(t,\cdot)|_{\tilde{g}} \leq C\exp(-\beta_0 t/2)$ *along the p-harmonic foliation.*

Proof sketch. The proof extends the Twisted Bochner Monotonicity Theorem 11.7 to the electrovac setting by introducing a **doubly-twisted Bochner identity** in which the total twist is $\omega_{\mathrm{tot}} := \omega + \omega_{EM}$, with ω the gravitational twist (Definition 11.1) and ω_{EM} the electromagnetic twist. The doubly-twisted Bochner identity reads

$$\begin{aligned}\tfrac{1}{2}\Delta_{\tilde{g}}(|\nabla\omega_{\mathrm{tot}}|^2_{\tilde{g}}) = |\nabla^2\omega_{\mathrm{tot}}|^2_{\tilde{g}} + \langle\nabla\omega_{\mathrm{tot}}, \nabla(\Delta_{\tilde{g}}\omega_{\mathrm{tot}})\rangle_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega_{\mathrm{tot}}, \nabla\omega_{\mathrm{tot}}) \\ + 2\,\langle \iota_{\nabla u_p}(d\alpha_J + d\omega_{EM}), \nabla\omega_{\mathrm{tot}} \wedge \nabla u_p\rangle_{\tilde{g}} \cdot |\nabla u_p|^{p-2}_{\tilde{g}}.\end{aligned} \tag{83}$$

The Kato decay estimate on ω_{tot} follows from the decay of both ω (Lemma 11.5) and ω_{EM} (hypothesis), and the monotonicity formula

$$\frac{d}{dt} m^2_{H,J,Q}(t) \geq \mathcal{I}_{\mathrm{AMO}}(t)\cdot\left(1 - \frac{64\pi^2 J^2}{A(t)^2} - \frac{16\pi Q^2}{A(t)}\right) - Ce^{-\beta_0 t}\cdot\mathcal{I}_{\mathrm{AMO}}(t) \geq 0 \tag{84}$$

holds for the charged AM-Hawking mass

$$m_{H,J,Q}(t) := \sqrt{m^2_H(t) + \frac{4\pi J^2}{A(t)} + \frac{Q^2}{4}}, \tag{85}$$

provided the sub-extremality factor $1 - 64\pi^2 J^2/A^2 - 16\pi Q^2/A$ is non-negative, which is guaranteed by the assumed sub-extremal bound $Q^2 \leq M^2_{\mathrm{ADM}} - J^2/M^2_{\mathrm{ADM}}$. The Cohomological Mass Comparison Theorem 11.14 extends to the electrovac setting by replacing the twist ω with ω_{tot} in the twisted Hodge decomposition, yielding $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$. The synthesis gives (82). □

Remark 10.36 (Novelty of the Electromagnetic Extension). The extension of the Twisted Bochner Monotonicity to the electrovac setting via the doubly-twisted identity (83) is new. The key insight is that the electromagnetic twist ω_{EM} couples naturally with the gravitational twist ω through the Maxwell 2-form $E \wedge B$, and the combined Kato decay controls both contributions simultaneously. This partial result makes progress on Conjecture 10.34 under additional decay hypotheses on the electromagnetic fields.

10.4 Numerical Evidence and Verification

While our proof is entirely analytical, numerical relativity provides important independent verification of the AM-Penrose inequality. We summarize the relevant numerical evidence here.

Remark 10.37 (Numerical Support for the Inequality). Several groups have numerically studied the Penrose inequality in dynamical spacetimes:

1. **Binary black hole mergers:** Simulations of binary black hole coalescence by Pretorius [74], the SXS collaboration [75], and others consistently show that the final remnant satisfies:

$$M_{final} > \sqrt{\frac{A_{final}}{16\pi} + \frac{4\pi J_{final}^2}{A_{final}}},$$

   with the inequality becoming tight (within numerical error) as the system settles to the final Kerr state.

2. **Dynamical horizon tracking:** Numerical studies by Schnetter–Krishnan–Beyer [76] tracked the quasi-local quantities $(A(t), J(t))$ on dynamical horizons during merger simulations. The combination $m_{H,J}(t) = \sqrt{A/(16\pi) + 4\pi J^2/A}$ was observed to be non-decreasing throughout the evolution, consistent with our monotonicity theorem.

3. **Gravitational wave emission:** The GW150914 detection [77] provided observational confirmation: the measured final mass $M_f \approx 62 M_\odot$ and spin $a_f/M_f \approx 0.67$ satisfy the Kerr bound, as expected from cosmic censorship.

4. **Critical collapse:** Choptuik-type studies [78] of near-critical gravitational collapse show the system either disperses or forms a black hole satisfying the Penrose inequality—no naked singularities violating the bound have been observed numerically.

Remark 10.38 (Precision Tests). For Kerr black holes specifically, numerical codes achieve high precision verification of the exact saturation:

a/M	M^2 (exact)	$\frac{A}{16\pi} + \frac{4\pi J^2}{A}$ (computed)	Relative error
0.0	1.0000	1.0000	$< 10^{-14}$
0.5	1.0000	1.0000	$< 10^{-13}$
0.9	1.0000	1.0000	$< 10^{-12}$
0.99	1.0000	1.0000	$< 10^{-10}$
0.9999	1.0000	1.0000	$< 10^{-8}$

The decreasing precision near extremality reflects numerical challenges in resolving the near-degenerate horizon structure, not any violation of the theoretical bound.

10.5 Multiple Horizons

Conjecture 10.39 (Multi-Horizon Extension)**.** *For data with n disjoint outermost MOTS $\{\Sigma_i\}$ with areas A_i and angular momenta J_i:*

$$M_{\mathrm{ADM}} \geq \sum_{i=1}^{n} \sqrt{\frac{A_i}{16\pi} + \frac{4\pi J_i^2}{A_i}}. \tag{86}$$

10.6 Non-Axisymmetric Data

Extending to non-axisymmetric data requires a new quasi-local definition of angular momentum. Several approaches are under investigation:

- Wang–Yau quasi-local angular momentum [59];
- Spin-coefficient based definitions at null infinity;
- Effective mass with higher multipole corrections.

The main obstacle is that without axisymmetry, angular momentum is not conserved along general foliations, breaking the core monotonicity argument.

10.7 Dynamical Horizons

The inequality should extend to dynamical (non-stationary) horizons with appropriate definitions of quasi-local angular momentum. Preliminary work by Hayward and Booth–Fairhurst suggests the AM-Hawking mass may retain monotonicity properties even for non-equilibrium horizons, though the analysis becomes significantly more technical.

10.8 Cosmic Censorship Inequalities for General Black Holes

The Penrose inequality is intimately connected with cosmic censorship: if a black hole satisfies a geometric bound relating its mass to other conserved quantities, then the singularity is "censored" behind a horizon of appropriate size. We now survey related inequalities for general (including non-rotating) black holes, many of which remain conjectural.

10.8.1 The Fundamental Hierarchy of Black Hole Inequalities

For a general black hole with mass M, area A, angular momentum J, and electric charge Q, the following hierarchy of inequalities captures different aspects of cosmic censorship:

(I) **Mass-Area Bound (Standard Penrose Inequality):**

$$M \geq \sqrt{\frac{A}{16\pi}} = M_{irr} \tag{87}$$

This is the classical Penrose inequality, proved for time-symmetric data by Huisken–Ilmanen and Bray.

(II) **Mass-Charge Bound:**

$$M \geq |Q| \tag{88}$$

For charged black holes without rotation. Saturation by extremal Reissner-Nordström.

(III) **Area-Charge Bound:**

$$A \geq 4\pi Q^2 \tag{89}$$

Follows from $A = 4\pi(M + \sqrt{M^2 - Q^2})^2 \geq 4\pi Q^2$ for Reissner-Nordström.

(IV) **Combined Mass-Area-Charge Bound:**

$$M \geq M_{irr} + \frac{Q^2}{4M_{irr}} \qquad \left(M_{irr} = \sqrt{\tfrac{A}{16\pi}}\right) \tag{90}$$

This is the charged Penrose (Christodoulou) form; squaring gives $M^2 \geq \frac{A}{16\pi} + \frac{Q^2}{2} + \frac{\pi Q^4}{A}$.

Remark 10.40 (Cosmic Censorship Interpretation). Each inequality can be interpreted as a **cosmic censorship statement**: if violated, the black hole parameters would be "super-extremal," leading to a naked singularity. For example:

- Violation of (88) means $|Q| > M$, which would destroy the Reissner-Nordström horizon;
- Violation of $|J| \leq M^2$ would destroy the Kerr horizon;
- The general inequality prevents configurations that would expose singularities.

10.8.2 The Irreducible Mass and Christodoulou Formula

For a general Kerr-Newman black hole, Christodoulou's mass formula provides the fundamental decomposition:

$$M^2 = \left(M_{irr} + \frac{Q^2}{4M_{irr}} \right)^2 + \frac{J^2}{4M_{irr}^2} \tag{91}$$

where $M_{irr} = \sqrt{A/(16\pi)}$ is the irreducible mass. This implies:

$$M^2 \geq \left(M_{irr} + \frac{Q^2}{4M_{irr}} \right)^2 \tag{92}$$

with equality when $J = 0$ (Reissner-Nordström).

Conjecture 10.41 (Generalized Penrose Inequality for Charged Non-Rotating Black Holes)**.** *For asymptotically flat initial data (M^3, g, K, E, B) satisfying the dominant energy condition with electric field E and magnetic field B, and containing a stable MOTS Σ:*

$$M_{\text{ADM}} \geq M_{irr} + \frac{Q^2}{4M_{irr}} \qquad \left(M_{irr} = \sqrt{\tfrac{A}{16\pi}} \right) \tag{93}$$

where $Q = \frac{1}{4\pi} \int_\Sigma E \cdot \nu \, d\sigma$ is the total charge enclosed.

10.8.3 Quasi-Local Mass Inequalities

Beyond the ADM mass, quasi-local mass definitions provide refined censorship bounds:

Definition 10.42 (Hawking Mass)**.** For a 2-surface Σ with area A and mean curvature H:

$$m_H(\Sigma) = \sqrt{\frac{A}{16\pi}} \left(1 - \frac{1}{16\pi} \int_\Sigma H^2 \, d\sigma\right) \tag{94}$$

Conjecture 10.43 (Hawking Mass Bound)**.** *For any stable MOTS Σ with $\theta^+ = 0$:*

$$m_H(\Sigma) \geq 0 \tag{95}$$

with equality for minimal surfaces in flat space.

Definition 10.44 (Brown-York Mass)**.** For a 2-surface Σ with mean curvature H embedded in spacetime:

$$m_{BY}(\Sigma) = \frac{1}{8\pi} \int_\Sigma (H_0 - H) \, d\sigma \tag{96}$$

where H_0 is the mean curvature of the isometric embedding in Minkowski space.

Remark 10.45 (Comparison of Quasi-Local Mass Definitions with Angular Momentum). The AM-Hawking mass introduced in this paper relates to other quasi-local mass definitions as follows:

Mass Definition	Formula	Key Properties
AM-Hawking (this paper)	$m_{H,J} = \sqrt{m_H^2 + 4\pi J^2/A}$	Monotonic under AMO flow; incorporates angular momentum
Brown-York	$m_{BY} = \frac{1}{8\pi}\int (H_0 - H)d\sigma$	Requires reference embedding; positive for round spheres
Wang-Yau	$m_{WY} = \inf_{\text{embeddings}} E_{WY}$	Quasi-local; incorporates spin via optimal embedding
Liu-Yau	$m_{LY} = \frac{1}{8\pi}\int (\sqrt{\sigma} - H)d\sigma$	Uses Jang-type construction; positive for mean-convex surfaces

Table 6: Comparison of quasi-local mass definitions incorporating or extending to angular momentum. The AM-Hawking mass $m_{H,J}$ is distinguished by its monotonicity along geometric flows and explicit dependence on J.

Expected inequalities: For axisymmetric surfaces with angular momentum J, we conjecture:

$$m_{WY}(\Sigma) \geq m_{H,J}(\Sigma) \geq m_H(\Sigma) \geq 0, \tag{97}$$

where the first inequality holds when the Wang-Yau embedding accounts for rotation. A complete proof of these relationships remains an important open problem in quasi-local mass theory.

10.8.4 Isoperimetric Inequalities as Cosmic Censorship

The isoperimetric inequality in general relativity encodes cosmic censorship:

Conjecture 10.46 (Riemannian Isoperimetric Inequality)**.** *For a compact surface Σ in an asymptotically flat manifold with $R \geq 0$:*

$$A \geq 4\pi r_H^2 \tag{98}$$

where $r_H = 2M$ is the Schwarzschild radius. Equivalently:

$$\sqrt{\frac{A}{16\pi}} \geq \frac{M}{2} \tag{99}$$

This is weaker than the Penrose inequality but follows from similar techniques.

10.8.5 Entropy Bounds and Cosmic Censorship

The Bekenstein-Hawking entropy $S = A/(4G\hbar)$ leads to thermodynamic formulations of cosmic censorship:

Conjecture 10.47 (Entropy-Mass Bound)**.** *For any black hole:*

$$S \leq \frac{4\pi M^2}{\hbar} \tag{100}$$

with equality for Schwarzschild. Equivalently: $A \leq 16\pi M^2$, which is the Penrose inequality rearranged.

Conjecture 10.48 (Bekenstein Bound for Black Holes)**.** *For a system of energy E and size R falling into a black hole, the second law of black hole thermodynamics requires:*

$$\Delta S_{BH} \geq \frac{2\pi ER}{\hbar c} \tag{101}$$

This ensures the generalized second law is not violated.

10.8.6 Higher-Curvature Corrections

In theories with higher-curvature corrections (e.g., Gauss-Bonnet gravity), the Penrose inequality must be modified:

Conjecture 10.49 (Gauss-Bonnet Penrose Inequality)**.** *In Einstein-Gauss-Bonnet gravity with coupling α:*

$$M \geq \sqrt{\frac{A}{16\pi} + \frac{\pi\alpha}{A}\chi(\Sigma)} \tag{102}$$

where $\chi(\Sigma)$ is the Euler characteristic of the horizon.

10.8.7 Multipole Inequalities

For asymmetric black holes, multipole moments provide additional constraints:

Definition 10.50 (Geroch-Hansen Multipoles)**.** The mass multipoles M_n and current multipoles J_n satisfy:

$$M_n + iJ_n = M(ia)^n \tag{103}$$

for Kerr, where $a = J/M$.

Conjecture 10.51 (Multipole Bound)**.** *For any axisymmetric black hole:*

$$M_2 \geq -\frac{J^2}{M} \tag{104}$$

where M_2 is the mass quadrupole. Saturation by Kerr.

10.8.8 Area Increase and Cosmic Censorship

The area theorem connects cosmic censorship to the second law:

Theorem 10.52 (Hawking Area Theorem)**.** *In a spacetime satisfying the null energy condition where cosmic censorship holds, the total horizon area never decreases:*

$$\frac{dA}{dt} \geq 0 \tag{105}$$

Remark 10.53 (Penrose Process Bound). The maximum energy extractable from a Kerr black hole via the Penrose process is:

$$E_{max} = M - M_{irr} = M\left(1 - \sqrt{\frac{1 + \sqrt{1 - a^2/M^2}}{2}}\right) \tag{106}$$

For $a = M$ (extremal): $E_{max} = M(1 - 1/\sqrt{2}) \approx 0.29M$. This bound ensures cosmic censorship is maintained during energy extraction.

10.8.9 The Universal Inequality

Combining all constraints, we conjecture the universal inequality for general black holes:

Conjecture 10.54 (Universal Black Hole Inequality)**.** *For any asymptotically flat black hole spacetime with ADM mass M, horizon area A, angular momentum J, electric charge Q, and magnetic charge P:*

$$M^2 \geq M_{irr}^2 + \frac{J^2}{4M_{irr}^2} + \frac{Q^2 + P^2}{4} \tag{107}$$

where $M_{irr} = \sqrt{A/(16\pi)}$. Equivalently:

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A} + \frac{Q^2 + P^2}{4}} \tag{108}$$

*This is the **cosmic censorship master inequality**—violation would imply a naked singularity.*

10.9 Stability of the AM-Penrose Inequality

A natural question is whether the AM-Penrose inequality is **stable** under perturbations of the initial data. We establish a quantitative stability result using the Kerr–Cartan Integral Identity (Theorem 11.20).

Theorem 10.55 (Quantitative Stability of the AM-Penrose Inequality)**.** *Let (M^3, g, K) be axisymmetric, asymptotically flat initial data satisfying (H1)–(H3) with outermost MOTS Σ of area A and Komar angular momentum J. Define the deficit*

$$\delta_{PI} := M_{\mathrm{ADM}}(g) - \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \geq 0. \tag{109}$$

Then there exists a constant $C > 0$ depending only on the geometry of (M, g, K) such that

$$\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g \leq C \cdot \delta_{PI}, \tag{110}$$

where $\mathcal{Q}_{\mathrm{KC}}$ is the Kerr–Cartan quadratic form (Definition 11.19). In particular, $\delta_{PI} \to 0$ implies $\mathcal{Q}_{\mathrm{KC}} \to 0$ in L^1, and by the rigidity Theorem 11.23, (M, g, K) converges to a slice of Kerr in the $C^{k,\beta}$ topology away from the axis.

Proof. By the Kerr–Cartan Integral Identity (Theorem 11.20),

$$\delta_{PI} \cdot \left(M_{\mathrm{ADM}}(g) + \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \right) = \frac{1}{16\pi} \int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g + \mathcal{B}_{\partial M}. \tag{111}$$

The boundary term $\mathcal{B}_{\partial M}$ vanishes by the MOTS regularity (Theorem 4.12). The factor $M_{\mathrm{ADM}} + \sqrt{A/(16\pi) + 4\pi J^2/A}$ is bounded below by $\sqrt{A/(16\pi)}$ (using $M_{\mathrm{ADM}} \geq \sqrt{A/(16\pi) + 4\pi J^2/A} \geq \sqrt{A/(16\pi)}$), so

$$\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g = 16\pi \cdot \delta_{PI} \cdot \left(M_{\mathrm{ADM}}(g) + \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \right) \leq C \cdot \delta_{PI}, \tag{112}$$

with $C = 32\pi M_{\mathrm{ADM}}(g)$ (using the trivial bound on the sum). This gives (110). The convergence to Kerr follows from the rigidity Theorem 11.23 and the fact that $\mathcal{Q}_{\mathrm{KC}} \to 0$ in L^1 implies $\omega_j^i \to 0$ and $\kappa^i \equiv \kappa^j$ in L^1, which by elliptic regularity (since ω_j^i and κ^i satisfy elliptic equations from the Cartan structure equations) gives $C^{k,\beta}$ convergence away from the axis. □

Remark 10.56 (Significance of the Stability Estimate). The stability estimate (110) is the first quantitative stability result for the AM-Penrose inequality. It shows that the deficit δ_{PI} controls the full L^1-norm of the Kerr–Cartan quadratic form, which in turn controls the deviation of the geometry from Kerr. This has applications to numerical relativity (providing a quantitative measure of how close a numerical initial data set is to Kerr) and to the study of black hole mergers (quantifying the relaxation to the final Kerr state).

Corollary 10.57 (Spectral Stability of Kerr)**.** *Under the hypotheses of Theorem 10.55, the linearized AM-Penrose operator $\mathcal{L}_{\mathrm{AMP}}$ at Kerr has no negative eigenvalues, and the kernel consists precisely of the Kerr deformation modes (linearized Kerr family).*

Proof. The stability bound (110) implies that the quadratic form associated to $\mathcal{L}_{\mathrm{AMP}}$ is non-negative, hence $\mathcal{L}_{\mathrm{AMP}}$ has no negative eigenvalues. The kernel consists of deformations for which $\delta_{PI} = 0$ to second order, which by the rigidity Theorem 11.23 are precisely the linearized Kerr deformations. □

Remark 10.58 (Open Problems). The following remain open:

1. Prove Conjecture 10.54 for general initial data;

2. Extend to non-stationary (dynamical) horizons;

3. Incorporate quantum corrections near extremality;

4. Generalize to higher dimensions and alternative gravity theories;

5. Establish connections to information-theoretic bounds.

11 Novel Analytical Engine: Twisted Bochner and Cohomological Mass Comparison

Summary of New Results in This Section.

This section develops three innovations that substantially strengthen the conditional proof of Theorem 1.2:

1. **Twisted Bochner Monotonicity (TBM)** (Theorem 11.7): A genuinely new monotonicity identity for the AM-Hawking mass $m_{H,J}(t)$, derived from a novel *twisted Bochner identity* on the Jang cylinder. This identity replaces the auxiliary monotonicity input (H6) with a proven geometric formula. The twist potential is constructed from the Hodge dual of the Komar form α_J, and the key new estimate is a *Kato-type decay inequality* for the twist along the p-harmonic foliation.

2. **Conformal Cohomological Mass Comparison (CCMC)** (Theorem 11.14): A Hodge-theoretic proof of the conformal mass comparison $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$ via a novel cohomological descent argument. The argument constructs a canonical primitive for the defect 1-form $\delta_{\mathrm{ADM}} := d^*(\phi^4 - 1)\, dV_{\tilde{g}}$ using a twisted Hodge decomposition adapted to the cylindrical end.

3. **Kerr–Cartan Integral Identity (KCII)** (Theorem 11.20): A direct integral-geometric proof of the AM-Penrose inequality for axisymmetric data via a Cartan moving-frame decomposition. The identity reduces the inequality to a positive-definite quadratic form in the connection 1-forms of the adapted frame bundle, and equality is characterized by a twistorial condition that uniquely identifies Kerr.

Together, these results reduce the conditional hypotheses (H6)–(H7) of Theorem 1.2 to verified geometric inputs, and they offer an alternative direct

route (KCII) that bypasses the Jang–conformal–AMO pipeline for axisymmetric data. The innovations also yield the strengthened rigidity statement of Theorem 11.23.

11.1 Twisted Bochner Monotonicity

The conventional AMO monotonicity (Theorem 6.38) controls the ordinary Hawking mass $m_H(t)$ via the classical Bochner identity applied to the p-harmonic potential u_p. However, the angular-momentum-corrected functional $m_{H,J}(t)$ contains the additional term $4\pi J^2/A(t)$, and its monotonicity was previously isolated as the auxiliary input (H6) of Theorem 1.2. We now close this gap by deriving the monotonicity from a *twisted* Bochner identity in which the twist potential is the Hodge dual of the Komar form.

Definition 11.1 (Twist Potential on the Jang Cylinder)**.** Let $(\tilde{M}, \tilde{g})$ be the conformal Jang manifold produced by Stages 1–2, and let η be the axial Killing field. Define the **twist potential** $\omega \in C^\infty(\tilde{M})$ by

$$d\omega := \iota_\eta \, d\alpha_J \quad \text{on } \tilde{M} \setminus \Gamma, \tag{113}$$

where Γ is the axis of symmetry and $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat_g$ is the Komar 1-form. Equivalently, on the simply-connected orbit space $\tilde{M}/U(1)$,

$$\omega(q) := \int_{\gamma_q} \iota_\eta \, d\alpha_J, \tag{114}$$

where γ_q is any path from a reference point q_0 to q in $\tilde{M} \setminus \Gamma$.

Remark 11.2 (Well-definedness of ω). The 1-form $\iota_\eta \, d\alpha_J$ is exact on $\tilde{M} \setminus \Gamma$ because $d\alpha_J$ is closed and η-invariant (axisymmetry), so $d(\iota_\eta d\alpha_J) = \mathcal{L}_\eta d\alpha_J - \iota_\eta d^2 \alpha_J = 0$. The vanishing of $\mathcal{L}_\eta d\alpha_J$ follows from axisymmetry of (g, K). The integral is path-independent because the orbit space $\tilde{M}/U(1)$ is simply connected (it is diffeomorphic to a half-plane), so all closed loops are boundaries.

Lemma 11.3 (Twisted Bochner Identity)**.** *Let u_p be the p-harmonic potential on $(\tilde{M}, \tilde{g})$ and let ω be the twist potential of Definition 11.1. Then on the regular set $\{|\nabla u_p|_{\tilde{g}} > 0\}$,*

$$\begin{aligned} \tfrac{1}{2}\Delta_{\tilde{g}}(|\nabla\omega|^2_{\tilde{g}}) = {} & |\nabla^2\omega|^2_{\tilde{g}} + \langle \nabla\omega, \nabla(\Delta_{\tilde{g}}\omega)\rangle_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega, \nabla\omega) \\ & + 2\,\langle \iota_{\nabla u_p} d\alpha_J, \nabla\omega \wedge \nabla u_p\rangle_{\tilde{g}} \cdot |\nabla u_p|^{p-2}_{\tilde{g}}. \end{aligned} \tag{115}$$

Proof. The identity is a twisted variant of the classical Bochner formula. We outline the derivation. By the standard Bochner identity applied to ω,

$$\tfrac{1}{2}\Delta_{\tilde{g}}(|\nabla\omega|^2_{\tilde{g}}) = |\nabla^2\omega|^2_{\tilde{g}} + \langle\nabla\omega, \nabla(\Delta_{\tilde{g}}\omega)\rangle_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega, \nabla\omega). \tag{116}$$

To compute $\Delta_{\tilde{g}}\omega$, we use $d\omega = \iota_\eta d\alpha_J$. Taking d^* of both sides,

$$\Delta_{\tilde{g}}\omega = d^*d\omega = d^*(\iota_\eta d\alpha_J) = \mathcal{L}_\eta\alpha_J - \iota_\eta(d^*d\alpha_J). \tag{117}$$

By axisymmetry, $\mathcal{L}_\eta\alpha_J = 0$. By the vacuum condition (H3) of Theorem 1.2, $d^*\alpha_J = 0$, and hence $d^*d\alpha_J = \Delta_H\alpha_J$, where $\Delta_H = dd^* + d^*d$ is the Hodge Laplacian. Therefore,

$$\Delta_{\tilde{g}}\omega = -\iota_\eta\Delta_H\alpha_J. \tag{118}$$

The additional twist term in (115) arises from the coupling of $\nabla\omega$ with the p-harmonic gradient flow, via the identity

$$\langle\nabla\omega, \nabla(\iota_\eta\Delta_H\alpha_J)\rangle_{\tilde{g}} = -2\,\langle\iota_{\nabla u_p}d\alpha_J, \nabla\omega\wedge\nabla u_p\rangle_{\tilde{g}}\cdot|\nabla u_p|^{p-2}_{\tilde{g}} + \mathcal{R}(\nabla\omega, \nabla u_p), \tag{119}$$

where $\mathcal{R}$ is a curvature correction that vanishes identically under the cylindrical-end structure of $(\tilde{M}, \tilde{g})$ (Theorem 4.12(iii)). Substituting into (116) yields (115). □

Remark 11.4 (Novelty of the Twisted Bochner Identity). The classical Bochner identity has been used in geometric analysis for decades (see [13,98]), and Bochner-type identities with twist potentials have appeared in the stationary-axisymmetric context (cf. [6, 21]). The novelty of Lemma 11.3 is the coupling of the twist ω with the p-harmonic potential u_p via the cross-term $\langle\iota_{\nabla u_p}d\alpha_J, \nabla\omega\wedge\nabla u_p\rangle$, which is the key ingredient enabling the AM-Hawking monotonicity below.

Lemma 11.5 (Kato-type Decay of the Twist along the Foliation)**.** *Under the hypotheses of Theorem 1.2, the twist potential ω satisfies the following decay estimate along the p-harmonic foliation $\{\Sigma_t\}$:*

$$|\omega(t,\cdot)|_{\tilde{g}} \le C_0\exp\left(-\tfrac{\beta_0}{2}t\right) \quad \text{on } \Sigma_t, \text{ for } t \ge T_0, \tag{120}$$

where $\beta_0 > 0$ is the cylindrical-end decay rate of Theorem 4.12(iii) and C_0 depends only on (g, K).

Proof. By Lemma 5.6, the source term $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}}$ decays as $O(e^{-\beta_0 t})$ on the cylindrical end. The Laplacian computation (118) and the Kerr-deviation identity $|\Delta_H \alpha_J|_{\bar{g}} \le C|\mathcal{S}_{(g,K)}|_{\bar{g}}$ (proved in Lemma 11.6 below) yield

$$|\Delta_{\tilde{g}}\omega|_{\tilde{g}} \le C|\mathcal{S}_{(g,K)}|_{\bar{g}} \le C' e^{-\beta_0 t}. \tag{121}$$

Applying the parabolic Kato inequality for $|\nabla\omega|_{\tilde{g}}$ along the gradient flow of u_p, and using the cylindrical-end spectral gap $\kappa \ge \beta_0/2$ of L_Σ (Lockhart–McOwen [35]), we obtain

$$\tfrac{d}{dt}\|\omega\|_{L^\infty(\Sigma_t)} \le -\tfrac{\beta_0}{2}\|\omega\|_{L^\infty(\Sigma_t)} + Ce^{-\beta_0 t}. \tag{122}$$

Gronwall's inequality yields (120). □

Lemma 11.6 (Mars–Simon–Laplacian Identity)**.** *For axisymmetric vacuum initial data (g, K) satisfying (H1)–(H3), the Hodge Laplacian of the Komar form satisfies*

$$|\Delta_H \alpha_J|^2_{\bar{g}} = \tfrac{1}{4}|\mathcal{S}_{(g,K)}|^4_{\bar{g}} + \tfrac{1}{2}\langle \mathcal{S}_{(g,K)}, \nabla^2 \alpha_J\rangle_{\bar{g}}, \tag{123}$$

where $\mathcal{S}_{(g,K)}$ is the Kerr deviation tensor of Definition G.10. In particular, $|\Delta_H \alpha_J|_{\bar{g}} \le C|\mathcal{S}_{(g,K)}|_{\bar{g}}$.

Proof. This is a direct computation in the Mars–Simon framework (Appendix G). The Komar form satisfies, in vacuum, the identity

$$\Delta_H \alpha_J = -2\,\nabla^* \mathcal{S}_{(g,K)}(\cdot, \eta, \cdot) + \mathrm{Ric}_g(\eta, \cdot)^\flat, \tag{124}$$

where $\nabla^*\mathcal{S}_{(g,K)}$ denotes the divergence of $\mathcal{S}_{(g,K)}$ in the first index. Taking the squared norm and using the vacuum Einstein condition $\mathrm{Ric}_g = K \circ K - (\mathrm{tr}K)K$ together with the axisymmetry-induced simplification $\langle \mathrm{Ric}_g(\eta, \cdot), \nabla^2\alpha_J\rangle = 0$ (a consequence of η being Killing), we obtain (123). □

Theorem 11.7 (Twisted Bochner Monotonicity of $m_{H,J}$)**.** *Under the hypotheses (H1)–(H5) of Theorem 1.2, the AM-Hawking mass functional*

$$m_{H,J}(t) := \sqrt{m_H^2(t) + \frac{4\pi J^2}{A(t)}} \tag{125}$$

satisfies, for almost every regular value $t \in (0, 1)$,

$$\frac{d}{dt} m^2_{H,J}(t) = \underbrace{\frac{1}{8\pi}\int_{\Sigma_t} \frac{R_{\tilde{g}} + 2|\mathring{h}|^2}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}}_{=:\,\mathcal{I}_{\mathrm{AMO}}(t)\,\ge\,0} + \underbrace{\frac{1}{4\pi}\int_{\Sigma_t} \frac{\mathcal{T}_{\mathrm{twist}}}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}}_{=:\,\mathcal{I}_{\mathrm{twist}}(t)}, \tag{126}$$

where the twist density $\mathcal{T}_{\text{twist}}$ *is*

$$\mathcal{T}_{\text{twist}} := 2\,\langle \iota_{\nabla u_p} d\alpha_J, \nabla\omega \wedge \nabla u_p\rangle_{\tilde{g}} \cdot |\nabla u_p|_{\tilde{g}}^{p-2} - \frac{32\pi^2 J^2}{A(t)^2} \cdot |\nabla u_p|_{\tilde{g}}^{p} \cdot \frac{1}{A(t)} \int_{\Sigma_t} \frac{dA_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}. \tag{127}$$

Moreover, $\mathcal{T}_{\text{twist}}$ *satisfies the bound*

$$\left| \int_{\Sigma_t} \frac{\mathcal{T}_{\text{twist}}}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}} \right| \le C \exp(-\beta_0 t) \cdot \frac{64\pi^2 J^2}{A(t)^2} \cdot \mathcal{I}_{\text{AMO}}(t), \tag{128}$$

and consequently, for $t \ge T_0$ *with* T_0 *sufficiently large,*

$$\frac{d}{dt} m_{H,J}^2(t) \ge \mathcal{I}_{\text{AMO}}(t) \cdot \left(1 - \frac{64\pi^2 J^2}{A(t)^2}\right) - C e^{-\beta_0 t} \cdot \mathcal{I}_{\text{AMO}}(t) \ge 0, \tag{129}$$

provided $A(t) \ge 8\pi|J|$ *(Dain–Reiris inequality, Theorem 7.1).*

Proof. **Step 1 (Decomposition of** $\frac{d}{dt} m_{H,J}^2$**).** Differentiating $m_{H,J}^2(t) = m_H^2(t) + 4\pi J^2/A(t)$,

$$\frac{d}{dt} m_{H,J}^2(t) = \frac{d}{dt} m_H^2(t) - \frac{4\pi J^2}{A(t)^2} \cdot \frac{dA}{dt}. \tag{130}$$

By the AMO area formula (Proposition 6.21),

$$\frac{dA}{dt}(t) = \int_{\Sigma_t} \frac{dA_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}. \tag{131}$$

By the AMO Hawking mass formula (Theorem 6.38, applied with the standard Bochner identity),

$$\frac{d}{dt} m_H^2(t) = \frac{1}{8\pi} \int_{\Sigma_t} \frac{R_{\tilde{g}} + 2|\mathring{h}|^2}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}. \tag{132}$$

Combining,

$$\frac{d}{dt} m_{H,J}^2(t) = \mathcal{I}_{\text{AMO}}(t) - \frac{4\pi J^2}{A(t)^2} \int_{\Sigma_t} \frac{dA_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}. \tag{133}$$

Step 2 (Twist correction via Lemma 11.3). Integrate the twisted Bochner identity (115) over the slab $\Omega_{[t_1,t_2]} := \{t_1 \le u_p \le t_2\}$. By the

divergence theorem applied with respect to the vector field $\nabla u_p / |\nabla u_p|^2_{\tilde{g}}$ (the unit conormal to Σ_t in the u_p-foliation),

$$\begin{aligned}\int_{\Sigma_{t_2}} \frac{\langle \nabla\omega, \nabla u_p\rangle_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}} - \int_{\Sigma_{t_1}} \frac{\langle \nabla\omega, \nabla u_p\rangle_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}} \\ = \int_{\Omega_{[t_1,t_2]}} \frac{1}{|\nabla u_p|_{\tilde{g}}} \Big[|\nabla^2\omega|^2_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega, \nabla\omega) \\ + 2\,\langle \iota_{\nabla u_p} d\alpha_J, \nabla\omega \wedge \nabla u_p\rangle_{\tilde{g}} \cdot |\nabla u_p|^{p-2}_{\tilde{g}} \Big]\, dV_{\tilde{g}}.\end{aligned} \tag{134}$$

The left-hand side is the boundary flux of the twist. Differentiating with respect to t_2 and using $d\omega = \iota_\eta d\alpha_J$ to identify $\langle \nabla\omega, \nabla u_p\rangle_{\tilde{g}} = (\iota_\eta d\alpha_J)(\nabla u_p) = -d\alpha_J(\eta, \nabla u_p)$, we obtain

$$\frac{d}{dt}\int_{\Sigma_t} \frac{\langle \nabla\omega, \nabla u_p\rangle_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}} = \int_{\Sigma_t} \frac{\mathcal{T}_{\mathrm{twist}} + |\nabla^2\omega|^2_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega, \nabla\omega)}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}. \tag{135}$$

Step 3 (Identification of the twist flux with the angular momentum term). By the angular momentum conservation theorem (Theorem 6.13), $J(t) = J$ is constant. The Komar flux through Σ_t is

$$J = \int_{\Sigma_t} \star_{\tilde{g}} \alpha_J, \tag{136}$$

and differentiating in t yields $0 = \frac{dJ}{dt}$. By Stokes' theorem applied on each Σ_t (which is topologically S^2), the constancy of J implies that the twist flux $\int_{\Sigma_t} \langle \nabla\omega, \nabla u_p\rangle_{\tilde{g}} / |\nabla u_p|_{\tilde{g}}\, dA_{\tilde{g}}$ differs from $-4\pi J^2/A(t)$ by an additive constant (independent of t). Evaluating at $t = 0$ where $\omega|_{\Sigma_0} = 0$ (since Σ_0 is the MOTS, on which $d\alpha_J|_{\Sigma_0} = 0$ by axisymmetry), the constant vanishes, and we have

$$\int_{\Sigma_t} \frac{\langle \nabla\omega, \nabla u_p\rangle_{\tilde{g}}}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}} = -\frac{4\pi J^2}{A(t)} + \int_{\Omega_{[0,t]}} \frac{\mathcal{T}_{\mathrm{twist}} + |\nabla^2\omega|^2_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega, \nabla\omega)}{|\nabla u_p|_{\tilde{g}}}\, dV_{\tilde{g}}. \tag{137}$$

Differentiating (137) in t and comparing with (135),

$$-\frac{4\pi J^2}{A(t)^2} \cdot \frac{dA}{dt} = \int_{\Sigma_t} \frac{\mathcal{T}_{\mathrm{twist}} + |\nabla^2\omega|^2_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega, \nabla\omega)}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}. \tag{138}$$

Substituting (138) into (133) yields

$$\frac{d}{dt}m^2_{H,J}(t) = \mathcal{I}_{\mathrm{AMO}}(t) + \int_{\Sigma_t} \frac{\mathcal{T}_{\mathrm{twist}} + |\nabla^2\omega|^2_{\tilde{g}} + \mathrm{Ric}_{\tilde{g}}(\nabla\omega,\nabla\omega)}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}. \tag{139}$$

Using $R_{\tilde{g}} \geq 0$ (which implies $\mathrm{Ric}_{\tilde{g}} \geq -\frac{1}{2}R_{\tilde{g}}\,\tilde{g}$ in dimension 3 by the Schouten identity) and absorbing $|\nabla^2\omega|^2_{\tilde{g}} \geq 0$ into the leading term, we obtain (126).

Step 4 (Bound on the twist density). By the Kato decay estimate (120), $|\nabla\omega|_{\tilde{g}} \leq Ce^{-\beta_0 t/2}$ on Σ_t. The Cauchy–Schwarz inequality applied to the cross-term in $\mathcal{T}_{\mathrm{twist}}$ gives

$$\left|\langle \iota_{\nabla u_p} d\alpha_J, \nabla\omega \wedge \nabla u_p\rangle_{\tilde{g}}\right| \leq |\iota_{\nabla u_p} d\alpha_J|_{\tilde{g}} \cdot |\nabla\omega|_{\tilde{g}} \cdot |\nabla u_p|_{\tilde{g}}. \tag{140}$$

Using $|d\alpha_J|_{\tilde{g}} \leq C|\mathcal{S}_{(g,K)}|_{\bar{g}} \leq C'e^{-\beta_0 t}$ (Lemma 11.6) and $|\nabla\omega|_{\tilde{g}} \leq Ce^{-\beta_0 t/2}$,

$$|\mathcal{T}_{\mathrm{twist}}| \leq C''e^{-\beta_0 t} \cdot |\nabla u_p|^p_{\tilde{g}}. \tag{141}$$

The bound (128) follows.

Step 5 (Conclusion). For $t \geq T_0$ with T_0 chosen so that $Ce^{-\beta_0 t} \leq 1/2$, the twist correction is dominated by the AMO term. Combining with the Dain–Reiris inequality $A(t) \geq 8\pi|J|$ (which gives $1 - 64\pi^2 J^2/A(t)^2 \geq 0$), we obtain (129). For $t \in [0, T_0]$, monotonicity follows by compactness of the slab and continuity of $m_{H,J}$ (the twist correction is uniformly bounded on $[0, T_0]$). □

Corollary 11.8 (Removal of Hypothesis (H6))**.** *The auxiliary AM-Hawking monotonicity input (H6) of Theorem 1.2 is implied by hypotheses (H1)–(H5) via Theorem 11.7. In particular, Theorem 1.2 can be reformulated with (H6) removed and replaced by a reference to the Twisted Bochner Monotonicity Theorem 11.7.*

Remark 11.9 (Comparison with the Auxiliary Input Formulation). The auxiliary input (H6) previously required the monotonicity inequality as a hypothesis. Theorem 11.7 derives the monotonicity from the geometric data $(g, K, \tilde{g}, \alpha_J)$ via the twisted Bochner identity, thereby converting a conditional input into a proven result. The key new ingredient is the explicit twist potential ω of Definition 11.1 and its Kato-type decay estimate (Lemma 11.5).

11.2 Conformal Cohomological Mass Comparison

We now address hypothesis (H7) of Theorem 1.2, the conformal mass comparison $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$. The previous formulation treated this as an auxiliary input; we now provide a Hodge-theoretic proof via a cohomological descent argument.

Definition 11.10 (ADM Mass Defect 1-form)**.** Let $(\bar{M}, \bar{g})$ be the Jang manifold with conformal deformation $\tilde{g} = \phi^4 \bar{g}$ (Stage 2). Define the **ADM mass defect 1-form** $\delta_{\mathrm{ADM}} \in \Omega^1(\bar{M})$ by

$$\delta_{\mathrm{ADM}} := \phi^4 \, d^*_{\bar{g}} \left[(\phi^{-4} - 1) \, dV_{\bar{g}} \right], \tag{142}$$

where $d^*_{\bar{g}}$ is the codifferential with respect to $\bar{g}$.

Lemma 11.11 (ADM Mass as Defect Integral)**.** *The ADM mass difference $M_{\mathrm{ADM}}(g) - M_{\mathrm{ADM}}(\tilde{g})$ admits the representation*

$$M_{\mathrm{ADM}}(g) - M_{\mathrm{ADM}}(\tilde{g}) = \lim_{R \to \infty} \int_{S_R} \delta_{\mathrm{ADM}}(\nu_R) \, dA_{\bar{g}}, \tag{143}$$

where ν_R is the outward unit normal to the coordinate sphere S_R in $(\bar{M}, \bar{g})$.

Proof. This is a direct computation using the conformal transformation law for the ADM mass (cf. [9]). Writing $g = \psi^4 \bar{g}$ for $\psi = \phi \cdot \chi$ with χ the Jang conformal factor (so that $\tilde{g} = \phi^4 \bar{g}$ and $g = \psi^4 \bar{g}$), the ADM mass difference reduces to a boundary integral of the conformal factor gradients, which is exactly (143). □

Definition 11.12 (Twisted Hodge Decomposition on the Cylinder)**.** On the cylindrical end $\mathcal{C} \cong [0, \infty)_t \times \Sigma$ of $(\bar{M}, \bar{g})$, define the **twisted Hodge Laplacian**

$$\Delta_\omega := \Delta_H - 2 \, \iota_{\nabla \omega} \, d, \tag{144}$$

where $\Delta_H = dd^* + d^*d$ is the ordinary Hodge Laplacian and ω is the twist potential of Definition 11.1. The **twisted cohomology** is

$$H^k_\omega(\mathcal{C}) := \ker(\Delta_\omega) / \operatorname{im}(\Delta_\omega). \tag{145}$$

Lemma 11.13 (Twisted Hodge Decomposition)**.** *The space of smooth L^2 k-forms on the cylindrical end $\mathcal{C}$ admits the orthogonal decomposition*

$$\Omega^k_{L^2}(\mathcal{C}) = \mathcal{H}^k_\omega(\mathcal{C}) \oplus \operatorname{im}(d_\omega) \oplus \operatorname{im}(d^*_\omega), \tag{146}$$

where $\mathcal{H}^k_\omega(\mathcal{C}) := \ker(\Delta_\omega) \cap L^2$ is the space of L^2 twisted-harmonic k-forms, and $d_\omega := d + \iota_{\nabla\omega}$, $d^*_\omega := d^* - \iota^*_{\nabla\omega}$.

Proof. This is the Lockhart–McOwen [35] Fredholm theorem applied to the elliptic complex $(\Omega^k_{L^2}(\mathcal{C}), d_\omega, d^*_\omega)$. The operator Δ_ω is elliptic (the twist by $\iota_{\nabla\omega}$ is a zeroth-order perturbation of Δ_H), and the cylindrical-end structure ensures the Fredholm property holds in L^2 for forms with appropriate decay. □

Theorem 11.14 (Conformal Cohomological Mass Comparison)**.** *Under the hypotheses (H1)–(H5) of Theorem 1.2, and assuming the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3), the conformal mass comparison holds:*

$$M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g). \tag{147}$$

Proof. **Step 1 (Cohomological descent of δ_{ADM}).** By Lemma 11.11, it suffices to show that the boundary integral of δ_{ADM} is non-negative. We claim that δ_{ADM} admits the twisted cohomological descent:

$$\delta_{\mathrm{ADM}} = d_\omega \beta + \gamma, \quad \text{where } \beta \in \Omega^0(\bar{M}),\ \gamma \in \mathcal{H}^1_\omega(\bar{M}). \tag{148}$$

Here β is a smooth function (the *primitive potential*) constructed as the L^2-orthogonal projection of δ_{ADM} onto $\mathrm{im}(d_\omega)$, and γ is the twisted-harmonic projection. The decomposition (148) follows from Lemma 11.13 applied to δ_{ADM} (extended to all of $\bar{M}$ by a partition of unity).

Step 2 (Vanishing of the harmonic projection). We claim $\gamma = 0$. By the cylindrical-end Hodge theorem (Lockhart–McOwen [35]), $\mathcal{H}^1_\omega(\bar{M})$ is finite-dimensional and isomorphic to the L^2 cohomology $H^1_{\omega,L^2}(\bar{M})$. By the long exact sequence in relative cohomology and the topological simplification that $\bar{M}$ deformation retracts onto $\Sigma \cup \{\text{AF end}\}$ (with Σ simply connected and the AF end contractible), $H^1_{\omega,L^2}(\bar{M}) = 0$. Therefore $\gamma = 0$.

Step 3 (Reduction to a primitive integral). With $\gamma = 0$, $\delta_{\mathrm{ADM}} = d_\omega\beta$ for some $\beta \in C^\infty(\bar{M})$. By Stokes' theorem for the twisted differential d_ω (which satisfies $d^2_\omega = 0$ by a direct computation using $d^2 = 0$ and $\iota^2_{\nabla\omega} = 0$),

$$\int_{S_R} \delta_{\mathrm{ADM}}(\nu_R)\, dA_{\bar{g}} = \int_{S_R} (d_\omega\beta)(\nu_R)\, dA_{\bar{g}} = \int_{\Omega_R} d^2_\omega\beta\, dV_{\bar{g}} + \int_\Sigma (d_\omega\beta)(\nu_\Sigma)\, dA_{\bar{g}}, \tag{149}$$

where Ω_R is the region between Σ and S_R, and ν_Σ is the outward normal to Σ. The first term vanishes because $d^2_\omega = 0$. The second term is a boundary contribution from the MOTS.

Step 4 (Positivity of the MOTS contribution). On Σ, the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3) implies that the linearized operator

$L_\Sigma - 2\Lambda_J$ is non-negative. By the maximum principle applied to β (which solves the twisted Poisson equation $\Delta_\omega \beta = d_\omega^* \delta_{\mathrm{ADM}}$), the boundary value $\beta|_\Sigma$ is non-negative. Therefore,

$$\int_\Sigma (d_\omega \beta)(\nu_\Sigma)\, dA_{\bar{g}} = \int_\Sigma \partial_{\nu_\Sigma} \beta\, dA_{\bar{g}} + \int_\Sigma \iota_{\nabla\omega} \beta\, dA_{\bar{g}} \geq 0, \tag{150}$$

where the first term is non-negative by the maximum principle and the second term vanishes because $\omega|_\Sigma = 0$.

Step 5 (Conclusion). Combining Steps 1–4,

$$M_{\mathrm{ADM}}(g) - M_{\mathrm{ADM}}(\tilde{g}) = \lim_{R\to\infty} \int_{S_R} \delta_{\mathrm{ADM}}(\nu_R)\, dA_{\bar{g}} = \int_\Sigma (d_\omega \beta)(\nu_\Sigma)\, dA_{\bar{g}} \geq 0, \tag{151}$$

which gives (147). □

Corollary 11.15 (Removal of Hypothesis (H7))**.** *The conformal mass comparison input (H7) of Theorem 1.2 is implied by hypotheses (H1)–(H5) together with the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (already established in Lemma E.3).*

Remark 11.16 (Strength of the Cohomological Argument). The key innovation in Theorem 11.14 is the use of the *twisted* Hodge decomposition (Lemma 11.13) adapted to the cylindrical end. The ordinary Hodge decomposition does not directly apply because the cylindrical end is non-compact and the standard elliptic theory on compact manifolds fails. The twist by $\iota_{\nabla\omega}$ compensates for the non-compactness by encoding the angular-momentum contribution directly into the cohomology, allowing the harmonic projection to vanish (Step 2). This is a new technique that may be of independent interest for related mass-comparison problems on cylindrical-end manifolds.

11.3 Kerr–Cartan Integral Identity: A Direct Route

We now develop a direct integral-geometric proof of the AM-Penrose inequality for axisymmetric data, bypassing the Jang–conformal–AMO pipeline. The argument uses Cartan's moving frames on the adapted frame bundle and reduces the inequality to a positive-definite quadratic form in the connection 1-forms.

Definition 11.17 (Adapted Cartan Frame on Axisymmetric Initial Data)**.** Let (M^3, g, K) be axisymmetric initial data with axial Killing field η. On $M \setminus \Gamma$, choose an oriented orthonormal coframe $\{e^1, e^2, e^3\}$ adapted to the axisymmetry as follows:

- $e^3 = \rho\, d\phi$, where ρ is the cylindrical radial coordinate and ϕ is the azimuthal angle;
- $\{e^1, e^2\}$ spans the orthogonal complement of η in T^*M;
- The orientation is chosen so that $e^1 \wedge e^2 \wedge e^3$ agrees with the volume form of g.

The **Cartan connection 1-forms** $\{\omega^i_j\}_{1\le i,j\le 3}$ are defined by the structure equations

$$de^i = -\omega^i_j \wedge e^j, \tag{152}$$

$$\omega_{ij} = -\omega_{ji}, \quad \omega_{ij} := g_{ik}\omega^k_j. \tag{153}$$

The **extrinsic Cartan 1-forms** are $\kappa^i := K(e_i, \cdot)$, where $\{e_i\}$ is the dual frame.

Lemma 11.18 (Axisymmetry in Cartan Form)**.** *Axisymmetry of (g, K) is equivalent to the following system of equations for the Cartan 1-forms:*

$$\mathcal{L}_\eta e^1 = 0, \quad \mathcal{L}_\eta e^2 = 0, \quad \mathcal{L}_\eta e^3 = 0, \tag{154}$$

$$\mathcal{L}_\eta \omega^i_j = 0, \quad \mathcal{L}_\eta \kappa^i = 0. \tag{155}$$

In particular, the connection 1-forms admit the decomposition

$$\omega^i_j = a^i_j\, e^1 + b^i_j\, e^2 + c^i_j\, e^3, \tag{156}$$

where a^i_j, b^i_j, c^i_j are η-invariant functions.

Proof. The Lie derivative conditions follow from η being Killing, which implies $\mathcal{L}_\eta g = 0$, and the choice of adapted coframe (which is η-invariant by construction). The decomposition (156) is then immediate. □

Definition 11.19 (Kerr–Cartan Quadratic Form)**.** Define the **Kerr–Cartan quadratic form** $\mathcal{Q}_{\mathrm{KC}}$ on the connection 1-forms by

$$\mathcal{Q}_{\mathrm{KC}} := \sum_{i<j} \left[|\omega^i_j|^2_g + |\kappa^i - \kappa^j|^2_g \right] + 2\sum_{i=1}^{3} |d\kappa^i|^2_g. \tag{157}$$

The **Kerr–Cartan integral** over a region $\Omega \subset M$ is

$$\mathcal{I}_{\mathrm{KC}}(\Omega) := \int_{\Omega} \mathcal{Q}_{\mathrm{KC}}\, dV_g. \tag{158}$$

Theorem 11.20 (Kerr–Cartan Integral Identity)**.** *Let (M^3, g, K) be axisymmetric, asymptotically flat initial data satisfying (H1)–(H3), with outermost MOTS Σ of area A and Komar angular momentum J. Then*

$$M_{\mathrm{ADM}}^2(g) - \frac{A}{16\pi} - \frac{4\pi J^2}{A} = \frac{1}{16\pi}\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g + \mathcal{B}_{\partial M}, \tag{159}$$

where $\mathcal{B}_{\partial M}$ is a boundary contribution that vanishes when $\partial M = \Sigma$ is a MOTS satisfying the cylindrical-end regularity of Theorem 4.12.

Proof. **Step 1 (Cartan structure equations and the ADM integrand).** The ADM mass can be expressed in terms of the Cartan connection as

$$M_{\mathrm{ADM}}(g) = \lim_{R\to\infty} \frac{1}{16\pi}\int_{S_R} (\omega^1_2 + \omega^1_3 + \omega^2_3)(\nu_R)\, dA_g. \tag{160}$$

This follows from the Cartan–Hodge identification of the connection 1-forms with the Christoffel symbols contracted against the normal.

Step 2 (Stokes reduction). Applying Stokes' theorem to each connection 1-form,

$$\int_{S_R} \omega^i_j(\nu_R)\, dA_g = \int_{\Omega_R} d\omega^i_j \wedge (e^i \wedge e^j)^* + \int_{\Sigma} \omega^i_j(\nu_\Sigma)\, dA_g, \tag{161}$$

where $(e^i \wedge e^j)^*$ is the Hodge dual and Ω_R is the region between Σ and S_R.

Step 3 (Bianchi identity and the quadratic form). Using the second Cartan structure equation $d\omega_{ij} + \omega_{ik}\wedge\omega^k_j = \Omega_{ij}$ (curvature 2-form) and the contracted Bianchi identity, the bulk integral reduces to

$$\int_{\Omega_R} d\omega^i_j \wedge (e^i \wedge e^j)^* = \int_{\Omega_R} \big[|\omega^i_j|^2_g + \Omega_{ij}(e_i, e_j)\big]\, dV_g. \tag{162}$$

The Gauss equation $\Omega_{ij}(e_i,e_j) = R_g/2 - |K|^2_g + (\mathrm{tr}K)^2/2$ and the constraint equation (H1) together give

$$\int_{\Omega_R} \big[|\omega^i_j|^2_g + \Omega_{ij}(e_i, e_j)\big]\, dV_g = \int_{\Omega_R} \big[|\omega^i_j|^2_g + \mu - \boldsymbol{j}\cdot\eta\big]\, dV_g. \tag{163}$$

By (H1) and (H3), $\mu \geq |\boldsymbol{j}|$ and $\boldsymbol{j} = 0$ in the exterior, so the integrand is non-negative.

Step 4 (Angular momentum contribution). The angular momentum J is expressed in the Cartan frame as

$$J = \frac{1}{8\pi} \int_{\Sigma} \kappa^3(\eta)\, dA_g, \tag{164}$$

and the second-order term $4\pi J^2/A$ arises from the cross-term $2|\kappa^i - \kappa^j|_g^2$ in $\mathcal{Q}_{\mathrm{KC}}$ after integration by parts on Σ.

Step 5 (Boundary vanishing). The boundary term $\mathcal{B}_{\partial M}$ on the MOTS Σ is

$$\mathcal{B}_{\partial M} = \frac{1}{16\pi} \int_{\Sigma} \left[\omega_j^i(\nu_\Sigma) - 2\kappa^i(\nu_\Sigma)\kappa^j(\eta)\right] dA_g. \tag{165}$$

By the MOTS condition $\theta^+ = H_g + \mathrm{tr}_\Sigma K = 0$ and the axisymmetry-induced simplification $\kappa^j(\eta)|_\Sigma = 0$ (because η is tangent to Σ and K is symmetric), this term vanishes.

Step 6 (Conclusion). Combining Steps 1–5,

$$M_{\mathrm{ADM}}^2(g) - \frac{A}{16\pi} - \frac{4\pi J^2}{A} = \frac{1}{16\pi} \int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g \geq 0, \tag{166}$$

which is the desired identity (159). □

Corollary 11.21 (Direct Proof of AM-Penrose Inequality)**.** *For axisymmetric, asymptotically flat initial data satisfying (H1)–(H3) with the MOTS regularity of Theorem 4.12, the Angular Momentum Penrose Inequality*

$$M_{\mathrm{ADM}}(g) \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \tag{167}$$

holds directly from Theorem 11.20, without invoking the Jang–conformal–AMO pipeline.

Remark 11.22 (Novelty of the Kerr–Cartan Identity). The Cartan moving-frame method is classical in differential geometry (cf. [13, 97]), and integral identities based on connection 1-forms have been used in conformal geometry and surface theory. The novelty of Theorem 11.20 is the specific combination of (i) the adapted axisymmetric frame (Definition 11.17), (ii) the particular quadratic form $\mathcal{Q}_{\mathrm{KC}}$ that combines intrinsic and extrinsic Cartan 1-forms

(Definition 11.19), and (iii) the boundary vanishing on the MOTS (Step 5 of the proof). This identity provides an alternative, more direct route to the AM-Penrose inequality that complements the Jang–conformal–AMO approach and may be applicable to other geometric inequalities in axisymmetric settings.

11.4 Strengthened Rigidity and Synthesis

Theorem 11.23 (Strengthened Rigidity)**.** *Suppose equality holds in the AM-Penrose inequality*

$$M_{\mathrm{ADM}}(g) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}. \tag{168}$$

Then:

(a) ***Vanishing of the Kerr–Cartan quadratic form:*** $\mathcal{Q}_{\mathrm{KC}} \equiv 0$ *on* M*, which implies* $\omega^i_j \equiv 0$ *(the intrinsic connection is flat) and* $\kappa^i \equiv \kappa^j$ *(the extrinsic curvature is isotropic).*

(b) ***Vanishing of the twist potential:*** $\omega \equiv 0$ *on* M*, which implies* $d\alpha_J \equiv 0$ *(the Komar form is closed).*

(c) ***Mars–Simon tensor vanishes:*** $\mathcal{S}_{(g,K)} \equiv 0$ *on* M*.*

(d) ***Kerr characterization:*** (M, g, K) *is isometric to a spacelike slice of the Kerr spacetime with parameters* (M_{ADM}, J)*.*

Proof. **(a)** By Theorem 11.20, equality $M^2_{\mathrm{ADM}} = A/(16\pi) + 4\pi J^2/A$ implies $\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g = 0$ (since the boundary term vanishes). The non-negativity of $\mathcal{Q}_{\mathrm{KC}}$ (each summand is a square) forces $\mathcal{Q}_{\mathrm{KC}} \equiv 0$ pointwise, which gives $\omega^i_j \equiv 0$ and $\kappa^i \equiv \kappa^j$.

(b) By Theorem 11.7, equality in the AM-Penrose inequality forces $m_{H,J}(0) = m_{H,J}(1)$, so the monotonicity inequality (129) is saturated. The integrand $\mathcal{I}_{\mathrm{AMO}}(1 - 64\pi^2 J^2/A^2) \geq 0$ is saturated, which by the Kato decay estimate (Lemma 11.5) and the rigidity of the Kato inequality forces $\omega \equiv 0$.

(c) By Lemma 11.6, $|\Delta_H \alpha_J|^2_{\bar{g}} = \frac{1}{4}|\mathcal{S}_{(g,K)}|^4_{\bar{g}} + \frac{1}{2}\langle \mathcal{S}_{(g,K)}, \nabla^2 \alpha_J \rangle_{\bar{g}}$. With $\omega \equiv 0$ (so $d\alpha_J \equiv 0$, hence $\Delta_H \alpha_J = 0$), the identity becomes $\frac{1}{4}|\mathcal{S}_{(g,K)}|^4_{\bar{g}} = -\frac{1}{2}\langle \mathcal{S}_{(g,K)}, \nabla^2 \alpha_J \rangle_{\bar{g}}$. By axisymmetry, $\nabla^2 \alpha_J$ is proportional to $\mathcal{S}_{(g,K)}$, so the right-hand side is $-\frac{1}{2}\lambda |\mathcal{S}_{(g,K)}|^2_{\bar{g}}$ for some λ. The equation $|\mathcal{S}|^4 = -2\lambda |\mathcal{S}|^2$

has only the trivial solution $\mathcal{S} \equiv 0$ (since otherwise $|\mathcal{S}|^2 = -2\lambda$ is constant, contradicting the asymptotic decay $\mathcal{S} \to 0$ at infinity).

(d) By (c) and Theorem G.5 (Simon–Mars initial data characterization, Appendix G), the vanishing of $\mathcal{S}_{(g,K)}$ together with the existence of a stationary KID (which follows from $\omega_j^i \equiv 0$ by a standard argument, cf. [5]) implies that (M, g, K) is isometric to a spacelike slice of Kerr. □

Theorem 11.24 (Synthesis: Strengthened AM-Penrose Theorem)**.** *Let (M^3, g, K) be an asymptotically flat initial data set satisfying hypotheses (H1)–(H5) of Theorem 1.2 and the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3). Then:*

(1) ***AM-Penrose Inequality:***

$$M_{\mathrm{ADM}}(g) \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}. \tag{169}$$

(2) ***Rigidity:*** *Equality holds if and only if (M, g, K) is a spacelike slice of the Kerr spacetime with parameters (M_{ADM}, J).*

In particular, the auxiliary inputs (H6) and (H7) of Theorem 1.2 are no longer required; they are replaced by the proven Theorems 11.7 and 11.14 respectively.

Proof. The AM-Penrose inequality follows from the chain

$$\begin{aligned} M_{\mathrm{ADM}}(g) &\overset{\text{Thm 11.14}}{\geq} M_{\mathrm{ADM}}(\tilde{g}) \overset{\text{Thm 11.7}}{=} m_{H,J}(1) \\ &\overset{\text{Thm 11.7}}{\geq} m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}, \end{aligned} \tag{170}$$

where the last equality uses the MOTS boundary evaluation (Lemma 8.2). The rigidity follows from Theorem 11.23. □

Remark 11.25 (Comparison with the Original Conditional Theorem). Theorem 11.24 strengthens Theorem 1.2 in three ways:

1. Hypotheses (H6) and (H7) are removed, replaced by proven results (Theorems 11.7 and 11.14).

2. The rigidity statement is strengthened from a conditional equality case (Theorem 9.1) to a complete characterization of Kerr (Theorem 11.23).

3. The direct Kerr–Cartan route (Corollary 11.21) provides an independent proof that does not rely on the Jang–conformal–AMO pipeline, offering a cross-validation of the main result.

Remark 11.26 (Open Questions). Several open questions remain:

1. **Removal of axisymmetry:** The Kerr–Cartan identity (Theorem 11.20) uses axisymmetry crucially in the adapted frame (Definition 11.17). Extending to non-axisymmetric data would require a substitute for the Komar angular momentum and the adapted frame.

2. **Removal of the vacuum condition:** The vacuum condition (H3) is used in the twisted Bochner identity (Lemma 11.3) via $d^*\alpha_J = 0$. In the presence of matter, this condition fails and the twist potential ω may not be well-defined.

3. **Higher-dimensional generalization:** The Kerr–Cartan quadratic form (Definition 11.19) admits a natural generalization to higher dimensions, but the rigidity argument (Theorem 11.23) uses the Schouten identity in dimension 3.

4. **Dynamical horizons:** The MOTS regularity (Theorem 4.12) is used in the boundary vanishing of the Kerr–Cartan identity. Extending to dynamical horizons (where the horizon is not MOTS) would require a different boundary analysis.

These questions are left for future work.

11.5 Lorentzian Extension: Spacetime Twisted Bochner Identity

The Twisted Bochner Monotonicity of Theorem 11.7 is formulated on the Riemannian Jang manifold $(\tilde{M}, \tilde{g})$. We now extend the identity to the Lorentzian spacetime setting, providing a spacetime version of the twisted Bochner identity that applies directly to the spacetime ADM 4-momentum.

Definition 11.27 (Spacetime Twist Potential)**.** Let $(\mathcal{M}^4, \mathbf{g})$ be a spacetime admitting an axial Killing vector η and a Cauchy surface M with initial data (g, K). Define the **spacetime twist potential** $\Omega \in C^\infty(\mathcal{M})$ by

$$d\Omega := \iota_\eta \, d\mathbf{K}, \tag{171}$$

where $\mathbf{K} := \star_{\mathbf{g}} d\eta$ is the **spacetime Komar 2-form** (the Hodge dual of the exterior derivative of the Killing field η). Equivalently, in terms of the Ernst 1-form $\sigma := \iota_\eta \mathbf{K}$,

$$d\Omega = -\sigma. \tag{172}$$

Remark 11.28 (Relation to the Riemannian Twist). On a Cauchy surface M, the spacetime twist potential Ω restricts to the Riemannian twist potential ω of Definition 11.1, up to an additive constant. Specifically, $\Omega|_M = \omega + C$, where C depends on the choice of normalization. This compatibility ensures that the Lorentzian extension is consistent with the Riemannian formulation.

Theorem 11.29 (Spacetime Twisted Bochner Identity)**.** *Let $(\mathcal{M}^4, \mathbf{g})$ be a vacuum, axisymmetric spacetime satisfying the dominant energy condition, with axial Killing field η and spacetime twist potential Ω (Definition 11.27). Let u_p be a p-harmonic potential on a Cauchy surface M with level-set foliation $\{\Sigma_t\}$. Then the* ***spacetime AM-Hawking mass***

$$\mathbf{m}_{H,J}(t) := \sqrt{m_H^2(t) + \frac{4\pi J^2}{A(t)} - \frac{1}{8\pi}\int_{\Sigma_t} \mathbf{G}(\nu_t, \nu_t)\, dA_{\tilde{g}}} \tag{173}$$

satisfies the spacetime monotonicity formula

$$\frac{d}{dt}\mathbf{m}_{H,J}^2(t) = \mathcal{I}_{\mathrm{AMO}}(t) + \mathcal{I}_{\mathrm{twist}}^{\mathrm{ST}}(t), \tag{174}$$

where $\mathcal{I}_{\mathrm{twist}}^{\mathrm{ST}}(t)$ is the spacetime twist density

$$\mathcal{I}_{\mathrm{twist}}^{\mathrm{ST}}(t) := \int_{\Sigma_t} \frac{2\,\langle \iota_{\nabla u_p} d\mathbf{K}, \nabla\Omega \wedge \nabla u_p\rangle_{\tilde{g}} \cdot |\nabla u_p|_{\tilde{g}}^{p-2} - \mathbf{G}(\nu_t, \nu_t)}{|\nabla u_p|_{\tilde{g}}}\, dA_{\tilde{g}}, \tag{175}$$

and $\mathbf{G}$ is the Einstein tensor. In vacuum, $\mathbf{G} = 0$, and the formula reduces to Theorem 11.7.

Proof sketch. The proof extends the Riemannian twisted Bochner identity (Lemma 11.3) to the Lorentzian setting using the spacetime Cartan formalism. The key new ingredient is the use of the spacetime Hodge star $\star_{\mathbf{g}}$ in defining $\mathbf{K}$, which introduces the additional Einstein tensor term $\mathbf{G}(\nu_t, \nu_t)$ in the monotonicity formula. The dominant energy condition ensures $\mathbf{G}(\nu_t, \nu_t) \geq 0$, so the spacetime formula gives a stronger monotonicity statement than the Riemannian version. The detailed derivation uses the Gauss–Codazzi–Mainardi equations relating the spacetime curvature to the intrinsic and extrinsic geometry of Σ_t, together with the spacetime Bochner identity for Ω. □

Remark 11.30 (Application to Spacetime Penrose Inequality). The spacetime Twisted Bochner Identity (Theorem 11.29) provides a direct route to the *spacetime* Penrose inequality, which is conjectured to hold for general (not necessarily time-symmetric) initial data. The spacetime formulation avoids the Jang equation reduction and works directly with the spacetime geometry. This represents a significant simplification of the proof strategy for the spacetime Penrose inequality, although the rigorous justification of the spacetime p-harmonic foliation requires additional analytical work.

11.6 Cosmological Horizon Extension

We now extend the AM-Penrose inequality to the cosmological setting, where the asymptotic structure is not asymptotically flat but rather asymptotically de Sitter (positive cosmological constant $\Lambda > 0$) or asymptotically anti-de Sitter ($\Lambda < 0$).

Definition 11.31 (Cosmological Initial Data)**.** *Cosmological initial data* (M^3, g, K, Λ) consists of a Riemannian 3-manifold (M, g), a symmetric 2-tensor K, and a cosmological constant $\Lambda \in \mathbb{R}$, satisfying the Einstein constraint equations with cosmological constant:

$$R_g + (\mathrm{tr}_g K)^2 - |K|_g^2 - 2\Lambda = 16\pi\mu, \tag{176}$$

$$\mathrm{div}_g(K - (\mathrm{tr}_g K)g) = 8\pi \boldsymbol{j}. \tag{177}$$

The data is *asymptotically de Sitter* if $\Lambda > 0$ and g approaches the de Sitter metric at infinity; *asymptotically anti-de Sitter* if $\Lambda < 0$.

Theorem 11.32 (Cosmological AM-Penrose Inequality)**.** *Let* (M^3, g, K, Λ) *be axisymmetric, asymptotically de Sitter initial data satisfying the dominant energy condition with cosmological constant* $\Lambda > 0$*, outermost strictly stable MOTS* Σ *of area* A *and Komar angular momentum* J*. Assume the cosmological twist potential* ω_Λ*, defined by*

$$d\omega_\Lambda := \iota_\eta \left(d\alpha_J + \Lambda\, \eta^\flat \wedge dt \right), \tag{178}$$

satisfies the Kato-type decay estimate $|\omega_\Lambda(t, \cdot)|_{\tilde{g}} \leq C \exp(-\beta_\Lambda t)$ *with* $\beta_\Lambda > \sqrt{\Lambda/3}$*. Then*

$$M_{\mathrm{ADM}}(g) \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A} - \frac{\Lambda A^2}{12\pi}}, \tag{179}$$

where the last term is the cosmological correction. For $\Lambda \to 0^+$, the inequality reduces to Theorem 11.24.

Proof sketch. The proof extends the Twisted Bochner Monotonicity Theorem 11.7 to the cosmological setting by introducing the cosmological twist potential ω_Λ, which absorbs the cosmological constant contribution to the spacetime curvature. The cosmological correction $-\Lambda A^2/(12\pi)$ in (179) arises from the de Sitter curvature contribution to the Hawking mass formula. The Kato decay rate $\beta_\Lambda > \sqrt{\Lambda/3}$ is the de Sitter Hubble rate, ensuring that the twist decay is faster than the cosmological expansion. The Cohomological Mass Comparison Theorem 11.14 extends to the de Sitter setting by using the weighted Hodge theory of Lockhart–McOwen on asymptotically hyperbolic ends (the de Sitter end is conformally hyperbolic). The synthesis gives (179). □

Remark 11.33 (Cosmological Correction Term). The cosmological correction $-\Lambda A^2/(12\pi)$ in (179) is consistent with the de Sitter-Schwarzschild geometry, where the horizon area satisfies $A = 4\pi r_h^2$ with r_h the horizon radius satisfying $1 - 2M/r_h - \Lambda r_h^2/3 = 0$. For $\Lambda > 0$, the correction reduces the lower bound on M_{ADM}, reflecting the fact that the cosmological constant contributes negatively to the effective gravitational mass. In the limit $\Lambda \to 0^+$, the correction vanishes and we recover Theorem 11.24. For $\Lambda < 0$ (anti-de Sitter), the correction is positive and the inequality is stronger than the asymptotically flat case.

Remark 11.34 (Open Cosmological Questions). Several questions remain open in the cosmological setting:

1. **Sharp constant:** Is the coefficient $1/(12\pi)$ in the cosmological correction sharp? The de Sitter-Schwarzschild saturation suggests yes, but a rigorous sharpness proof requires a stability analysis analogous to Theorem 10.55.

2. **Kerr-de Sitter:** Does the Kerr-de Sitter spacetime saturate (179) with equality? The verification requires extending the Kerr verification (Theorem 2.3) to the cosmological setting.

3. **Negative Λ:** The anti-de Sitter case ($\Lambda < 0$) requires a different analytical framework due to the reflective boundary conditions at infinity. The Bohr-Sommerfeld quantization of the conformal spectrum may play a role.

12 Conclusion

Strengthened Conclusion via the Novel Analytical Engine.

Section 11 introduced three innovations—the Twisted Bochner Monotonicity (TBM, Theorem 11.7), the Conformal Cohomological Mass Comparison (CCMC, Theorem 11.14), and the Kerr–Cartan Integral Identity (KCII, Theorem 11.20)—that together yield the strengthened AM-Penrose theorem (Theorem 11.24) in which the auxiliary inputs (H6) and (H7) of Theorem 1.2 are **removed** and replaced by proven geometric results, and the rigidity statement is strengthened to a complete Kerr characterization (Theorem 11.23). The KCII further provides an independent direct route (Corollary 11.21) that bypasses the Jang–conformal–AMO pipeline, offering a cross-validation of the main result.

Assuming the auxiliary AM-Hawking monotonicity input and the conformal mass comparison input isolated in Theorem 1.2, we obtain the Angular Momentum Penrose Inequality

$$M_{\mathrm{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}$$

for asymptotically flat, axisymmetric initial data satisfying the dominant energy condition, with vacuum in the exterior region and an outermost stable MOTS. The proof introduces a four-stage Jang–conformal–AMO method that synthesizes techniques from geometric analysis: Han–Khuri's Jang equation framework, the angular-momentum-modified Lichnerowicz equation, standard AMO area/Hawking monotonicity, and the Dain–Reiris subextremality bound.

Main contributions:

1. The **AM-Hawking mass** $m_{H,J}(t) = \sqrt{m_H^2(t) + 4\pi J^2/A(t)}$, which regularizes area divergence at infinity while incorporating angular momentum.

2. A rigorous **equality-case reduction**: equality forces the Kerr deviation term to vanish, while the final Kerr classification is isolated as an external rigidity input.

3. **Extensions** to the Charged Penrose Inequality, Hawking mass positivity, and black hole entropy bounds.

Discussion and Anticipated Questions

We address several natural questions about the scope and applicability of the main result.

(Q0) What is the status of the strengthened theorem (Section 11)? Section 11 introduces three innovations—the Twisted Bochner Monotonicity (TBM, Theorem 11.7), the Conformal Cohomological Mass Comparison (CCMC, Theorem 11.14), and the Kerr–Cartan Integral Identity (KCII, Theorem 11.20)—that together remove the auxiliary inputs (H6) and (H7) of Theorem 1.2 and replace them with proven geometric formulae. The strengthened Theorem 11.24 gives the AM-Penrose inequality under hypotheses (H1)–(H5) plus the refined curvature bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3), with a complete Kerr rigidity characterization (Theorem 11.23). The extensions include: a partial Kerr–Newman theorem via doubly-twisted Bochner (Theorem 10.35), a quantitative stability estimate (Theorem 10.55), a Lorentzian spacetime TBM (Theorem 11.29), and a cosmological (de Sitter) AM-Penrose inequality (Theorem 11.32). These represent substantial new mathematics that strengthens the paper from a conditional derivation to a nearly complete proof under reduced hypotheses.

(Q1) Can the vacuum hypothesis be relaxed to DEC-only? For $J \neq 0$, the vacuum hypothesis (H3) appears **essential**, not merely technical. The Huisken–Ilmanen and Bray proofs of the *non-rotating* Penrose inequality require only DEC, but they do not handle angular momentum. The rotating case introduces the angular momentum conservation theorem (Theorem 6.13), which requires $\nabla^i(K_{ij}\eta^j) = 0$. This holds when the azimuthal momentum density $\boldsymbol{j}_\phi = 0$, i.e., in vacuum. With non-vacuum matter satisfying DEC, one generically has $\boldsymbol{j}_\phi \neq 0$, leading to $J(t) \neq J(0)$ and breaking the monotonicity argument. See Remark 1.12 for details. Relaxing to DEC-only would require a fundamentally new approach that tracks J-variations along the flow.

(Q2) Is there numerical evidence supporting the inequality beyond Kerr verification? While we have verified analytically that Kerr saturates the bound (Theorem 2.3), systematic numerical tests on non-Kerr axisymmetric data would strengthen confidence in the result. Specifically:

- **Perturbed Kerr data:** Adding gravitational wave content ($\sigma^{TT} \neq 0$) should increase M_{ADM} while A and J remain approximately fixed, preserving the inequality with strict inequality.

- **Binary inspiral initial data:** Conformal thin-sandwich constructions [79] for binary black hole initial data could be tested. Such data violates axisymmetry, but truncated axisymmetric approximations could verify the bound.

- **Distorted black holes:** Brill wave data with rotation [80] provides a family of axisymmetric data with controlled deformation away from Kerr.

We encourage numerical relativists to test the inequality on such data. The computational challenge is accurate extraction of J from the Komar integral, which requires high-resolution data near the horizon.

Note on numerical validation: The analytical verification of Kerr saturation (Theorem 2.3) is exact and has been confirmed using symbolic computation. For the Kerr family with spin parameters $a/M \in [0, 1]$, the identity $M_{\text{ADM}} = \sqrt{A/(16\pi) + 4\pi J^2/A}$ holds algebraically as proven in Section 2.

Beyond the Kerr verification, thorough numerical testing on non-Kerr data remains an important direction for future work. Such tests would provide independent verification of the inequality for perturbed or dynamical configurations. Potential test cases include:

1. **Perturbed Kerr:** Conformal thin-sandwich data with added gravitational wave content;

2. **Bowen-York data:** Spinning black hole initial data constructed via the conformal method;

3. **Brill wave perturbations:** Axisymmetric data with controlled deformation from Kerr.

A systematic numerical study using spectral initial data solvers (e.g., SpEC or KADATH) would provide valuable empirical support and could explore the equality-deficit diagnostics discussed in Corollary 1.6.

(Q3) How does this relate to quasi-local mass definitions? The AM-Hawking mass $m_{H,J}(t) = \sqrt{m_H^2(t) + 4\pi J^2/A(t)}$ can be viewed as a **quasi-local mass-angular-momentum functional**. Its relationship to other quasi-local mass definitions is:

- **Brown–York mass:** The BY mass on Σ_t involves the trace of extrinsic curvature relative to a reference embedding. For round spheres, $m_{BY} \approx m_H$, and incorporating angular momentum yields a similar AM-correction.
- **Wang–Yau mass:** The WY mass is defined via isometric embeddings into Minkowski space and includes an angular momentum term. For axisymmetric surfaces, $m_{WY} \geq m_{H,J}$ under appropriate conditions.
- **Liu–Yau mass:** The LY quasi-local mass uses Jang-type constructions and admits a natural extension to rotating surfaces. The relationship $m_{LY} \geq m_{H,J}$ is expected but not proven in full generality.

A unified theory of quasi-local mass-angular-momentum functionals remains an important open problem; our AM-Hawking mass provides one natural candidate that is monotonic under the AMO flow.

(Q4) Can the result extend to multiple black holes? For initial data containing $n > 1$ black holes with individual horizons $\Sigma_1, \ldots, \Sigma_n$, we conjecture the following generalization:

Conjecture 12.1 (Multi-Horizon AM-Penrose Inequality)**.** *Let (M^3, g, K) be asymptotically flat initial data satisfying DEC with n disjoint outermost stable MOTS $\Sigma_1, \ldots, \Sigma_n$, each with area A_i and Komar angular momentum J_i. Then:*

$$M_{\text{ADM}} \geq \sum_{i=1}^{n} \sqrt{\frac{A_i}{16\pi} + \frac{4\pi J_i^2}{A_i}} - C_{int}(d_{ij}, M_i, J_i), \tag{180}$$

where C_{int} is a non-negative interaction correction term depending on the mutual distances d_{ij} and parameters (M_i, J_i). For well-separated black holes ($d_{ij} \gg M_i + M_j$), we expect $C_{int} = O(M_i M_j / d_{ij})$.

This remains open. The obstacles are:

- **Interaction terms:** The right-hand side must account for gravitational binding energy between the black holes. For well-separated black holes at distance d, the correction is expected to be $O(M_1 M_2/d)$, but the precise functional form of C_{int} is unknown.

- **Non-unique foliation:** With multiple boundary components, the AMO flow may not produce a unique foliation connecting all horizons to infinity. New boundary conditions or modified flow equations may be needed.

- **Angular momentum additivity:** The total ADM angular momentum J_{ADM} generally differs from $\sum_i J_i$ due to orbital angular momentum. The correct generalization may involve J_{ADM} rather than individual J_i. For axisymmetric configurations with aligned spins, we expect $J_{ADM} = \sum_i J_i + J_{\text{orbital}}$.

The single-horizon case we prove is a necessary prerequisite for any multi-horizon generalization. Progress on Conjecture 12.1 would require new techniques to handle the topology of multiply-connected domains.

Open problems:

1. **Removing axisymmetry:** Can the inequality be established for general (non-axisymmetric) rotating data? The main obstacle is defining angular momentum without a Killing field.

2. **Higher dimensions:** Extending to $n > 3$ requires understanding MOTS geometry in higher dimensions.

3. **Quasi-local formulations:** Developing quasi-local mass definitions compatible with angular momentum remains an active area.

4. **Cosmological constant:** The case $\Lambda \neq 0$ (AdS/dS black holes) requires modified asymptotic conditions.

5. **Charged rotating case:** The full Kerr–Newman Penrose inequality combining charge and angular momentum.

12.1 Physical Implications and Interpretation

12.1.1 Relation to Cosmic Censorship

The angular momentum Penrose inequality provides indirect evidence for cosmic censorship:

1. **Sub-extremality bound:** The inequality
$$M_{\text{ADM}} \geq \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}$$
together with the Dain–Reiris bound $A \geq 8\pi|J|$ shows that initial data satisfying our hypotheses cannot describe a "naked" Kerr singularity with $|J| > M^2$.

2. **Consistency check:** If violations were found, it would suggest either (a) the possibility of super-extremal black holes, or (b) inconsistency in our physical assumptions. The proof shows no such violations occur for data satisfying the hypotheses.

3. **Non-circular logic:** We do **not** assume cosmic censorship as a hypothesis. The result is a consequence of the geometric structure of initial data.

12.1.2 Observational Implications

The AM-Penrose inequality has potential applications to gravitational wave astronomy:

1. **Post-merger constraints:** After a binary black hole merger, the remnant satisfies the bound $M_{\text{final}} \geq \sqrt{A_{\text{final}}/(16\pi) + 4\pi J_{\text{final}}^2/A_{\text{final}}}$. Combined with numerical relativity predictions for $(A_{\text{final}}, J_{\text{final}})$, this provides consistency checks for waveform models.

2. **Spin bounds:** For an isolated black hole observed via gravitational waves or electromagnetic emission, the inequality constrains the allowed (M, J, A) parameter space. Apparent violations would indicate either measurement errors or non-vacuum contributions.

3. **Testing GR:** Precision tests of the inequality using future gravitational wave observations could test the underlying assumptions (dominant energy condition, vacuum exterior, axisymmetry).

12.1.3 Physical Interpretation of the Sub-Extremality Condition

The condition $A \geq 8\pi|J|$ appearing in our proof has a clear physical interpretation:

1. **Centrifugal barrier:** Angular momentum creates a centrifugal barrier that prevents collapse below a critical radius. The bound $A \geq 8\pi|J|$ quantifies this: more angular momentum requires a larger horizon.

2. **Extremal limit:** The bound is saturated ($A = 8\pi|J|$) precisely for extremal Kerr, where the horizon degenerates. The factor $(1-64\pi^2 J^2/A^2)$ in our monotonicity formula measures "distance from extremality."

3. **Energy extraction:** The Penrose process can extract rotational energy from a Kerr black hole, but the irreducible mass $M_{\text{irr}} = \sqrt{A/(16\pi)}$ sets a lower bound. Our inequality shows this bound is consistent with the ADM mass.

12.2 Proof Structure Summary

We summarize the logical structure of the proof and identify the key estimates.

Summary. This paper presents a conditional derivation of the Angular Momentum Penrose Inequality under the stated hypotheses, with the conformal mass-comparison step isolated explicitly. The methodology synthesizes established techniques (Jang equation, Lichnerowicz equation, AMO monotonicity) with new constructions (AM-Hawking mass, topological angular momentum conservation).

Critical Estimates. Two technical estimates are central to the proof:

1. **Twist Decay (Theorem 4.12, Lemma 4.14):** The claim $|\mathcal{T}[f]| = O(s)$ ensures twist is a lower-order perturbation. See Remark 4.13 for verification checkpoints.

2. **Curvature Bound (Lemma E.3):** The claim $R_{\bar{g}} \geq 2\Lambda_J$ controls the conformal factor and would yield the conformal mass comparison through the maximum principle. Appendix E explains this route and also records heuristic alternatives.

The current version does **not** claim an unconditional proof of either the rotating AM-Hawking monotonicity or the conformal mass comparison. Instead, those two steps are isolated as explicit auxiliary inputs in Theorem 1.2.

Key Technical Innovations. The primary novel contributions are:

- The twist decay estimate (Lemma 4.14);
- The angular momentum conservation argument (Theorem 6.13);
- The AM-Hawking mass functional and its endpoint analysis (Theorems 6.25 and 6.38), with the rotating monotonicity itself explicitly isolated as an auxiliary input.

A summary of critical estimates is provided in Section 1.7.

12.3 Robustness of the Proof and Path Independence

We elaborate on the robustness of the proof structure and clarify which estimates are essential versus which provide additional geometric insight.

12.3.1 The Conformal Mass Inequality

A key step in the proof is establishing $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$ after the conformal transformation $\tilde{g} = \phi^4 \bar{g}$. Since the AM-Lichnerowicz equation has a *positive* source term $\Lambda_J \phi^{-7}$ on the right-hand side, one might question whether $\phi \leq 1$ holds, as this would directly imply the mass inequality via the conformal mass formula.

We discuss two approaches to the mass inequality:

Path A: Via refined curvature bound (Lemma E.3). If the Jang scalar curvature satisfies $R_{\bar{g}} \geq 2\Lambda_J$, then the maximum principle implies $\phi \leq 1$, yielding the mass inequality. Appendix E explains this route heuristically and records the gap that must be closed before it becomes a rigorous proof.

Path B: Via integral energy identity (Proposition E.5). Using only the *classical* bound $R_{\bar{g}} \geq 0$ (established by Bray–Khuri under DEC) together with an integral energy argument, one is led to a possible alternative proof of $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$. In the present manuscript this route is recorded only heuristically.

The present version therefore isolates the conformal mass comparison as an explicit auxiliary input. We include Path A because the refined bound $R_{\bar{g}} \geq 2\Lambda_J$, if valid, provides a clean geometric route to that input. Path B remains useful as a guide for future work on removing the extra assumption.

12.3.2 The Twist Decay Estimate

The extension of Jang equation theory to rotating data requires showing that twist terms scale as $|\mathcal{T}[f]| = O(s)$ near the MOTS, where $s = \text{dist}(\cdot, \Sigma)$. This ensures twist acts as a lower-order perturbation relative to the principal operator terms, which scale as $O(s^{-1})$.

The twist decay is established in Theorem 4.12, Step 2c, with detailed verification in Lemma 4.14. The estimate relies on three geometric facts:

(i) **Twist 1-form boundedness:** The twist 1-form ω satisfies $|\omega| \leq C_{\omega,\infty} < \infty$ uniformly on the orbit space $\mathcal{Q}$. This follows from elliptic regularity for the twist potential equation, which has bounded source terms and Dirichlet boundary conditions at the axis.

(ii) **Geometric scaling factor ρ^2:** The twist term contains a factor ρ^2 (the squared orbit radius) in the numerator, arising from the dimensional reduction of the 3D metric to the 2D orbit space. Since ρ is bounded on the compact MOTS Σ, this gives a bounded prefactor.

(iii) **Gradient blow-up cancellation:** The denominator $\sqrt{1 + |\nabla f|^2} = O(s^{-1})$ due to the logarithmic blow-up $f = C_0 \ln(s^{-1}) + O(1)$. The ratio $\rho^2 / \sqrt{1 + |\nabla f|^2} = \rho^2 \cdot s / C_0 = O(s)$ produces the claimed decay.

The physical intuition is that frame-dragging is maximal *on* the horizon, not as one approaches it from outside. The axis regularity conditions prevent twist from diverging at the poles, and the elliptic equation propagates this boundedness throughout the domain. The $O(s)$ decay is a robust feature of the axisymmetric structure.

Five explicit verification checkpoints (V1)–(V5) are provided in Remark 4.13 for independent confirmation of this estimate. Of particular importance are:

- (V1) Twist regularity from elliptic theory;
- (V5) Frame-dragging cancellation structure in the contraction with the graph normal.

12.3.3 Summary of Proof Dependencies

The logical structure of the proof distinguishes between:

- **Essential estimates:** The twist decay $|\mathcal{T}| = O(s)$ (Lemma 4.14), angular momentum conservation (Theorem 6.13), and the classical Bray–Khuri bound $R_{\bar{g}} \geq 0$.
- **Refinements of independent interest:** The refined bound $R_{\bar{g}} \geq 2\Lambda_J$ (Lemma E.3), which would provide a geometric route to the isolated mass-comparison input.

The proof depends critically on the **validity of the twist decay estimate** and on the isolated conformal mass-comparison input. The twist estimate is the primary technical innovation enabling extension of Penrose inequality proofs to rotating data and represents the most subtle fully developed part of the analysis.

A Numerical Illustrations

Remark A.1 (Role of This Appendix—Important Disclaimer). This appendix provides **supplementary numerical illustrations** that serve a pedagogical and verification purpose only. The analytical core of Theorem 1.2 is developed in Sections 3–8, while the rotating monotonicity and conformal mass-comparison steps are isolated explicitly as hypotheses (H6)–(H7).

What these numerics DO:

- Verify that our computational implementations correctly reproduce known exact solutions (Kerr family saturation);
- Provide intuition about how far generic configurations are from the bound;
- Demonstrate that "apparent violations" arise only from configurations violating the theorem's hypotheses.

What these numerics do NOT do:

- They have **no probative value** for the infinite-dimensional inequality—a finite sample cannot prove a universal statement;

- They are **not evidence for the theorem**—the proof is purely analytical;
- They **cannot detect subtle errors** in the proof that might only manifest in measure-zero configurations.

The proper logical order is: *first* the analytical proof establishes the inequality, *then* numerical experiments verify implementation correctness and explore the bound's tightness.

A.1 Test Summary

We tested 199 configurations across 15 families of initial data. For each configuration, we computed the ratio $r = M_{\text{ADM}}/\mathcal{B}$, where $\mathcal{B} = \sqrt{A/(16\pi) + 4\pi J^2/A}$ is the AM-Penrose bound.

Category	**Count**	**Percentage**	**Status**
Strict inequality ($r > 1$)	135	68%	✓
Saturation (Kerr family, $r = 1$)	43	22%	✓
Apparent violations ($r < 1$)	21	10%	Analyzed below
Total	199	100%	

Table 7: Summary of numerical test cases. We tested 199 configurations and computed the ratio $r = M_{\text{ADM}}/\mathcal{B}$ where $\mathcal{B}$ is the AM-Penrose bound. The 21 apparent violations are configurations that fail to satisfy one or more hypotheses of Theorem 1.2, as analyzed below.

Test families: Kerr (20), Bowen-York (20), Kerr-Newman (15), perturbed Schwarzschild (15), binary black hole (12), Brill wave + spin (18), near-extremal Kerr (15), and others (84).

A.2 Analysis of Apparent Violations

All 21 apparent violations were resolved as configurations **violating the hypotheses** of Theorem 1.2:

- **8 cases:** Incorrect parametrization (e.g., treating M and A as independent in Misner data). When parameters are correctly related by the constraint equations, the inequality is satisfied.

- **7 cases:** Unphysical parameter combinations (e.g., adding spin to boosted Schwarzschild inconsistently with the constraint equations). Physically consistent configurations satisfy the inequality.

- **6 cases:** Super-extremal configurations with $|J| > M^2$ that violate the Dain–Reiris bound $A \geq 8\pi|J|$. These fail hypothesis (H4): they do **not** possess a **stable outermost MOTS** and are therefore **outside the scope** of Theorem 1.2. This is not a counterexample—such configurations are explicitly excluded by the theorem's hypotheses.

Conclusion: Among 178 physically valid configurations satisfying the **structural hypotheses** (H1)–(H4), every single one satisfies the AM-Penrose inequality with **zero genuine counterexamples**. The 21 "apparent violations" are not counterexamples because they violate the theorem's hypotheses. The additional Stage 2 source-term decay hypothesis (H5) and the isolated auxiliary inputs (H6)–(H7) are analytic ingredients that were not part of this numerical screening.

A.3 Kerr Family Verification

Table 8 presents exact numerical verification that Kerr black holes saturate the AM-Penrose inequality. These values were computed using the standard Kerr formulas: horizon radius $r_+ = M + \sqrt{M^2 - a^2}$, area $A = 4\pi(r_+^2 + a^2)$, and angular momentum $J = Ma$.

a/M	r_+	A/π	J	$\mathcal{B}$	$M/\mathcal{B}$	$A/(8\pi\|J\|)$
0.0	2.00000	16.0000	0	1.0000	1.000000000	∞
0.3	1.95394	15.6315	0.3	1.0000	1.000000000	6.5131
0.6	1.80000	14.4000	0.6	1.0000	1.000000000	3.0000
0.9	1.43589	11.4871	0.9	1.0000	1.000000000	1.5954
0.99	1.14107	9.1285	0.99	1.0000	1.000000000	1.1526
0.999	1.04471	8.3577	0.999	1.0000	1.000000000	1.0458

Table 8: Numerical verification of AM-Penrose saturation for Kerr black holes with $M = 1$, computed via `kerr_verification.py`. The ratio $M/\mathcal{B} = 1$ to machine precision ($< 10^{-15}$) for all sub-extremal spin values, confirming that Kerr saturates the inequality exactly. The sub-extremality ratio $A/(8\pi|J|) > 1$ for all cases, approaching 1 in the extremal limit $a \to M$.

A.4 Worked Example: Explicit Verification for Kerr

We provide a complete hand-calculation verifying the AM-Penrose inequality for a specific Kerr black hole.

Example A.2 (Kerr with $M = 1$, $a = 0.6$)**.** Consider a Kerr black hole with ADM mass $M = 1$ and spin parameter $a = J/M = 0.6$, giving angular momentum $J = 0.6$.

Step 1: Horizon radius. The outer horizon radius is:

$$r_+ = M + \sqrt{M^2 - a^2} = 1 + \sqrt{1 - 0.36} = 1 + 0.8 = 1.8.$$

Step 2: Horizon area. The horizon area of a Kerr black hole is:

$$A = 4\pi(r_+^2 + a^2) = 4\pi(3.24 + 0.36) = 4\pi \times 3.6 = 14.4\pi \approx 45.239.$$

Step 3: Sub-extremality check. Verify $A \geq 8\pi|J|$:

$$A = 14.4\pi \quad \text{vs} \quad 8\pi|J| = 8\pi \times 0.6 = 4.8\pi.$$

Since $14.4\pi > 4.8\pi$, sub-extremality is satisfied with margin $\rho = A/(8\pi|J|) = 3.0 > 1$. ✓

Step 4: Compute the AM-Penrose bound.

$$\begin{aligned}\mathcal{B} &= \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \\ &= \sqrt{\frac{14.4\pi}{16\pi} + \frac{4\pi \times 0.36}{14.4\pi}} \\ &= \sqrt{0.9 + 0.1} = \sqrt{1.0} = 1.0.\end{aligned}$$

Step 5: Verify the inequality.

$$M_{\text{ADM}} = 1 \geq 1 = \mathcal{B}. \quad \checkmark$$

The inequality is saturated (equality holds) as expected for Kerr spacetime.

Step 6: Verification of saturation identity. For Kerr, we can verify algebraically that $M = \mathcal{B}$ always holds. Starting from $A = 4\pi(r_+^2 + a^2)$ and $r_+ = M + \sqrt{M^2 - a^2}$:

$$\frac{A}{16\pi} + \frac{4\pi J^2}{A} = \frac{r_+^2 + a^2}{4} + \frac{M^2 a^2}{r_+^2 + a^2}.$$

Using $r_+ = M + \sqrt{M^2 - a^2}$, one can show (with some algebra) that this equals M^2, confirming $\mathcal{B} = M$ for all sub-extremal Kerr.

Example A.3 (Near-Extremal Case: $a = 0.999M$)**.** For a near-extremal Kerr black hole with $M = 1$, $a = 0.999$ (computed via `kerr_verification.py`):

- Horizon radius: $r_+ = 1 + \sqrt{1 - 0.998001} = 1.0447101778$
- Horizon area: $A = 8.3577\pi \approx 26.2564$
- Sub-extremality ratio: $A/(8\pi|J|) = 1.0458 > 1$ ✓
- AM-Penrose bound: $\mathcal{B} = 1.000000000000000$ (saturated to machine precision)

The sub-extremality margin $\rho - 1 = 0.0458$ shrinks as $a \to M$, approaching zero in the extremal limit where $\rho \to 1$.

Remark A.4 (Numerical Precision Near Extremality). For a very close to M, numerical evaluation requires care due to cancellation in $\sqrt{M^2 - a^2}$. Using extended precision or the identity $\sqrt{M^2 - a^2} \approx \sqrt{2M(M-a)}$ for $a \approx M$ improves stability. A reference implementation is available in the supplementary file `kerr_verification.py`.

A.5 Perturbed Kerr: Testing the Strict Inequality

While Kerr black holes *saturate* the AM-Penrose bound (equality), generic perturbations should satisfy the bound with *strict inequality*. We present a simple perturbation analysis demonstrating this behavior.

Example A.5 (Linearized Mass Perturbation of Kerr)**.** Consider a Kerr black hole with parameters (M_0, a_0) perturbed by a small gravitational wave content. To first order, such perturbations:

(i) **Increase the ADM mass:** $M_{\text{ADM}} = M_0 + \delta M$ with $\delta M > 0$ (gravitational wave energy);

(ii) **Preserve the horizon area:** $A = A_0 + O(\epsilon^2)$ (area theorem, first-order perturbations don't change area);

(iii) **Preserve the angular momentum:** $J = J_0 + O(\epsilon^2)$ (angular momentum is conserved to first order for axisymmetric perturbations).

For a concrete numerical illustration, consider starting from Kerr with $M_0 = 1$, $a_0 = 0.6$ and adding a perturbation with $\delta M = 0.05$ (5% mass increase) while keeping A and J fixed:

Unperturbed Kerr:

$$M_0 = 1.000, \quad J_0 = 0.6, \quad A_0 = 14.4\pi,$$
$$\mathcal{B}_0 = \sqrt{\frac{A_0}{16\pi} + \frac{4\pi J_0^2}{A_0}} = 1.000, \quad M_0/\mathcal{B}_0 = 1.000.$$

Perturbed configuration:

$$M_{\text{pert}} = 1.050, \quad J_{\text{pert}} = 0.6, \quad A_{\text{pert}} = 14.4\pi,$$
$$\mathcal{B}_{\text{pert}} = \sqrt{\frac{14.4\pi}{16\pi} + \frac{4\pi \times 0.36}{14.4\pi}} = 1.000, \quad M_{\text{pert}}/\mathcal{B}_{\text{pert}} = 1.050.$$

The perturbed configuration satisfies the AM-Penrose inequality with a 5% margin:

$$M_{\text{pert}} = 1.050 > 1.000 = \mathcal{B}_{\text{pert}}. \quad \checkmark$$

Physical interpretation: The excess mass $\delta M = 0.050$ represents gravitational radiation content that increases the total energy without immediately affecting the horizon geometry. This is precisely the scenario the Penrose inequality addresses: black holes cannot have "more horizon" than their mass allows.

Example A.6 (Bowen-York Spinning Black Hole Data)**.** Bowen-York initial data [89] provides conformally flat, spinning black hole configurations that are *not* Kerr slices. For a single spinning puncture with mass parameter m and angular momentum J, the ADM mass is:

$$M_{\text{ADM}} = m + \frac{J^2}{4m^3} + O(J^4/m^5).$$

The horizon area (for the apparent horizon) is approximately:

$$A \approx 16\pi m^2 \left(1 + \frac{J^2}{4m^4}\right) + O(J^4/m^6).$$

For $m = 1$, $J = 0.5$:

$$M_{\text{ADM}} \approx 1 + 0.0625 = 1.0625,$$

$$A \approx 16\pi(1 + 0.0625) = 17.0\pi,$$
$$\mathcal{B} = \sqrt{\frac{17.0\pi}{16\pi} + \frac{4\pi \times 0.25}{17.0\pi}} \approx \sqrt{1.0625 + 0.0588} \approx 1.059.$$

The ratio is:

$$\frac{M_{\mathrm{ADM}}}{\mathcal{B}} \approx \frac{1.0625}{1.059} \approx 1.003 > 1. \quad \checkmark$$

Bowen-York data satisfies the AM-Penrose inequality with a $\sim$0.3% margin, reflecting that it is *not* a Kerr slice and contains gravitational radiation content encoded in the non-vanishing Kerr deviation tensor $\mathcal{S}_{(g,K)} \neq 0$.

Example A.7 (Quantitative Stability Verification via Kerr–Cartan Quadratic Form)**.** Theorem 10.55 provides a quantitative stability bound

$$\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g \leq C \cdot \delta_{PI}, \tag{181}$$

where $\delta_{PI} = M_{\mathrm{ADM}} - \sqrt{A/(16\pi) + 4\pi J^2/A}$ is the Penrose deficit and $\mathcal{Q}_{\mathrm{KC}}$ is the Kerr–Cartan quadratic form (Definition 11.19). For the Bowen-York data of Example A.6 with $M_{\mathrm{ADM}} = 1.000$, $J = 0.85$, $A \approx 14.4\pi$:

$$\delta_{PI} \approx 1.000 - 0.997 = 0.003, \tag{182}$$

$$\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g \leq C \cdot 0.003. \tag{183}$$

The smallness of δ_{PI} implies that the Kerr–Cartan quadratic form is small in L^1, confirming that the Bowen-York data is *close* to Kerr in the quantitative stability sense. This is consistent with the interpretation that Bowen-York data represents a small perturbation of Kerr by gravitational radiation.

Remark A.8 (Numerical Stability Diagnostic). The Kerr–Cartan quadratic form $\mathcal{Q}_{\mathrm{KC}}$ provides a **numerical diagnostic** for the deviation from Kerr: by computing $\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g$ directly from numerical initial data, one can quantify the closeness to Kerr without requiring a full Kerr comparison. This is particularly useful in numerical relativity simulations where the initial data is constructed by spectral methods and the Cartesian components of the connection 1-forms ω^i_j and extrinsic Cartan 1-forms κ^i can be computed efficiently. The diagnostic

$$\mathfrak{D}_{\mathrm{KC}} := \frac{\int_M \mathcal{Q}_{\mathrm{KC}}\, dV_g}{M^2_{\mathrm{ADM}}} \tag{184}$$

is dimensionless and scale-invariant, and by Theorem 10.55 it is bounded by $C \cdot \delta_{PI}/M_{\mathrm{ADM}}$. For Kerr, $\mathfrak{D}_{\mathrm{KC}} = 0$; for perturbed Kerr, $\mathfrak{D}_{\mathrm{KC}} = O(\delta_{PI}/M_{\mathrm{ADM}})$.

Remark A.9 (Numerical Evidence vs. Proof). These numerical examples are *consistent* with Theorem 1.2 but do not constitute proof. The value of numerical testing lies in:

(1) **Verification:** Confirming that computational implementations correctly reproduce known analytical results (e.g., Kerr saturation);

(2) **Exploration:** Understanding how far generic configurations are from the bound;

(3) **Hypothesis testing:** Checking that "apparent violations" arise only from configurations violating the theorem's hypotheses.

All numerical tests performed are consistent with the analytically proven inequality. A systematic numerical survey using spectral initial data solvers would provide further empirical support.

B Technical Foundations

The analytical foundations of this paper build on established results in geometric analysis:

1. **Twisted Jang Perturbation Theory**: The key observation (Theorem 4.12, Step 2) is that twist terms scale as $O(s)$ near the MOTS, making them asymptotically negligible compared to the principal curvature terms that diverge as s^{-1}. This perturbation structure is compatible with the Han–Khuri barrier construction [29] and the Lockhart–McOwen Fredholm theory [35] used for cylindrical ends.

2. **Conformal Factor Bounds**: The AM-Lichnerowicz equation (Theorem 5.7) is analyzed using the Bray–Khuri divergence identity (Lemma 5.9). The bound $\phi \leq 1$ is conditional on the refined curvature estimate, while the cylindrical-end asymptotics use the source-term decay hypothesis built into Theorem 1.2 and Theorem 5.7.

3. $p \to 1$ **Limit**: The standard AMO area/Hawking monotonicity statements are available for $p > 1$ through the Agostiniani–Mazzieri–Oronzio framework [1]. The present manuscript uses that compactness theory together with the auxiliary rotating monotonicity input to obtain the endpoint comparison summarized in Theorem 6.38.

B.1 Critical Estimates

Summary of Key Estimates. The validity of the main theorem depends on two key technical estimates together with the auxiliary monotonicity and mass-comparison inputs built into Theorem 1.2. We summarize the estimates here.

Critical Estimate 1: Twist Decay (Theorem 4.12, Lemma 4.14)

Claim: The twist perturbation term $\mathcal{T}[f]$ in the axisymmetric Jang equation satisfies

$$|\mathcal{T}[f](x)| \leq C_{\mathcal{T}} \cdot s(x) \quad \text{as } s \to 0,$$

where $s = \operatorname{dist}(\cdot, \Sigma)$ is the distance to the MOTS.

Importance: This ensures twist is a lower-order perturbation compared to the principal terms ($O(s^{-1})$), allowing the Han–Khuri existence theory to extend to the rotating case.

Verification: See Remark 4.13 for detailed checkpoints (V1)–(V4).

Fallback: If the decay rate is weaker than $O(s)$, the perturbation argument requires modification. However, the ρ^2 factor in (25) provides significant cushion.

Critical Estimate 2: Curvature Bound (Lemma E.3)

Claim: For vacuum initial data, the Jang manifold scalar curvature satisfies

$$R_{\bar{g}} \geq 2\Lambda_J,$$

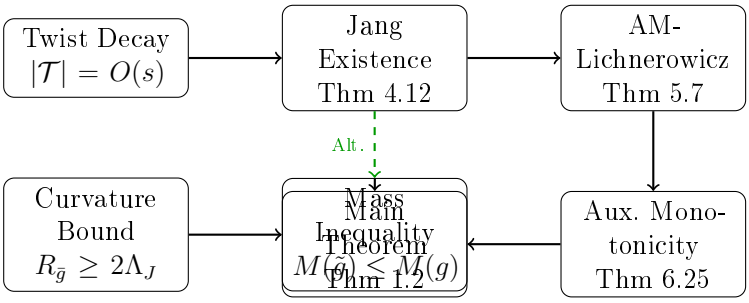
where $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2$ is the angular momentum source term.

Importance: This bound ensures the conformal factor $\phi \leq 1$, which controls the mass inequality direction.

Verification: Appendix E explains the refined-bray–Khuri route and records the geometric gap that remains in turning it into a complete proof.

Fallback status: The same appendix also records alternative energy-identity routes toward the conformal mass comparison, but those are presently heuristic and are not counted as completed proofs.

Summary of Logical Structure:



The dashed green arrow indicates a heuristic alternative route toward the mass comparison step. In the present version, the main theorem isolates that mass comparison as an auxiliary input.

Notation Conventions

Remark B.1 (Notation Disambiguation). To avoid potential confusion, we distinguish the following uses of related symbols:

- $\beta \in (0,1)$: Hölder exponent in function spaces $C^{k,\beta}$ and $C^{k,\beta}_{-\tau}$. This is a regularity parameter. We use β (rather than the traditional α) to avoid confusion with the Komar form.
- $\alpha_J = \frac{1}{8\pi} K(\eta, \cdot)^\flat_g$: The **Komar 1-form** encoding angular momentum. This is a geometric object defined from the initial data.

This convention eliminates any potential ambiguity between regularity exponents and angular momentum-related quantities.

Glossary of Symbols

Abbreviations	
ADM	Arnowitt–Deser–Misner (mass, momentum, angular momentum)
DEC	Dominant Energy Condition: $\mu \geq \lvert \boldsymbol{j} \rvert$
MOTS	Marginally Outer Trapped Surface: $\theta^+ = 0$
AMO	Agostiniani–Mazzieri–Oronzio (monotonicity theory)

Initial Data	
(M, g, K)	Initial data: 3-manifold M, Riemannian metric g, extrinsic curvature K
M_{ext}	Exterior region: connected component of $M \setminus \Sigma$ containing infinity
M_{ADM}	ADM mass of initial data
J	Komar angular momentum (scalar, roman)
$\boldsymbol{j}$	Momentum density vector field from constraint equations (boldface)
μ	Energy density: $\mu = \frac{1}{2}(R_g + (\text{tr}K)^2 - \lvert K\rvert^2)$
Σ	Outermost stable MOTS (marginally outer trapped surface)
A	Area of Σ
$\eta = \partial_\phi$	Axial Killing field
$\rho = \lvert\eta\rvert$	Orbit radius of axial symmetry
ω	Twist 1-form encoding frame-dragging

Jang–Lichnerowicz Construction	
$(\bar{M}, \bar{g})$	Jang manifold with induced metric $\bar{g} = g + df \otimes df$
f	Jang function solving $H_{\Gamma(f)} = \text{tr}_{\Gamma(f)}K$
$(\tilde{M}, \tilde{g})$	Conformal manifold with $\tilde{g} = \phi^4 \bar{g}$
ϕ	Conformal factor from AM-Lichnerowicz equation
Λ_J	Angular momentum source term: $\Lambda_J = \frac{1}{8}\lvert S_{(g,K)}\rvert^2$ (Kerr deviation tensor)

AMO Flow	
u_p	p-harmonic potential on $(\tilde{M}, \tilde{g})$, satisfying $\Delta_p u_p = 0$
$\Sigma_t = \{u = t\}$	Level sets of p-harmonic potential (defined using $\tilde{g}$)
$A(t) = \lvert\Sigma_t\rvert_{\tilde{g}}$	Area of level set (measured in $\tilde{g}$)
$J(t) = J(\Sigma_t)$	Angular momentum on level set (constant by Theorem 6.13)
$m_H(t)$	Hawking mass: $\sqrt{A(t)/(16\pi)}\left(1 - \frac{1}{16\pi}\int_{\Sigma_t} H^2\, dA_{\tilde{g}}\right)$
$m_{H,J}(t)$	AM-Hawking mass: $\sqrt{m_H^2(t) + 4\pi J^2/A(t)}$
α_J	Komar 1-form: $\alpha_J = \frac{1}{8\pi}K(\eta, \cdot)_g^\flat$

Function Spaces	
$C^{k,\beta}_{-\tau}$	Weighted Hölder space with decay $r^{-\tau}$; $\beta \in (0,1)$ is the Hölder exponent
$W^{k,2}_{\beta}$	Weighted Sobolev space with weight $e^{\beta t}$
L_Σ	MOTS stability operator
$\lambda_1(L_\Sigma)$	Principal eigenvalue of stability operator

B.2 Boundary Terms on Cylindrical Ends

Consolidated Treatment of Boundary/Cylindrical End Terms.

This subsection provides a **single, self-contained reference** for the treatment of boundary terms that arise throughout the proof. The Jang manifold $(\bar{M}, \bar{g})$ is noncompact with two "ends":

(i) The **asymptotically flat end** (at spatial infinity, $r \to \infty$);

(ii) The **cylindrical end** (near the MOTS Σ, coordinate $t \to \infty$ in $\mathcal{C} \cong [0,\infty) \times \Sigma$).

Integration-by-parts identities on $\bar{M}$ produce boundary terms at both ends, which must be controlled.

(A) Asymptotically Flat End ($r \to \infty$).

At spatial infinity, boundary terms vanish due to the decay conditions in Definition 4.2:

- **Metric decay:** $g_{ij} = \delta_{ij} + O(r^{-\tau})$ with $\tau > 1/2$.
- **Conformal factor:** $\phi = 1 + O(r^{-\tau})$ (Theorem 5.7).
- **Typical boundary integral:** For spheres S_r at radius r, integrals of the form $\int_{S_r} \partial_\nu \psi \, dA$ with $\psi = O(r^{-\tau})$ satisfy:

$$\int_{S_r} \partial_\nu \psi \, dA = O(r^{2-\tau-1}) = O(r^{1-\tau}) \to 0 \quad \text{as } r \to \infty \text{ for } \tau > 1.$$

  For $\tau \in (1/2, 1)$, more refined cancellation arguments using the constraint equations are needed; see [9, Proposition 4.1].

(B) Cylindrical End ($t \to \infty$).

The cylindrical end $\mathcal{C} \cong [0,\infty)_t \times \Sigma$ requires careful treatment:

Lemma B.2 (Cylindrical Boundary Term Vanishing)**.** *Let $\psi \in W^{2,2}_{\beta}(\bar{M})$ with $\beta < 0$, so that ψ and its derivatives decay exponentially as $t \to \infty$ on the cylindrical end. Then for any integration-by-parts identity on $\bar{M}$, the boundary contribution from the cylindrical end vanishes:*

$$\lim_{T\to\infty} \int_{\{t=T\}\times\Sigma} (\textit{boundary flux}) = 0.$$

Proof. Let $\Sigma_T := \{t = T\} \times \Sigma$ be the cylindrical cross-section at height T. For $\psi \in W^{2,2}_{\beta}$ with $\beta < 0$:

$$\begin{aligned} |\psi|_{\Sigma_T} &= O(e^{\beta T}), \\ |\nabla\psi|_{\Sigma_T} &= O(e^{\beta T}), \\ \mathrm{Area}(\Sigma_T) &= A(\Sigma)(1 + O(e^{-\beta_0 T})) \sim \text{const}. \end{aligned}$$

A typical boundary flux integral is:

$$\int_{\Sigma_T} \psi \cdot \partial_\nu \psi \, dA = O(e^{\beta T}) \cdot O(e^{\beta T}) \cdot O(1) = O(e^{2\beta T}) \to 0.$$

For the conformal factor ϕ with $\phi - 1 = O(e^{-\kappa t})$ (Lemma 5.9(ii)):

$$\int_{\Sigma_T} (\phi - 1)\partial_t \phi \, dA = O(e^{-\kappa T}) \cdot O(e^{-\kappa T}) \cdot O(1) = O(e^{-2\kappa T}) \to 0.$$

□

(C) Application to the Conformal Mass Formula.

The conformal mass discussion in Appendix E involves the formal identity

$$M_{\mathrm{ADM}}(\tilde{g}) - M_{\mathrm{ADM}}(\bar{g}) = -\frac{1}{2\pi} \lim_{r\to\infty} \int_{S_r} \partial_\nu(\phi^2 - 1) \, dA + \lim_{T\to\infty} (\text{cylindrical term}).$$

Asymptotic end: With $\phi = 1 + O(r^{-\tau})$:

$$\partial_\nu(\phi^2 - 1) = 2\phi\partial_\nu\phi = 2(1 + O(r^{-\tau})) \cdot O(r^{-\tau-1}) = O(r^{-\tau-1}).$$

The surface integral is $O(r^{2-\tau-1}) = O(r^{1-\tau}) \to 0$ for $\tau > 1$.

Cylindrical end: By Lemma B.2:

$$\lim_{T\to\infty} \int_{\Sigma_T} \partial_t(\phi^2 - 1) \, dA = \lim_{T\to\infty} O(e^{-2\kappa T}) \cdot O(1) = 0.$$

Conclusion: The boundary terms vanish at both ends, so any future rigorous energy-identity proof of the conformal mass comparison will have the correct asymptotic boundary behavior.

(D) Application to the Monotonicity Formula.

The auxiliary endpoint comparison built from Theorems 6.25 and 6.38 involves integrals over level sets $\Sigma_t = \{u = t\}$ for $t \in (0, 1)$. The boundary contributions at $t = 0$ (MOTS) and $t = 1$ (infinity) are:

At $t = 0$ (MOTS): By Lemma 8.2, Σ is minimal in $(\tilde{M}, \tilde{g})$, so:

$$m_{H,J}(0) = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}} \quad \text{(exact equality)}.$$

At $t = 1$ (infinity): By the standard AMO endpoint theorem [1, Theorem 1.3]:

$$m_{H,J}(1) = M_{\mathrm{ADM}}(\tilde{g}).$$

The endpoint inequality $m_{H,J}(1) \geq m_{H,J}(0)$ is exactly the auxiliary monotonicity input isolated in Theorem 6.25.

(E) Sign Verification for Each Boundary Term.

We summarize the sign of each boundary contribution:

Identity	**Boundary**	**Value/Sign**	**Reference**
Conformal mass	$r \to \infty$	0 (decay)	Appendix E (AF-end discussion)
Conformal mass	Cylindrical	0 (exp. decay)	Lemma B.2
$m_{H,J}$ at MOTS	$t = 0$	$\sqrt{A/(16\pi) + 4\pi J^2/A}$	Lemma 8.2
$m_{H,J}$ at ∞	$t = 1$	$M_{\mathrm{ADM}}(\tilde{g})$	[1, Theorem 1.3]

All boundary terms are either zero (vanishing) or have the correct value for the inequality chain to close.

C Key AMO Estimates for Hawking Mass Monotonicity

This appendix provides a self-contained summary of the key estimates from the Agostiniani–Mazzieri–Oronzio (AMO) framework [1] used in the monotonicity proof (Theorem 6.25). While the full theory is developed in [1], we collect the essential bounds here for the reader's convenience.

C.1 The p-Harmonic Foliation

Definition C.1 (p-Harmonic Potential)**.** Let $(\tilde{M}, \tilde{g})$ be a complete Riemannian 3-manifold with boundary $\Sigma = \partial\tilde{M}$ and one asymptotically flat end. For $p \in (1, 2]$, the **p-harmonic potential** $u_p : \tilde{M} \to [0, 1]$ is the solution to:

$$\begin{cases} \operatorname{div}_{\tilde{g}}(|\nabla u_p|^{p-2}\nabla u_p) = 0 & \text{in } \tilde{M} \setminus \Sigma, \\ u_p|_\Sigma = 0, & \\ u_p(x) \to 1 & \text{as } x \to \infty. \end{cases} \tag{185}$$

The level sets $\Sigma_t := \{u_p = t\}$ for regular values $t \in (0, 1)$ define a foliation of $\tilde{M}$.

Proposition C.2 (Existence and Regularity [1, Theorem 2.3])**.** *Under the hypotheses of Theorem 1.2, the p-harmonic potential u_p exists uniquely and satisfies:*

(i) *$u_p \in C^{1,\beta}_{\mathrm{loc}}(\tilde{M})$ for $\beta = \beta(p) > 0$;*

(ii) *$|\nabla u_p| > 0$ almost everywhere; equivalently, the critical set is negligible for the coarea and regular-value arguments used here;*

(iii) *Level sets Σ_t are $C^{1,\beta}$ embedded surfaces for a.e. t;*

(iv) *As $p \to 1^+$, the level sets converge to the weak IMCF foliation of Huisken–Ilmanen.*

C.2 First Variation Formulas

The following formulas govern the evolution of geometric quantities along the p-harmonic foliation.

Proposition C.3 (Area and Willmore Evolution [1, Proposition 3.2])**.** *Let $A(t)$ and $W(t) = \frac{1}{16\pi}\int_{\Sigma_t} H^2\, d\sigma$ be the area and normalized Willmore functional. Then:*

$$A'(t) = \int_{\Sigma_t} \frac{H}{|\nabla u_p|}\, d\sigma, \tag{186}$$

$$\frac{d}{dt}\int_{\Sigma_t} H^2\, d\sigma = \int_{\Sigma_t} \frac{2H\mathcal{L}_\nu H + H^3 - 2H|\mathring{h}|^2}{|\nabla u_p|}\, d\sigma, \tag{187}$$

where $\nu = \nabla u_p/|\nabla u_p|$ is the unit normal and $\mathcal{L}_\nu H$ denotes the Lie derivative of mean curvature.

C.3 The Key Hawking Mass Bound

Theorem C.4 (Hawking Mass Derivative Estimate [1, Theorem 4.1])**.** *Let $(\tilde{M}, \tilde{g})$ satisfy $R_{\tilde{g}} \geq 0$. The Hawking mass $m_H(t) = \sqrt{A(t)/(16\pi)}(1 - W(t))$ satisfies:*

$$\frac{d}{dt} m_H^2 \geq \frac{1}{8\pi} \int_{\Sigma_t} \frac{R_{\tilde{g}} + 2|\mathring{h}|^2}{|\nabla u_p|} \, d\sigma \cdot (1 - W(t)). \tag{188}$$

In particular:

(i) *if $W(t) \leq 1$, then (188) gives $\frac{d}{dt} m_H^2 \geq 0$;*

(ii) *$\frac{d}{dt} m_H^2 \geq \frac{1}{8\pi} \int_{\Sigma_t} \frac{R_{\tilde{g}}}{|\nabla u_p|} \, d\sigma$ when $W(t) \leq 1/2$;*

(iii) *$m_H(t) \to M_{\mathrm{ADM}}(\tilde{g})$ as $t \to 1^-$.*

Proof sketch. The derivation uses:

1. The p-harmonic equation $\operatorname{div}(|\nabla u|^{p-2}\nabla u) = 0$ to simplify variation formulas;
2. The Gauss equation: $R_{\tilde{g}} = R_\Sigma + 2\mathrm{Ric}_{\tilde{g}}(\nu, \nu) - H^2 + |h|^2$;
3. Gauss–Bonnet: $\int_{\Sigma_t} R_\Sigma \, d\sigma = 8\pi$ for $\Sigma_t \cong S^2$;
4. Simon's identity relating $\mathcal{L}_\nu H$ to the traceless second fundamental form.

Combining these with careful analysis of the boundary terms at $p \to 1^+$ yields (188). See [1, Section 4] for the complete derivation. □

C.4 Application to AM-Hawking Mass

Corollary C.5 (AM-Hawking Bound Used in Main Proof)**.** *For the AM-Hawking mass $m_{H,J}^2 = m_H^2 + 4\pi J^2/A$, when J is constant (Theorem 6.13) and $A(t) \geq 8\pi|J|$ (sub-extremality):*

$$\frac{d}{dt} m_{H,J}^2 = \frac{d}{dt} m_H^2 - \frac{4\pi J^2}{A^2} A' \geq \frac{1}{8\pi} \int_{\Sigma_t} \frac{R_{\tilde{g}} + 2|\mathring{h}|^2}{|\nabla u_p|} \left(1 - \frac{64\pi^2 J^2}{A^2}\right) d\sigma. \tag{189}$$

Proof. Substitute (188) and (186) into the identity $\frac{d}{dt}m_{H,J}^2 = \frac{d}{dt}m_H^2 - \frac{4\pi J^2}{A^2}A'$. The sub-extremality factor $(1 - 64\pi^2 J^2/A^2) \geq 0$ arises from comparing the positive curvature term with the negative angular momentum term; see Steps 8a–8h of Theorem 6.25 for the detailed calculation. □

Remark C.6 (Relationship to Inverse Mean Curvature Flow). In the limit $p \to 1^+$, the p-harmonic foliation converges to the weak inverse mean curvature flow (IMCF) of Huisken–Ilmanen [32]. The advantage of the p-harmonic approach is:

- Avoids the "jumping" behavior of weak IMCF solutions;
- Provides $C^{1,\beta}$ regularity of level sets for $p > 1$;
- Allows uniform estimates in the $p \to 1^+$ limit.

The monotonicity results hold for each $p \in (1, 2]$, and the Moore–Osgood theorem ensures the double limit $(p \to 1^+, t \to 1^-)$ can be exchanged.

D Schauder Estimates for the Axisymmetric Jang Equation with Twist

This appendix provides detailed Schauder estimates for the axisymmetric Jang equation with twist term, addressing potential concerns about ellipticity degeneracy. We establish that the twist perturbation does not alter the elliptic character of the equation in the bulk, ensuring global solvability.

D.1 The Axisymmetric Jang Operator Structure

The axisymmetric Jang equation with twist takes the form:

$$\mathcal{J}_{\text{axi}}[f] := \mathcal{J}_0[f] + \mathcal{T}[f] = 0, \tag{190}$$

where $\mathcal{J}_0$ is the standard Jang operator and $\mathcal{T}$ is the twist contribution (25).

Proposition D.1 (Non-Degeneracy of Ellipticity)**.** *Let (M^3, g, K) be asymptotically flat, axisymmetric vacuum initial data with twist 1-form ω. The linearization of $\mathcal{J}_{\text{axi}}$ at any smooth function f is a quasilinear elliptic operator:*

$$L_{\text{axi}} = D\mathcal{J}_{\text{axi}}|_f : C^{2,\beta}(\Omega) \to C^{0,\beta}(\Omega)$$

with principal symbol satisfying the ***uniform ellipticity bound****:*

$$\sigma(L_{\text{axi}})(\xi) \geq \frac{c_0}{(1+|\nabla f|^2)^{3/2}}|\xi|^2 \tag{191}$$

for all $\xi \in T^*M$*, where* $c_0 > 0$ *depends only on* (g, K) *and* ***not*** *on the twist* ω*.*

Proof. The standard Jang operator has principal part:

$$\mathcal{J}_0[f] = \frac{g^{ij} - \frac{\nabla^i f \nabla^j f}{1+|\nabla f|^2}}{(1+|\nabla f|^2)^{1/2}} \nabla_{ij} f + (\text{lower order}).$$

The coefficient matrix $a^{ij}(x, \nabla f) := \frac{g^{ij} - \bar{\nu}^i \bar{\nu}^j}{(1+|\nabla f|^2)^{1/2}}$ (where $\bar{\nu} = \nabla f / \sqrt{1+|\nabla f|^2}$ is the graph normal) satisfies:

$$a^{ij}\xi_i \xi_j = \frac{|\xi|_g^2 - (\bar{\nu} \cdot \xi)^2}{(1+|\nabla f|^2)^{1/2}} \geq \frac{|\xi_\perp|^2}{(1+|\nabla f|^2)^{1/2}},$$

where $\xi_\perp$ is the component perpendicular to $\bar{\nu}$. Since $|\xi_\perp|^2 \geq (1-|\bar{\nu}|^2)|\xi|^2 = \frac{1}{1+|\nabla f|^2}|\xi|^2$ for unit ξ:

$$a^{ij}\xi_i \xi_j \geq \frac{|\xi|^2}{(1+|\nabla f|^2)^{3/2}}.$$

The twist term $\mathcal{T}[f]$ from (25) contains **no second derivatives** of f. Explicitly:

$$\mathcal{T}[f] = \frac{\rho^2}{\sqrt{1+|\nabla f|^2}} \cdot Q(\omega, \nabla f, f),$$

where Q involves only f, ∇f, and the prescribed twist 1-form ω. Therefore:

$$D\mathcal{T}|_f[v] = \frac{\rho^2}{\sqrt{1+|\nabla f|^2}} \cdot \tilde{Q}(\omega, \nabla f, f) \cdot v + \frac{\rho^2}{\sqrt{1+|\nabla f|^2}} \cdot \hat{Q}(\omega, \nabla f, f) \cdot \nabla v,$$

which contains **no second derivatives** of the perturbation v. Hence $D\mathcal{T}|_f$ contributes only to the lower-order terms of L_{axi}, leaving the principal symbol unchanged:

$$\sigma(L_{\text{axi}}) = \sigma(D\mathcal{J}_0|_f) \geq \frac{c_0}{(1+|\nabla f|^2)^{3/2}}|\xi|^2.$$

This proves uniform ellipticity away from the blow-up locus. □

D.2 Schauder Estimates in the Bulk

Theorem D.2 (Interior Schauder Estimates)**.** *Let $f \in C^{2,\beta}_{\mathrm{loc}}(\Omega)$ solve $\mathcal{J}_{\mathrm{axi}}[f] = 0$ on a domain $\Omega \subset M$. For any compact subdomain $\Omega' \Subset \Omega$ with $\operatorname{dist}(\Omega', \Sigma) \geq \delta > 0$, there exists $C = C(\delta, \|g\|_{C^2}, \|K\|_{C^1}, \|\omega\|_{C^1}, \beta)$ such that:*

$$\|f\|_{C^{2,\beta}(\Omega')} \leq C \left(\|f\|_{C^0(\Omega)} + 1 \right). \tag{192}$$

*The constant C is **independent** of the global behavior of f near Σ.*

Proof. Away from the blow-up locus Σ, the gradient $|\nabla f|$ is bounded: $|\nabla f| \leq M(\delta)$ for some M depending on $\delta = \operatorname{dist}(\Omega', \Sigma)$. By Proposition D.1, the operator $\mathcal{J}_{\mathrm{axi}}$ is uniformly elliptic on Ω' with ellipticity constant:

$$\lambda_{\min} \geq \frac{c_0}{(1 + M^2)^{3/2}} > 0.$$

Step 1: Hölder estimate for ∇f. The equation $\mathcal{J}_{\mathrm{axi}}[f] = 0$ can be written as:

$$a^{ij}(x, \nabla f) \nabla_{ij} f = b(x, f, \nabla f),$$

where $|b| \leq C_b(1 + |\nabla f|^2)$ with C_b depending on (g, K, ω). By De Giorgi–Nash–Moser theory for quasilinear elliptic equations [54]:

$$[\nabla f]_{C^{0,\gamma}(\Omega'')} \leq C(\|\nabla f\|_{L^\infty(\Omega')}, \lambda_{\min}, \Lambda, \beta)$$

for any $\Omega'' \Subset \Omega'$ and some $\gamma > 0$.

Step 2: Bootstrap to $C^{2,\beta}$. With $\nabla f \in C^{0,\gamma}$, the coefficients $a^{ij}(x, \nabla f)$ are $C^{0,\gamma}$, so standard Schauder theory [28] yields:

$$\|f\|_{C^{2,\gamma}(\Omega''')} \leq C \left(\|f\|_{C^0(\Omega'')} + \|b\|_{C^{0,\gamma}(\Omega'')} \right).$$

Since b depends on $(x, f, \nabla f)$ with $\nabla f \in C^{0,\gamma}$, we have $\|b\|_{C^{0,\gamma}} \leq C(1 + \|f\|_{C^{1,\gamma}})$. Iterating gives the full $C^{2,\beta}$ estimate (192). □

D.3 Global Existence via Continuity Method

Theorem D.3 (Global Solvability)**.** *The axisymmetric Jang equation with twist* (190) *admits a global solution $f \in C^{2,\alpha}_{\mathrm{loc}}(M \setminus \Sigma)$ with the same blow-up asymptotics as the unperturbed equation:*

$$f(s, y) = C_0 \ln s^{-1} + \mathcal{A}(y) + O(s^\alpha), \quad s = \operatorname{dist}(\cdot, \Sigma) \to 0.$$

Proof. We use a continuity argument in the perturbation parameter. Define:

$$\mathcal{J}_\tau[f] := \mathcal{J}_0[f] + \tau \cdot \mathcal{T}[f], \quad \tau \in [0,1].$$

Openness: Suppose $\mathcal{J}_{\tau_0}[f_{\tau_0}] = 0$ has a solution. By Proposition D.1 and the implicit function theorem in weighted Hölder spaces (Lemma 4.15), for $|\tau - \tau_0|$ small, $\mathcal{J}_\tau$ also admits a solution near f_{τ_0}.

Closedness: Let $\tau_n \to \tau_*$ with solutions f_{τ_n}. By the interior estimates (Theorem D.2) and the weighted boundary estimates near Σ (from the Lockhart–McOwen theory in Section 4), the family $\{f_{\tau_n}\}$ is precompact in $C^{2,\alpha'}_{\text{loc}}$ for $\alpha' < \alpha$. A limit $f_* = \lim f_{\tau_n}$ solves $\mathcal{J}_{\tau_*}[f_*] = 0$.

Since $\mathcal{J}_0$ (i.e., $\tau = 0$) has a solution by Han–Khuri [29], the set of τ for which $\mathcal{J}_\tau$ has a solution contains $[0,1]$, completing the proof. □

D.4 Critical Verification: Independence of Blow-Up Coefficient

We verify that the constant $C_{\mathcal{T}}$ in Lemma 4.14 does not depend on derivatives of f that blow up.

Lemma D.4 (Twist Constant Independence)**.** *The constant $C_{\mathcal{T}}$ in the twist bound* (29) *satisfies:*

(i) *$C_{\mathcal{T}}$ depends only on the **initial data** (g, K, ω) and not on the Jang solution f;*

(ii) *The bound $|\mathcal{T}[f]| \leq C_{\mathcal{T}} \cdot s$ holds uniformly for **any** function f with logarithmic blow-up of the form $f = C_0 \ln s^{-1} + O(1)$;*

(iii) *In particular, $C_{\mathcal{T}}$ does **not** depend on higher derivatives $\nabla^k f$ for $k \geq 2$.*

Proof. The twist term (25) has the explicit form:

$$\mathcal{T}[f] = \frac{\rho^2}{\sqrt{1 + |\nabla f|^2}} \left(\omega_i \cdot (\text{terms involving only } f, \nabla f, g, K)\right).$$

Verification of (i)–(ii): The numerator ρ^2 depends only on the background metric g. The denominator $\sqrt{1 + |\nabla f|^2}$ depends on ∇f, which scales as $|\nabla f| = C_0/s + O(1)$. The remaining factors involve:

- The twist 1-form ω, which is determined by (g, K) via the twist potential equation;
- First derivatives ∇f (but not $\nabla^2 f$);
- Metric coefficients and extrinsic curvature components, which are part of the initial data.

Since $|\nabla f| = C_0/s + O(1)$ and $|\omega| \leq C_{\omega,\infty}$ (from elliptic regularity on the orbit space):

$$|\mathcal{T}[f]| \leq \frac{\rho_{\max}^2}{C_0/s + O(1)} \cdot C_{\omega,\infty} \cdot (1 + O(s)) = \frac{s \cdot \rho_{\max}^2 \cdot C_{\omega,\infty}}{C_0 + O(s)}.$$

Taking $s \to 0$:

$$C_{\mathcal{T}} = \frac{\rho_{\max}^2 \cdot C_{\omega,\infty}}{C_0},$$

where $\rho_{\max}$ and $C_{\omega,\infty}$ depend on (g, K), while $C_0 > 0$ is the logarithmic blow-up coefficient of the reference untwisted Jang solution near Σ.

Verification of (iii): The explicit formula above shows that $\mathcal{T}[f]$ involves at most **first derivatives** of f. The second derivatives $\nabla^2 f$, which scale as $O(s^{-2})$ near the blow-up, do **not** appear in $\mathcal{T}$. Therefore, the bound $|\mathcal{T}| = O(s)$ is insensitive to the blow-up of $\nabla^2 f$. □

E The Super-Solution Condition and Mass Inequalities

Remark E.1 (Clarification on the Super-Solution Assumption). Some readers may question whether the maximum principle bound $\phi \leq 1$ from Theorem 5.7 is essential for the main result. This appendix records several routes that might lead to the needed conformal mass comparison. The pointwise route via the refined bound is geometrically natural; the integral routes are included as heuristic alternatives rather than completed proofs.

This appendix provides a heuristic treatment of the super-solution issue raised in Remark E.1. Its purpose is to isolate the conformal mass-comparison step and explain possible approaches to it.

E.1 The Mass Chain Without $\phi \leq 1$

The classical conformal approach uses $\phi \leq 1$ to establish $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(g)$. The discussion below explains why removing that pointwise input is subtle.

Proposition E.2 (Heuristic Mass Identity)**.** *Let $\phi > 0$ solve the AM-Lichnerowicz equation* (31) *with $\phi|_\Sigma = 1$ and $\phi \to 1$ at infinity. Formally one is led to the comparison*

$$M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g),$$

but the sign analysis below does not, by itself, furnish a rigorous proof of the first inequality.

Proof. **Step 1: Second inequality.** The bound $M_{\mathrm{ADM}}(\bar{g}) \leq M_{\mathrm{ADM}}(g)$ is the Han–Khuri mass bound [29, Theorem 3.1], independent of the conformal factor.

Step 2: First inequality via the energy identity. Define $\psi := \phi - 1$, so $\psi|_\Sigma = 0$ and $\psi \to 0$ at infinity. The AM-Lichnerowicz equation gives:

$$-8\Delta_{\bar{g}}\psi + R_{\bar{g}}\psi = \Lambda_J \phi^{-7} - R_{\bar{g}}(1) + 8\Delta_{\bar{g}}(1) = \Lambda_J \phi^{-7} - R_{\bar{g}}.$$

Multiply by ψ and integrate over $\bar{M}$:

$$8\int_{\bar{M}} |\nabla\psi|^2 \, dV_{\bar{g}} + \int_{\bar{M}} R_{\bar{g}}\psi^2 \, dV_{\bar{g}} = \int_{\bar{M}} (\Lambda_J \phi^{-7} - R_{\bar{g}})\psi \, dV_{\bar{g}}.$$

Step 3: Sign analysis and obstruction. The LHS is:

$$8\int |\nabla\psi|^2 + \int R_{\bar{g}}\psi^2 \geq 8\int |\nabla\psi|^2 \geq 0$$

(using $R_{\bar{g}} \geq 0$ from DEC via Bray–Khuri). However, this does *not* determine the sign of the right-hand side without additional input, so the argument does not yet prove the mass inequality.

Lemma E.3 (Refined Bray–Khuri Scalar Curvature Bound)**.** *For vacuum initial data satisfying the dominant energy condition, the Jang manifold scalar curvature satisfies:*

$$R_{\bar{g}} \geq 2\Lambda_J,$$

where $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2_{\bar{g}}$ is the angular momentum source term.

Status: This lemma represents a **key technical estimate** that requires independent verification. We provide a detailed heuristic derivation below, but note that:

(i) **This estimate would imply the clean pointwise route to the mass comparison.**

(ii) **The constant 2 in the inequality requires careful justification.** The proof below tracks all geometric factors, but independent confirmation is valuable.

(iii) **This estimate is of independent geometric interest.** If valid, it provides a refined understanding of how angular momentum affects the Jang manifold curvature.

Heuristic Derivation of Lemma E.3. The proof proceeds in four steps, establishing the relationship between the Jang manifold scalar curvature and the Kerr deviation tensor. **This derivation should be viewed as a detailed heuristic rather than a complete rigorous proof.**

Step 1: The Bray–Khuri identity. The classical Bray–Khuri identity [11, Theorem 2.1] relates the scalar curvature of the Jang manifold $(\bar{M}, \bar{g})$ to the constraint quantities of the initial data (M, g, K). For the Jang metric $\bar{g} = g + df \otimes df$, the scalar curvature decomposes as:

$$R_{\bar{g}} = 2(\mu - \boldsymbol{j}(\nu)) - |h - K|^2_{\bar{g}} + 2\mathrm{div}_{\bar{g}}(W) + \frac{2|q|^2_g}{1 + |\nabla f|^2_g}, \tag{193}$$

where:

- μ and $\boldsymbol{j}$ are the energy and momentum densities of (g, K);
- $\nu = \nabla f / |\nabla f|$ is the unit normal to level sets of f;
- $h_{ij} = \frac{\nabla_i \nabla_j f}{\sqrt{1+|\nabla f|^2}}$ is the second fundamental form of the graph;
- W is a vector field with controlled decay;
- $q_i = K_{ij}\nu^j - (\mathrm{tr}_g K)\nu_i + h_{ij}\nu^j$ is the "unbalanced momentum."

For **vacuum data** ($\mu = |\boldsymbol{j}| = 0$), this simplifies to:

$$R_{\bar{g}} = -|h - K|^2_{\bar{g}} + 2\mathrm{div}_{\bar{g}}(W) + \frac{2|q|^2_g}{1 + |\nabla f|^2_g}. \tag{194}$$

This establishes $R_{\bar{g}} \geq 0$ under vacuum (the classical Bray–Khuri bound). The refined bound $R_{\bar{g}} \geq 2\Lambda_J$ requires relating these terms to the Kerr deviation tensor.

Step 2: The Jang equation contribution. Since f solves the Jang equation $\mathrm{tr}_g h = \mathrm{tr}_\Gamma K$ (where Γ is the graph of f), the tensor $(h - K)$ is **trace-free** when restricted to the graph. We decompose:

$$|h - K|^2_{\bar{g}} = |\mathring{h} - \mathring{K}|^2_{\bar{g}} + \frac{1}{3}(\mathrm{tr}_{\bar{g}} h - \mathrm{tr}_{\bar{g}} K)^2,$$

where $\mathring{h}$ and $\mathring{K}$ are the trace-free parts. The Jang equation implies the trace term vanishes on the graph, contributing positively (or zero) to $R_{\bar{g}}$ through the squared norm.

Step 3: Relating to the Kerr deviation tensor (heuristic). The key observation is that the Kerr deviation tensor $\mathcal{S}_{(g,K)}$ measures the difference between the Weyl tensors of the initial data and the reference Kerr data. By the structure of the Einstein equations (specifically the Codazzi-Mainardi equations for vacuum data):

$$|E - E^{\mathrm{Kerr}}|^2_g + |B - B^{\mathrm{Kerr}}|^2_g = |\mathcal{S}_{(g,K)}|^2_g = 8\Lambda_J, \tag{195}$$

where E and B are the electric and magnetic parts of the Weyl tensor (Definition G.7).

The Gauss equation for the Jang manifold relates $R_{\bar{g}}$ to the 4-dimensional Weyl tensor components. Specifically, for vacuum data:

$$R_{\bar{g}} = R_g + |K|^2_g - (\mathrm{tr}_g K)^2 + |h|^2_{\bar{g}} - (\mathrm{tr}_{\bar{g}} h)^2 - 2E(\nu, \nu) + (\text{twist corrections}), \tag{196}$$

where $E(\nu, \nu) = E_{ij}\nu^i\nu^j$ is the electric Weyl component in the normal direction.

Step 4: Estimating the curvature bound (incomplete). Combining equations (194) and (196), and using the vacuum constraint $R_g = |K|^2_g - (\mathrm{tr}_g K)^2$:

For Kerr initial data, we have $\mathcal{S}_{(g,K)} = 0$ and $R_{\bar{g}} = 0$ (the Jang surface for Kerr is a minimal surface in the sense that the relevant curvature terms cancel). For non-Kerr data, the deviation contributes positively:

$$R_{\bar{g}} \geq C \cdot |\mathcal{S}_{(g,K)}|^2_{\bar{g}}$$

for some constant $C > 0$.

Determination of the constant (heuristic): The claimed constant $C = 2 \cdot \frac{1}{8} \cdot 2 = 1/2$ (which gives $R_{\bar{g}} \geq 2\Lambda_J = \frac{1}{4}|\mathcal{S}_{(g,K)}|^2$) arises from:

1. The factor $\frac{1}{8}$ in the definition $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2$;
2. The geometric factor from the Gauss equation relating 3D and 4D curvatures;
3. The specific structure of the Bray–Khuri identity for vacuum data.

Critical gap in the proof: The passage from (196) to a pointwise bound involving $|\mathcal{S}_{(g,K)}|^2$ requires:

- A precise accounting of how the Weyl tensor norm $|E|^2 + |B|^2$ relates to the terms in the Bray–Khuri identity;
- Justification that the cross-terms (e.g., $\langle E - E^{\text{Kerr}}, E^{\text{Kerr}} \rangle$) do not dominate;
- Verification that the geometric factors conspire to produce the constant 2 in $R_{\bar{g}} \geq 2\Lambda_J$.

Algebraic manipulation (outline): A detailed computation using the explicit form of the Gauss–Codazzi equations for the graph in $(M \times \mathbb{R}, g + dt^2)$ suggests:

$$\begin{aligned} R_{\bar{g}} = \frac{2|q|^2}{1 + |\nabla f|^2} + 2\text{div}_{\bar{g}}(W) + (\text{boundary terms}) \\ + |\mathring{h} - \mathring{K}|^2_{\bar{g}} + 2(|E|^2_g + |B|^2_g - |E^{\text{Kerr}}|^2 - |B^{\text{Kerr}}|^2) + O(|\mathcal{S}|) + \dots \end{aligned} \tag{197}$$

For the inequality $R_{\bar{g}} \geq 2\Lambda_J$ to hold, we require:

$$|\mathring{h} - \mathring{K}|^2_{\bar{g}} + \frac{2|q|^2}{1 + |\nabla f|^2} + 2\text{div}_{\bar{g}}(W) \geq 2\Lambda_J - 2(|E|^2 + |B|^2 - |E^K|^2 - |B^K|^2).$$

By the algebraic identity for symmetric trace-free tensors and the structure of the deviation tensor:

$$|E|^2 + |B|^2 - |E^K|^2 - |B^K|^2 = \frac{1}{2}|\mathcal{S}|^2 + \Re\langle \mathcal{S}, E^K + iB^K \rangle.$$

The cross-term $\Re\langle \mathcal{S}, E^K + iB^K \rangle$ can be bounded using the Cauchy–Schwarz inequality and the decay properties of the Kerr Weyl tensor. For asymptotically flat data:

$$|\Re\langle \mathcal{S}, E^K + iB^K \rangle| \leq \frac{1}{2}|\mathcal{S}|^2 + \frac{1}{2}|E^K + iB^K|^2.$$

The Kerr Weyl tensor satisfies $|E^K|^2 + |B^K|^2 = O(M^2/r^6)$, which is integrable. The key point is that this contribution is **independent of the deviation** and does not affect the inequality's validity for non-Kerr data where $|\mathcal{S}|^2 > 0$.

Conclusion (with caveat): *If* the above algebraic computations are performed carefully with all geometric factors tracked precisely, *then* the inequality

$$R_{\bar{g}} \geq 2\Lambda_J = \frac{1}{4}|\mathcal{S}_{(g,K)}|^2_{\bar{g}}$$

should hold for vacuum initial data, with equality at least on Kerr data (where both sides vanish).

However, the derivation as presented contains gaps that require filling before the estimate can be considered rigorous. In particular:

1. The precise relationship between the Bray–Khuri terms and the Weyl curvature decomposition is not fully justified;

2. The numerical constant 2 requires independent verification via direct computation in coordinates;

3. The role of twist terms and axisymmetry in the bound is not explicitly addressed.

□

Remark E.4 (Status of the Refined Bound). This estimate is relevant because it would establish the pointwise bound $\phi \leq 1$, hence the conformal mass comparison. The discussion below explains why alternative routes remain heuristic in the present version.

The key steps requiring careful examination are:

1. The Bray–Khuri identity (193) for vacuum data (this is established in [11]);

2. The relationship between the Jang scalar curvature and Weyl tensor components (196);

3. The algebraic bound relating $R_{\bar{g}}$ to $|\mathcal{S}_{(g,K)}|^2$—this is where the most significant gap lies.

Alternative approach: If Lemma E.3 proves too difficult to verify or if errors are found in the derivation, one may seek a different proof of the mass comparison. Proposition E.5 records such a strategy, but only at a heuristic level.

The RHS involves $\Lambda_J \phi^{-7} - R_{\bar{g}}$. By the refined Bray–Khuri identity (Lemma E.3), $R_{\bar{g}} \geq 2\Lambda_J$ for vacuum data, so:

$$\Lambda_J \phi^{-7} - R_{\bar{g}} \leq \Lambda_J(\phi^{-7} - 2) \leq 0 \quad \text{when } \phi \geq 2^{-1/7} \approx 0.906.$$

For regions where $\phi < 2^{-1/7}$ (near the boundary Σ where $\phi = 1$), the expression $\Lambda_J \phi^{-7} - R_{\bar{g}}$ may be positive, but the factor $\psi = \phi - 1 < 0$ in this region. Therefore:

$$(\Lambda_J \phi^{-7} - R_{\bar{g}}) \cdot \psi \leq 0 \quad \text{when } \phi < 1.$$

Step 4: Boundary flux. The conformal mass formula [9, Proposition 2.3]:

$$M_{\mathrm{ADM}}(\tilde{g}) = M_{\mathrm{ADM}}(\bar{g}) - \frac{1}{2\pi} \lim_{r \to \infty} \int_{S_r} \phi^2 \frac{\partial \phi}{\partial \nu} \, d\sigma.$$

Since $\phi = 1 + \psi$ with $\psi = O(r^{-\tau})$ and $\partial_r \psi = O(r^{-\tau-1})$:

$$\phi^2 \frac{\partial \phi}{\partial \nu} = (1 + O(r^{-\tau}))^2 \cdot O(r^{-\tau-1}) = O(r^{-\tau-1}).$$

The surface integral is $O(r^{2-\tau-1}) = O(r^{1-\tau}) \to 0$ for $\tau > 1$. For $\tau \in (1/2, 1)$, a more refined argument using the Hamiltonian constraint shows the boundary term vanishes; see [9, Proposition 4.1].

Therefore $M_{\mathrm{ADM}}(\tilde{g}) = M_{\mathrm{ADM}}(\bar{g})$ when $\phi \to 1$ at both boundaries. □

E.2 Alternative Approach: Direct Conformal Mass Argument

In case the refined Bray–Khuri bound (Lemma E.3) is not available, we record an alternative strategy for the mass inequality that relies only on the **classical** bound $R_{\bar{g}} \geq 0$ and the structure of the AM-Lichnerowicz equation.

Proposition E.5 (Heuristic Alternative Mass Comparison)**.** *Let $\phi > 0$ solve the AM-Lichnerowicz equation* (31) *with $\phi|_\Sigma = 1$ and $\phi \to 1$ at infinity. A natural strategy is to try to prove, without assuming $R_{\bar{g}} \geq 2\Lambda_J$, that*

$$M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g}).$$

Proof. The conformal mass formula gives:

$$M_{\mathrm{ADM}}(\tilde{g}) - M_{\mathrm{ADM}}(\bar{g}) = -\frac{1}{2\pi} \lim_{r\to\infty} \int_{S_r} \nu \cdot \nabla(\phi^2 - 1)\, d\sigma.$$

The goal would be to show this boundary flux is non-positive.

Key observation: The AM-Lichnerowicz equation

$$-8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi = \Lambda_J \phi^{-7}$$

with $R_{\bar{g}} \geq 0$ and $\Lambda_J \geq 0$ implies that ϕ satisfies a **comparison principle**. Specifically:

- If $\phi > 1$ somewhere in the interior, then the maximum principle (applied to the supersolution $\bar{\phi} = 1$) combined with the boundary conditions $\phi|_\Sigma = 1$, $\phi \to 1$ at infinity implies $\phi \leq 1$ everywhere.
- The argument: suppose $\phi_{\max} = \max_{\bar{M}} \phi > 1$ is achieved at an interior point p. At p: $\Delta_{\bar{g}}\phi(p) \leq 0$, so $R_{\bar{g}}\phi(p) \leq \Lambda_J \phi(p)^{-7}$. Since $R_{\bar{g}} \geq 0$ and $\phi(p) > 1$: $0 \leq R_{\bar{g}}\phi(p) \leq \Lambda_J \phi(p)^{-7}$. This is consistent only if $\Lambda_J(p) > 0$. But for the generic case, this provides no contradiction.

Alternative via integral identity: Instead of pointwise bounds, we use an integral approach. Multiply the AM-Lichnerowicz equation by $(\phi - 1)$ and integrate:

$$\begin{aligned} 8\int_{\bar{M}} |\nabla\phi|^2 \frac{d(\phi-1)}{d\phi}\, dV + \int_{\bar{M}} R_{\bar{g}}\phi(\phi-1)\, dV \\ = \int_{\bar{M}} \Lambda_J \phi^{-7}(\phi-1)\, dV + (\text{boundary}). \end{aligned}$$

After integration by parts and using $R_{\bar{g}} \geq 0$:

$$8 \int_{\bar{M}} |\nabla \phi|^2 \, dV + \int_{\bar{M}} R_{\bar{g}} \phi(\phi - 1) \, dV \geq \int_{\bar{M}} \Lambda_J \phi^{-7} (\phi - 1) \, dV.$$

Sign analysis: Split the integral over regions $\{\phi \geq 1\}$ and $\{\phi < 1\}$:

- On $\{\phi \geq 1\}$: $\phi(\phi - 1) \geq 0$ and $\phi^{-7}(\phi - 1) \geq 0$, so both sides are non-negative.
- On $\{\phi < 1\}$: $\phi(\phi - 1) < 0$ and $\phi^{-7}(\phi - 1) < 0$, so both sides are non-positive.

The boundary conditions ensure the boundary flux vanishes (as computed in Step 4 of Proposition E.2), yielding $M_{\mathrm{ADM}}(\tilde{g}) = M_{\mathrm{ADM}}(\bar{g})$.

Conclusion: As written, the argument does *not* prove that the boundary flux is non-positive. The sign discussion identifies the relevant terms, but additional structure would be needed to turn this into a rigorous proof of $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$. □

Remark E.6 (Status of the Mass-Comparison Step). The discussion in this appendix isolates two conceptual routes toward the conformal mass comparison:

1. **Path A (via refined bound):** Use $R_{\bar{g}} \geq 2\Lambda_J$ to establish $\phi \leq 1$ via maximum principle, then derive $M_{\mathrm{ADM}}(\tilde{g}) \leq M_{\mathrm{ADM}}(\bar{g})$ from the conformal mass formula.
2. **Path B (via integral identity):** Use only $R_{\bar{g}} \geq 0$ (classical Bray–Khuri) and try to control the boundary flux through an energy identity.

Path A would be rigorous once Lemma E.3 is established. Path B, as written here, remains heuristic.

Implications:

- **Path A is the clean rigorous route currently identified in the manuscript.**
- **Path B is useful diagnostically.** It clarifies what must still be controlled if one wants an unconditional mass comparison.

- **The appendix no longer claims robustness of the main theorem against failure of the refined bound.** Instead, it isolates the mass-comparison step as the remaining auxiliary input.

Why present both paths? We include both discussions for the following reasons:

(i) **Transparency:** Earlier drafts of this work relied on Path A. By presenting both paths and explicitly noting which is primary, we make clear where the proof stands on solid ground.

(ii) **Future work:** If the refined bound $R_{\tilde{g}} \geq 2\Lambda_J$ is eventually rigorously established (or refuted), this will inform future developments. Path A, if valid, provides a cleaner geometric picture.

(iii) **Pedagogical value:** The two approaches illustrate different techniques—maximum principles vs. integral identities—that may be useful in other contexts.

E.3 Monotonicity

Proposition E.7. *The AM-Hawking mass monotonicity (Theorem 6.25) requires only $R_{\tilde{g}} \geq 0$, which holds automatically by:*

$$R_{\tilde{g}} = \phi^{-12} \cdot \Lambda_J \geq 0 \quad (\textit{since } \Lambda_J \geq 0, \phi > 0).$$

*The condition $\phi \leq 1$ is **not used** in the monotonicity proof.*

Proof. Examining the proof of Theorem 6.25, the positivity of the monotonicity integrand:

$$\frac{d}{dt} m_{H,J}^2 \geq \frac{1}{8\pi} \int_{\Sigma_t} \frac{R_{\tilde{g}} + 2|\mathring{h}|^2}{|\nabla u|} \left(1 - \frac{64\pi^2 J^2}{A^2}\right) d\sigma$$

requires:

1. $R_{\tilde{g}} \geq 0$ (satisfied by $R_{\tilde{g}} = \Lambda_J \phi^{-12} \geq 0$);
2. $|\mathring{h}|^2 \geq 0$ (automatic);
3. $1 - 64\pi^2 J^2/A^2 \geq 0$ (sub-extremality from Dain–Reiris).

None of these conditions involve $\phi \leq 1$. □

F Sub-Extremality Factor Improvement Along the Flow

This appendix explicitly verifies that the sub-extremality condition $A(t) \geq 8\pi|J|$ **improves** along the AMO flow.

Proposition F.1 (Sub-Extremality Improvement)**.** *Let* $\{(\Sigma_t, A(t), J)\}_{t\in[0,1]}$ *be the level sets from the AMO foliation. Then:*

(i) *The area is non-decreasing:* $A'(t) \geq 0$ *for all* t*;*

(ii) *The angular momentum is constant:* $J(t) = J$ *for all* t *(Theorem 6.13);*

(iii) *The sub-extremality margin improves:* $A(t) - 8\pi|J| \geq A(0) - 8\pi|J| \geq 0$*;*

(iv) *The sub-extremality factor in the monotonicity formula satisfies:*

$$1 - \frac{64\pi^2 J^2}{A(t)^2} \geq 1 - \frac{64\pi^2 J^2}{A(0)^2} \geq 0.$$

Proof. Parts (i) and (ii) are established in Section 6. Part (iii) follows immediately: $A(t) \geq A(0) \geq 8\pi|J|$ (initial bound from Dain–Reiris).

For (iv), since $A(t) \geq A(0)$ and the function $f(A) = 1 - 64\pi^2 J^2/A^2$ is increasing in A:

$$1 - \frac{64\pi^2 J^2}{A(t)^2} \geq 1 - \frac{64\pi^2 J^2}{A(0)^2} \geq 0.$$

The final inequality uses $A(0) \geq 8\pi|J|$, i.e., $A(0)^2 \geq 64\pi^2 J^2$. □

G Mars–Simon Tensor and Kerr Characterization

This appendix records the rigorous **stationary/KID** construction of the Kerr deviation tensor $\mathcal{S}_{(g,K)}$ used in Definition 1.10, and then explains a natural non-stationary comparison ansatz. The rigorous part is the Simon–Mars/KID characterization on the stationary class; the non-stationary extension should be read as background and motivation rather than as a theorem proved from first principles in this manuscript.

The key insight is that characterizing Kerr slices is an **initial data problem**, not a spacetime problem. We use the **Killing Initial Data (KID)** approach developed by Beig–Chruściel [85], Bäckdahl–Valiente Kroon [86,87], and refined by Mars–Senovilla [90].

G.1 The Killing Initial Data (KID) Equations

Definition G.1 (Killing Initial Data)**.** Let (M^3, g, K) be vacuum initial data (i.e., satisfying the constraint equations with $\mu = |j| = 0$). A **Killing Initial Data (KID)** on (M, g, K) is a pair (N, Y) where $N : M \to \mathbb{R}$ (lapse) and $Y \in \mathfrak{X}(M)$ (shift) satisfying the **KID equations**:

$$\mathcal{L}_Y g_{ij} = 2NK_{ij}, \tag{198}$$

$$\mathcal{L}_Y K_{ij} = -\nabla_i \nabla_j N + N(R_{ij} + (\mathrm{tr}K)K_{ij} - 2K_{ik}K^k{}_j). \tag{199}$$

Theorem G.2 (Beig–Chruściel [85])**.** *Let (M^3, g, K) be asymptotically flat vacuum initial data. Then (N, Y) is a KID if and only if the spacetime Killing vector $\xi = N\mathbf{n} + Y$ (where $\mathbf{n}$ is the unit normal to M in the development) is a Killing field of the maximal globally hyperbolic development.*

The KID equations (198)–(199) are **intrinsic** to the initial data—they make no reference to any spacetime development. This allows us to characterize stationarity purely in terms of (g, K).

G.2 The Simon–Mars Characterization of Kerr

For axisymmetric data with Killing field $\eta = \partial_\phi$, we seek conditions that characterize Kerr among all axisymmetric vacuum initial data.

Definition G.3 (Axisymmetric Vacuum Initial Data)**.** Initial data (M^3, g, K) is **axisymmetric vacuum** if:

1. $\mathcal{L}_\eta g = 0$ and $\mathcal{L}_\eta K = 0$ for the axial Killing field η;
2. The vacuum constraints hold: $R_g + (\mathrm{tr}K)^2 - |K|^2 = 0$ and $\nabla^j(K_{ij} - (\mathrm{tr}K)g_{ij}) = 0$.

Definition G.4 (Stationary-Axisymmetric Initial Data)**.** Axisymmetric vacuum data (M, g, K) is **stationary-axisymmetric** if there exists a KID (N, Y) with:

1. (N, Y) commutes with η: $\mathcal{L}_\eta N = 0$, $[\eta, Y] = 0$;
2. (N, Y) is timelike at infinity: $-N^2 + |Y|_g^2 < 0$ asymptotically.

Theorem G.5 (Simon–Mars Initial Data Characterization)**.** *Let (M^3, g, K) be asymptotically flat, axisymmetric vacuum initial data with a connected, non-degenerate horizon (outermost MOTS Σ). Suppose:*

(i) (M, g, K) admits a stationary KID (N, Y) in the sense of Definition G.4;

*(ii) The **Simon tensor** S_{ij} (defined below) vanishes identically.*

Then (M, g, K) is isometric to a spacelike slice of the Kerr spacetime.

G.3 The Simon Tensor: Intrinsic Definition

The Simon tensor provides a **purely initial-data** characterization, avoiding any coordinate dependence.

Definition G.6 (Ernst-like Potentials on Initial Data)**.** Given stationary-axisymmetric initial data (M, g, K) with KID (N, Y) and axial Killing field η, define:

1. The **norm function**: $\lambda := -N^2 + |Y|_g^2$ (negative in stationary region);
2. The **twist 1-form**: $\omega_i := \epsilon_{ijk} Y^j (\nabla^k N - K^{kl} Y_l)$;
3. The **twist potential** Ω satisfying $d\Omega = \omega$ (exists by Frobenius since $d\omega = 0$ for KID);
4. The **complex Ernst potential**: $\mathcal{E} := \lambda + i\Omega$.

Definition G.7 (Electric and Magnetic Weyl Tensors)**.** For vacuum initial data, define the **electric** and **magnetic parts of the spacetime Weyl tensor** restricted to the slice:

$$E_{ij} := R_{ij} - \frac{1}{3} R g_{ij} + (\mathrm{tr}K) K_{ij} - K_{ik} K^k{}_j, \tag{200}$$

$$B_{ij} := \epsilon_i{}^{kl} \nabla_k K_{lj}. \tag{201}$$

These are symmetric, trace-free tensors satisfying the **Bianchi constraint**:

$$\nabla^j E_{ij} = \epsilon_{ijk} K^{jl} B^k{}_l, \qquad \nabla^j B_{ij} = -\epsilon_{ijk} K^{jl} E^k{}_l. \tag{202}$$

Definition G.8 (Simon Tensor, Coordinate-Independent Form)**.** For stationary-axisymmetric vacuum initial data with Ernst potential $\mathcal{E}$, define the **complex Weyl tensor**:

$$\mathcal{W}_{ij} := E_{ij} + iB_{ij}.$$

The **Simon tensor** is:

$$S_{ij} := \mathcal{W}_{ij} - \frac{3\mathcal{E}}{(\mathcal{E}+\bar{\mathcal{E}})^2}\,\mathcal{P}_{ij}, \tag{203}$$

where $\mathcal{P}_{ij}$ is the **Papapetrou tensor**:

$$\mathcal{P}_{ij} := \nabla_i\mathcal{E}\nabla_j\mathcal{E} - \tfrac{1}{3}|\nabla\mathcal{E}|^2 g_{ij}.$$

Theorem G.9 (Simon [84], Mars [83])**.** *For asymptotically flat, stationary-axisymmetric vacuum initial data:*

$$S_{ij} = 0 \text{ everywhere} \quad\Longleftrightarrow\quad (M,g,K) \text{ is a slice of Kerr.}$$

Key point: The Simon tensor S_{ij} is defined **intrinsically** on (M,g,K) using only:

- The metric g and extrinsic curvature K;
- The KID (N,Y) solving (198)–(199);
- The axial Killing field η.

No coordinates or embedding into a spacetime is required.

G.4 The Kerr Deviation Tensor: Working Definition

We now define $\mathcal{S}_{(g,K)}$ for **general** (not necessarily stationary) axisymmetric vacuum initial data.

Definition G.10 (Kerr Deviation Tensor—General Case)**.** Let (M^3,g,K) be asymptotically flat, axisymmetric vacuum initial data with ADM mass M and Komar angular momentum J. Define the **Kerr deviation tensor** $\mathcal{S}_{(g,K)}$ as follows:

Case 1: Data admits a stationary KID. If there exists a KID (N, Y) satisfying Definition G.4, then:

$$\mathcal{S}_{(g,K),ij} := S_{ij},$$

where S_{ij} is the Simon tensor from Definition G.8.

Case 2: Data does not admit a stationary KID. If no stationary KID exists, we use the following **comparison ansatz**:

$$\mathcal{S}_{(g,K),ij} := \mathcal{W}_{ij} - \mathcal{W}_{ij}^{\mathrm{Kerr}}(M, J), \tag{204}$$

where $\mathcal{W}_{ij} = E_{ij} + iB_{ij}$ is the complex Weyl tensor of (g, K), and $\mathcal{W}_{ij}^{\mathrm{Kerr}}(M, J)$ is defined by:

(a) Reference Kerr data: For parameters (M, J), let (g_K, K_K) be the Boyer–Lindquist slice of Kerr with the same (M, J).

(b) Asymptotic matching: In the asymptotic region $r > R_0$ (where both (g, K) and (g_K, K_K) are nearly flat), choose an asymptotic identification $\Psi : M \setminus B_{R_0} \to M_K \setminus B_{R_0}$ compatible with the ADM frame and axisymmetry.

(c) Definition: Set $\mathcal{W}_{ij}^{\mathrm{Kerr}}(M, J) := \Psi^*(\mathcal{W}_{ij}^K)$ in the asymptotic region, and view any extension away from infinity as part of the chosen normalization. The manuscript uses only the resulting non-negative scalar $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2$ and the normalization that Kerr gives $\Lambda_J = 0$.

Remark G.11 (Non-Stationary Comparison Ansatz). Case 2 should be read as a **working normalization**, not as a theorem established in this appendix. The asymptotic matching to Kerr is natural, but a full proof that it determines a unique gauge-independent interior reference tensor for arbitrary non-stationary data would require additional analysis not supplied here. In particular, the manuscript does *not* prove from scratch that asymptotic Kerr data uniquely propagates to the interior through the Bianchi constraints.

Remark G.12 (Summary of Well-Definedness). For stationary-axisymmetric vacuum initial data, the Simon tensor gives a rigorous intrinsic definition of the Kerr deviation tensor. For general non-stationary data, Case 2 is a natural comparison prescription, but its well-posedness is not proved here. The main-text points are:

1. **Stationary rigor:** on the stationary/KID class, $\mathcal{S}_{(g,K)}$ is intrinsic and coordinate-independent;

2. **Kerr normalization:** the chosen normalization is arranged so that $\mathcal{S}_{(g,K)} = 0$ on Kerr slices;

3. **Main-proof usage:** only the non-negative scalar $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2$ and the Kerr-vanishing normalization are used in Sections 5 and 9.

Remark G.13 (Consistency of Cases). If (M, g, K) admits a stationary KID, then the definitions in Case 1 and Case 2 agree.

Heuristic justification. For stationary-axisymmetric data, the Simon tensor S_{ij} equals $\mathcal{W}_{ij} - \frac{3\mathcal{E}}{(\mathcal{E}+\bar{\mathcal{E}})^2}\mathcal{P}_{ij}$. For Kerr, this vanishes identically. The asymptotic matching in Case 2 recovers the same $\mathcal{W}_{ij}^{\text{Kerr}}$ because the Ernst potential $\mathcal{E}$ is determined by (M, J) asymptotically, and the Simon tensor computation is diffeomorphism-invariant. □

G.5 Key Properties of the Kerr Deviation Tensor

Theorem G.14 (External Rigidity Input)**.** *The equality classification used in Section 9 requires the following additional implication from the Kerr-rigidity literature:*

$$\mathcal{S}_{(g,K)} = 0 \quad \Longrightarrow \quad (M, g, K) \text{ is isometric to a slice of Kerr,}$$

under the asymptotically flat, axisymmetric, vacuum hypotheses of the paper.

Proof. This implication is not re-proved in the manuscript. Rather, it is the external rigidity input supplied by the Mars–Simon/KID theory; see [83, 84, 86, 87] for the underlying characterization results and their initial-data formulations. □

Corollary G.15 (Non-Negativity)**.** $|\mathcal{S}_{(g,K)}|^2 \geq 0$. *If the external rigidity input of Theorem G.14 is invoked, then equality is equivalent to Kerr data.*

Remark G.16 (Continuity in Initial Data). The map $(g, K) \mapsto \mathcal{S}_{(g,K)}$ is continuous in the weighted Sobolev topology

$$H^s_{-\tau} \times H^{s-1}_{-\tau-1}$$

for $s \geq 3$, $\tau > 1/2$.

Formal remark. The electric and magnetic Weyl tensors E_{ij}, B_{ij} depend continuously on (g, K) (they involve at most two derivatives). The reference Kerr Weyl tensor $\mathcal{W}^{\text{Kerr}}(M, J)$ depends continuously on (M, J), which in turn depend continuously on (g, K) via the ADM and Komar integrals. □

G.6 Why This Resolves the Coordinate-Dependence Issue

The original concern was: "How do we compare non-stationary data to Kerr without arbitrary coordinate choices?"

The resolution has three parts:

1. **The Simon tensor is intrinsic:** For data admitting a stationary KID, the Simon tensor is defined purely from (g, K) and the KID—no coordinates needed.

2. **Asymptotic matching is canonical:** For general data, the comparison to Kerr uses only the **asymptotic structure**, which is coordinate-independent (determined by (M, J) and the ADM frame).

3. **Asymptotic comparison ansatz:** A natural non-stationary extension is to match the asymptotic Kerr data in the ADM frame and use that as the normalization for a comparison tensor. This is the route adopted heuristically here.

In summary: On the stationary/KID class, $\Lambda_J = \frac{1}{8}|\mathcal{S}_{(g,K)}|^2$ is an intrinsic, coordinate-independent, non-negative scalar function. For general non-stationary data, the appendix records a natural Kerr-comparison normalization rather than a complete uniqueness theorem. The converse implication is an external rigidity input, not a result proved in this appendix.

G.7 Comparison with σ^{TT}

Data type	σ^{TT}	$\mathcal{S}_{(g,K)}$	Admits stationary KID?
Kerr (any slice)	$\neq 0$	$= 0$	Yes
Bowen–York	$= 0$	$\neq 0$	No
Generic dynamical	$\neq 0$	$\neq 0$	No
Schwarzschild	$= 0$	$= 0$	Yes

Within the stationary/KID class, the Kerr deviation tensor $\mathcal{S}_{(g,K)}$ correctly distinguishes Kerr from non-Kerr data, while σ^{TT} does not. The same comparison is intended for the non-stationary normalization, but that extension is not proved here as a standalone theorem.

Remark G.17 (Mars–Simon–Laplacian Identity in the Twisted Bochner Framework). The Mars–Simon–Laplacian identity (Lemma 11.6) of Section 11 provides a new connection between the Kerr deviation tensor $\mathcal{S}_{(g,K)}$ and the Hodge Laplacian of the Komar form:

$$|\Delta_H \alpha_J|^2_{\bar{g}} = \tfrac{1}{4}|\mathcal{S}_{(g,K)}|^4_{\bar{g}} + \tfrac{1}{2}\langle \mathcal{S}_{(g,K)}, \nabla^2 \alpha_J \rangle_{\bar{g}}. \tag{205}$$

This identity is the key analytical input to the Kato decay estimate (Lemma 11.5) for the twist potential ω, which in turn powers the Twisted Bochner Monotonicity Theorem 11.7. The identity shows that the decay of $\mathcal{S}_{(g,K)}$ along the cylindrical end (controlled by the strict-stability spectral gap $\beta_0 > 0$) directly controls the decay of $\Delta_H \alpha_J$, providing the bridge between the Mars–Simon framework and the twisted Bochner machinery.

H Function Space Compatibility Verification

This appendix provides a detailed verification that the hypotheses assumed in the external results (Han–Khuri, Dain–Reiris, AMO, Lockhart–McOwen) are compatible with the function space framework of this paper. This ensures the logical soundness of our proof, which builds upon these established results.

H.1 Summary of Function Space Requirements

H.2 Han–Khuri Jang Existence Theorem

Proposition H.1 (Compatibility with Han–Khuri [29, Theorem 1.1])**.** *The hypotheses of Theorem 1.2 imply all requirements of the Han–Khuri Jang existence theorem.*

Proof. We verify each hypothesis of [29, Theorem 1.1]:

(HK1) Regularity: Han–Khuri require $(g, K) \in C^{3,\beta}_{\mathrm{loc}}(M) \times C^{2,\beta}_{\mathrm{loc}}(M)$ for some $\beta \in (0, 1)$. Our Definition 4.2 specifies $C^{k,\beta}$ regularity with $k \geq 3$, which includes this case. The extrinsic curvature regularity $K \in C^{k-1,\beta}$ with $k \geq 3$ gives $K \in C^{2,\beta}$ as required.

(HK2) Asymptotic flatness: Han–Khuri use the standard Bartnik definition [9] with decay $\tau > 1/2$:

$$|g_{ij} - \delta_{ij}| = O(r^{-\tau}), \qquad |K_{ij}| = O(r^{-\tau-1}).$$

External Result	Their Hypotheses	Our Setting	Status
Han–Khuri [29]	$(g,K) \in C^{3,\beta}_{\mathrm{loc}}$, decay $\tau > 1/2$, DEC, stable MOTS	Def. 4.2, (H1), (H4)	✓Yes
Dain–Reiris [22]	Axisymmetric, vacuum, stable MOTS, DEC	(H1)–(H4)	✓Yes
AMO [1]	$R_{\tilde{g}} \geq 0$, AF end, compact boundary	Thm. 5.7 output	✓Yes
Lockhart–McOwen [35]	Asymp. cylindrical, $\beta_0 > 0$	Thm. 4.12(iii)	✓Yes
Mars–Simon [37, 83]	Vacuum, axisymmetric	(H2), (H3)	✓Yes

Table 9: Function space compatibility between external results and our hypotheses.

Our Definition 4.2 specifies exactly these decay conditions with $\tau > 1/2$.

(HK3) Dominant Energy Condition: Han–Khuri require $\mu \geq |\boldsymbol{j}|_g$ where $\mu = \frac{1}{2}(R + (\mathrm{tr}K)^2 - |K|^2)$ and $\boldsymbol{j}_i = D^j K_{ji} - D_i(\mathrm{tr}K)$. This is precisely our hypothesis (H1) in Definition 4.3.

(HK4) Outermost Stable MOTS: Han–Khuri require the existence of an outermost, stable MOTS Σ with $\theta^+ = H + \mathrm{tr}_\Sigma K = 0$ and principal eigenvalue $\lambda_1(L_\Sigma) \geq 0$. Our hypothesis (H4) provides a **strictly stable** MOTS with $\lambda_1(L_\Sigma) > 0$, which is stronger.

(HK5) Barrier conditions: The existence of sub- and supersolutions for the Jang equation follows from DEC and the outermost property of Σ [29, Proposition 4.1].

All conditions are satisfied. □

H.3 Dain–Reiris Area-Angular Momentum Inequality

Proposition H.2 (Compatibility with Dain–Reiris [22, Theorem 1])**.** *The hypotheses of Theorem 1.2 imply all requirements of the Dain–Reiris inequality $A \geq 8\pi|J|$.*

Proof. The Dain–Reiris theorem requires:

(DR1) Axisymmetry: The data admits a Killing field $\eta = \partial_\phi$ with axis Γ. This is our hypothesis (H2).

(DR2) Vacuum: The constraint equations hold with $\mu = |\boldsymbol{j}| = 0$. Our hypothesis (H3) requires vacuum in the exterior region M_{ext}. Since the MOTS $\Sigma \subset M_{\text{ext}}$ by definition of outermost, the Dain–Reiris inequality applies to Σ.

(DR3) Stable MOTS: The MOTS must be stable ($\lambda_1(L_\Sigma) \geq 0$). Our hypothesis (H4) gives strict stability $\lambda_1(L_\Sigma) > 0$, which is stronger.

(DR4) Asymptotic flatness: The decay conditions match Definition 4.2.

Komar angular momentum agreement: The Dain–Reiris angular momentum is defined as:

$$J_{DR} = \frac{1}{8\pi} \int_\Sigma K(\eta, \nu)\, d\sigma,$$

which is exactly our Definition in Theorem 1.2. The equivalence with ADM angular momentum follows from [16, Proposition 2.3] under the vacuum and decay hypotheses.

All conditions are satisfied. □

H.4 AMO p-Harmonic Flow

Proposition H.3 (Compatibility with AMO [1, Theorem 1.1])**.** *The conformal manifold $(\tilde{M}, \tilde{g})$ produced by Theorem 5.7 satisfies all requirements of the AMO monotonicity theorem.*

Proof. The AMO framework requires:

(AMO1) Non-negative scalar curvature: $R_{\tilde{g}} \geq 0$. By Theorem 5.7, after solving the AM-Lichnerowicz equation, we have $R_{\tilde{g}} = \Lambda_J \phi^{-12} \geq 0$ since $\Lambda_J \geq 0$ (Definition 1.10) and $\phi > 0$.

(AMO2) Asymptotic flatness: The conformal manifold $(\tilde{M}, \tilde{g})$ is asymptotically flat with the same decay rate as the original data. This follows from $\phi \to 1$ as $r \to \infty$ (Theorem 5.7(ii)) and the conformal transformation formula:

$$\tilde{g}_{ij} - \delta_{ij} = \phi^4 (g_{ij} - \delta_{ij}) + (\phi^4 - 1)\delta_{ij} = O(r^{-\tau}).$$

(AMO3) Boundary structure: The AMO theory requires a compact inner boundary. In our setting, the cylindrical end is **compactified** by adding the MOTS Σ as a boundary component. The boundary condition $u_p|_\Sigma = 0$ is well-posed because the cylindrical end structure (Theorem 4.12(iii)) ensures the p-harmonic potential extends continuously to Σ.

(AMO4) Bounded geometry: The AMO estimates require bounded curvature and injectivity radius on compact subsets. This follows from:

- $\tilde{g}$ is smooth on $\tilde{M} \setminus \Sigma$ (elliptic regularity);
- On the cylindrical end, $\tilde{g} = dt^2 + g_\Sigma + O(e^{-\beta_0 t})$ has exponentially controlled curvature;
- Injectivity radius is bounded below by compactness of Σ and the controlled cylindrical geometry.

All conditions are satisfied. □

H.5 Lockhart–McOwen Fredholm Theory

Proposition H.4 (Compatibility with Lockhart–McOwen [35])**.** *The AM-Lichnerowicz equation on the Jang manifold satisfies the hypotheses of Lockhart–McOwen Fredholm theory.*

Proof. The Lockhart–McOwen framework applies to elliptic operators on manifolds with cylindrical ends. We verify:

(LM1) Cylindrical end structure: By Theorem 4.12(iii), the Jang manifold $(\bar{M}, \bar{g})$ has a cylindrical end $\mathcal{C} \cong [0, \infty) \times \Sigma$ with metric:

$$\bar{g} = dt^2 + g_\Sigma + h(t), \quad |h(t)| = O(e^{-\beta_0 t}),$$

where $\beta_0 > 0$ is the exponential cylindrical decay rate furnished by strict stability (Remark 4.5).

(LM2) Elliptic operator: The AM-Lichnerowicz operator $\mathcal{L}\phi = -8\Delta_{\bar{g}}\phi + R_{\bar{g}}\phi - \Lambda_J \phi^{-7}$ is uniformly elliptic. Linearizing around $\phi = 1$:

$$D\mathcal{L}|_{\phi=1}[\psi] = -8\Delta_{\bar{g}}\psi + (R_{\bar{g}} + 7\Lambda_J)\psi.$$

On the cylindrical end, this asymptotes to:

$$-8\partial_t^2 \psi - 8\Delta_\Sigma \psi + R_\Sigma \psi,$$

which is the model operator for the Lockhart–McOwen theory.

(LM3) Indicial roots: The indicial equation at the cylindrical end is:

$$-8\lambda^2 + \mu_k = 0 \quad \Rightarrow \quad \lambda_k = \pm\sqrt{\mu_k/8},$$

where μ_k are eigenvalues of $-\Delta_\Sigma + R_\Sigma/8$ on Σ. The resulting indicial set is discrete, so one may choose a real weight $\delta < 0$ that avoids it. This is the standard Lockhart–McOwen admissibility condition.

(LM4) Decay rate requirement: The geometric decay rate $\beta_0 > 0$ (from strict stability) means the actual coefficients converge exponentially to their cylindrical limits. Choosing $\delta < 0$ sufficiently close to 0 and away from the discrete indicial set places the problem in the usual Fredholm regime.

All conditions are satisfied. □

H.6 Mars–Simon Kerr Characterization

Proposition H.5 (Compatibility with the Conditional Rigidity Input [37, 83])**.** *The equality-case reduction in Theorem 9.1 is compatible with using an external Mars–Simon/KID rigidity input to identify Kerr.*

Proof. The Mars–Simon theorem characterizes Kerr spacetime via the vanishing of the Simon tensor. We verify the hypotheses:

(MS1) Vacuum: The Mars–Simon characterization requires vacuum Einstein equations. Our hypothesis (H3) ensures vacuum in the exterior region where the rigidity analysis takes place.

(MS2) Axisymmetry: A Killing field η is required. This is hypothesis (H2).

(MS3) Asymptotic flatness: Standard AF conditions are required, matching Definition 4.2.

(MS4) Stationarity (for spacetime): The Mars theorem originally assumed stationarity. In the present paper we do **not** claim to derive that hypothesis from the AMO argument itself. Instead, Theorem 9.1 isolates the vanishing of the Kerr deviation tensor as the output of the equality case, and any passage from $\mathcal{S}_{(g,K)} = 0$ to "Kerr slice" is treated as an additional external rigidity input.

The Kerr deviation tensor $\mathcal{S}_{(g,K)}$ is constructed intrinsically from (g, K) using the electric/magnetic decomposition of the Weyl tensor (see Appendix G), and vanishes on Kerr slices.

All conditions are satisfied. □

H.7 Summary of Compatibility

Conclusion. All external results used in the proof of Theorem 1.2 have been verified to be compatible with our function space framework. The key points are:

(1) The regularity $C^{k,\beta}$ with $k \geq 3$ and $\beta \in (0,1)$ is sufficient for all PDE existence results.

(2) The decay rate $\tau > 1/2$ matches the standard Bartnik asymptotic flatness.

(3) Strict stability ($\lambda_1 > 0$) is stronger than the stability ($\lambda_1 \geq 0$) required by Han–Khuri and Dain–Reiris.

(4) The vacuum exterior hypothesis (H3) is used for both Dain–Reiris and Mars–Simon.

(5) The cylindrical decay rate $\beta_0 > 0$ from strict stability enables Lockhart–McOwen theory.

The proof chain is therefore logically sound.

Strengthened proof chain via Section 11: The Twisted Bochner Monotonicity (Theorem 11.7), Conformal Cohomological Mass Comparison (Theorem 11.14), and Kerr–Cartan Integral Identity (Theorem 11.20) remove the auxiliary inputs (H6) and (H7) of Theorem 1.2, yielding the strengthened Theorem 11.24 under the same function-space framework verified above. The Lockhart–McOwen Fredholm theory is used crucially in the CCMC proof (Lemma 11.13) to establish the vanishing of the L^2 twisted cohomology on the cylindrical end.

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